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(54) **METHODS OF ACTIVATING REGULATORY T CELLS**

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(57) **ABSTRACT**

Provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell, compositions including a population of T_{reg} cells generated by these methods, and methods of treating a subject using these compositions.

14 Claims, 14 Drawing Sheets

Specification includes a Sequence Listing.

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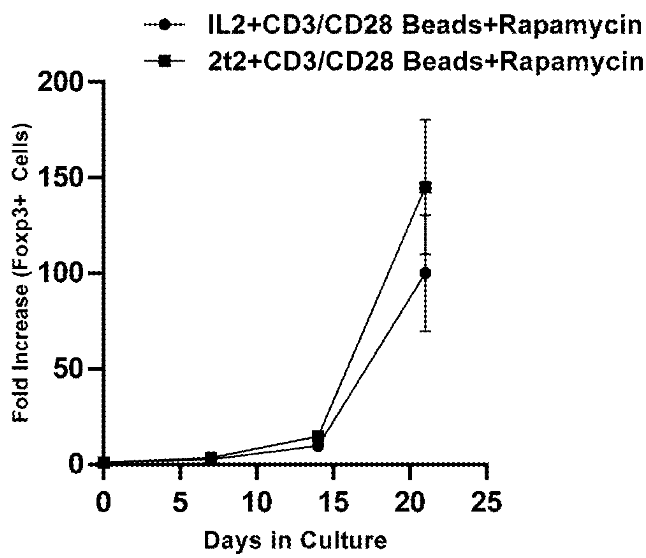

IL2+CD3/CD28 Beads+Rapamycin
2t2+CD3/CD28 Beads+Rapamycin
Fold Increase (Foxp3+ Cells)
200
150
100
50
0
0
5
10
15
20
25
Days in Culture

FIG. 1

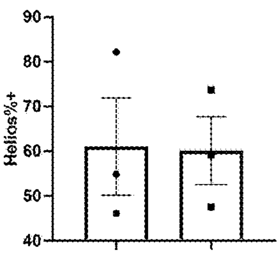


FIG. 2A

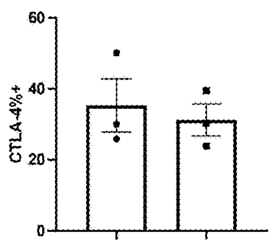


FIG. 2B

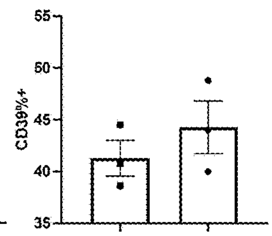


FIG. 2C

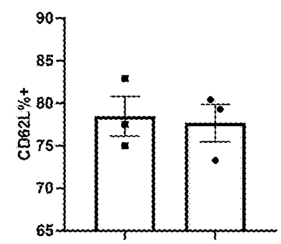


FIG. 2D

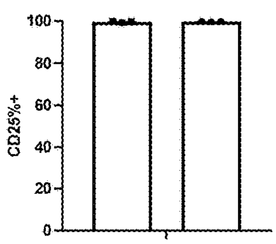


FIG. 2E

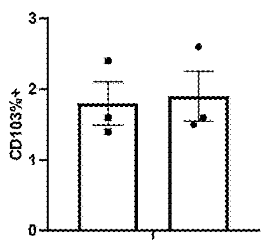


FIG. 2F

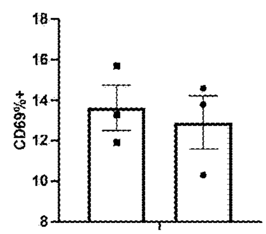


FIG. 2G

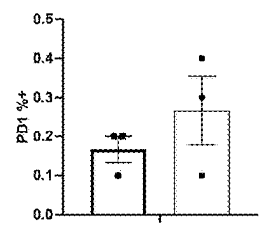


FIG. 2H

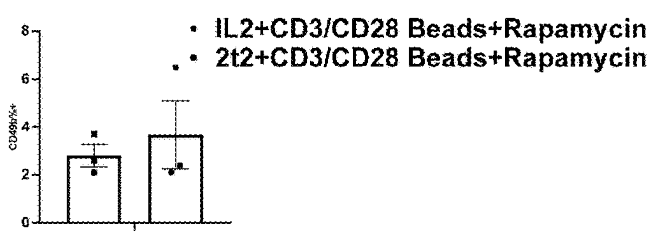


FIG. 2I

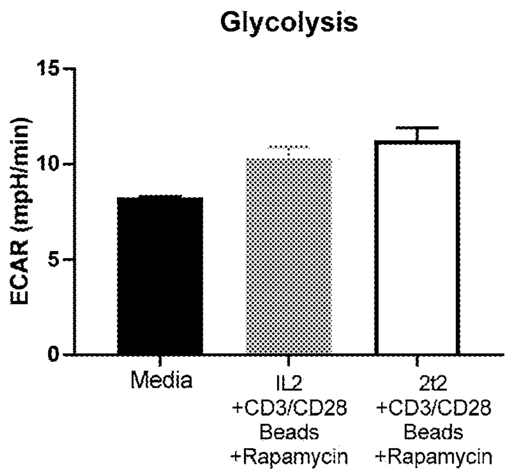


FIG. 3A

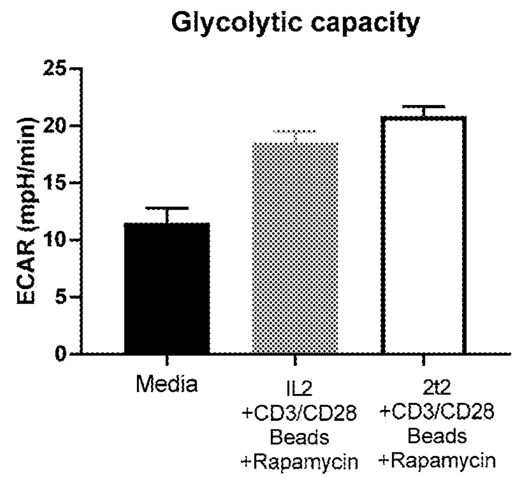


FIG. 3B

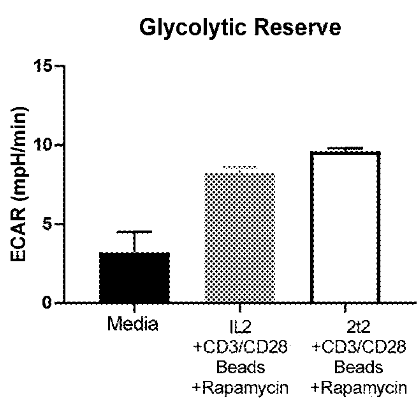


FIG. 3C

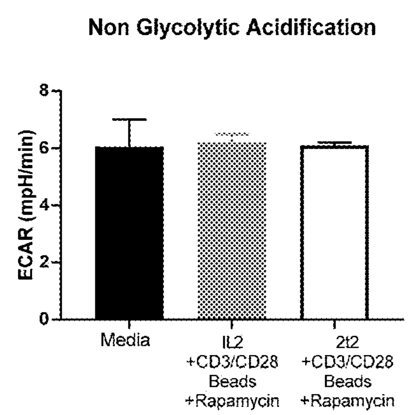


FIG. 3D

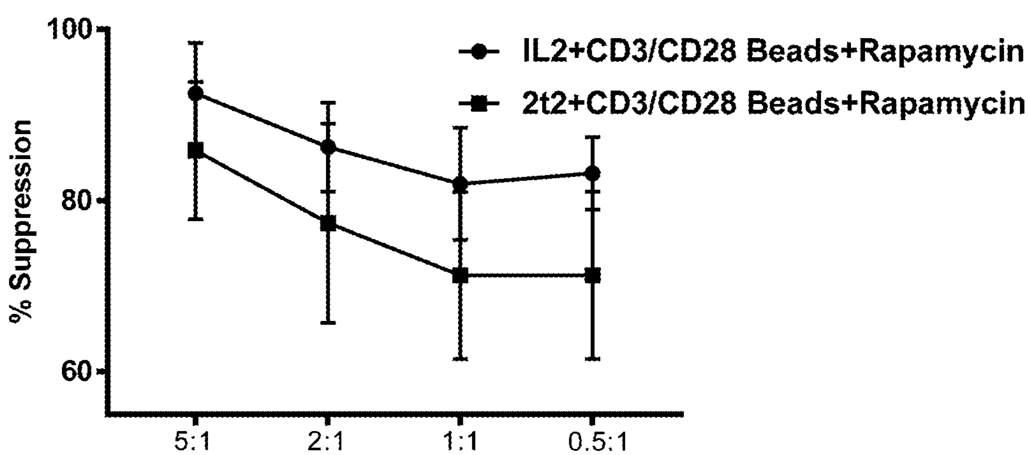

100
80
60
% Suppression
5:1
2:1
1:1
0.5:1
IL2+CD3/CD28 Beads+Rapamycin
2t2+CD3/CD28 Beads+Rapamycin

FIG. 4

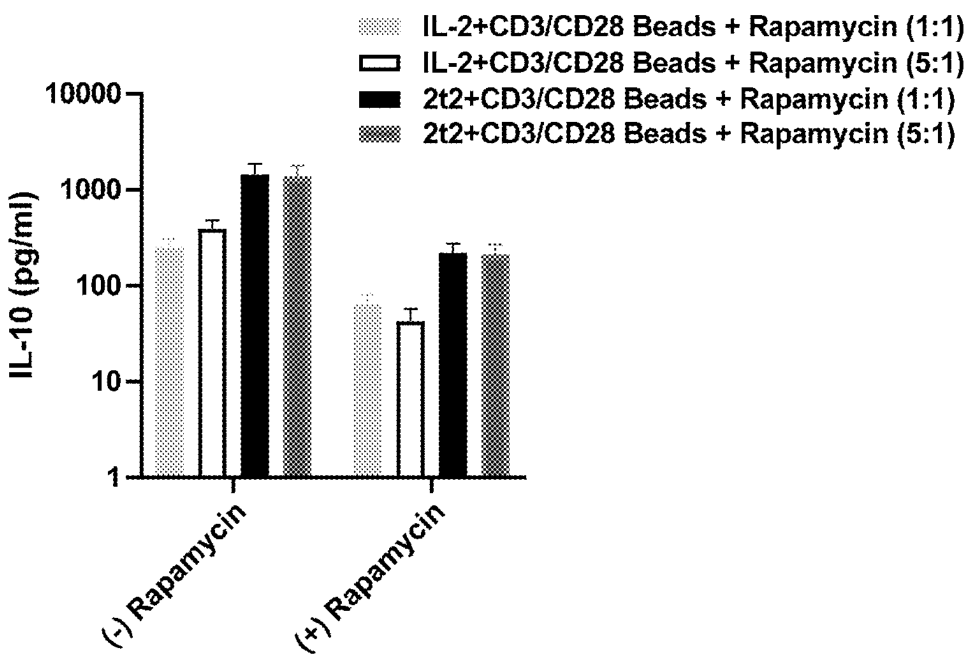

IL-2+CD3/CD28 Beads + Rapamycin (1:1)
IL-2+CD3/CD28 Beads + Rapamycin (5:1)
2t2+CD3/CD28 Beads + Rapamycin (1:1)
2t2+CD3/CD28 Beads + Rapamycin (5:1)
IL-10 (pg/ml)
10000
1000
100
10
1
(-) Rapamycin
(+) Rapamycin

FIG. 5

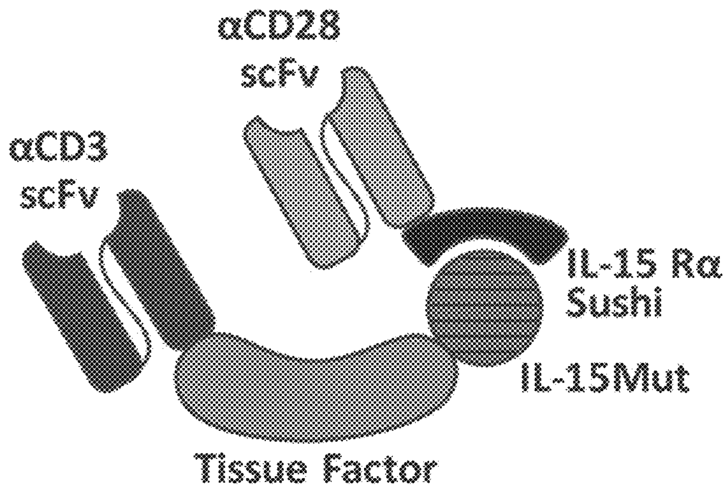

αCD28
scFv
αCD3
scFv
IL-15 Rα
Sushi
IL-15Mut
Tissue Factor

FIG. 6A

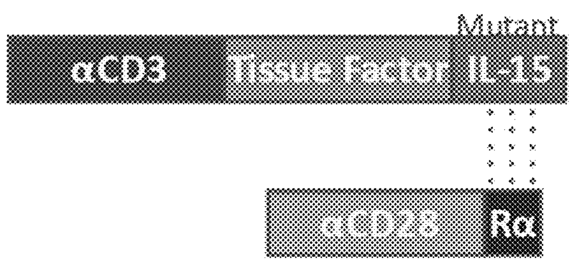

Mutant
αCD3
Tissue Factor
IL-15
αCD28
Rα

FIG. 6B

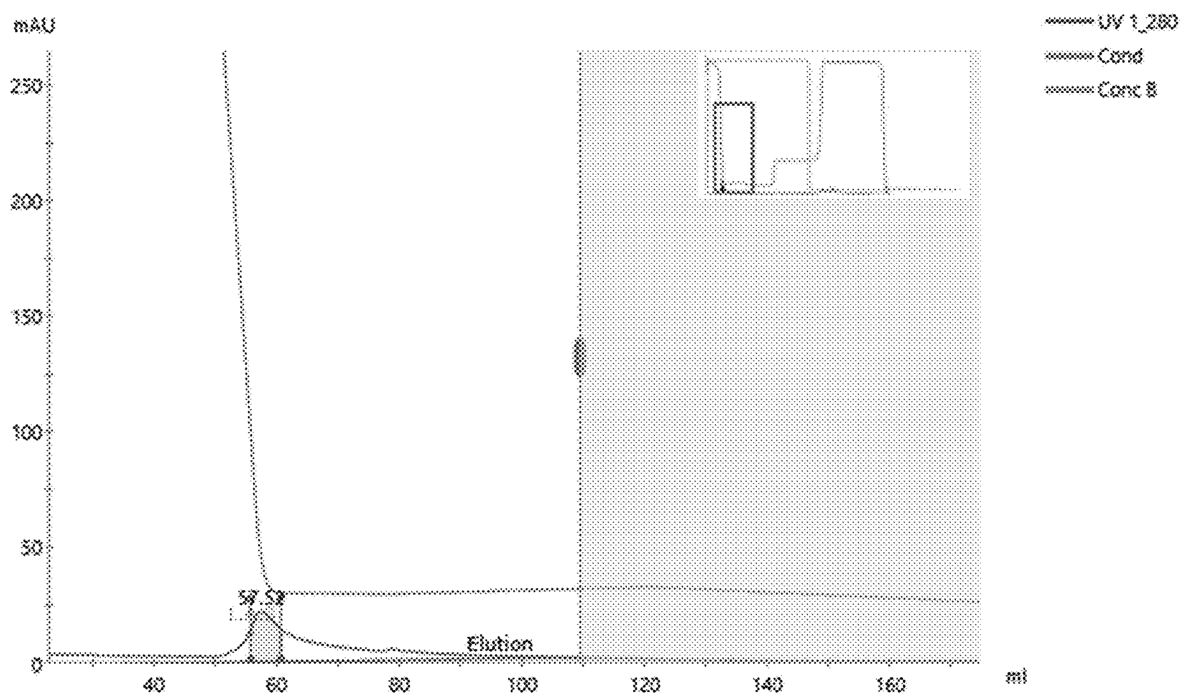
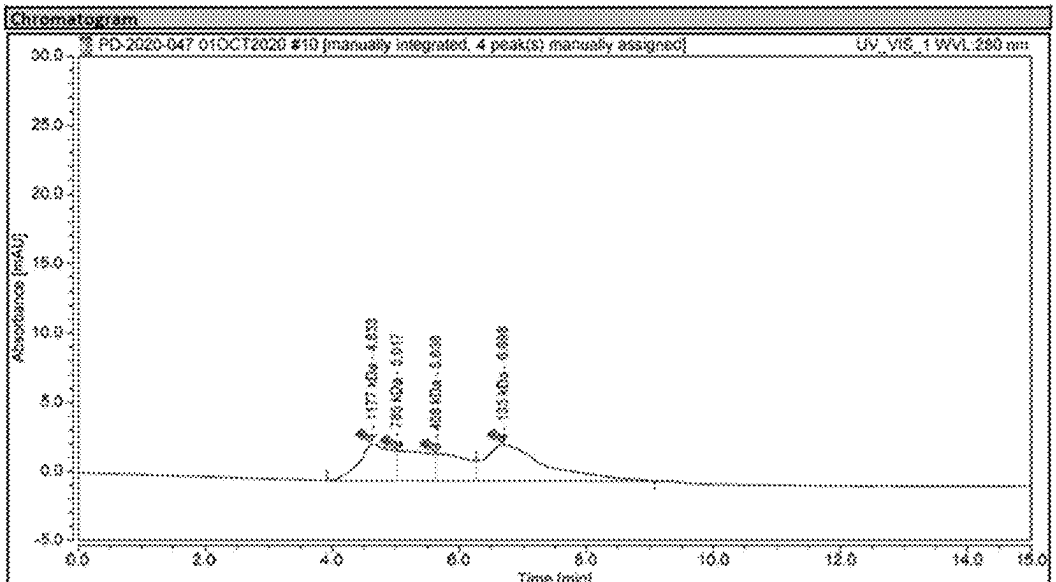

mAU
250
200
150
100
50
0
40
60
80
100
120
140
160
ml
57.53
Elution
UV 1_280
Cond
Conc B

FIG. 7

Injection Name:	58-77 9213	Run Time (min):	15.00
Vial Number:	RB2	Injection Volume:	50.00
Injection Type:	Unknown	Channel:	UV_VIS_1
Calibration Level:		Wavelength:	280
Instrument Method:	Agilent AdvanceBioSEC 4.6 x 300 with Guard	Bandwidth:	n.a.
Processing Method:	HCW9218 SEC Processing Method	Dilution Factor:	1.0000
Injection Date/Time:	01/Oct/20 19:14	Sample Weight:	1.0000

Chromatogram



Integration Results

No.	Peak Name	Retention Time min	Area mAU*min	Height mAU	Relative Area %	Relative Height %	Amount n.a.
1	1177 kDa	4.633	1.587	2.681	22.89	28.14	n.a.
2	785 kDa	5.017	1.312	2.201	18.92	23.10	n.a.
3	408 kDa	5.638	1.107	1.985	15.96	20.84	n.a.
4	133 kDa	6.698	2.928	3.661	42.23	37.92	n.a.
Total:			6.934	9.629	100.00	100.00	

FIG. 8

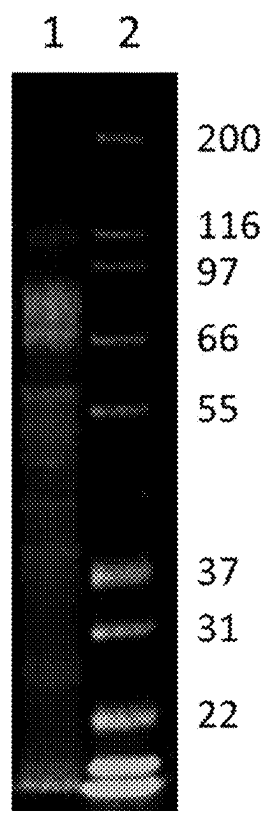


FIG. 9

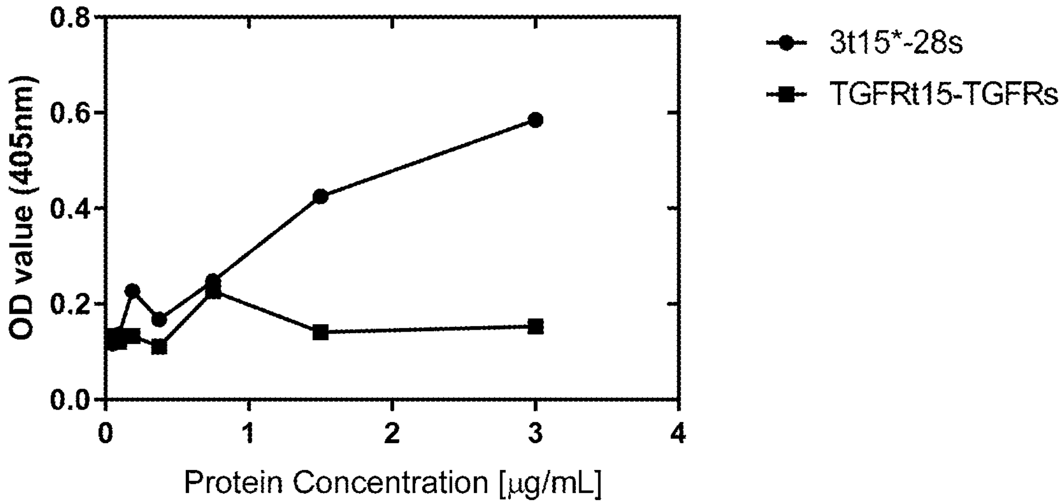


FIG. 10A

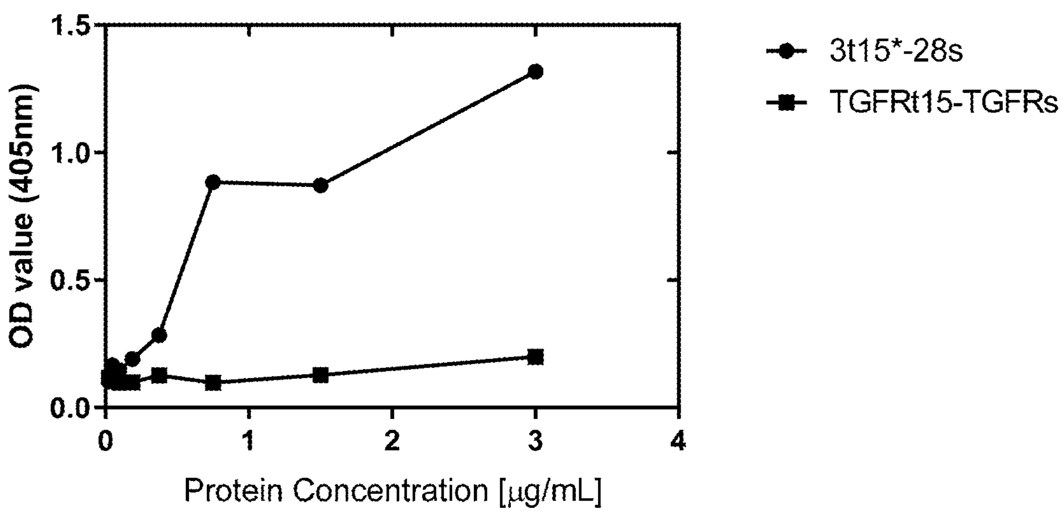
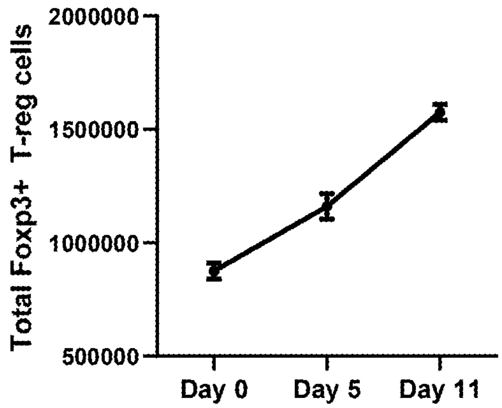
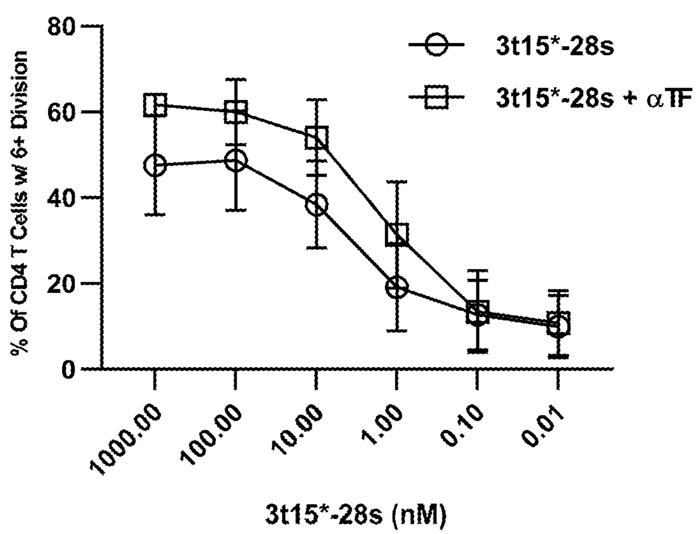


FIG. 10B



Total Foxp3+ T-reg cells
2000000
1500000
1000000
500000
Day 0
Day 5
Day 11

FIG. 11



% Of CD4 T Cells w/ 6+ Division
80
60
40
20
0
3t15*-28s
3t15*-28s + αTF
1000.00
100.00
10.00
1.00
0.10
0.01
3t15*-28s (nM)

FIG. 12

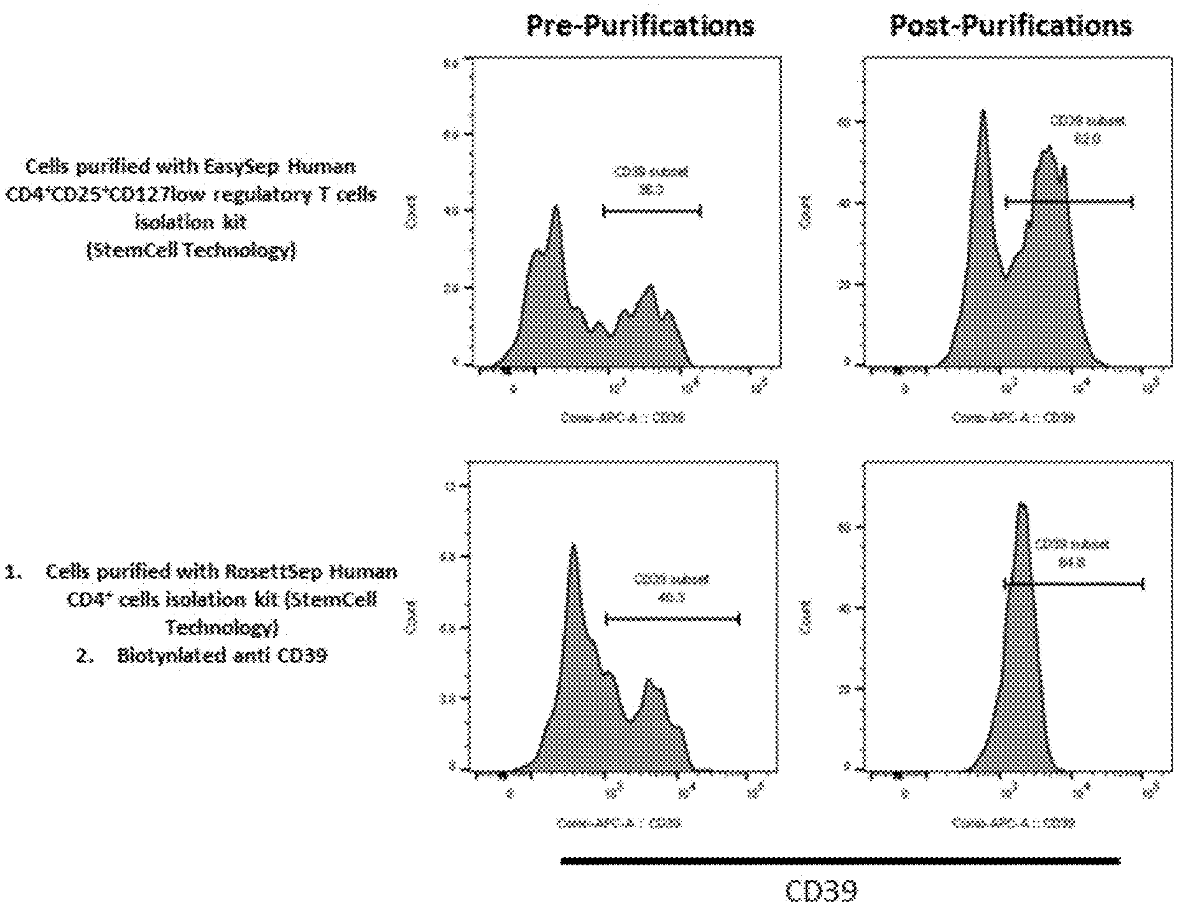

Pre-Purifications
Post-Purifications
Cells purified with EasySep Human CD4+CD25+CD127low regulatory T cells isolation kit (StemCell Technology)
1. Cells purified with RosettSep Human CD4+ cells isolation kit (StemCell Technology)
2. Biotynlated anti CD39
CD39

FIG. 13A

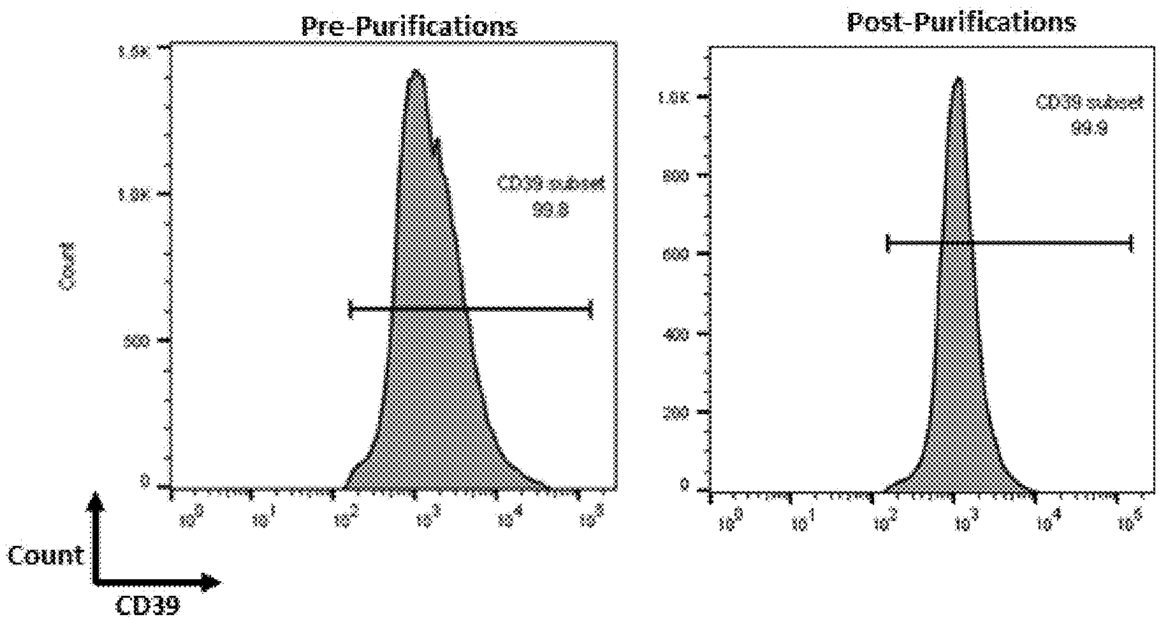

Pre-Purifications
Post-Purifications
CD39 subset
99.8
CD39 subset
99.9
Count
Count
CD39

FIG. 13B

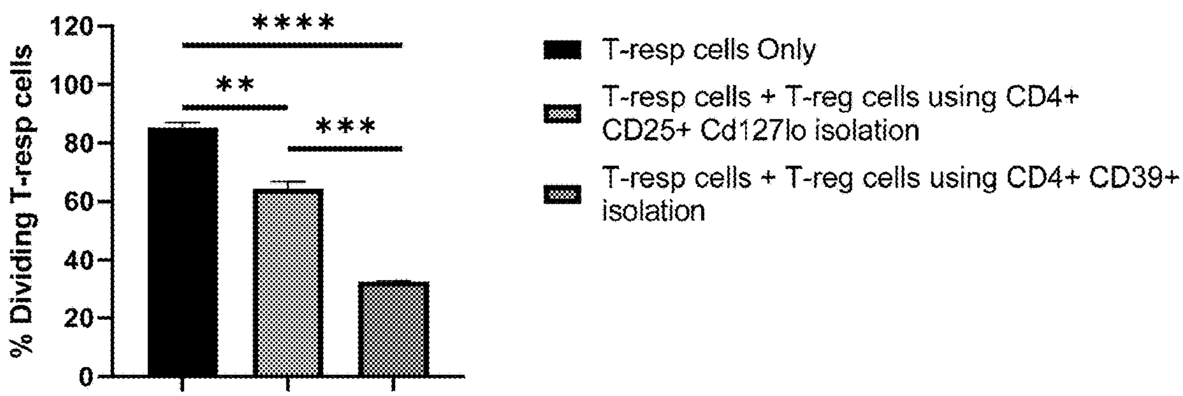

% Dividing T-resp cells
120
100
80
60
40
20
0
****
**
***
T-resp cells Only
T-resp cells + T-reg cells using CD4+ CD25+ Cd127lo isolation
T-resp cells + T-reg cells using CD4+ CD39+ isolation

FIG. 14

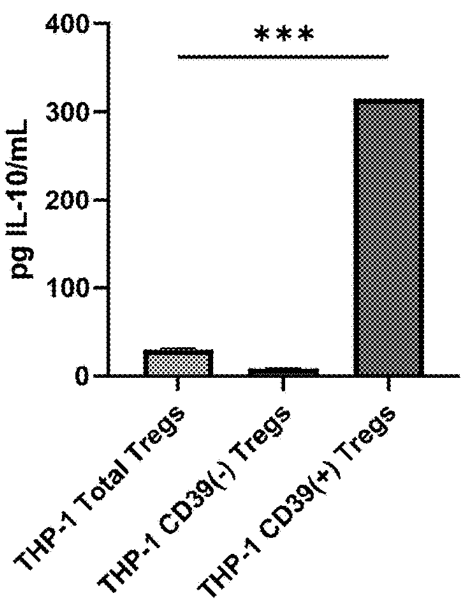

pg IL-10/mL
400
300
200
100
0
***
THP-1 Total Tregs
THP-1 CD39(-) Tregs
THP-1 CD39(+) Tregs

FIG. 15

METHODS OF ACTIVATING REGULATORY T CELLS

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to U.S. Provisional Patent Application Ser. No. 62/981,944, filed on Feb. 26, 2020, and U.S. Provisional Patent Application Ser. No. 62/975,141, filed on Feb. 11, 2020, which is incorporated herein by reference in its entirety.

SEQUENCE LISTING

This application contains a Sequence Listing that has been submitted electronically as an ASCII text file named 47039-0019001 ST25.txt. The ASCII text file, created on Jul. 10, 2023, is 147,000 bytes in size. The material in the ASCII text file is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to the fields of biotechnology and immunology, and more specifically, to methods of stimulating or increasing the proliferation of a T_{reg} cell.

BACKGROUND

Tissue factor (TF), a 263 amino acid integral membrane glycoprotein with a molecular weight of ~46 kDa and the trigger protein of the extrinsic blood coagulation pathway, is the primary initiator of coagulation in vivo. Tissue factor, normally not in contact with circulating blood, initiates the coagulation cascade upon exposure to the circulating coagulation serine protease factors. Vascular damage exposes sub-endothelial cells expressing tissue factor, resulting in the formation of a calcium-dependent, high-affinity complex with pre-existing plasma factor VIIa (FVIIa). Binding of the serine protease FVIIa to tissue factor promotes rapid cleavage of FX to FXa and FIX to FIXa. The proteolytic activity of the resulting FXa and an active membrane surface then inefficiently converts a small amount of prothrombin to thrombin. The thrombin generated by FXa initiates platelet activation and activates minute amounts of the pro-cofactors factor V (FV) and factor VIII (FVIII) to become active cofactors, factor Va (FVa) and factor VIIIa (FVIIIa). FIXa complexes with FVIIIa on the platelet surface forming the intrinsic tenase complex, which results in rapid generation of FXa. FXa complexes with FVa to form the pro-thrombinase complex on the activated platelet surface which results in rapid cleavage of prothrombin to thrombin.

In addition to the tissue factor-FVIIa complex, a recent study showed that the tissue factor-FVIIa-FXa complex can activate FVIII, which would provide additional levels of FVIIIa during the initiation phase. The extrinsic pathway is paramount in initiating coagulation via the activation of limited amounts of thrombin, whereas the intrinsic pathway maintains coagulation by dramatic amplification of the initial signal.

Much of the tissue factor expressed on a cell surface is "encrypted," which must be "decrypted" for full participation in coagulation. The mechanism of "decryption" of cell-surface tissue factor is still unclear at this time, however, exposure of anionic phospholipids plays a major role in this process. Healthy cells actively sequester anionic phospholipids such as phosphatidyl serine (PS) to the inner leaflet of the plasma membrane. Following cellular damage, activation, or increased levels of cytosolic Ca^{2+}, this bilayer asymmetry is lost, resulting in increased PS exposure on the outer leaflet, which increases the specific activity of cell-surface tissue factor-FVIIa complexes. PS exposure is known to decrease the apparent Km for activation of FIX and FX by tissue factor-FVIIa complexes, but additional mechanisms could include conformational rearrangement of tissue factor or tissue factor-FVIIa and subsequent exposure of substrate binding sites.

Adoptive immunotherapy or cellular therapy requires the culture of immune cells obtained from a subject in vivo (and optionally genetic manipulation of the immune cells to express a chimeric antigen receptor or a T-cell receptor) before administration back into the subject. A sufficient number of immune cells is necessary in order to provide a therapeutic effect in the subject. In many examples, immune cells obtained from a subject need to be cultured for three or more weeks before a therapeutically effective number of immune cells can be obtained. In addition, many methods of culturing immune cells obtained from a subject in vitro require a layer of feeder cells, which requires subsequent purification or isolation of the immune cells before administration back to the subject.

SUMMARY

Excellent safety profiles have been shown in patients receiving Treg cells (Esensten et al., *J Allergy Clin Immunol* 142(6): 1710-1718, 2018). Early clinical studies showed encouraging results in using Treg cells to prevent and treat acute and chronic Graft versus Host Diseases (Brunstein et al., *Blood* 117(3): 1061-1070, 2011; Di Ianni et al., *Blood* 117(14): 3921-3928, 2011; Martelli et al., *Blood* 124(4): 638-644, 2014; Theil et al., *Cytotherapy* 17(4): 473-486, 2015; Brunstein et al., *Blood* 127(8): 1044-1051, 2016), autoimmune and neurodegenerative diseases (Thonhoff et al., *Neurol Neuroimmunol Neuroinflamm* 5(4): e465, 2018; Dall'Era et al., *Arthritis Rheumatol* 71(3): 431-440, 2019).

To use T cell-based adoptive therapies, including Treg cells, ex-vivo cell stimulation step is required for GMP manufacturing setting. Anti-CD3/anti-CD28 antibody-coated magnetic beads (Dynabeads, Thermo Fisher) is most commonly used for cell stimulation and expansion in combination with IL-2 (Highfill et al., *Curr Hematol Malig Rep* 14(4): 269-277, 2019). If using this approach, methods must be developed to ensure T cells are "bead free" at the culture harvest and prior to infusion into patients. In this study, we describe a method of employing 2t2 and 3t15*-28s to effectively substitute IL-2 and anti-CD3/antiCD28-coated magnetic beads, respectively, for T cell stimulation and expansion without using rapamycin. 3t15*-28s is soluble fusion protein complex that can be easily removed from the culture with simple wash steps which eliminates the need for bead removal, saving time and processing cost, and enhancing cell recovery. In addition, we also demonstrated that the inclusion of a separation step using an anti-CD39 antibody in the isolation or the purification of Treg cells before or after cell stimulation and expansion could enrich the $CD4^{+}$ $CD25^{hi}$ $CD127^{lo}$ and $CD39^{hi}$ Tregs, resulting in T_{reg} cells with increased the purity, consistency, and the effector functions.

Aging in humans is associated with elevated systemic inflammation (Ferrucci et al., *Blood* 105(6): 2294-2299, 2005; Dinarello, *Am J Clin Nutr* 83(2): 447S-455S, 2006). The process that connects inflammation with aging has been termed inflamm-aging (Franceschi et al., *Ann N Y Acad Sci* 908:244-254, 2000). The process of aging is connected with

major changes that affect the immune system and results in a variety aging-associated pathologies.

The function of the immune system is to detect and respond to damage to tissues or to the invasion of pathogenic microorganisms. The innate immune cells, the first line of defense against infection, express distinct germ-line encoded pattern recognition receptors (PRRs) that recognize conserved pathogen-associated molecular patterns (PAMPs) unique to microbes (Janeway, *Cold Spring Harb Symp Quant Biol* 54 Pt 1:1-13, 1989; Gong et al., *Nat Rev Immunol* 20(2): 95-112, 2020). Danger signals released by distressed or damaged cells are also recognized by the receptors of damage-associated molecular patterns (DAMPs) (Gong et al., *Nat Rev Immunol* 20(2): 95-112, 2020). Both PAMPs and DAMPs can initiate innate immune responses through the activation of classical PRRs, such as Toll-like receptors (TLRs), and multiple germ-line-encoded receptors, such as NOD-like receptors (NLRs), retinoic acid-inducible gene I (RIG-I)-like receptors (RLRs), C-type lectin receptors (CLRs) and intracellular DNA sensors (Cao, *Nat Rev Immunol* 16(1): 35-50, 2016). DAMPs can also be sensed by several other receptors. These include receptor for advanced glycation end products (RAGE) (Hudson et al., *Annu Rev Med* 69: 349-364, 2018; Teissier et al., *Biogerontology* 20(3): 279-301, 2019), triggering receptors expressed on myeloid cells (TREMs) (Ford et al., *Curr Opin Immunol* 21(1): 38-46, 2009), several G-protein-coupled receptors (GPCRs) (Heng et al., *Annu Rev Pharmacol Toxicol* 54: 227-249, 2014; Weiss et al., *Trends Immunol* 39(10): 815-829, 2018) and ion channels (Eisenhut et al., *Pflugers Arch* 461(4): 401-421, 2011).

DAMPs-initiated inflammatory responses are referred as sterile inflammation because they are independent of pathogen infection (Chen et al., *Nat Rev Immunol* 10(12): 826-837, 2010). DAMPs can activate both non-immune cells and innate immune cells (Chen et al., *Nat Rev Immunol* 10(12): 826-837, 2010). Activation of these cells leads to the production of cytokines and chemokines, which in turn recruit inflammatory cells and activate adaptive immune responses (Chen et al., *Nat Rev Immunol* 10(12): 826-837, 2010). Some DAMPs are also known to directly activate adaptive immune cells (Lau et al., *J Exp Med* 202(9): 1171-1177, 2005; Qin et al., *J Immunol* 199(1): 72-81, 2017). Although sterile inflammation plays an essential role in tissue repair and regeneration to reestablish the tissue hemostasis after injurious insults, unresolved chronic inflammation due to repeated tissue damage or in response to an overabundance of innate-immune triggers present in tissue is detrimental to the host and may lead to sterile inflammatory diseases, including cancer, metabolic disorders (e.g., diabetes), neurodegenerative diseases (e.g., Alzheimer's disease, Parkinson's disease), and autoimmune diseases (e.g., multiple sclerosis) (Roh et al., *Immune Netw* 18(4): e27, 2018).

Inflammasomes are large, multimeric protein complexes comprising a cytosolic pattern-recognition receptor, the adaptor protein apoptosis-associated Speck-like protein containing a caspase recruitment domain (ASC), and a caspase-1 (Lamkanfi et al., *Cell* 157(5): 1013-1022, 2014). Their assembly in innate immune cells and other cells is triggered by a variety of stimuli and culminates in the activation of caspase-1 which then cleaves pro-IL-1β to IL-1β (Latz et al., *Nat Rev Immunol* 13(6): 397-411, 2013; Walsh et al., *Nat Rev Neurosci* 15(2): 84-97, 2014). To date, diverse inflammasomes have been discovered. Among the various inflammasomes identified, the nucleotide-binding oligomerization domain, leucine-rich repeat-containing receptor (NLR) family pyrin domain-containing 3 (NLRP3) inflammasome is best characterized (Swanson et al., *Nat Rev Immunol* 19(8): 477-489, 2019). The NLRs are recognized as the key sensors of pathogens and danger signals. The NLRP3 inflammasome has a two-step activation mechanism: "priming", which entails induction of Pro-IL-1β and NLRP3, and "activation", wherein a functional inflammasome complex is assembled following uptake of PAMPs or DAMPs. The pathology of various diseases, including Alzheimer's disease (Heneka et al., *Nature* 493(7434): 674-678, 2013), Parkinson's disease (Heneka et al., *Nat Rev Neurosci* 19(10): 610-621, 2018), and atherosclerosis (Jin et al., *J Am Heart Assoc* 8(12): e012219, 2019) has been linked to hyperactivation of the NLRP3 inflammasome.

Sterile inflammation can also result from the accumulation of senescent cells. Cellular senescence is defined as an irreversible cell cycle arrest that occurs in responses to cellular stress and prevents transmission of defects to the next generation (Collado et al., *Nat Rev Cancer* 10(1): 51-57, 2010; McHugh et al., *J Cell Biol* 217(1): 65-77, 2018). Cellular senescence plays a major protective role in the process of development, tissue hemostasis, and wound healing (Munoz-Espin et al., *Cell* 155(5): 1104-1118, 2013; Storer et al., *Cell* 155(5): 1119-1130, 2013; Demaria et al., *Dev Cell* 31(6): 722-733, 2014; Yun et al., *Elife* 4, 2015). Cellular senescent is accompanied by a pro-inflammatory phenotype. The phenotype is referred to the senescence-associated secretory phenotype (SASP) (McHugh et al., *J Cell Biol* 217(1): 65-77, 2018). The SASP is characterized by the release of inflammatory cytokines, chemokines, growth factors and proteases. This reinforces cellular senescence through autocrine and paracrine signaling, and recruits and instructs immune cells to clear senescence cells. Thus, cellular senescence and SASP are a vital physiological response that maintains homeostasis at the cellular level, tissue level and organ level. However, upon persistent damage or during aging, senescent cell clearance is compromised, and dysfunctional cells accumulate. The SASP from these uncleared and accumulated senescent cells is a protracted and chronic source of inflammatory factors that create an inflammatory microenvironment that results in a diverse range of pathological manifestations (Munoz-Espin et al., *Nat Rev Mol Cell Biol* 15(7): 482-496, 2014; van Deursen, *Nature* 509(7501): 439-446, 2014; McHugh et al., *J Cell Biol* 217(1): 65-77, 2018). In addition, some of the SASP factors prime the inflammasome-containing cells which increases the risk of fueling chronic inflammasome activation to induce sterile inflammation.

Regulatory T (Treg) cells are essential mediators of peripheral tolerance and the global immunoregulatory potential in hosts to self and non-self-antigens (Sakaguchi et al., *Cell* 133(5): 775-787, 2008; Sakaguchi et al., *Annu Rev Immunol* 38:541-566, 2020). Treg cells achieve this immunoregulatory control through multiple suppressive mechanisms. These include IL-2 deprivation, the secretion of inhibitory cytokines (i.e., IL-10 and TGF-β) and the acquisition of co-stimulatory molecules from antigen-presenting cells via high-affinity binding to CTLA-4 (Oberle et al., *J Immunol* 179(6): 3578-3587, 2007; Tang et al., *Nat Immunol* 9(3): 239-244, 2008; Zheng et al., *J Immunol* 181(3): 1683-1691, 2008) The adenosine triphosphate (ATP)-adenosine pathway is also utilized by regulatory Treg as a key modulator of innate and adaptive immunity. CD39 is the dominant ecto-nucleotidase broadly expressed on immune cells (e.g., Tregs), endothelial cells and tumor cell, that hydrolyses ATP and adenosine diphosphate (ADP) to adenosine monophosphate (AMP) (Moesta et al., *Nat Rev Immunol* 20(12): 739-755, 2020). AMP is then hydrolyzed by CD73 to adenosine. Adenosine binds to its receptors A1, A2A, A2B, and A3 displayed on immune cells. A2A and A2B receptors (A2AR and A2BR) are Gs-coupled receptors that increase intracellular cAMP and PKA levels, playing dominant roles in adenosine-induced immunosuppression in a cAMP-dependent manner. A1 and A3 receptors (AIR and A3R) are Gi/o-coupled receptors that decrease intracellular cAMP favoring cell activation, and therefore are generally viewed as immune-promoting adenosine receptors. In humans, AIR, A2AR and A3R display high affinity for adenosine whereas A2BR has a significantly lower affinity. A2A and A2BR are expressed on immune cells (Feng et al., *Cancer Cell Int* 20: 110, 2020). Recently, it has been shown that human CD39^{hi} regulatory T cells exhibits stronger stability, higher Foxp3 expression, and suppressive ability under inflammatory conditions (Gu et al., *Cell Mol Immunol* 14(6): 521-528, 2017). Alternations in Treg cell development, homeostasis or function can predispose these cells to a variety of disease conditions including allergy, autoimmunity, graft rejection, cancer, and response to immunotherapies (Sakaguchi et al., *Annu Rev Immunol* 38: 541-566, 2020). Current research is focused on developing novel therapies to enhance Treg cell functions in vivo through use of cytokines and small molecular weight drugs to support endogenous Treg cell proliferation or activation, ex-vivo manipulated Treg cells in autologous adoptive cell therapy (ACT) to promote immunoregulation in settings of autoimmunity, or antigen-specific Treg cells, including chimeric antigen receptor Treg (CAR-Treg) cells, to strengthen tolerance in allergic inflammation (Ferreira et al., *Nat Rev Drug Discov* 18(10): 749-769, 2019). The present invention is a method that uses Treg cells to deactivate the inflammasome and then subsequently uses immune cells activated by one or more immunotherapeutics to reduce the accumulated senescent cells in an individual to treat inflamm-aging and/or any aging-associated pathologies. Provided is a method of inhibiting inflammasome related diseases and inhibiting senescent related diseases. The Treg cells can be in-vivo enhanced endogenous Treg cells or ex-vivo manipulated Treg cells used in an ACT setting. The immune cells for senescent-cell clearance could be activated by immunotherapeutics in vivo or generated by ex-vivo stimulation and expansion methods to support an ACT administration. The present invention is based on the discovery that single-chain chimeric polypeptides that include a soluble tissue factor domain or multi-chain chimeric polypeptides that include a soluble tissue factor domain, in combination with CD3/CD28-binding agents or IL-2 receptor-activating agents are effective in stimulating or increasing the proliferation of a T_{reg} cell. Based on this discovery provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell that include culturing a T_{reg} cell in a liquid culture medium over a period of time, where at the beginning of the period of time, the liquid culture medium includes: a CD3/CD28-binding agent; and a single-chain chimeric polypeptide including a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where the first target-binding domain and the second target-binding domain bind to a receptor for IL-2.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the soluble tissue factor domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain and the second target-binding domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain comprise the same amino acid sequence. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain comprise a soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble IL-2 is a human soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain each comprise a sequence that is at least 80% identical to SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain each comprise a sequence that is at least 90% identical to SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain each comprise a sequence of SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises or consists of a sequence from a wildtype soluble human tissue factor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 4. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 4. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 4. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 500 nM of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 150 nM of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises the anti-CD3/anti-CD28 bead at a ratio of about 1:1 to about 6:1 beads/cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises the anti-CD3/anti-CD28 bead at a ratio of about 3:1 to about 5:1 beads/cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3/CD28-binding agent is an additional single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein: the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide and the soluble tissue factor domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a

T_{reg} cell described herein, the soluble tissue factor domain and the second target-binding domain of the additional single-chain chimeric polypeptide directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3 is human CD3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD28 is human CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the additional single-chain chimeric polypeptide is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the single-chain chimeric polypeptide comprises or consists of a sequence from a wildtype soluble human tissue factor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the additional single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about

10 nM to about 1000 nM of the additional single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 300 nM of the additional single-chain chimeric polypeptide.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD3/CD28 binding agent is a multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide and the soluble tissue factor domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain and the second target-binding domain of the second chimeric polypeptide directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD3 is human CD3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide comprises a sequence of SEQ ID NO: 5.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide comprises or consists of a sequence from a wildtype soluble human tissue factor.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence of SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 104. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 104. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide comprises a sequence of SEQ ID NO: 104.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD28 is human CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide comprises a sequence of SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence of SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 108. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 108. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide comprises a sequence of SEQ ID NO: 108.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the multi-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium comprises about 10 nM to about 1000 nM of the multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium comprises about 50 nM to about 300 nM of the multi-chain chimeric polypeptide.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further comprises an mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the mTOR inhibitor is rapamycin. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 500 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 150 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium does not comprise an mTOR inhibitor.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each 36 hours to about 60 hours after the start of the period of time, the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the single-chain chimeric polypeptide is performed to result in a concentration of about 10 nM to about 500 nM of the single-chain chimeric polypeptide in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the single-chain chimeric polypeptide is performed to result in a concentration of about 50 nM to about 150 nM of the single-chain chimeric polypeptide in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each about 6 days to about 8 days after the start of the period of time, the CD3/CD28-binding agent. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of time is about 7 days to about 56 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of claim is about 15 days to about 25 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the method further comprises, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the method further comprises, after the culturing step, a step of isolating the Treg cell from a sample obtained from a subject.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the isolating comprises the use of fluorescence-assorted cell sorting. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is an autologous T_{reg} cell, a haploidentical T_{reg} cell, or an allogeneic T_{reg} cell isolated from peripheral blood or umbilical cord blood. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+Foxp3^+$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+CD127^{dim}$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cells comprises a chimeric antigen receptor.

In another aspect, provided herein are populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein. Provided herein are compositions comprising any of the population of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. Also provided herein are methods of treating a subject in need thereof, the methods include administering to the subject a therapeutically effective amount of any of the compositions comprising any of the population of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. In some embodiments of any of the methods of treating a subject in need thereof, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some embodiments of any of the methods of treating a subject in need thereof, the aging-related disease is selected from the group consisting of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction. In some embodiments of any of the methods of treating a subject in need thereof, the inflammatory disease is selected from the group consisting of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, nonalcoholic steatohepatitis, and mood disorders.

In another aspect, provided herein are kits that include: (i) a CD3/CD28-binding agent; (ii) a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein the first target-binding domain and the second target-binding domain bind to a receptor for IL-2; and (iii) an mTOR inhibitor. In some embodiments of any of the kits described herein, the kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide. In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead. In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is an additional single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein: the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28. In some embodiments of any of the kits described herein, the kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional single-chain chimeric polypeptide.

In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is a multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein: the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3. In some embodiments of any of the kits described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide.

In another aspect, provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell, the methods include: culturing a T_{reg} cell in a liquid culture medium over a period of time, wherein at the beginning of the period of time, the liquid culture medium comprises: an IL-2 receptor-activating agent; and a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein: the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the soluble tissue factor domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain and the second target-binding domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3 is human CD3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain comprises a sequence that is at least 80% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain comprises a sequence that is at least 90% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain comprises a sequence of SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD28 is human CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain comprises a sequence that is at least 80% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain comprises a sequence that is at least 90% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain comprises a sequence of SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises or consists of a sequence from a wildtype soluble human tissue factor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 1000 nM of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 300 nM of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IL-2 receptor-activating agent is a soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble IL-2 is a human soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the human soluble IL-2 comprises a sequence that is at least 80% identical to SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the human soluble IL-2 comprises a sequence that is at least 90% identical to SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the human soluble IL-2 comprises a sequence of SEQ ID NO: 1. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 100 IU/mL to about 800 IU/mL of the human soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 400 IU/mL to about 600 IU/mL of the human soluble IL-2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further comprises an mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the mTOR inhibitor is rapamycin. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 500 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 150 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium does not comprise an mTOR inhibitor.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each 36 hours to about 60 hours after the start of the period of time, the IL-2 receptor-activating agent. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the IL-2 receptor-activating agent is performed to result in a concentration of about 10 nM to about 1000 nM of the IL-2 receptor-activating agent in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the IL-2 receptor activating agent is performed to result in a concentration of about 50 nM to about 150 nM of the IL-2 receptor-activating agent in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each about 6 days to about 8 days after the start of the period of time, the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of time is about 7 days to about 56 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of claim is about 15 days to about 25 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the method further comprises, after the culturing step, a step of isolating the Treg cell from a sample obtained from a subject.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the isolating comprises the use of fluorescence-assorted cell sorting. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is an autologous T_{reg} cell, a haploidentical T_{reg} cell, or an allogeneic T_{reg} cell isolated from peripheral blood or umbilical cord blood. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+Foxp3^+$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+CD127^{dim}$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cells comprises a chimeric antigen receptor.

Provided herein are populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein. Provided herein are compositions comprising any populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. Also provided herein are methods of treating a subject in need thereof, the methods include administering to the subject a therapeutically effective amount of any of the compositions comprising any populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. In some embodiments of any of the methods of treating a subject in need thereof described herein, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some embodiments of any of the methods of treating a subject in need thereof described herein, the aging-related disease is selected from the group consisting of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction. In some embodiments of any of the methods of treating a subject in need thereof described herein, the inflammatory disease is selected from the group consisting of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, nonalcoholic steatohepatitis, and mood disorders.

In another aspect, provided herein are kits that include: (i) an interleukin-2 receptor-activating agent; (ii) a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein: the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28; and (iii) an mTOR inhibitor. In some embodiments of any of the kits described herein, the kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide.

In another aspect, provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell, the methods include: culturing a T_{reg} cell in a liquid culture medium over a period of time, wherein at the beginning of the period of time, the liquid culture medium comprises: a CD3/CD28-binding agent; and a multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the soluble tissue factor domain directly abut each other in the first chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain in the first chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain and the first domain of the pair of affinity domains directly abut each other in the first chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the first domain of the pair of affinity domains in the first chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second domain of the pair of affinity domains and the second target-binding domain directly abut each other in the second chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, second chimeric polypeptide further comprises a linker sequence between the second domain of the pair of affinity domains and the second target-binding domain in the second chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain bind specifically to the same antigen. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain and the second target-binding domain bind specifically to different antigens. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, one or both of the first target-binding domain and the second target-binding domain is an antigen-binding domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, one or both of the first target-binding domain and the second target-binding domain bind specifically to a target selected from the group consisting of: CD16a, CD28, CD3, CD33, CD20, CD19, CD22, CD123, IL-1R, IL-1, VEGF, IL-6R, IL-4, IL-10, PDL-1, TIGIT, PD-1, TIM3, CTLA4, MICA, MICB, IL-6, IL-8, TNFα, CD26, CD36, ULBP2, CD30, CD200, IGF-1R, MUC4AC, MUC5AC, Trop-2, CMET, EGFR, HER1, HER2, HER3, PSMA, CEA, B7H3, EPCAM, BCMA, P-cadherin, CEACAM5, a UL16-binding protein, HLA-DR, DLL4, TYRO3, AXL, MER, CD122, CD155, PDGF-DD, a ligand of TGF-β receptor II (TGF-βRII), a ligand of TGF-βRIII, a ligand of DNAM-1, a ligand of NKp46, a ligand of NKp44, a ligand of NKG2D, a ligand of NKp30, a ligand for a scMHCI, a ligand for a scMHCII, a ligand for a scTCR, a receptor for IL-1, a receptor for IL-2, a receptor for IL-3, a receptor for IL-7, a receptor for IL-8, a receptor for IL-10, a receptor for IL-12, a receptor for IL-15, a receptor for IL-17, a receptor for IL-18, a receptor for IL-21, a receptor for PDGF-DD, a receptor for stem cell factor (SCF), a receptor for stem cell-like tyrosine kinase 3 ligand (FLT3L), a receptor for MICA, a receptor for MICB, a receptor for a ULP16-binding protein, a receptor for CD155, a receptor for CD122, and a receptor for CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, one or both of the first target-binding domain and the second target-binding domain is a soluble interleukin or cytokine protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble interleukin, cytokine, or ligand protein is selected from the group consisting of: IL-1, IL-2, IL-3, IL-7, IL-8, IL-10, IL-12, IL-15, IL-17, IL-18, IL-21, PDGF-DD, SCF, and FLT3L. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, one or both of the first target-binding domain and the second target-binding domain is a soluble interleukin or cytokine receptor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble receptor is a soluble TGF-β receptor II (TGF-βRII), a soluble TGF-βRIII, a soluble NKG2D, a soluble NKp30, a soluble NKp44, a soluble NKp46, a soluble DNAM-1, a scMHCI, a scMHCII, a scTCR, a soluble CD155, or a soluble CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first chimeric polypeptide further comprises one or more additional target-binding domain(s). In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second chimeric polypeptide further comprises one or more additional target-binding domains. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the pair of affinity domains is a sushi domain from an alpha chain of human IL-15 receptor (IL15Ra) and a soluble IL-15. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble IL-15 has a D8N or D8A amino acid substitution. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the pair of affinity domains is selected from the group consisting of: barnase and barnstar, a PKA and an AKAP, adapter/docking tag modules based on mutated RNase I fragments, and SNARE modules based on interactions of the proteins syntaxin, synaptotagmin, synaptobrevin, and SNAP25. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises or consists of a sequence from a wildtype soluble human tissue factor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the multi-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises the anti-CD3/anti-CD28 bead at a ratio of about 1:1 to about 6:1 beads/cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises the anti-CD3/anti-CD28 bead at a ratio of about 3:1 to about 5:1 beads/cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3/CD28-binding agent is an single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein: the first target-binding domain of the single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the single-chain chimeric polypeptide binds to CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the single-chain chimeric polypeptide and the soluble tissue factor domain directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain and the second target-binding domain of the single-chain chimeric polypeptide directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD3 is human CD3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the first target-binding domain of the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the CD28 is human CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the second target-binding domain of the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the single-chain chimeric polypeptide is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble human tissue factor domain of the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the single-chain chimeric polypeptide comprises or consists of a sequence from a wildtype soluble human tissue factor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the soluble tissue factor domain of the single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 7. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 8. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 1000 nM of the single-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 300 nM of the single-chain chimeric polypeptide.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD3/CD28 binding agent is an additional multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide and the soluble tissue factor domain of the additional multi-chain chimeric polypeptide directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain and the second target-binding domain of the second chimeric polypeptide of the additional multi-chain chimeric polypeptide directly abut each other. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD3 is human CD3. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first target-binding domain of the additional single-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 5.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide is a soluble human tissue factor domain. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble human tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 2. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises or consists of a sequence from a wildtype soluble human tissue factor.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide does not comprise any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the soluble tissue factor domain of the first chimeric polypeptide of the additional multi-chain chimeric polypeptide does not initiate blood coagulation.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 102. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 104. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 104. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the first chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 104.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the CD28 is human CD28. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second target-binding domain of the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 6.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 106. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 80% identical to SEQ ID NO: 108. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 108. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the second chimeric polypeptide of the additional multi-chain chimeric polypeptide comprises a sequence of SEQ ID NO: 108.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the additional multi-chain chimeric polypeptide does not initiate blood coagulation. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium comprises about 10 nM to about 1000 nM of the additional multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the liquid culture medium comprises about 50 nM to about 300 nM of the additional multi-chain chimeric polypeptide.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium further comprises an mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg}

cell described herein, the mTOR inhibitor is rapamycin. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 10 nM to about 500 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium comprises about 50 nM to about 150 nM of the mTOR inhibitor. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each 36 hours to about 60 hours after the start of the period of time, the multi-chain chimeric polypeptide. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the multi-chain chimeric polypeptide is performed to result in a concentration of about 10 nM to about 500 nM of the multi-chain chimeric polypeptide in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the periodic addition of the multi-chain chimeric polypeptide is performed to result in a concentration of about 50 nM to about 150 nM of the multi-chain chimeric polypeptide in the liquid culture medium. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the methods further include: periodically adding to the liquid culture medium, after each about 6 days to about 8 days after the start of the period of time, the CD3/CD28-binding agent. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of time is about 7 days to about 56 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the period of claim is about 15 days to about 25 days. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the method further comprises, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject. In some embodiments of any of the methods of stimulating or increasing the proliferation of a Treg cell described herein, the method further comprises, after the culturing step, a step of isolating the Treg cell from a sample obtained from a subject.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the isolating comprises the use of fluorescence-assorted cell sorting. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is an autologous T_{reg} cell, a haploidentical T_{reg} cell, or an allogeneic T_{reg} cell isolated from peripheral blood or umbilical cord blood. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+Foxp3^+$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cell is a $CD4^+CD25^+CD127^{dim}$ cell. In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the T_{reg} cells comprises a chimeric antigen receptor.

Provided herein are populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein. Also provided herein are compositions comprising any of the populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. Provided herein are methods of treating a subject in need thereof, the methods include administering to the subject a therapeutically effective amount of any of the compositions comprising any of the populations of T_{reg} cells generated by any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein and a pharmaceutically acceptable carrier. In some embodiments of any of the methods of treating a subject in need thereof described herein, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some embodiments of any of the methods of treating a subject in need thereof described herein, the aging-related disease is selected from the group consisting of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction. In some embodiments of any of the methods of treating a subject in need thereof described herein, the inflammatory disease is selected from the group consisting of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, non-alcoholic steatohepatitis, and mood disorders.

In another aspect, provided herein are kits that include: a CD3/CD28-binding agent; a multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and an mTOR inhibitor. In some embodiments of any of the kits described herein, the kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide. In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead. In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is an single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28. In some embodiments of any of the kits described herein, the kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide.

In some embodiments of any of the kits described herein, the CD3/CD28-binding agent is an additional multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3.

In some embodiments of any of the kits described herein, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional multi-chain chimeric polypeptide.

In another aspect, provided herein are methods of isolating a Treg cell from a sample obtained from a subject, wherein the method comprises separating the Treg cell from the sample based on its expression of CD39, thereby isolating the Treg cell.

In some embodiments of any of the methods of isolating a Treg cell described herein, the method comprises: mixing the sample with an antibody or ligand capable of binding CD39 under conditions that allow binding of the antibody or ligand to Treg cells expressing CD39; separating the Treg cell bound to the antibody or ligand from other components in the sample, thereby isolating the Treg cell.

In some embodiments of any of the methods of isolating a Treg cell described herein, the antibody is a mouse, a humanized, or a human antibody or antigen-binding fragment thereof; and/or the antibody or the ligand is labeled with at least one of biotin, avidin, streptavidin, or a fluorochrome, or is bound to a particle, bead, resin, or solid support. In some embodiments of any of the methods of isolating a Treg cell described herein, separating comprises the use of flow cytometry, fluorescence-activated cell sorting (FACS), centrifugation, or column, plate, particle, or bead-based methods.

In some embodiments of any of the methods of isolating a Treg cell described herein, the method further comprises: mixing the sample with a biotinylated antibody or ligand capable of binding CD39 under conditions that allow binding of the antibody or the ligand to the Treg cell; capturing the Treg cell bound to the biotinylated antibody or the ligand using streptavidin-coated magnetic particles; separating the magnetic particle-bound Treg cell using a magnet; washing the magnet particle-bound Treg cell to remove other components in the sample; and releasing the magnetic particle-bound Treg cell from the magnet into a solution, thereby isolating the Treg cell.

In some embodiments of any of the methods of isolating a Treg cell described herein, the Treg cell is an autologous Treg cell, a haploidentical Treg cell, or an allogeneic Treg cell isolated from a sample comprising fresh or frozen peripheral blood, umbilical cord blood, peripheral blood mononuclear cells, lymphocytes, $CD4^+$ T cells or Treg cells. In some embodiments of any of the methods of isolating a Treg cell described herein, the Treg cell is a $CD4^+CD25^+$ $Foxp3^+$ cell. In some embodiments of any of the methods of isolating a Treg cell described herein, the Treg cell is a $CD4^+CD25^+CD127^{dim}$ cell. In some embodiments of any of the methods of isolating a Treg cell described herein, the Treg cells comprises a chimeric antigen receptor. In some embodiments of any of the methods of isolating a Treg cell described herein, the Treg cells are immunosuppressive in vitro and in vivo.

In another aspect, provided herein are populations of isolated Treg cells generated by any one of the methods described herein. In some embodiments of any of the populations of isolated Treg cells described herein, the isolated population of Treg cells are greater than 70% $CD39^+$ cells. Also provided are method of expanding any of the populations of isolated Treg cells described herein that include culturing the population of isolated Treg cells under conditions that allow for proliferation of the isolated Treg cells.

Also provided herein are compositions comprising any of the populations of isolated Treg cells described herein and a pharmaceutically acceptable carrier.

Also provided herein are methods of treating a subject in need thereof, comprising administering to the subject a therapeutically effective amount of any of the compositions described herein. In some embodiments of any of the methods of treating a subject in need described herein, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some embodiments of any of the methods of treating a subject in need described herein, the aging-related disease is selected from the group consisting of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction. In some embodiments of any of the methods of treating a subject in need described herein, the inflammatory disease is selected from the group consisting of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, nonalcoholic steatohepatitis, and mood disorders.

As used herein, the term "chimeric" refers to a polypeptide that includes amino acid sequences (e.g., domains) originally derived from two different sources (e.g., two different naturally-occurring proteins, e.g., from the same or different species). For example, a chimeric polypeptide can include domains from at least two different naturally occurring human proteins. In some examples, a chimeric polypeptide can include a domain that is a synthetic sequence (e.g., a scFv) and a domain that is derived from a naturally-occurring protein (e.g., a naturally-occurring human protein). In some embodiments, a chimeric polypeptide can include at least two different domains that are synthetic sequences (e.g., two different scFvs).

An "antigen-binding domain" is one or more protein domain(s) (e.g., formed from amino acids from a single polypeptide or formed from amino acids from two or more polypeptides (e.g., the same or different polypeptides) that is capable of specifically binding to one or more different antigen(s). In some examples, an antigen-binding domain can bind to an antigen or epitope with specificity and affinity similar to that of naturally-occurring antibodies. In some embodiments, the antigen-binding domain can be an antibody or a fragment thereof. In some embodiments, an antigen-binding domain can include an alternative scaffold. Non-limiting examples of antigen-binding domains are described herein. Additional examples of antigen-binding domains are known in the art.

A "soluble tissue factor domain" refers to a polypeptide having at least 70% identity (e.g., at least 75% identity, at least 80% identity, at least 85% identity, at least 90% identity, at least 95% identity, at least 99% identity, or 100% identical) to a segment of a wildtype mammalian tissue factor protein (e.g., a wildtype human tissue factor protein) that lacks the transmembrane domain and the intracellular domain. Non-limiting examples of soluble tissue factor domains are described herein.

The term "antibody" is used herein in its broadest sense and includes certain types of immunoglobulin molecules that include one or more antigen-binding domains that specifically bind to an antigen or epitope. An antibody specifically includes, e.g., intact antibodies (e.g., intact immunoglobulins), antibody fragments, and multi-specific antibodies. One example of an antigen-binding domain is an antigen-binding domain formed by a VH-VL dimer. Additional examples of an antibody are described herein. Additional examples of an antibody are known in the art.

"Affinity" refers to the strength of the sum total of non-covalent interactions between an antigen-binding site and its binding partner (e.g., an antigen or epitope). Unless indicated otherwise, as used herein, "affinity" refers to intrinsic binding affinity, which reflects a 1:1 interaction between members of an antigen-binding domain and an antigen or epitope. The affinity of a molecule X for its partner Y can be represented by the dissociation equilibrium constant (K_D). The kinetic components that contribute to the dissociation equilibrium constant are described in more detail below. Affinity can be measured by common methods known in the art, including those described herein. Affinity can be determined, for example, using surface plasmon resonance (SPR) technology (e.g., BIACORE®) or biolayer interferometry (e.g., FORTEBIO®). Additional methods for determining the affinity for an antigen-binding domain and its corresponding antigen or epitope are known in the art.

A "single-chain polypeptide" as used herein refers to a single protein chain.

A "multi-chain polypeptide" as used herein to refers to a polypeptide comprising two or more (e.g., three, four, five, six, seven, eight, nine, or ten) protein chains (e.g., at least a first chimeric polypeptide and a second polypeptide), where the two or more proteins chains associate through non-covalent bonds to form a quaternary structure.

The term "epitope" means a portion of an antigen that specifically binds to an antigen-binding domain. Epitopes can, e.g., consist of surface-accessible amino acid residues and/or sugar side chains and may have specific three-dimensional structural characteristics, as well as specific charge characteristics. Conformational and non-conformational epitopes are distinguished in that the binding to the former but not the latter may be lost in the presence of denaturing solvents. An epitope may comprise amino acid residues that are directly involved in the binding, and other amino acid residues, which are not directly involved in the binding. Methods for identifying an epitope to which an antigen-binding domain binds are known in the art.

The term "treatment" means to ameliorate at least one symptom of a disorder. In some examples, the disorder being treated is cancer and to ameliorate at least one symptom of cancer includes reducing aberrant proliferation, gene expression, signaling, translation, and/or secretion of factors. Generally, the methods of treatment include administering a therapeutically effective amount of composition that reduces at least one symptom of a disorder to a subject who is in need of, or who has been determined to be in need of such treatment.

The term "CD3/CD28-binding agent" means an agent that binds specifically to both CD3 and CD28 on the surface of a mammalian cell (e.g., a T cell, such as a T_{reg} cell) and induces downstream signaling in the mammalian cell. Non-limiting examples of CD3/CD28-binding agents are described herein.

The term "IL-2 receptor activating agent" means an agent that binds specifically to an IL-2 receptor present on a surface of a mammalian cell (e.g., a T cell, such as a T_{reg} cell) and induces downstream signaling in the mammalian cell. Non-limiting examples of IL-2 receptor activating agents are described herein.

The term "IgG1 antibody construct" means a single chain polypeptide or a multi-chain polypeptide that includes at least one antigen-binding domain that binds to a soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and at least one IgG1 Fc domain. Non-limiting examples of IgG1 antibody constructs are described herein.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Methods and materials are described herein for use in the present invention; other, suitable methods and materials known in the art can also be used. The materials, methods, and examples are illustrative only and not intended to be limiting. All publications, patent applications, patents, sequences, database entries, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control.

Other features and advantages of the invention will be apparent from the following detailed description and figures, and from the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a diagram showing expansion of Foxp3$^+$ T_{reg} cells using either (1) 2t2, CD3/CD28 beads, and rapamycin or (2) recombinant human IL-2, CD3/CD28 beads, and rapamycin.

FIGS. **2**A-**2**I are diagrams showing an increase in Helios$^+$, CTLA-4$^+$, CD39$^+$, CD62L$^+$, CD25$^+$, CD103$^+$, CD69$^+$, PD1$^+$, and CD49b$^+$ markers after stimulation of T_{reg} cells with (1) 2t2, CD3/CD28 beads, and rapamycin or (2) recombinant human IL-2, CD3/CD28 beads, and rapamycin.

FIGS. **3**A-**3**D are diagrams showing glucose metabolism levels upon activation of human T_{reg} cells with (1) 2t2, CD3/CD28 beads, and rapamycin, or (2) recombinant human IL-2, CD3/CD28 beads, and rapamycin. FIG. **3**A shows levels of glycolysis of T_{reg} cells. FIG. **3**B shows levels of glycolytic capacity of T_{reg} cells. FIG. **3**C shows levels of glycolytic reserve of T_{reg} cells. FIG. **3**D shows levels of non-glycolytic acidification of T_{reg} cells.

FIG. **4** is a diagram showing the suppression of autologous T responder cells by T_{reg} cells previously stimulated with (1) 2t2, CD3/CD28 beads, and rapamycin, or (2) recombinant human IL-2, CD3/CD28 beads, and rapamycin.

FIG. **5** is a diagram showing the levels of IL-10 secretion by T_{reg} cells in a T responder cell suppression assay, where the T_{reg} cells were previously stimulated with either (1) 2t2 and CD3/CD28 beads in the presence or absence of rapamycin, or (2) recombinant human IL-2 and CD3/CD28 beads in the presence or absence of rapamycin.

FIGS. **6**A-**6**B show exemplary diagrams for a fusion protein generated including anti-CD3scFv/TF/IL15D8N mutant (3t15*) and anti-CD28scFv/IL15RαSu (28s) fusion proteins.

FIG. **7** is a graph showing 3t15*-28s purification elution by immunoaffinity chromatography from an anti-TF antibody affinity column.

FIG. **8** shows size exclusion chromatography analysis of 3t15*-28s.

FIG. **9** shows an SDS gel of 3t15*-28s purified from anti-TF antibody affinity chromatography. Lane 1: 3t15*-28s; Lane 2: SeeBlue Plus 2 prestained standards (numbers on the left side indicate molecular weights in kilodaltons).

FIG. **10**A is a graph showing interaction between 3t15*-28s and CD28.

FIG. **10**B is a graph showing interaction between 3t15*-28s and CD3.

FIG. **11** is a graph showing the expansion of $Foxp3^+$ regulatory T cells in media containing 2t2 and 3t15*-28s.

FIG. **12** is a graph showing the level of proliferating human immune cells following incubation in media containing 2t2 and 3t15*-28s with or without anti-TF antibody.

FIG. **13**A shows flow cytometry histograms indicating the purity of $CD39^+$ regulatory T cells isolated with different purification schemes.

FIG. **13**B shows a flow cytometry histogram indicating the purity of $CD39^+$ regulatory T cells isolated using an anti-CD39 Ab reagent.

FIG. **14** shows a bar graph depicting the level of T responder cell proliferation following incubation with regulatory T cells isolated with different methods.

FIG. **15** shows a graph showing T responder cell proliferation (as assessed by levels of IL-10) following incubation with T regulatory cells isolated using different methods.

DETAILED DESCRIPTION

Provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell (e.g., any of the T_{reg} cells described herein), a population of T_{reg} cells produced using any of the methods described herein, a composition including any of these populations of T_{reg} cells, and methods of treating a subject that include administering any of these populations of T_{reg} cells or compositions.

Tissue Factor

Human tissue factor is a 263 amino-acid transmembrane protein containing three domains: (1) a 219-amino acid N-terminal extracellular domain (residues 1-219); (2) a 22-amino acid transmembrane domain (residues 220-242); and (3) a 21-amino acid cytoplasmic C-terminal tail (residues 242-263) ((UniProtKB Identifier Number: P13726). The cytoplasmic tail contains two phosphorylation sites at Ser253 and Ser258, and one S-palmitoylation site at Cys245. Deletion or mutation of the cytoplasmic domain was not found to affect tissue factor coagulation activity. Tissue factor has one S-palmitoylation site in the intracellular domain of the protein at Cys245. The Cys245 is located at the amino acid terminus of the intracellular domain and close to the membrane surface. The tissue factor transmembrane domain is composed of a single-spanning α-helix.

The extracellular domain of tissue factor, composed of two fibronectin type III domains, is connected to the transmembrane domain through a six-amino acid linker. This linker provides conformational flexibility to decouple the tissue factor extracellular domain from its transmembrane and cytoplasmic domains. Each tissue factor fibronectin type III module is composed of two overlapping β sheets with the top sheet domain containing three antiparallel β-strands and the bottom sheet containing four β-strands. The β-strands are connected by β-loops between strand βA and βB, βC and βD, and βE and βF, all of which are conserved in conformation in the two modules. There are three short α-helix segments connecting the β-strands. A unique feature of tissue factor is a 17-amino acid β-hairpin between strand β10 and strand β11, which is not a common element of the fibronectin superfamily. The N-terminal domain also contains a 12 amino acid loop between β6F and β7G that is not present in the C-terminal domain and is unique to tissue factor. Such a fibronectin type III domain structure is a feature of the immunoglobulin-like family of protein folds and is conserved among a wide variety of extracellular proteins.

The zymogen FVII is rapidly converted to FVIIa by limited proteolysis once it binds to tissue to form the active tissue factor-FVIIa complex. The FVIIa, which circulates as an enzyme at a concentration of approximately 0.1 nM (1% of plasma FVII), can also bind directly to tissue factor. The allosteric interaction between tissue factor and FVIIa on the tissue factor-FVIIa complex greatly increases the enzymatic activity of FVIIa: an approximate 20- to 100-fold increase in the rate of hydrolysis of small, chromogenic peptidyl substrates, and nearly a million-fold increase in the rate of activation of the natural macromolecular substrates FIX and FX. In concert with allosteric activation of the active site of FVIIa upon binding to tissue factor, the formation of tissue factor-FVIIa complex on phospholipid bilayer (i.e., upon exposure of phosphatidyl-L-serine on membrane surfaces) increases the rate of FIX or FX activation, in a Ca^{2+}-dependent manner, an additional 1,000-fold. The roughly million-fold overall increase in FX activation by tissue factor-FVIIa-phospholipid complex relative to free FVIIa is a critical regulatory point for the coagulation cascade.

FVII is a ~50 kDa, single-chain polypeptide consisting of 406 amino acid residues, with an N-terminal γ-carboxyglutamate-rich (GLA) domain, two epidermal growth factor-like domains (EGF1 and EFG2), and a C-terminal serine protease domain. FVII is activated to FVIIa by a specific proteolytic cleavage of the $Ile\text{-}^{154}$-Arg^{152} bond in the short linker region between the EGF2 and the protease domain. This cleavage results in the light and heavy chains being held together by a single disulfide bond of Cys^{135} and Cys^{262}. FVIIa binds phospholipid membrane in a Ca^{2+}-dependent manner through its N-terminal GLA-domain. Immediately C-terminal to the GLA domain is an aromatic stack and two EGF domains. The aromatic stack connects the GLA to EGF1 domain which binds a single Ca^{2+} ion. Occupancy of this Ca^{2+}-binding site increases FVIIa amidolytic activity and tissue factor association. The catalytic triad consist of His^{193}, Asp^{242}, and Ser^{344}, and binding of a single Ca^{2+} ion within the FVIIa protease domain is critical for its catalytic activity. Proteolytic activation of FVII to FVIIa frees the newly formed amino terminus at Ile^{153} to fold back and be inserted into the activation pocket forming a salt bridge with the carboxylate of Asp^{343} to generate the oxyanion hole. Formation of this salt bridge is critical for FVIIa activity. However, oxyanion hole formation does not occur in free FVIIa upon proteolytic activation. As a result, FVIIa circulates in a zymogen-like state that is poorly recognized by plasma protease inhibitors, allowing it to circulate with a half-life of approximately 90 minutes.

Tissue factor-mediated positioning of the FVIIa active site above the membrane surface is important for FVIIa towards cognate substrates. Free FVIIa adopts a stable, extended structure when bound to the membrane with its active site positioned ~80 Å above the membrane surface. Upon FVIIa binding to tissue factor, the FVa active site is repositioned ~6 Å closer to the membrane. This modulation may aid in a proper alignment of the FVIIa catalytic triad with the target substrate cleavage site. Using GLA-domainless FVIIa, it has been shown that the active site was still positioned a similar distance above the membrane, demonstrating that tissue factor is able to fully support FVIIa active site positioning even in the absence of FVIIa-membrane interaction. Additional data showed that tissue factor supported full FVIIa proteolytic activity as long as the tissue factor extracellular domain was tethered in some way to the membrane surface. However, raising the active site of FVIIa greater than 80 Å above the membrane surface greatly reduced the ability of the tissue factor-FVIIa complex to activate FX but did not diminish tissue factor-FVIIa amidolytic activity.

Alanine scanning mutagenesis has been used to assess the role of specific amino acid side chains in the tissue factor extracellular domain for interaction with FVIIa (Gibbs et al., *Biochemistry* 33(47): 14003-14010, 1994; Schullek et al., *J Biol Chem* 269(30): 19399-19403, 1994). Alanine substitution identified a limited number of residue positions at which alanine replacements cause 5- to 10-fold lower affinity for FVIIa binding. Most of these residue side chains were found to be well-exposed to solvent in the crystal structure, concordant with macromolecular ligand interaction. The FVIIa ligand-binding site is located over an extensive region at the boundary between the two modules. In the C-module, residues Arg^{135} and Phe^{140} located on the protruding B-C loop provide an independent contact with FVIIa. Leu^{133} is located at the base of the fingerlike structure and packed into the cleft between the two modules. This provides continuity to a major cluster of important binding residues consisting of Lys^{20}, Thr^{60}, Asp^{58}, and Ile^{22}. Thr^{60} is only partially solvent-exposed and may play a local structural role rather than making a significant contact with ligand. The binding site extends onto the concave side of the intermodule angle involving Glu^{24} and Gln^{110}, and potentially the more distant residue Val^{207}. The binding region extends from Asp^{58} onto a convex surface area formed by Lys^{48}, Lys^{46}, Gln^{37}, Asp^{44}, and Trp^{45}. Trp^{45} and Asp^{44} do not interact independently with FVIIa, indicating that the mutational effect at the Trp^{45} position may reflect a structural importance of this side chain for the local packing of the adjacent Asp^{44} and Gln^{37} side chain. The interactive area further includes two surface-exposed aromatic residues, Phe^{76} and Tyr^{78}, which form part of the hydrophobic cluster in the N-module.

The known physiologic substrates of tissue factor-FVIIa are FVII, FIX, and FX and certain proteinase-activated receptors. Mutational analysis has identified a number of residues that, when mutated, support full FVIIa amidolytic activity towards small peptidyl substrates but are deficient in their ability to support macromolecular substrate (i.e., FVII, FIX, and FX) activation (Ruf et al., *J Biol Chem* 267(31): 22206-22210, 1992; Ruf et al., *J Biol Chem* 267(9): 6375-6381, 1992; Huang et al., *J Biol Chem* 271(36): 21752-21757, 1996; Kirchhofer et al., *Biochemistry* 39(25): 7380-7387, 2000). The tissue factor loop region at residues 159-165, and residues in or adjacent to this flexible loop have been shown to be critical for the proteolytic activity of the tissue factor-FVIIa complex. This defines the proposed substrate-binding exosite region of tissue factor that is quite distant from the FVIIa active site. A substitution of the glycine residue by a marginally bulkier residue alanine, significantly impairs tissue factor-FVIIa proteolytic activity. This suggests that the flexibility afforded by glycine is critical for the loop of residues 159-165 for tissue factor macromolecular substrate recognition.

The residues Lys^{165} and Lys^{166} have also been demonstrated to be important for substrate recognition and binding. Mutation of either of these residues to alanine results in a significant decrease in the tissue factor co-factor function. Lys^{165} and Lys^{166} face away from each other, with Lys^{165} pointing towards FVIIa in most tissue factor-FVIIa structures, and Lys^{166} pointing into the substrate binding exosite region in the crystal structure. Putative salt bridge formation between Lys^{165} of and Gla^{35} of FVIIa would support the notion that tissue factor interaction with the GLA domain of FVIIa modulates substrate recognition. These results suggest that the C-terminal portion of the tissue factor ectodomain directly interacts with the GLA-domain, the possible adjacent EGF1 domains, of FIX and FX, and that the presence of the FVIIa GLA-domain may modulate these interactions either directly or indirectly.

Multi-Chain Chimeric Polypeptides

Non-limiting examples of cell activating agents are multi-chain chimeric polypeptides that include: (a) a first chimeric polypeptide including: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide including: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, where the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the first target-binding domain (e.g., any of the first target-binding domains described herein) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) directly abut each other in the first chimeric polypeptide. In some embodiments of any of the multi-chain chimeric polypeptides described herein, the first chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the first target-binding domain (e.g., any of the exemplary first target-binding domains described herein) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) in the first chimeric polypeptide.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein) directly abut each other in the first chimeric polypeptide. In some embodiments of any of the multi-chain chimeric polypeptides described herein, the first chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the second domain of the pair of affinity domains (e.g., any of the exemplary second domains of any of the exemplary pairs of affinity domains described herein) and the second target-binding domain (e.g., any of the exemplary second target-binding domains described herein) directly abut each other in the second chimeric polypeptide. In some embodiments of any of the multi-chain chimeric polypeptides described herein, the second chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the second domain of the pair of affinity domains (e.g., any of the exemplary second domains of any of the exemplary pairs of affinity domains described herein) and the second target-binding domain (e.g., any of the exemplary second target-binding domains described herein) in the second chimeric polypeptide.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) bind specifically to a target selected from the group consisting of: CD16a, CD28, CD3, CD33, CD20, CD19, CD22, CD123, IL-1R, IL-1, VEGF, IL-6R, IL-4, IL-10, PDL-1, TIGIT, PD-1, TIM3, CTLA4, MICA, MICB, IL-6, IL-8, TNFα, CD26, CD36, ULBP2, CD30, CD200, IGF-1R, MUC4AC, MUC5AC, Trop-2, CMET, EGFR, HER1, HER2, HER3, PSMA, CEA, B7H3, EPCAM, BCMA, P-cadherin, CEACAM5, a UL16-binding protein, HLA-DR, DLL4, TYRO3, AXL, MER, CD122, CD155, PDGF-DD, a ligand of TGF-β receptor II (TGF-βRII), a ligand of TGF-βRIII, a ligand of DNAM1, a ligand of NKp46, a ligand of NKp44, a ligand of NKG2D, a ligand of NKp30, a ligand for a scMHCI, a ligand for a scMHCII, a ligand for a scTCR, a receptor for IL-1, a receptor for IL-2, a receptor for IL-3, a receptor for IL-7, a receptor for IL-8, a receptor for IL-10, a receptor for IL-12, a receptor for IL-15, a receptor for IL-17, a receptor for IL-18, a receptor for IL-21, a receptor for PDGF-DD, a receptor for stem cell factor (SCF), a receptor for stem cell-like tyrosine kinase 3 ligand (FLT3L), a receptor for MICA, a receptor for MICB, a receptor for a ULP16-binding protein, a receptor for CD155, a receptor for CD122, and a receptor for CD28.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) is a soluble interleukin or cytokine protein. Non-limiting examples of soluble interleukin proteins and soluble cytokine proteins include: IL-1, IL-2, IL-3, IL-7, IL-8, IL-10, IL-12, IL-15, IL-17, IL-18, IL-21, PDGF-DD, and SCF.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) is a soluble interleukin or cytokine receptor. Non-limiting examples of soluble interleukin receptors and soluble cytokine receptors include: a soluble TGF-β receptor II (TGF-βRII), a soluble TGF-βRIII, a soluble NKG2D, a soluble NKp30, a soluble NKp44, a soluble NKp46, a soluble DNAM-1, a scMHCI, a scMHCII, a scTCR, a soluble CD155, a soluble CD122, a soluble CD3, or a soluble CD28.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain and the second target-binding domain is a soluble interleukin or cytokine receptor. In some embodiments of any of the multi-chain chimeric polypeptides described herein, the soluble receptor is a soluble TGF-β receptor II (TGF-βRII) a soluble TGF-βRIII, a soluble receptor for TNFα, a soluble receptor for IL-4, or a soluble receptor for IL-10.

Soluble Tissue Factor Domains

In some embodiments, the soluble tissue factor domain can be a wildtype tissue factor polypeptide lacking the signal sequence, the transmembrane domain, and the intracellular domain. In some examples, the soluble tissue factor domain can be a tissue factor mutant, wherein a wildtype tissue factor polypeptide lacking the signal sequence, the transmembrane domain, and the intracellular domain, and has been further modified at selected amino acids. In some examples, the soluble tissue factor domain can be a soluble human tissue factor domain. In some examples, the soluble tissue factor domain can be a soluble mouse tissue factor domain. In some examples, the soluble tissue factor domain can be a soluble rat tissue factor domain. Non-limiting examples of soluble human tissue factor domains, a mouse soluble tissue factor domain, a rat soluble tissue factor domain, and mutant soluble tissue factor domains are shown below.

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Exemplary Soluble Human Tissue Factor Domain
                                                     (SEQ ID NO: 2)
SGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTD
TECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEFTPYLETNLGQ
PTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKDLIYTLYYWKSSSSGK
KTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVECMGQEKGEFRE
Exemplary Nucleic Acid Encoding Soluble Human Tissue
Factor Domain
                                                    (SEQ ID NO: 13)
AGCGGCACAACCAACACAGTCGCTGCCTATAACCTCACTTGGAAGAGCACCA
ACTTCAAAACCATCCTCGAATGGGAACCCAAACCCGTTAACCAAGTTTACACC
GTGCAGATCAGCACCAAGTCCGGCGACTGGAAGTCCAAATGTTTCTATACCAC
CGACACCGAGTGCGATCTCACCGATGAGATCGTGAAAGATGTGAAACAGACC
TACCTCGCCCGGGTGTTTAGCTACCCCGCCGGCAATGTGGAGAGCACTGGTTC
CGCTGGCGAGCCTTTATACGAGAACAGCCCCGAATTTACCCCTTACCTCGAGA
CCAATTTAGGACAGCCCACCATCCAAAGCTTTGAGCAAGTTGGCACAAAGGT
GAATGTGACAGTGGAGGACGAGCGGACTTTAGTGCGGCGGAACAACACCTTT
CTCAGCCTCCGGGATGTGTTCGGCAAAGATTTAATCTACACACTGTATTACTGG
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AAGTCCTCTTCCTCCGGCAAGAAGACAGCTAAAACCAACACAAACGAGTTTT
TAATCGACGTGGATAAAGGCGAAAACTACTGTTTCAGCGTGCAAGCTGTGATC
CCCTCCCGGACCGTGAATAGGAAAAGCACCGATAGCCCCGTTGAGTGCATGG
GCCAAGAAAAGGGCGAGTTCCGGGAG

Exemplary Soluble Mouse Tissue Factor Domain

(SEQ ID NO: 14)

agipekafnltwistdfktilewqpkptnytytvqisdrsrnwknkcfstt
dtecdltdeivkdvtwayeakvlsvprrnsvhgdgdqlvihgeeppftnap
kflpyrdtnlgqpviqqfeqdgrklnvvvkdsltlvrkngtfltlrqvfgk
dlgyiityrkgsstgkktnitntnefsidveegvsycffvqamifsrktnq
nspgsstvcteqwksflge

Exemplary Soluble Rat Tissue Factor Domain

(SEQ ID NO: 15)

Agtppgkafnltwistdfktilewqpkptnytytvqisdrsrnwkykctgt
tdtecdltdeivkdvnwtyearvlsvpwinsthgketlfgthgeeppftna
rkflpyrdtkigqpviqkyeqggtklkvtvkdsftlvrkngtfltlrqvfg
ndlgyiltyrkdsstgrktntthtneflidvekgvsycffaqavifsrktn
hkspesitkcteqwksvlge

Exemplary Mutant Soluble Human Tissue Factor Domain

(SEQ ID NO: 16)

SGTTNTVAAYNLTWKSTNFATALEWEPKPVNQVYTVQISTKSGDWKSKCFYTT
DTECALTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEFTPYLETNL
GQPTIQSFEQVGTKVNVTVEDERTLVARNNTALSLRDVFGKDLIYTLYYWKSSSS
GKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVECMGQEKGEF
RE

Exemplary Mutant Soluble Human Tissue Factor Domain

(SEQ ID NO: 17)

SGTTNTVAAYNLTWKSTNFATALEWEPKPVNQVYTVQISTKSGDAKSKCFYTTD
TECALTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLAENSPEFTPYLETNLG
QPTIQSFEQVGTKVNVTVEDERTLVARNNTALSLRDVFGKDLIYTLYYWKSSSSG
KKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVECMGQEKGEFR
E

In some embodiments, a soluble tissue factor domain can include a sequence that is at least 70% identical, at least 72% identical, at least 74% identical, at least 76% identical, at least 78% identical, at least 80% identical, at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical to SEQ ID NO: 2, 14, 15, 16, or 17. In some embodiments, a soluble tissue factor domain can include a sequence of SEQ ID NO: 2, 14, 15, 16, or 17 with one to twenty amino acids (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20) amino acids removed from its N-terminus and/or one to twenty amino acids (e.g., 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20) amino acids removed from its C-terminus.

As can be appreciated in the art, one skilled in the art would understand that mutation of amino acids that are conserved between different mammalian species is more likely to decrease the activity and/or structural stability of the protein, while mutation of amino acids that are not conserved between different mammalian species is less likely to decrease the activity and/or structural stability of the protein.

In some examples, the soluble tissue factor domain is not capable of binding to Factor VIIa. In some examples, the soluble tissue factor domain does not convert inactive Factor X into Factor Xa. In some embodiments, any of the single-chain chimeric polypeptides or multi-chain chimeric polypeptides described herein does not stimulate blood coagulation in a mammal.

In some examples, the soluble tissue factor domain can be a soluble human tissue factor domain. In some embodiments, the soluble tissue factor domain can be a soluble mouse tissue factor domain. In some embodiments, the soluble tissue factor domain can be a soluble rat tissue factor domain.

In some examples, the soluble tissue factor domain does not include one or more (e.g., two, three, four, five, six or seven) of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some examples, the soluble tissue factor domain does not include any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some examples, the soluble tissue factor domain includes one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments, the mutant soluble tissue factor possesses the amino acid sequence of SEQ ID NO: 16 or SEQ ID NO: 17.

In some examples, the soluble tissue factor domain can be encoded by a nucleic acid including a sequence that is at least 70% identical, at least 72% identical, at least 74% identical, at least 76% identical, at least 78% identical, at least 80% identical, at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical to SEQ ID NO: 13.

In some embodiments, the soluble tissue factor domain can have a total length of about 20 amino acids to about 220 amino acids, about 20 amino acids to about 215 amino acids, about 20 amino acids to about 210 amino acids, about 20 amino acids to about 205 amino acids, about 20 amino acids to about 200 amino acids, about 20 amino acids to about 195 amino acids, about 20 amino acids to about 190 amino acids, about 20 amino acids to about 185 amino acids, about 20 amino acids to about 180 amino acids, about 20 amino acids to about 175 amino acids, about 20 amino acids to about 170 amino acids, about 20 amino acids to about 165 amino acids, about 20 amino acids to about 160 amino acids, about 20 amino acids to about 155 amino acids, about 20 amino acids to about 150 amino acids, about 20 amino acids to about 145 amino acids, about 20 amino acids to about 140 amino acids, about 20 amino acids to about 135 amino acids, about 20 amino acids to about 130 amino acids, about 20 amino acids to about 125 amino acids, about 20 amino acids to about 120 amino acids, about 20 amino acids to about 115 amino acids, about 20 amino acids to about 110 amino acids, about 20 amino acids to about 105 amino acids, about 20 amino acids to about 100 amino acids, about 20 amino acids to about 95 amino acids, about 20 amino acids to about 90 amino acids, about 20 amino acids to about 85 amino acids, about 20 amino acids to about 80 amino acids, about 20 amino acids to about 75 amino acids, about 20 amino acids to about 70 amino acids, about 20 amino acids to about 60 amino acids, about 20 amino acids to about 50 amino acids, about 20 amino acids to about 40 amino acids, about 20 amino acids to about 30 amino acids, about 30 amino acids to about 220 amino acids, about 30 amino acids to about 215 amino acids, about 30 amino acids to about 210 amino acids, about 30 amino acids to about 205 amino acids, about 30 amino acids to about 200 amino acids, about 30 amino acids to about 195 amino acids, about 30 amino acids to about 190 amino acids, about 30 amino acids to about 185 amino acids, about 30 amino acids to about 180 amino acids, about 30 amino acids to about 175 amino acids, about 30 amino acids to about 170 amino acids, about 30 amino acids to about 165 amino acids, about 30 amino acids to about 160 amino acids, about 30 amino acids to about 155 amino acids, about 30 amino acids to about 150 amino acids, about 30 amino acids to about 145 amino acids, about 30 amino acids to about 140 amino acids, about 30 amino acids to about 135 amino acids, about 30 amino acids to about 130 amino acids, about 30 amino acids to about 125 amino acids, about 30 amino acids to about 120 amino acids, about 30 amino acids to about 115 amino acids, about 30 amino acids to about 110 amino acids, about 30 amino acids to about 105 amino acids, about 30 amino acids to about 100 amino acids, about 30 amino acids to about 95 amino acids, about 30 amino acids to about 90 amino acids, about 30 amino acids to about 85 amino acids, about 30 amino acids to about 80 amino acids, about 30 amino acids to about 75 amino acids, about 30 amino acids to about 70 amino acids, about 30 amino acids to about 60 amino acids, about 30 amino acids to about 50 amino acids, about 30 amino acids to about 40 amino acids, about 40 amino acids to about 220 amino acids, about 40 amino acids to about 215 amino acids, about 40 amino acids to about 210 amino acids, about 40 amino acids to about 205 amino acids, about 40 amino acids to about 200 amino acids, about 40 amino acids to about 195 amino acids, about 40 amino acids to about 190 amino acids, about 40 amino acids to about 185 amino acids, about 40 amino acids to about 180 amino acids, about 40 amino acids to about 175 amino acids, about 40 amino acids to about 170 amino acids, about 40 amino acids to about 165 amino acids, about 40 amino acids to about 160 amino acids, about 40 amino acids to about 155 amino acids, about 40 amino acids to about 150 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In some embodiments, the soluble tissue factor domain can comprise or consist of a soluble wildtype human tissue factor (or any sequence therefrom).

Linker Sequences

In some embodiments, the linker sequence can be a flexible linker sequence. Non-limiting examples of linker sequences that can be used are described in Klein et al., Protein Engineering, Design & Selection Vol. 27, No. 10, pp. 325-330, 2014; Priyanka et al., Protein Sci., 2013 February; 22(2): 153-167. In some examples, the linker sequence is a synthetic linker sequence.

In some embodiments of any of the single-chain chimeric polypeptides described herein can include one, two, three, four, five, six, seven, eight, nine, or ten linker sequence(s) (e.g., the same or different linker sequences, e.g., any of the exemplary linker sequences described herein or known in the art). In some embodiments of any of the single-chain chimeric polypeptides described herein can include one, two, three, four, five, six, seven, eight, nine, or ten linker sequence(s) (e.g., the same or different linker sequences, e.g., any of the exemplary linker sequences described herein or known in the art).

In some embodiments of any of the first chimeric polypeptides and/or the second chimeric polypeptides described herein can include one, two, three, four, five, six, seven, eight, nine, or ten linker sequence(s) (e.g., the same or different linker sequences, e.g., any of the exemplary linker sequences described herein or known in the art). In some embodiments of any of the first chimeric polypeptides and/or the second chimeric polypeptides described herein can include one, two, three, four, five, six, seven, eight, nine, or ten linker sequence(s) (e.g., the same or different linker sequences, e.g., any of the exemplary linker sequences described herein or known in the art).

In some embodiments, a linker sequence can have a total length of 1 amino acid to about 100 amino acids, 1 amino acid to about 90 amino acids, 1 amino acid to about 80 amino acids, 1 amino acid to about 70 amino acids, 1 amino acid to about 60 amino acids, 1 amino acid to about 50 amino acids, 1 amino acid to about 45 amino acids, 1 amino acid to about 40 amino acids, 1 amino acid to about 35 amino acids, 1 amino acid to about 30 amino acids, 1 amino acid to about 25 amino acids, 1 amino acid to about 24 amino acids, 1 amino acid to about 22 amino acids, 1 amino acid to about 20 amino acids, 1 amino acid to about 18 amino acids, 1 amino acid to about 16 amino acids, 1 amino acid to about 14 amino acids, 1 amino acid to about 12 amino acids, 1 amino acid to about 10 amino acids, 1 amino acid to about 8 amino acids, 1 amino acid to about 6 amino acids, 1 amino acid to about 4 amino acids, about 2 amino acids to about 100 amino acids, about 2 amino acids to about 90 amino acids, about 2 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In some embodiments, the linker is rich in glycine (Gly or G) residues. In some embodiments, the linker is rich in serine (Ser or S) residues. In some embodiments, the linker is rich in glycine and serine residues. In some embodiments, the linker has one or more glycine-serine residue pairs (GS), e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more GS pairs. In some embodiments, the linker has one or more Gly-Gly-Gly-Ser (GGGS) (SEQ ID NO: 111) sequences, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more GGGS (SEQ ID NO: 111) sequences. In some embodiments, the linker has one or more Gly-Gly-Gly-Gly-Ser (GGGGS) (SEQ ID NO: 110) sequences, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more GGGGS (SEQ ID NO: 110) sequences. In some embodiments, the linker has one or more Gly-Gly-Ser-Gly (GGSG) (SEQ ID NO: 112) sequences, e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or more GGSG (SEQ ID NO: 112) sequences.

In some embodiments, the linker sequence can comprise or consist of GGGGSGGGGSGGGGS (SEQ ID NO: 18). In some embodiments, the linker sequence can be encoded by a nucleic acid comprising or consisting of: GGCGGTGGAGGATCCGGAGGAGGTGGCTCCGGCGGCGGAGGATCT (SEQ ID NO: 19). In some embodiments, the linker sequence can comprise or consist of: GGGSGGGS (SEQ ID NO: 20).

Target-Binding Domains

In some embodiments of any of the single- or multi-chain chimeric polypeptides described herein, the first target-binding domain, the second target-binding domain, and/or the additional one or more target-binding domains can be an antigen-binding domain (e.g., any of the exemplary antigen-binding domains described herein or known in the art), a soluble interleukin or cytokine protein (e.g., any of the exemplary soluble interleukin proteins or soluble cytokine proteins described herein), and a soluble interleukin or cytokine receptor (e.g., any of the exemplary soluble interleukin receptors or soluble cytokine receptors described herein).

In some embodiments of any of the single- or multi-chain chimeric polypeptides described herein, the first target-binding domain, the second target-binding domain, and/or the one or more additional target-binding domains can each independent have a total number of amino acids of about 5 amino acids to about 1000 amino acids, about 5 amino acids to about 950 amino acids, about 5 amino acids to about 900 amino acids, about 5 amino acids to about 850 amino acids, about 5 amino acids to about 800 amino acids, about 5 amino acids to about 750 amino acids, about 5 amino acids to about 700 amino acids, about 5 amino acids to about 650 amino acids, about 5 amino acids to about 600 amino acids, about 5 amino acids to about 550 amino acids, about 5 amino acids to about 500 amino acids, about 5 amino acids to about 450 amino acids, about 5 amino acids to about 400 amino acids, about 5 amino acids to about 350 amino acids, about 5 amino acids to about 300 amino acids, about 5 amino acids to about 280 amino acids, about 5 amino acids to about 260 amino acids, about 5 amino acids to about 240 amino acids, about 5 amino acids to about 220 amino acids, about 5 amino acids to about 200 amino acids, about 5 amino acids to about 195 amino acids, about 5 amino acids to about 190 amino acids, about 5 amino acids to about 185 amino acids, about 5 amino acids to about 180 amino acids, about 5 amino acids to about 175 amino acids, about 5 amino acids to about 170 amino acids, about 5 amino acids to about 165 amino acids, about 5 amino acids to about 160 amino acids, about 5 amino acids to about 155 amino acids, about 5 amino acids to about 150 amino acids, about 5 amino acids to about 145 amino acids, about 5 amino acids to about 140 amino acids, about 5 amino acids to about 135 amino acids, about 5 amino acids to about 130 amino acids, about 5 amino acids to about 125 amino acids, about 5 amino acids to about 120 amino acids, about 5 amino acids to about 115 amino acids, about 5 amino acids to about 110 amino acids, about 5 amino acids to about 105 amino acids, about 5 amino acids to about 100 amino acids, about 5 amino acids to about 95 amino acids, about 5 amino acids to about 90 amino acids, about 5 amino acids to about 85 amino acids, about 5 amino acids to about 80 amino acids, about 5 amino acids to about 75 amino acids, about 5 amino acids to about 70 amino acids, about 5 amino acids to about 65 amino acids, about 5 amino acids to about 60 amino acids, about 5 amino acids to about 55 amino acids, about 5 amino acids to about 50 amino acids, about 5 amino acids to about 45 amino acids, about 5 amino acids to about 40 amino acids, about 5 amino acids to about 35 amino acids, about 5 amino acids to about 30 amino acids, about 5 amino acids to about 25 amino acids, about 5 amino acids to about 20 amino acids, about 5 amino acids to about 15 amino acids, about 5 amino acids to about 10 amino acids,

Any of the target-binding domains described herein can bind to its target with a dissociation equilibrium constant (K_D) of less than 1×10^{-7}M, less than 1×10^{-8} M, less than 1×10^{-9}M, less than 1×10^{-10}M, less than 1×10^{-11} M, less than 1×10^{-12} M, or less than 1×10^{-13} M. In some embodiments, the antigen-binding protein construct provided herein can bind to an identifying antigen with a K_D of about 1×10^{-3} M to about 1×10^{-5} M, about 1×10^{-4} M to about 1×10^{-6} M, about 1×10^{-5} M to about 1×10^{-7} M, about 1×10^{-6} M to about 1×10^{-8} M, about 1×10^{-7} M to about 1×10^{-9}M, about 1×10^{-8} M to about 1×10^{-10} M, or about 1×10^{-9} M to about 1×10^{-11} M (inclusive).

Any of the target-binding domains described herein can bind to its target with a K_D of between about 1 pM to about 30 nM (e.g., about 1 pM to about 25 nM, about 1 pM to about 20 nM, about 1 pM to about 15 nM, about 1 pM to about 10 nM, about 1 pM to about 5 nM, about 1 pM to about 2 nM, about 1 pM to about 1 nM, about 1 pM to about 950 pM, about 1 pM to about 900 pM, about 1 pM to about 850 pM, about 1 pM to about 800 pM, about 1 pM to about 750 pM, about 1 pM to about 700 pM, about 1 pM to about 650 pM, about 1 pM to about 600 pM, about 1 pM to about 550 pM, about 1 pM to about 500 pM, about 1 pM to about 450 pM, about 1 pM to about 400 pM, about 1 pM to about 350 pM, about 1 pM to about 300 pM, about 1 pM to about 250 pM, about 1 pM to about 200 pM, about 1 pM to about 150 pM, about 1 pM to about 100 pM, about 1 pM to about 90 pM, about 1 pM to about 80 pM, about 1 pM to about 70 pM, about 1 pM to about 60 pM, about 1 pM to about 50 pM, about 1 pM to about 40 pM, about 1 pM to about 30 pM, about 1 pM to about 20 pM, about 1 pM to about 10 pM, about 1 pM to about 5 pM, about 1 pM to about 4 pM, about 1 pM to about 3 pM, about 1 pM to about 2 pM).

Any of the target-binding domains described herein can bind to its target with a K_D of between about 1 nM to about 10 nM (e.g., about 1 nM to about 9 nM, about 1 nM to about 8 nM, about 1 nM to about 7 nM, about 1 nM to about 6 nM, about 1 nM to about 5 nM, about 1 nM to about 4 nM, about 1 nM to about 3 nM, about 1 nM to about 2 nM, about 2 nM to about 10 nM, about 2 nM to about 9 nM, about 2 nM to about 8 nM, about 2 nM to about 7 nM, about 2 nM to about 6 nM, about 2 nM to about 5 nM, about 2 nM to about 4 nM, about 2 nM to about 3 nM, about 3 nM to about 10 nM, about 3 nM to about 9 nM, about 3 nM to about 8 nM, about 3 nM to about 7 nM, about 3 nM to about 6 nM, about 3 nM to about 5 nM, about 3 nM to about 4 nM, about 4 nM to about 10 nM, about 4 nM to about 9 nM, about 4 nM to about 8 nM, about 4 nM to about 7 nM, about 4 nM to about 6 nM, about 4 nM to about 5 nM, about 5 nM to about 10 nM, about 5 nM to about 9 nM, about 5 nM to about 8 nM, about 5 nM to about 7 nM, about 5 nM to about 6 nM, about 6 nM to about 10 nM, about 6 nM to about 9 nM, about 6 nM to about 8 nM, about 6 nM to about 7 nM, about 7 nM to about 10 nM, about 7 nM to about 9 nM, about 7 nM to about 8 nM, about 8 nM to about 10 nM, about 8 nM to about 9 nM, and about 9 nM to about 10 nM).

A variety of different methods known in the art can be used to determine the K_D values of any of the antigen-binding protein constructs described herein (e.g., an electrophoretic mobility shift assay, a filter binding assay, surface plasmon resonance, and a biomolecular binding kinetics assay, etc.).

Antigen-Binding Domains

In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, the first target-binding domain and the second target-binding domain bind specifically to the same antigen. In some embodiments of these single-chain or multi-chain chimeric polypeptides, the first target-binding domain and the second target-binding domain bind specifically to the same epitope. In some embodiments of these single-chain or multi-chain chimeric polypeptides, the first target-binding domain and the second target-binding domain include the same amino acid sequence.

In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, the first target-binding domain and the second target-binding domain bind specifically to different antigens.

In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain and the second target-binding domain is an antigen-binding domain. In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, the first target-binding domain and the second target-binding domain are each antigen-binding domains. In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, the antigen-binding domain includes or is a scFv or a single domain antibody (e.g., a Va_HH or a V_{NAR} domain).

In some examples, an antigen-binding domain (e.g., any of the antigen-binding domains described herein) can bind specifically to any one of CD16a (see, e.g., those described in U.S. Pat. No. 9,035,026), CD28 (see, e.g., those described in U.S. Pat. No. 7,723,482), CD3 (see, e.g., those described in U.S. Pat. No. 9,226,962), CD33 (see, e.g., those described in U.S. Pat. No. 8,759,494), CD20 (see, e.g., those described in WO 2014/026054), CD19 (see, e.g., those described in U.S. Pat. No. 9,701,758), CD22 (see, e.g., those described in WO 2003/104425), CD123 (see, e.g., those described in WO 2014/130635), IL-1R (see, e.g., those described in U.S. Pat. No. 8,741,604), IL-1 (see, e.g., those described in WO 2014/095808), VEGF (see, e.g., those described in U.S. Pat. No. 9,090,684), IL-6R (see, e.g., those described in U.S. Pat. No. 7,482,436), IL-4 (see, e.g., those described in U.S. Patent Application Publication No. 2012/0171197), IL-10 (see, e.g., those described in U.S. Patent Application Publication No. 2016/0340413), PDL-1 (see, e.g., those described in Drees et al., *Protein Express. Purif.* 94:60-66, 2014), TIGIT (see, e.g., those described in U.S. Patent Application Publication No. 2017/0198042), PD-1 (see, e.g., those described in U.S. Pat. No. 7,488,802), TIM3 (see, e.g., those described in U.S. Pat. No. 8,552,156), CTLA4 (see, e.g., those described in WO 2012/120125), MICA (see, e.g., those described in WO 2016/154585), MICB (see, e.g., those described in U.S. Pat. No. 8,753,640), IL-6 (see, e.g., those described in Gejima et al., *Human Antibodies* 11(4): 121-129, 2002), IL-8 (see, e.g., those described in U.S. Pat. No. 6,117,980), TNFα (see, e.g., those described in Geng et al., *Immunol. Res.* 62(3): 377-385, 2015), CD26 (see, e.g., those described in WO 2017/189526), CD36 (see, e.g., those described in U.S. Patent Application Publication No. 2015/0259429), ULBP2 (see, e.g., those described in U.S. Pat. No. 9,273,136), CD30 (see, e.g., those described in Homach et al., *Scand. J. Immunol.* 48(5): 497-501, 1998), CD200 (see, e.g., those described in U.S. Pat. No. 9,085,623), IGF-1R (see, e.g., those described in U.S. Patent Application Publication No. 2017/0051063), MUC4AC (see, e.g., those described in WO 2012/170470), MUC5AC (see, e.g., those described in U.S. Pat. No. 9,238,084), Trop-2 (see, e.g., those described in WO 2013/068946), CMET (see, e.g., those described in Edwardraja et al., *Biotechnol. Bioeng.* 106(3): 367-375, 2010), EGFR (see, e.g., those described in Akbari et al., *Protein Expr. Purif.* 127:8-15, 2016), HER1 (see, e.g., those described in U.S. Patent Application Publication No. 2013/0274446), HER2 (see, e.g., those described in Cao et al., *Biotechnol. Lett.* 37(7): 1347-1354, 2015), HER3 (see, e.g., those described in U.S. Pat. No. 9,505,843), PSMA (see, e.g., those described in Parker et al., *Protein Expr. Purif.* 89(2): 136-145, 2013), CEA (see, e.g., those described in WO 1995/015341), B7H3 (see, e.g., those described in U.S. Pat. No. 9,371,395), EPCAM (see, e.g., those described in WO 2014/159531), BCMA (see, e.g., those described in Smith et al., *Mol. Ther.* 26(6): 1447-1456, 2018), P-cadherin (see, e.g., those described in U.S. Pat. No. 7,452,537), CEACAM5 (see, e.g., those described in U.S. Pat. No. 9,617,345), a UL16-binding protein (see, e.g., those described in WO 2017/083612), HLA-DR (see, e.g., Pistillo et al., *Exp. Clin. Immunogenet.* 14(2): 123-130, 1997), DLL4 (see, e.g., those described in WO 2014/007513), TYRO3 (see, e.g., those described in WO 2016/166348), AXL (see, e.g., those described in WO 2012/175692), MER (see, e.g., those described in WO 2016/106221), CD122 (see, e.g., those described in U.S. Patent Application Publication No. 2016/0367664), CD155 (see, e.g., those described in WO 2017/149538), or PDGF-DD (see, e.g., those described in U.S. Pat. No. 9,441,034).

The antigen-binding domains present in any of the single-chain or multi-chain chimeric polypeptides described herein are each independently selected from the group consisting of: a VHH domain, a VNAR domain, and a scFv. In some embodiments, any of the antigen-binding domains described herein is a BiTe, a (scFv) 2, a nanobody, a nanobody-HSA, a DART, a TandAb, a scDiabody, a scDiabody-CH3, scFv-CH-CL-scFv, a HSAbody, scDiabody-HAS, or a tandem-scFv. Additional examples of antigen-binding domains that can be used in any of the single-chain or multi-chain chimeric polypeptide are known in the art.

A VHH domain is a single monomeric variable antibody domain that can be found in camelids. A VNAR domain is a single monomeric variable antibody domain that can be found in cartilaginous fish. Non-limiting aspects of VHH domains and VNAR domains are described in, e.g., Cromie et al., *Curr. Top. Med. Chem.* 15:2543-2557, 2016; De Genst et al., *Dev. Comp. Immunol.* 30:187-198, 2006; De Meyer et al., *Trends Biotechnol.* 32:263-270, 2014; Kijanka et al., Nanomedicine 10:161-174, 2015; Kovaleva et al., *Expert. Opin. Biol. Ther.* 14:1527-1539, 2014; Krah et al., *Immunopharmacol. Immunotoxicol.* 38:21-28, 2016; Mujic-Delic et al., *Trends Pharmacol. Sci.* 35:247-255, 2014; Muyldermans, *J. Biotechnol.* 74:277-302, 2001; Muyldermans et al., *Trends Biochem. Sci.* 26:230-235, 2001; Muyldermans, *Ann. Rev. Biochem.* 82:775-797, 2013; Rahbarizadeh et al., *Immunol. Invest.* 40:299-338, 2011; Van Audenhove et al., *EBioMedicine* 8:40-48, 2016; Van Bockstaele et al., *Curr. Opin. Investig. Drugs* 10:1212-1224, 2009; Vincke et al., *Methods Mol. Biol.* 911:15-26, 2012; and Wesolowski et al., *Med. Microbiol. Immunol.* 198:157-174, 2009.

In some embodiments, each of the antigen-binding domains in the single-chain or multi-chain chimeric polypeptides described herein are both VHH domains, or at least one antigen-binding domain is a VHH domain. In some embodiments, each of the antigen-binding domains in the single-chain or multi-chain chimeric polypeptides described herein are both VNAR domains, or at least one antigen-binding domain is a VNAR domain. In some embodiments, each of the antigen-binding domains in the single-chain or multi-chain chimeric polypeptides described herein are both scFv domains, or at least one antigen-binding domain is a scFv domain.

In some embodiments, two or more of polypeptides present in the single-chain or multi-chain chimeric polypeptide can assemble (e.g., non-covalently assemble) to form any of the antigen-binding domains described herein, e.g., an antigen-binding fragment of an antibody (e.g., any of the antigen-binding fragments of an antibody described herein), a VHH-scAb, a VHH-Fab, a Dual scFab, a $F(ab')_2$, a diabody, a crossMab, a DAF (two-in-one), a DAF (four-in-one), a DutaMab, a DT-IgG, a knobs-in-holes common light chain, a knobs-in-holes assembly, a charge pair, a Fab-arm exchange, a SEEDbody, a LUZ-Y, a Fcab, a κλ-body, an orthogonal Fab, a DVD-IgG, a IgG(H)-scFv, a scFv-(H)IgG, IgG(L)-scFv, scFv-(L)IgG, IgG(L,H)-Fv, IgG(H)-V, V(H)-IgG, IgG (L)-V, V(L)-IgG, KIH IgG-scFab, 2scFv-IgG, IgG-2scFv, scFv4-Ig, Zybody, DVI-IgG, Diabody-CH3, a triple body, a miniantibody, a minibody, a TriBi minibody, scFv-CH3 KIH, Fab-scFv, a $F(ab')_2$-$scFv_2$, a scFv-KIH, a Fab-scFv-Fc, a tetravalent HCAb, a scDiabody-Fc, a Diabody-Fc, a tandem scFv-Fc, an Intrabody, a dock and lock, a 1 mm TAC, an IgG-IgG conjugate, a Cov-X-Body, and a scFv1-PEG-scFv2. See, e.g., Spiess et al., *Mol. Immunol.* 67:95-106, 2015, incorporated in its entirety herewith, for a description of these elements. Non-limiting examples of an antigen-binding fragment of an antibody include an Fv fragment, a Fab fragment, a $F(ab')_2$ fragment, and a Fab' fragment. Additional examples of an antigen-binding fragment of an antibody is an antigen-binding fragment of an IgG (e.g., an antigen-binding fragment of IgG1, IgG2, IgG3, or IgG4) (e.g., an antigen-binding fragment of a human or humanized IgG, e.g., human or humanized IgG1, IgG2, IgG3, or IgG4); an antigen-binding fragment of an IgA (e.g., an antigen-binding fragment of IgA1 or IgA2) (e.g., an antigen-binding fragment of a human or humanized IgA, e.g., a human or humanized IgA1 or IgA2); an antigen-binding fragment of an IgD (e.g., an antigen-binding fragment of a human or humanized IgD); an antigen-binding fragment of an IgE (e.g., an antigen-binding fragment of a human or humanized IgE); or an antigen-binding fragment of an IgM (e.g., an antigen-binding fragment of a human or humanized IgM).

An "Fv" fragment includes a non-covalently-linked dimer of one heavy chain variable domain and one light chain variable domain.

A "Fab" fragment includes, the constant domain of the light chain and the first constant domain (C_{H1}) of the heavy chain, in addition to the heavy and light chain variable domains of the Fv fragment.

A "$F(ab')_2$" fragment includes two Fab fragments joined, near the hinge region, by disulfide bonds.

A "dual variable domain immunoglobulin" or "DVD-Ig" refers to multivalent and multispecific binding proteins as described, e.g., in DiGiammarino et al., *Methods Mol. Biol.* 899:145-156, 2012; Jakob et al., *MABs* 5:358-363, 2013; and U.S. Pat. Nos. 7,612,181; 8,258,268; 8,586,714; 8,716,450; 8,722,855; 8,735,546; and 8,822,645, each of which is incorporated by reference in its entirety.

DARTs are described in, e.g., Garber, Nature Reviews Drug Discovery 13:799-801, 2014.

In some embodiments of any of the antigen-binding domains described herein can bind to an antigen selected from the group consisting of: a protein, a carbohydrate, a lipid, and a combination thereof.

Additional examples and aspects of antigen-binding domains are known in the art.

Soluble Interleukin or Cytokine Protein

In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain and the second target-binding domain can be a soluble interleukin protein or soluble cytokine protein. In some embodiments, the soluble interleukin or soluble cytokine protein is selected from the group of: IL-2, IL-3, IL-7, IL-8, IL-10, IL-12, IL-15, IL-17, IL-18, IL-21, PDGF-DD, SCF, and FLT3L. Non-limiting examples of soluble IL-2, IL-3, IL-7, IL-8, IL-10, IL-15, IL-17, IL-18, IL-21, PDGF-DD, SCF, and FLT3L are provided below.

Human Soluble IL-2

(SEQ ID NO: 75)

aptssstkkt qlqlehllld lqmilnginn yknpkltrml tfkfympkka
telkhlqcle eelkpleevl nlaqsknfhl rprdlisnin vivlelkgse
ttfmceyade tativeflnr witfcqsiis tlt

Human Soluble IL-3

(SEQ ID NO: 76)

apmtqttplkt swvncsnmid eiithlkqpp lplldinnln gedqdilmen
nlrrpnleaf nravkslqna saiesilknl lpclplataa ptrhpihikd
gdwnefrrkl tfylktlena qaqqttlsla if

Human Soluble IL-7

(SEQ ID NO: 77)

dcdiegkdgkqyesv lmvsidqlld smkeigsncl nnefnffkrh icdankegmf
lfraarklrq flkmnstgdf dlhllkvseg ttillnctgq vkgrkpaalg
eaqptkslee nkslkeqkkl ndlcflkrll qeiktcwnki Imgtkeh

Human Soluble IL-8

(SEQ ID NO: 78)

egavlprsak elrcqcikty skpfhpkfik elrviesgph canteiivkl
sdgrelcldp kenwvqrvve kflkraens

Human Soluble IL-10

(SEQ ID NO: 79)

spgqgtqsensc thfpgnlpnm Irdlrdafsr vktffqmkdq ldnlllkesl
ledfkgylgc qalsemiqfy leevmpqaen qdpdikahvn slgenlktlr
lrlrrchrfl pcenkskave qvknafnklq ekgiykamse fdifinyiea
ymtmkirn

Human Soluble IL-15

(SEQ ID NO: 80)

nwvnvisdlkki edliqsmhid atlytesdvh psckvtamkc fllelqvisl
esgdasihdt venliiilann slssngnvte sgckeceele eknikeflqs
fvhivqmfin ts

Human Soluble IL-15 D8N Mutant

(SEQ ID NO: 99)

NWVNVISNLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLELQVISLESGDASI
HDTVENLIILANNSLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS

Human Soluble IL-15 D8N Mutant

(SEQ ID NO: 100)

AACTGGGTGAACGTCATCAGCAATTTAAAGAAGATCGAAGATTTAATTCAGTC
CATGCATATCGACGCCACTTTATACACAGAATCCGACGTGCACCCCTCTTGTAA
GGTGACCGCCATGAAATGTTTTTTACTGGAGCTGCAAGTTATCTCTTTAGAGAG
CGGAGACGCTAGCATCCACGACACCGTGGAGAATTTAATCATTTTAGCCAATA
ACTCTTTATCCAGCAACGGCAACGTGACAGAGTCCGGCTGCAAGGAGTGCGA
AGAGCTGGAGGAGAAGAACATCAAGGAGTTTCTGCAATCCTTTGTGCACATTG
TCCAGATGTTCATCAATACCTCC

Human Soluble IL-17

(SEQ ID NO: 81)

gitiprn pgcpnsedkn fprtvmvnln ihnrntntnp krssdyynrs
tspwnlhrne dperypsviw eakcrhlgci nadgnvdyhm nsvpiqqeil
vlrrepphcp nsfrlekilv svgctcvtpi vhhva

Human Soluble IL-18

(SEQ ID NO: 82)

yfgklesklsvirn lndqvlfidq gnrplfedmt dsdcrdnapr tifiismykd
sqprgmavti svkcekistl scenkiisfk emnppdnikd tksdiiffqr
svpghdnkmq fesssyegyf lacekerdlf klilkkedel gdrsimftvq ned

Human Soluble PDGF-DD

(SEQ ID NO: 83)

rdtsatpqsasi kalrnanlrr desnhltdly rrdetiqvkg ngyvqsprfp
nsyprnlllt wrlhsqentr iqlvfdnqfg leeaendicr ydfvevedis
etstiirgrw cghkevppri ksrtnqikit fksddyfvak pgfkiyysll
edfqpaaase tnwesvtssi sgvsynspsv tdptliadal dkkiaefdtv
edllkyfnpe swqedlenmy ldtpryrgrs yhdrkskvdl drlnddakry
sctprnysvn ireelklanv vffprcllvq rcggncgcgt vnwrsctens
gktvkkyhev lqfepghikr rgraktmalv diqldhherc dcicssrppr

Human Soluble SCF

(SEQ ID NO: 84)

egicrnrvtnnvkdv tklvanlpkd ymitlkyvpg mdvlpshcwi semvvqlsds
ltdlldkfsn iseglsnysi idklvnivdd lvecvkenss kdlkksfksp
eprlftpeef frifnrsida fkdfvvaset sdcvvsstls pekdsrvsvt
kpfmlppvaa sslrndssss nrkaknppgd sslhwaamal palfsliigf
afgalywkkr qpsltraven iqineednei smlqekeref qev

-continued

Human Soluble FLT3L

(SEQ ID NO: 85)

tqdcsfqhspissd favkirelsd yllqdypvtv asnlqdeelc gglwrlvlaq
rwmerlktva gskmqgller vnteihfvtk cafqpppscl rfvqtnisrl
lqetseqlva lkpwitrqnf srclelqcqp dsstlpppws prpleatapt
apqppllll llpvgllla aawclhwqrt rrrtprpgeq vppvpspqdl
llveh

Non-limiting examples of soluble MICA, MICB, ULBP1, ULBP2, ULBP3, ULBP4, ULBP5, and ULBP6 are provided below.

Human Soluble MICA

(SEQ ID NO: 86)

ephslry nltvlswdgs vqsgfltevh ldgqpflrcd rqkcrakpqg
qwaedvlgnk twdretrdlt gngkdlrmtl ahikdqkegl hslqeirvce
ihednstrss qhfyydgelf lsqnletkew tmpqssraqt lamnvrnflk
edamktkthy hamhadclqe lrrylksgvv lrrtvppmvn vtrseasegn
itvtcrasgf ypwnitlswr qdgvslshdt qqwgdvlpdg ngtyqtwvat
ricqgeeqrf tcymehsgnh sthpvpsgkv lvlqshwqtf hvsavaaaai
fviiifyvrc ckkktsaaeg pelvslqvld qhpvgtsdhr datqlgfqpl
msdlgstgst ega

Human Soluble MICB

(SEQ ID NO: 87)

aephslry nlmvlsqdes vqsgflaegh ldgqpflryd rqkrrakpqg
qwaedvlgak twdtetedlt engqdlrrtl thikdqkggl hslqeirvce
ihedsstrgs rhfyydgelf lsqnletqes tvpqssraqt lamnvtnfwk
edamktkthy ramqadclqk lqrylksgva irrtvppmvn vtcsevsegn
itvtcrassf yprnitltwr qdgvslshnt qqwgdvlpdg ngtyqtwvat
rirqgeeqrf tcymehsgnh gthpvpsgkv lvlqsqrtdf pyvsaampcf
viiiilcvpc ckkktsaaeg pelvslqvld qhpvgtgdhr
daaqlgfqpl msatgstgst ega

Human Soluble ULBP1

(SEQ ID NO: 88)

wvdthclcydfiit pksrpepqwc evqglvderp flhydcvnhk akafaslgkk
vnvtktweeq tetlrdvvdf lkgqlldiqv enlipieplt lqarmscehe
ahghgrgswq flfngqkfll fdsnnrkwta lhpgakkmte kweknrdvtm
ffqkislgdc kmwleeflmy weqmldptkp pslapg

Human Soluble ULBP2

(SEQ ID NO: 89)

gradphslcyditvi pkfrpgprwc avqgqvdekt flhydcgnkt vtpvsplgkk
Invttawkaq npvlrevvdi lteqlrdiql enytpkeplt lqarmsceqk
aeghssgswq fsfdgqifll fdsekrmwtt vhpgarkmke kwendkvvam
sfhyfsmgdc igwledflmg mdstlepsag aplams

Human Soluble ULBP3

(SEQ ID NO: 90)

dahslwynfti ihlprhgqqw cevqsqvdqk nflsydcgsd kvlsmghlee
qlyatdawgk qlemlrevgq rlrleladte ledftpsgpl tlqvrmscec
eadgyirgsw qfsfdgrkfl lfdsnnrkwt vvhagarrmk ekwekdsglt
tffkmvsmrd ckswirdflm hrkkrlepta pptmapg

Human Soluble ULBP4

(SEQ ID NO: 91)

hslcfnftik slsrpgqpwc eaqvflnknl flqynsdnnm vkplgllgkk
vyatstwgel tqtlgevgrd lrmllcdikp qiktsdpstl qvemfcqrea
erctgaswqf atngeksllf damnmtwtvi nheaskiket wkkdrgleky
frklskgdcd hwlreflghw eampeptvsp vnasdihwss sslpdrwiil
gafillvlmg ivlicvwwqn gewqaglwpl rts

Human Soluble ULBP5

(SEQ ID NO: 92)

gladp hslcyditvi pkfrpgprwc avqgqvdekt flhydcgskt
vtpvsplgkk Invttawkaq npvlrevvdi lteqlldiql enyipkeplt
lqarmsceqk aeghgsgswq lsfdgqifll fdsenrmwtt vhpgarkmke
kwendkdmtm sfhyismgdc tgwledflmg mdstlepsag apptmssg

Human Soluble ULBP6

(SEQ ID NO: 93)

rrddp hslcyditvi pkfrpgprwc avqgqvdekt flhydcgnkt
vtpvsplgkk Invtmawkaq npvlrevvdi lteqlldiql enytpkeplt
lqarmsceqk aeghssgswq fsidgqtfll fdsekrmwtt vhpgarkmke
kwendkdvam sfhyismgdc igwledflmg mdstlepsag aplamssg

Additional examples of soluble interleukin proteins and soluble cytokine proteins are known in the art.

Soluble Receptor

In some embodiments of any of the single-chain or multi-chain chimeric polypeptides described herein, one or both of the first target-binding domain and the second target-binding domain is a soluble interleukin receptor or a soluble cytokine receptor. In some embodiments, the soluble receptor is a soluble TGF-β receptor II (TGF-β RII) (see, e.g., those described in Yung et al., *Am. J. Resp. Crit. Care*

Med. 194(9): 1140-1151, 2016), a soluble TGF-βRIII (see, e.g., those described in Heng et al., *Placenta* 57:320, 2017), a soluble NKG2D (see, e.g., Cosman et al., *Immunity* 14(2): 123-133, 2001; Costa et al., *Front. Immunol.*, Vol. 9, Article 1150 May 29, 2018; doi: 10.3389/fimmu.2018.01150), a soluble NKp30 (see, e.g., Costa et al., *Front. Immunol.*, Vol. 9, Article 1150 May 29, 2018; doi: 10.3389/fimmu.2018.01150), a soluble NKp44 (see, e.g., those described in Costa et al., *Front. Immunol.*, Vol. 9, Article 1150 May 29, 2018; doi: 10.3389/fimmu.2018.01150), a soluble NKp46 (see, e.g., Mandelboim et al., *Nature* 409: 1055-1060, 2001; Costa et al., *Front. Immunol.*, Vol. 9, Article 1150 May 29, 2018; doi: 10.3389/fimmu.2018.01150), a soluble DNAM1 (see, e.g., those described in Costa et al., *Front. Immunol.*, Vol. 9, Article 1150 May 29, 2018; doi: 10.3389/fimmu.2018.01150), a scMHCI (see, e.g., those described in Washburn et al., *PLOS One* 6(3): e18439, 2011), a scMHCII (see, e.g., those described in Bishwajit et al., *Cellular Immunol.* 170(1): 25-33, 1996), a scTCR (see, e.g., those described in Weber et al., *Nature* 356(6372): 793-796, 1992), a soluble CD155 (see, e.g., those described in Tahara-Hanaoka et al., *Int. Immunol.* 16(4): 533-538, 2004), or a soluble CD28 (see, e.g., Hebbar et al., *Clin. Exp. Immunol.* 136:388-392, 2004). Additional examples of soluble interleukin receptors and soluble cytokine receptors are known in the art.

Pairs of Affinity Domains

In some embodiments, a multi-chain chimeric polypeptide includes: 1) a first chimeric polypeptide that includes a first domain of a pair of affinity domains, and 2) a second chimeric polypeptide that includes a second domain of a pair of affinity domains such that the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains. In some embodiments, the pair of affinity domains is a sushi domain from an alpha chain of human IL-15 receptor (IL15Rα) and a soluble IL-15. A sushi domain, also known as a short consensus repeat or type 1 glycoprotein motif, is a common motif in protein-protein interaction. Sushi domains have been identified on a number of protein-binding molecules, including complement components C1r, C1s, factor H, and C2m, as well as the nonimmunologic molecules factor XIII and β2-glycoprotein. A typical Sushi domain has approximately 60 amino acid residues and contains four cysteines (Ranganathan, Pac. Symp Biocomput. 2000:155-67). The first cysteine can form a disulfide bond with the third cysteine, and the second cysteine can form a disulfide bridge with the fourth cysteine. In some embodiments in which one member of the pair of affinity domains is a soluble IL-15, the soluble IL15 has a D8N or D8A amino acid substitution. In some embodiments in which one member of the pair of affinity domains is an alpha chain of human IL-15 receptor (IL15Rα), the human IL15Rα is a mature full-length IL15Rα. In some embodiments, the pair of affinity domains is barnase and barnstar. In some embodiments, the pair of affinity domains is a PKA and an AKAP. In some embodiments, the pair of affinity domains is an adapter/docking tag module based on mutated RNase I fragments (Rossi, *Proc Natl Acad Sci USA.* 103: 6841-6846, 2006; Sharkey et al., *Cancer Res.* 68:5282-5290, 2008; Rossi et al., *Trends Pharmacol Sci.* 33:474-481, 2012) or SNARE modules based on interactions of the proteins syntaxin, synaptotagmin, synaptobrevin, and SNAP25 (Deyev et al., *Nat Biotechnol.* 1486-1492, 2003).

In some embodiments, a first chimeric polypeptide of a multi-chain chimeric polypeptide includes a first domain of a pair of affinity domains and a second chimeric polypeptide of the multi-chain chimeric polypeptide includes a second domain of a pair of affinity domains, wherein the first domain of the pair of affinity domains and the second domain of the pair of affinity domains bind to each other with a dissociation equilibrium constant (K_D) of less than 1×10^{-7} M, less than 1×10^{-8} M, less than 1×10^{-9} M, less than 1×10^{-10}M, less than 1×10^{-11} M, less than 1×10^{-12} M, or less than 1×10^{-13} M. In some embodiments, the first domain of the pair of affinity domains and the second domain of the pair of affinity domains bind to each other with a K_D of about 1×10^{-4} M to about 1×10^{-6} M, about 1×10^{-5} M to about 1×10^{-7} M, about 1×10^{-6} M to about 1×10^{-8} M, about 1×10^{-7} M to about 1×10^{-9} M, about 1×10^{-8} M to about 1×10^{-10} M, about 1×10^{-9} M to about 1×10^{-11} M, about 1×10^{-10} M to about 1×10^{-12} M, about 1×10^{-11} M to about 1×10^{-13} M, about 1×10^{-4} M to about 1×10^{-5} M, about 1×10^{-5} M to about 1×10^{-6} M, about 1×10^{-6} M to about 1×10^{-7} M, about 1×10^{-7} M to about 1×10^{-8} M, about 1×10^{-8} M to about 1×10^{-9} M, about 1×10^{-9} M to about 1×10^{-10} M, about 1×10^{-10} M to about 1×10^{-11} M, about 1×10^{-11} M to about 1×10^{-12} M, or about 1×10^{-12} M to about 1×10^{-13} M (inclusive). Any of a variety of different methods known in the art can be used to determine the K_D value of the binding of the first domain of the pair of affinity domains and the second domain of the pair of affinity domains (e.g., an electrophoretic mobility shift assay, a filter binding assay, surface plasmon resonance, and a biomolecular binding kinetics assay, etc.).

In some embodiments, a first chimeric polypeptide of a multi-chain chimeric polypeptide includes a first domain of a pair of affinity domains and a second chimeric polypeptide of the multi-chain chimeric polypeptide includes a second domain of a pair of affinity domains, wherein the first domain of the pair of affinity domains, the second domain of the pair of affinity domains, or both is about 10 to 100 amino acids in length. For example, a first domain of a pair of affinity domains, a second domain of a pair of affinity domains, or both can be about 10 to 100 amino acids in length, about 15 to 100 amino acids in length, about 20 to 100 amino acids in length, about 25 to 100 amino acids in length, about 30 to 100 amino acids in length, about 35 to 100 amino acids in length, about 40 to 100 amino acids in length, about 45 to 100 amino acids in length, about 50 to 100 amino acids in length, about 55 to 100 amino acids in length, about 60 to 100 amino acids in length, about 65 to 100 amino acids in length, about 70 to 100 amino acids in length, about 75 to 100 amino acids in length, about 80 to 100 amino acids in length, about 85 to 100 amino acids in length, about 90 to 100 amino acids in length, about 95 to 100 amino acids in length, about 10 to 95 amino acids in length, about 10 to 90 amino acids in length, about 10 to 85 amino acids in length, about 10 to 80 amino acids in length, about 10 to 75 amino acids in length, about 10 to 70 amino acids in length, about 10 to 65 amino acids in length, about 10 to 60 amino acids in length, about 10 to 55 amino acids in length, about 10 to 50 amino acids in length, about 10 to 45 amino acids in length, about 10 to 40 amino acids in length, about 10 to 35 amino acids in length, about 10 to 30 amino acids in length, about 10 to 25 amino acids in length, about 10 to 20 amino acids in length, about 10 to 15 amino acids in length, about 20 to 30 amino acids in length, about 30 to 40 amino acids in length, about 40 to 50 amino acids in length, about 50 to 60 amino acids in length, about 60 to 70 amino acids in length, about 70 to 80 amino acids in length, about 80 to 90 amino acids in length, about 90 to 100 amino acids in length, about 20 to 90 amino acids in length, about 30 to 80 amino acids in length, about 40 to 70 amino acids in length, about 50 to 60 amino acids in length, or any range in between. In some embodiments, a first domain of a pair of affinity domains, a second domain of a pair of affinity domains, or both is about 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 amino acids in length.

In some embodiments, any of the first and/or second domains of a pair of affinity domains disclosed herein can include one or more additional amino acids (e.g., 1, 2, 3, 5, 6, 7, 8, 9, 10, or more amino acids) at its N-terminus and/or C-terminus, so long as the function of the first and/or second domains of a pair of affinity domains remains intact. For example, a sushi domain from an alpha chain of human IL-15 receptor (IL15Rα) can include one or more additional amino acids at the N-terminus and/or the C-terminus, while still retaining the ability to bind to a soluble IL-15. Additionally or alternatively, a soluble IL-15 can include one or more additional amino acids at the N-terminus and/or the C-terminus, while still retaining the ability to bind to a sushi domain from an alpha chain of human IL-15 receptor (IL15Rα).

A non-limiting example of a sushi domain from an alpha chain of IL-15 receptor alpha (IL15Rα) can include a sequence that is at least 70% identical, at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical to

(SEQ ID NO: 94)
ITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSLTECVLNK
ATNVAHWTTPSLKCIR.

In some embodiments, a sushi domain from an alpha chain of IL15Rα can be encoded by a nucleic acid including

(SEQ ID NO: 95)
ATTACATGCCCCCCTCCCATGAGCGTGGAGCACGCCGACATCTGGGTGA
AGAGCTATAGCCTCTACAGCCGGGAGAGGTATATCTGTAACAGCGGCTT
CAAGAGGAAGGCCGGCACCAGCAGCCTCACCGAGTGCGTGCTGAATAAG
GCTACCAACGTGGCTCACTGGACAACACCCTCTTTAAAGTGCATCCGG.

In some embodiments, a soluble IL-15 can include a sequence that is at least 70% identical, at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical to

(SEQ ID NO: 96)
NWVNVISDLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLELQV
ISLESGDASIHDTVENLIILANNSLSSNGNVTESGCKECEELEEKNIKE
FLQSFVHIVQMFINTS.

In some embodiments, a soluble IL-15 can be encoded by a nucleic acid including the sequence of

(SEQ ID NO: 97)
AACTGGGTGAACGTCATCAGCGATTTAAAGAAGATCGAAGATTTAATTC
AGTCCATGCATATCGACGCCACTTTATACACAGAATCCGACGTGCACCC
CTCTTGTAAGGTGACCGCCATGAAATGTTTTTTACTGGAGCTGCAAGTT
ATCTCTTTAGAGAGCGGAGACGCTAGCATCCACGACACCGTGGAGAATT

-continued

TAATCATTTTAGCCAATAACTCTTTATCCAGCAACGGCAACGTGACAGA
GTCCGGCTGCAAGGAGTGCGAAGAGCTGGAGGAGAAGAACATCAAGGAG
TTTCTGCAATCCTTTGTGCACATTGTCCAGATGTTCATCAATACCTCC.

Single Chain Chimeric Polypeptides

A single-chain chimeric polypeptide includes: a first target-binding domain (e.g., any of the target-binding domains described herein or known in the art), a soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein or known in the art), and a second target-binding domain (e.g., any of the target-binding domains described herein or known in the art).

In some embodiments of any of the single-chain chimeric polypeptides described herein, the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) directly abut each other. In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein). In some embodiments of any of the single-chain chimeric polypeptides described herein, the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abut each other. In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art).

In some embodiments of any of the single-chain chimeric polypeptides described herein, the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abut each other. In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art). In some embodiments of any of the single-chain chimeric polypeptides described herein, the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) directly abut each other. In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein or known in the art).

In some examples of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide can have a total length of about 50 amino acids to about 3000 amino acids, about 50 amino acids to about 2500 amino acids, about 50 amino acids to about 2000 amino acids, about 50 amino acids to about 1500 amino acids, about 50 amino acids to about 1000 amino acids, about 50 amino acids to about 950 amino acids, about 50 amino acids to about 900 amino acids, about 50 amino acids to about 850 amino acids, about 50 amino acids to about 800 amino acids, about 50 amino acids to about 750 amino acids, about 50 amino acids to about 700 amino acids, about 50 amino acids to about 650 amino acids, about 50 amino acids to about 600 amino acids, about 50 amino acids to about 550 amino acids, about 50 amino acids to about 500 amino acids, about 50 amino acids to about 480 amino acids, about 50 amino acids to about 460 amino acids, about 50 amino acids to about 440 amino acids, 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about 750 amino acids, about 500 amino acids to about 700 amino acids, about 500 amino acids to about 650 amino acids, about 500 amino acids to about 600 amino acids, about 500 amino acids to about 550 amino acids, about 550 amino acids to about 3000 amino acids, about 550 amino acids to about 2500 amino acids, about 550 amino acids to about 2000 amino acids, about 550 amino acids to about 1500 amino acids, about 550 amino acids to about 1000 amino acids, about 550 amino acids to about 950 amino acids, about 550 amino acids to about 900 amino acids, about 550 amino acids to about 850 amino acids, about 550 amino acids to about 800 amino acids, about 550 amino acids to about 750 amino acids, about 550 amino acids to about 700 amino acids, about 550 amino acids to about 650 amino acids, about 550 amino acids to about 600 amino acids, about 600 amino acids to about 3000 amino acids, about 600 amino acids to about 2500 amino acids, about 600 amino acids to about 2000 amino acids, about 600 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acids, about 700 amino acids to about 2500 amino acids, about 700 amino acids to about 2000 amino acids, about 700 amino acids to about 1500 amino acids, about 700 amino acids to about 1000 amino acids, about 700 amino acids to about 950 amino acids, about 700 amino acids to about 900 amino acids, about 700 amino acids to about 850 amino acids, about 700 amino acids to about 800 amino acids, about 700 amino acids to about 750 amino acids, about 750 amino acids to about 3000 amino acids, about 750 amino acids to about 2500 amino acids, about 750 amino acids to about 2000 amino acids, about 750 amino acids to about 1500 amino acids, about 750 amino acids to about 1000 amino acids, about 750 amino acids to about 950 amino acids, about 750 amino acids to about 900 amino acids, about 750 amino acids to about 850 amino acids, about 750 amino acids to about 800 amino acids, about 800 amino acids to about 3000 amino acids, about 800 amino acids to about 2500 amino acids, about 800 amino acids to about 2000 amino acids, about 800 amino acids to about 1500 amino acids, about 800 amino acids to about 1000 amino acids, about 800 amino acids to about 950 amino acids, about 800 amino acids to about 900 amino acids, about 800 amino acids to about 850 amino acids, about 850 amino acids to about 3000 amino acids, about 850 amino acids to about 2500 amino acids, about 850 amino acids to about 2000 amino acids, about 850 amino acids to about 1500 amino acids, about 850 amino acids to about 1000 amino acids, about 850 amino acids to about 950 amino acids, about 850 amino acids to about 900 amino acids, about 900 amino acids to about 3000 amino acids, about 900 amino acids to about 2500 amino acids, about 900 amino acids to about 2000 amino acids, about 900 amino acids to about 1500 amino acids, about 900 amino acids to about 1000 amino acids, about 900 amino acids to about 950 amino acids, about 950 amino acids to about 3000 amino acids, about 950 amino acids to about 2500 amino acids, about 950 amino acids to about 2000 amino acids, about 950 amino acids to about 1500 amino acids, about 950 amino acids to about 1000 amino acids, about 1000 amino acids to about 3000 amino acids, about 1000 amino acids to about 2500 amino acids, about 1000 amino acids to about 2000 amino acids, about 1000 amino acids to about 1500 amino acids, about 1500 amino acids to about 3000 amino acids, about 1500 amino acids to about 2500 amino acids, about 1500 amino acids to about 2000 amino acids, about 2000 amino acids to about 3000 amino acids, about 2000 amino acids to about 2500 amino acids, or about 2500 amino acids to about 3000 amino acids.

Exemplary IL-2 Receptor Activating Agents that are Single-Chain Chimeric Polypeptides

In some embodiments, a single-chain chimeric polypeptide can be an IL-2 receptor activating agent. In such embodiments, one or both of the first target-binding domain and/or the second target-binding domain bind to a receptor for IL-2 (e.g., a human receptor for IL-2).

In some embodiments of any of the single-chain chimeric polypeptides that are an IL-2 receptor activating agent, the first target-binding domain and/or the second target-binding domain can be an agonistic antigen-binding domain that binds specifically to an IL-2 receptor (e.g., a human IL-2 receptor) (see, e.g., those described in Gaulton et al., *Clin. Immunol. Immunopathol.* 36(1): 18-29, 1985)).

In some embodiments of any of the single-chain chimeric polypeptides that are an IL-2 receptor activating agent, one or both of the first target-binding domain and the second target-binding domain can be a soluble interleukin 2 (IL-2) protein. An exemplary sequence of soluble IL-2 is provided below.

```
Human Soluble IL-2
                                        (SEQ ID NO: 1)
aptssstkkt qlqlehllld lqmilnginn yknpkltrml

tfkfympkka telkhlqcle eelkpleevl nlaqsknfhl

rprdlisnin vivlelkgse ttfmceyade tativeflnr

witfcqsiis tlt
```

In some embodiments of these single-chain chimeric polypeptides, the first target-binding domain and the second target-binding domain is a soluble human IL-2 protein. A non-limiting example of an IL-2 protein that binds specifically to an IL-2 receptor can include a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 1.

In some embodiments, a soluble human IL-2 protein can be encoded by a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

```
                                        (SEQ ID NO: 31)
GCACCTACTTCAAGTTCTACAAAGAAAACACAGCTACAACTGGAGCATT

TACTGCTGGATTTACAGATGATTTTGAATGGAATTAATAATTACAAGAA

TCCCAAACTCACCAGGATGCTCACATTTAAGTTTTACATGCCCAAGAAG

GCCACAGAACTGAAACATCTTCAGTGTCTAGAAGAAGAACTCAAACCTC

TGGAGGAAGTGCTAAATTTAGCTCAAAGCAAAAACTTTCACTTAAGACC

CAGGGACTTAATCAGCAATATCAACGTAATAGTTCTGGAACTAAAGGGA

TCTGAAACAACATTCATGTGTGAATATGCTGATGAGACAGCAACCATTG

TAGAATTTCTGAACAGATGGATTACCTTTTGTCAAAGCATCATCTCAAC

ACTAACT.
```

In some embodiments, a soluble human IL-2 protein can be encoded by a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 32)
GCCCCCACCTCCTCCTCCACCAAGAAGACCCAGCTGCAGCTGGAGCATT

TACTGCTGGATTTACAGATGATTTTAAACGGCATCAACAACTACAAGAA

CCCCAAGCTGACTCGTATGCTGACCTTCAAGTTCTACATGCCCAAGAAG

GCCACCGAGCTGAAGCATTTACAGTGTTTAGAGGAGGAGCTGAAGCCCC

TCGAGGAGGTGCTGAATTTAGCCCAGTCCAAGAATTTCCATTTAAGGCC

CCGGGATTTAATCAGCAACATCAACGTGATCGTTTTAGAGCTGAAGGGC

TCCGAGACCACCTTCATGTGCGAGTACGCCGACGAGACCGCCACCATCG

TGGAGTTTTTAAATCGTTGGATCACCTTCTGCCAGTCCATCATCTCCAC

TTTAACC

In some embodiments, a single-chain chimeric polypeptide that is an IL-2 receptor activating agent can include a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 3)
APTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRMLTFKFYMPKK

ATELKHLQCLEEELKPLEEVLNLAQSKNFHLRPRDLISNINVIVLELKG

SETTFMCEYADETATIVEFLNRWITFCQSIISTLTSGTTNTVAAYNLTW

KSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTDTECDLTDEI

VKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEFTPYLETNLGQPT

IQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKDLIYTLYYWKSS

SSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVECM

GQEKGEFREAPTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRML

TFKFYMPKKATELKHLQCLEEELKPLEEVLNLAQSKNFHLRPRDLISNI

NVIVLELKGSETTFMCEYADETATIVEFLNRWITFCQSIISTLT.

In some embodiments, a single-chain chimeric polypeptide that is an IL-2 receptor activating agent is encoded by a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 11)
GCCCCCACCTCCTCCTCCACCAAGAAGACCCAGCTGCAGCTGGAGCATT

TACTGCTGGATTTACAGATGATTTTAAACGGCATCAACAACTACAAGAA

CCCCAAGCTGACTCGTATGCTGACCTTCAAGTTCTACATGCCCAAGAAG

GCCACCGAGCTGAAGCATTTACAGTGTTTAGAGGAGGAGCTGAAGCCCC

TCGAGGAGGTGCTGAATTTAGCCCAGTCCAAGAATTTCCATTTAAGGCC

CCGGGATTTAATCAGCAACATCAACGTGATCGTTTTAGAGCTGAAGGGC

TCCGAGACCACCTTCATGTGCGAGTACGCCGACGAGACCGCCACCATCG

TGGAGTTTTTAAATCGTTGGATCACCTTCTGCCAGTCCATCATCTCCAC

-continued

TTTAACCAGCGGCACAACCAACACAGTCGCTGCCTATAACCTCACTTGG

AAGAGCACCAACTTCAAAACCATCCTCGAATGGGAACCCAAACCCGTTA

ACCAAGTTTACACCGTGCAGATCAGCACCAAGTCCGGCGACTGGAAGTC

CAAATGTTTCTATACCACCGACACCGAGTGCGATCTCACCGATGAGATC

GTGAAAGATGTGAAACAGACCTACCTCGCCCGGGTGTTTAGCTACCCCG

CCGGCAATGTGGAGAGCACTGGTTCCGCTGGCGAGCCTTTATACGAGAA

CAGCCCCGAATTTACCCCTTACCTCGAGACCAATTTAGGACAGCCCACC

ATCCAAAGCTTTGAGCAAGTTGGCACAAAGGTGAATGTGACAGTGGAGG

ACGAGCGGACTTTAGTGCGGCGGAACAACACCTTTCTCAGCCTCCGGGA

TGTGTTCGGCAAAGATTTAATCTACACACTGTATTACTGGAAGTCCTCT

TCCTCCGGCAAGAAGACAGCTAAAACCAACACAAACGAGTTTTTAATCG

ACGTGGATAAAGGCGAAAACTACTGTTTCAGCGTGCAAGCTGTGATCCC

CTCCCGGACCGTGAATAGGAAAAGCACCGATAGCCCCGTTGAGTGCATG

GGCCAAGAAAAGGGCGAGTTCCGGGAGGCACCTACTTCAAGTTCTACAA

AGAAAACACAGCTACAACTGGAGCATTTACTGCTGGATTTACAGATGAT

TTTGAATGGAATTAATAATTACAAGAATCCCAAACTCACCAGGATGCTC

ACATTTAAGTTTTACATGCCCAAGAAGGCCACAGAACTGAAACATCTTC

AGTGTCTAGAAGAAGAACTCAAACCTCTGGAGGAAGTGCTAAATTTAGC

TCAAAGCAAAAACTTTCACTTAAGACCCAGGGACTTAATCAGCAATATC

AACGTAATAGTTCTGGAACTAAAGGGATCTGAAACAACATTCATGTGTG

AATATGCTGATGAGACAGCAACCATTGTAGAATTTCTGAACAGATGGAT

TACCTTTTGTCAAAGCATCATCTCAACACTAACT.

In some embodiments, a single-chain chimeric polypeptide that is an IL-2 receptor activating agent can include a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 4)
MKWVTFISLLFLFSSAYSAPTSSSTKKTQLQLEHLLLDLQMILNGINNYK

NPKLTRMLTFKFYMPKKATELKHLQCLEEELKPLEEVLNLAQSKNFHLRP

RDLISNINVIVLELKGSETTFMCEYADETATIVEFLNRWITFCQSIISTL

TSGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKC

FYTTDTECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPE

FTPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGK

DLIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVN

RKSTDSPVECMGQEKGEFREAPTSSSTKKTQLQLEHLLLDLQMILNGINN

YKNPKLTRMLTFKFYMPKKATELKHLQCLEEELKPLEEVLNLAQSKNFHL

RPRDLISNINVIVLELKGSETTFMCEYADETATIVEFLNRWITFCQSIIS

TLT.

In some embodiments, a single-chain chimeric polypeptide that is an IL-2 receptor activating agent is encoded by a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 12)

```
ATGAAGTGGGTGACCTTCATCAGCCTGCTGTTCCTGTTCTCCAGCGCCTA
CTCCGCCCCCACCTCCTCCTCCACCAAGAAGACCCAGCTGCAGCTGGAGC
ATTTACTGCTGGATTTACAGATGATTTTAAACGGCATCAACAACTACAAG
AACCCCAAGCTGACTCGTATGCTGACCTTCAAGTTCTACATGCCCAAGAA
GGCCACCGAGCTGAAGCATTTACAGTGTTTAGAGGAGGAGCTGAAGCCCC
TCGAGGAGGTGCTGAATTTAGCCCAGTCCAAGAATTTCCATTTAAGGCCC
CGGGATTTAATCAGCAACATCAACGTGATCGTTTTAGAGCTGAAGGGCTC
CGAGACCACCTTCATGTGCGAGTACGCCGACGAGACCGCCACCATCGTGG
AGTTTTTAAATCGTTGGATCACCTTCTGCCAGTCCATCATCTCCACTTTA
ACCAGCGGCACAACCAACACAGTCGCTGCCTATAACCTCACTTGGAAGAG
CACCAACTTCAAAACCATCCTCGAATGGGAACCCAAACCCGTTAACCAAG
TTTACACCGTGCAGATCAGCACCAAGTCCGGCGACTGGAAGTCCAAATGT
TTCTATACCACCGACACCGAGTGCGATCTCACCGATGAGATCGTGAAAGA
TGTGAAACAGACCTACCTCGCCCGGGTGTTTAGCTACCCCGCCGGCAATG
TGGAGAGCACTGGTTCCGCTGGCGAGCCTTTATACGAGAACAGCCCCGAA
TTTACCCCTTACCTCGAGACCAATTTAGGACAGCCCACCATCCAAAGCTT
TGAGCAAGTTGGCACAAAGGTGAATGTGACAGTGGAGGACGAGCGGACTT
TAGTGCGGCGGAACAACACCTTTCTCAGCCTCCGGGATGTGTTCGGCAAA
GATTTAATCTACACACTGTATTACTGGAAGTCCTCTTCCTCCGGCAAGAA
GACAGCTAAAACCAACACAAACGAGTTTTTAATCGACGTGGATAAAGGCG
AAAACTACTGTTTCAGCGTGCAAGCTGTGATCCCCTCCCGGACCGTGAAT
AGGAAAAGCACCGATAGCCCCGTTGAGTGCATGGGCCAAGAAAAGGGCGA
GTTCCGGGAGGCACCTACTTCAAGTTCTACAAAGAAAACACAGCTACAAC
TGGAGCATTTACTGCTGGATTTACAGATGATTTTGAATGGAATTAATAAT
TACAAGAATCCCAAACTCACCAGGATGCTCACATTTAAGTTTTACATGCC
CAAGAAGGCCACAGAACTGAAACATCTTCAGTGTCTAGAAGAAGAACTCA
AACCTCTGGAGGAAGTGCTAAATTTAGCTCAAAGCAAAAACTTTCACTTA
AGACCCAGGGACTTAATCAGCAATATCAACGTAATAGTTCTGGAACTAAA
GGGATCTGAAACAACATTCATGTGTGAATATGCTGATGAGACAGCAACCA
TTGTAGAATTTCTGAACAGATGGATTACCTTTTGTCAAAGCATCATCTCA
ACACTAACT.
```

Exemplary CD3/CD38 Binding Agents that are Single-Chain Chimeric Polypeptides

In some embodiments where the single-chain chimeric polypeptide is an CD3/CD28 binding agent, the first target-binding domain binds specifically to CD3 (e.g., one or more of CD3δ (e.g., human CD3δ), CD3ε (e.g., human CD3ε), CD3γ (e.g., human CD3γ), and CD3ζ (e.g., human CD3ζ)) and the second target-binding domain binds specifically to CD28 (e.g., human CD28).

In some embodiments, the first target-binding domain that binds specifically to CD3 (e.g., one or more of CD3δ (e.g., human CD3δ), CD3ε (e.g., human CD3ε), CD3γ (e.g., human CD3γ), and CD3ζ (e.g., human CD3ζ)) is an antigen-binding domain (e.g., any of the exemplary antigen-binding domains described herein, e.g., an anti-CD3 scFv).

In some embodiments, the first target binding domain can be an anti-CD3 scFv. In some embodiments, the anti-CD3 scFv can include a heavy chain variable domain including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 21)

```
QVQLQQSGAELARPGASVKMSCKASGYTFTRYTMHWVKQRPGQGLEWIGY
INPSRGYTNYNQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAVYYCARYY
DDHYCLDYWGQGTTLTVSS
```

and/or a light chain variable domain including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 22)

```
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDT
SKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSG
TKLEINR.
```

In some embodiments, a scFv (e.g., any of the scFvs described herein) can include a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the heavy chain variable domain and the light chain variable domain.

In some embodiments, the anti-CD3 scFv can include a heavy chain variable domain encoded by a nucleic acid including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 23)

```
CAAGTTCAGCTCCAGCAAAGCGGCGCCGAACTCGCTCGGCCCGGCGCTTC
CGTGAAGATGTCTTGTAAGGCCTCCGGCTATACCTTCACCCGGTACACAA
TGCACTGGGTCAAGCAACGGCCCGGTCAAGGTTTAGAGTGGATTGGCTAT
ATCAACCCCTCCCGGGGCTATACCAACTACAACCAGAAGTTCAAGGACAA
AGCCACCCTCACCACCGACAAGTCCAGCAGCACCGCTTACATGCAGCTGA
GCTCTTTAACATCCGAGGATTCCGCCGTGTACTACTGCGCTCGGTACTAC
GACGATCATTACTGCCTCGATTACTGGGGCCAAGGTACCACCTTAACAGT
CTCCTCC,
```

and/or a light chain variable domain encoded by a nucleic acid including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 24)

CAGATCGTGCTGACCCAGTCCCCCGCTATTATGAGCGCTAGCCCCGGTGA

AAAGGTGACTATGACATGCAGCGCCAGCTCTTCCGTGAGCTACATGAACT

GGTATCAGCAGAAGTCCGGCACCAGCCCTAAAAGGTGGATCTACGACACC

AGCAAGCTGGCCAGCGGCGTCCCCGCTCACTTTCGGGGCTCCGGCTCCGG

AACAAGCTACTCTCTGACCATCAGCGGCATGGAAGCCGAGGATGCCGCTA

CCTATTACTGTCAGCAGTGGAGCTCCAACCCCTTCACCTTTGGATCCGGC

ACCAAGCTCGAGATTAATCGT.

In some embodiments, an anti-CD3 scFv can include a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 5)

QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDT

SKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSG

TKLEINRGGGGSGGGGSGGGGSQVQLQQSGAELARPGASVKMSCKASGYT

FTRYTMHWVKQRPGQGLEWIGYINPSRGYTNYNQKFKDKATLTTDKSSST

AYMQLSSLTSEDSAVYYCARYYDDHYCLDYWGQGTTLTVSS.

In some embodiments, an anti-CD3 scFv can include the six CDRs present in SEQ ID NO: 5.

In some embodiments, an anti-CD3 scFv can include a sequence encoded by a nucleic acid sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 25)

CAGATCGTGCTGACCCAGTCCCCCGCTATTATGAGCGCTAGCCCCGGTGA

AAAGGTGACTATGACATGCAGCGCCAGCTCTTCCGTGAGCTACATGAACT

GGTATCAGCAGAAGTCCGGCACCAGCCCTAAAAGGTGGATCTACGACACC

AGCAAGCTGGCCAGCGGCGTCCCCGCTCACTTTCGGGGCTCCGGCTCCGG

AACAAGCTACTCTCTGACCATCAGCGGCATGGAAGCCGAGGATGCCGCTA

CCTATTACTGTCAGCAGTGGAGCTCCAACCCCTTCACCTTTGGATCCGGC

ACCAAGCTCGAGATTAATCGTGGAGGCGGAGGTAGCGGAGGAGGCGGATC

CGGCGGTGGAGGTAGCCAAGTTCAGCTCCAGCAAAGCGGCGCCGAACTCG

CTCGGCCCGGCGCTTCCGTGAAGATGTCTTGTAAGGCCTCCGGCTATACC

TTCACCCGGTACACAATGCACTGGGTCAAGCAACGGCCCGGTCAAGGTTT

AGAGTGGATTGGCTATATCAACCCCTCCCGGGGCTATACCAACTACAACC

AGAAGTTCAAGGACAAAGCCACCCTCACCACCGACAAGTCCAGCAGCACC

GCTTACATGCAGCTGAGCTCTTTAACATCCGAGGATTCCGCCGTGTACTA

CTGCGCTCGGTACTACGACGATCATTACTGCCTCGATTACTGGGGCCAAG

GTACCACCTTAACAGTCTCCTCC.

In some embodiments, the second target-binding domain that binds specifically to CD28 (e.g., human CD28) is an antigen-binding domain (e.g., any of the exemplary antigen-binding domains described herein, e.g., an anti-CD28 scFv).

In some embodiments, the anti-CD28 scFv can include a heavy chain variable domain including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 26)

DIEMTQSPAIMSASLGERVTMTCTASSSVSSSYFHWYQQKPGSSPKLCIY

STSNLASGVPPRFSGSGSTSYSLTISSMEAEDAATYFCHQYHRSPTFGGG

TKLETKR

and/or a light chain variable domain including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 27)

VQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKPGQGLEWIGSI

NPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSALYYCARWGD

GNYWGRGTTLTVSS.

In some embodiments, the anti-CD28 scFv can include a heavy chain variable domain encoded by a nucleic acid including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 28)

GACATCGAGATGACACAGTCCCCCGCTATCATGAGCGCCTCTTTAGGAGA

ACGTGTGACCATGACTTGTACAGCTTCCTCCAGCGTGAGCAGCTCCTATT

TCCACTGGTACCAGCAGAAACCCGGCTCCTCCCCTAAACTGTGTATCTAC

TCCACAAGCAATTTAGCTAGCGGCGTGCCTCCTCGTTTTAGCGGCTCCGG

CAGCACCTCTTACTCTTTAACCATTAGCTCTATGGAGGCCGAAGATGCCG

CCACATACTTTTGCCATCAGTACCACCGGTCCCCTACCTTTGGCGGAGGC

ACAAAGCTGGAGACCAAGCGG,

and/or a light chain variable domain encoded by a nucleic acid including a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 29)

GTGCAGCTGCAGCAGTCCGGACCCGAACTGGTCAAGCCCGGTGCCTCCGT

GAAAATGTCTTGTAAGGCTTCTGGCTACACCTTTACCTCCTACGTCATCC

AATGGGTGAAGCAGAAGCCCGGTCAAGGTCTCGAGTGGATCGGCAGCATC

AATCCCTACAACGATTACACCAAGTATAACGAAAAGTTTAAGGGCAAGGC

CACTCTGACAAGCGACAAGAGCTCCATTACCGCCTACATGGAGTTTTCCT

CTTTAACTTCTGAGGACTCCGCTTTATACTATTGCGCTCGTTGGGGCGAT

GGCAATTATTGGGGCCGGGGAACTACTTTAACAGTGAGCTCC.

In some embodiments, an anti-CD28 scFv can include a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 6)
VQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKPGQGLEWIGSI
NPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSALYYCARWGD
GNYWGRGTTLTVSSGGGGSGGGGSGGGGSDIEMTQSPAIMSASLGERVTM
TCTASSSVSSSYFHWYQQKPGSSPKLCIYSTSNLASGVPPRFSGSGSTSY
SLTISSMEAEDAATYFCHQYHRSPTFGGGTKLETKR.

In some embodiments, an anti-CD28 scFv can include the six CDRs present in SEQ ID NO: 6.

In some embodiments, an anti-CD28 scFv can include a sequence encoded by a nucleic acid sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 30)
GTGCAGCTGCAGCAGTCCGGACCCGAACTGGTCAAGCCCGGTGCCTCCGT
GAAAATGTCTTGTAAGGCTTCTGGCTACACCTTTACCTCCTACGTCATCC
AATGGGTGAAGCAGAAGCCCGGTCAAGGTCTCGAGTGGATCGGCAGCATC
AATCCCTACAACGATTACACCAAGTATAACGAAAAGTTTAAGGGCAAGGC
CACTCTGACAAGCGACAAGAGCTCCATTACCGCCTACATGGAGTTTTCCT
CTTTAACTTCTGAGGACTCCGCTTTATACTATTGCGCTCGTTGGGGCGAT
GGCAATTATTGGGGCCGGGGAACTACTTTAACAGTGAGCTCCGGCGGCGG
CGGAAGCGGAGGTGGAGGATCTGGCGGTGGAGGCAGCGACATCGAGATGA
CACAGTCCCCCGCTATCATGAGCGCCTCTTTAGGAGAACGTGTGACCATG
ACTTGTACAGCTTCCTCCAGCGTGAGCAGCTCCTATTTCCACTGGTACCA
GCAGAAACCCGGCTCCTCCCCTAAACTGTGTATCTACTCCACAAGCAATT
TAGCTAGCGGCGTGCCTCCTCGTTTTAGCGGCTCCGGCAGCACCTCTTAC
TCTTTAACCATTAGCTCTATGGAGGCCGAAGATGCCGCCACATACTTTTG
CCATCAGTACCACCGGTCCCCTACCTTTGGCGGAGGCACAAAGCTGGAGA
CCAAGCGG.

In some embodiments, a single-chain chimeric polypeptide that is a CD3/CD28-binding agent can include a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 7)
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDT
SKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSG
TKLEINRGGGGSGGGGSGGGGSQVQLQQSGAELARPGASVKMSCKASGYT
FTRYTMHWVKQRPGQGLEWIGYINPSRGYTNYNQKFKDKATLTTDKSSST

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AYMQLSSLTSEDSAVYYCARYYDDHYCLDYWGQGTTLTVSSSGTTNTVAA
YNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTDTECDL
TDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEFTPYLETNLG
QPTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKDLIYTLYYWK
SSSSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVEC
MGQEKGEFREVQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKP
GQGLEWIGSINPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDS
ALYYCARWGDGNYWGRGTTLTVSSGGGGSGGGGSGGGGSDIEMTQSPAIM
SASLGERVTMTCTASSSVSSSYFHWYQQKPGSSPKLCIYSTSNLASGVPP
RFSGSGSTSYSLTISSMEAEDAATYFCHQYHRSPTFGGGTKLETKR.

In some embodiments, a single-chain chimeric polypeptide that is a CD3/CD28-binding agent is encoded by a nucleic acid that includes a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to

(SEQ ID NO: 9)
CAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCGGTGA
GAAGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATGAACT
GGTATCAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGACACC
AGCAAGCTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGATCCGG
CACCAGCTACTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCTGCCA
CCTACTATTGCCAGCAATGGAGCAGCAACCCCTTCACATTCGGATCTGGC
ACCAAGCTCGAAATCAATCGTGGAGGAGGTGGCAGCGGCGGCGGTGGATC
CGGCGGAGGAGGAAGCCAAGTTCAACTCCAGCAGAGCGGCGCTGAACTGG
CCCGGCCCGGCGCCTCCGTCAAGATGAGCTGCAAGGCTTCCGGCTATACA
TTTACTCGTTACACAATGCATTGGGTCAAGCAGAGGCCCGGTCAAGGTTT
AGAGTGGATCGGATATATCAACCCTTCCCGGGGCTACACCAACTATAACC
AAAAGTTCAAGGATAAAGCCACTTTAACCACTGACAAGAGCTCCTCCACC
GCCTACATGCAGCTGTCCTCTTTAACCAGCGAGGACTCCGCTGTTTACTA
CTGCGCTAGGTATTACGACGACCACTACTGTTTAGACTATTGGGGACAAG
GTACCACTTTAACCGTCAGCAGCTCCGGCACCACCAATACCGTGGCCGCT
TATAACCTCACATGGAAGAGCACCAACTTCAAGACAATTCTGGAATGGGA
ACCCAAGCCCGTCAATCAAGTTTACACCGTGCAGATCTCCACCAAATCCG
GAGACTGGAAGAGCAAGTGCTTCTACACAACAGACACCGAGTGTGATTTA
ACCGACGAAATCGTCAAGGACGTCAAGCAAACCTATCTGGCTCGGGTCTT
TTCCTACCCCGCTGGCAATGTCGAGTCCACCGGCTCCGCTGGCGAGCCTC
TCTACGAGAATTCCCCCGAATTCACCCCTTATTTAGAGACCAATTTAGGC
CAGCCTACCATCCAGAGCTTCGAGCAAGTTGGCACCAAGGTGAACGTCAC
CGTCGAGGATGAAAGGACTTTAGTGCGGCGGAATAACACATTTTTATCCC
TCCGGGATGTGTTCGGCAAAGACCTCATCTACACACTGTACTATTGGAAG
TCCAGCTCCTCCGGCAAAAAGACCGCTAAGACCAACACCAACGAGTTTTT

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AATTGACGTGGACAAAGGCGAGAACTACTGCTTCAGCGTGCAAGCCGTGA

TCCCTTCTCGTACCGTCAACCGGAAGAGCACAGATTCCCCCGTTGAGTGC

ATGGGCCAAGAAAAGGGCGAGTTCCGGGAGGTCCAGCTGCAGCAGAGCGG

ACCCGAACTCGTGAAACCCGGTGCTTCCGTGAAAATGTCTTGTAAGGCCA

GCGGATACACCTTCACCTCCTATGTGATCCAGTGGGTCAAACAGAAGCCC

GGACAAGGTCTCGAGTGGATCGGCAGCATCAACCCTTACAACGACTATAC

CAAATACAACGAGAAGTTTAAGGGAAAGGCTACTTTAACCTCCGACAAAA

GCTCCATCACAGCCTACATGGAGTTCAGCTCTTTAACATCCGAGGACAGC

GCTCTGTACTATTGCGCCCGGTGGGGCGACGGCAATTACTGGGGACGGGG

CACAACACTGACCGTGAGCAGCGGAGGCGGAGGCTCCGGCGGAGGCGGAT

CTGGCGGTGGCGGCTCCGACATCGAGATGACCCAGTCCCCCGCTATCATG

TCCGCCTCTTTAGGCGAGCGGGTCACAATGACTTGTACAGCCTCCTCCAG

CGTCTCCTCCTCCTACTTCCATTGGTACCAACAGAAACCCGGAAGCTCCC

CTAAACTGTGCATCTACAGCACCAGCAATCTCGCCAGCGGCGTGCCCCCT

AGGTTTTCCGGAAGCGGAAGCACCAGCTACTCTTTAACCATCTCCTCCAT

GGAGGCTGAGGATGCCGCCACCTACTTTTGTCACCAGTACCACCGGTCCC

CCACCTTCGGAGGCGGCACCAAACTGGAGACAAAGAGG.

In some embodiments, a single-chain chimeric polypeptide that is a CD3/CD28-binding agent can include a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 8)

MKWVTFISLLFLFSSAYSQIVLTQSPAIMSASPGEKVTMTCSASSSVSYM

NWYQQKSGTSPKRWIYDTSKLASGVPAHFRGSGSGTSYSLTISGMEAEDA

ATYYCQQWSSNPFTFGSGTKLEINRGGGGSGGGGSGGGGSQVQLQQSGAE

LARPGASVKMSCKASGYTFTRYTMHWVKQRPGQGLEWIGYINPSRGYTNY

NQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAVYYCARYYDDHYCLDYWG

QGTTLTVSSSGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTK

SGDWKSKCFYTTDTECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGE

PLYENSPEFTPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRRNNTFL

SLRDVFGKDLIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQA

VIPSRTVNRKSTDSPVECMGQEKGEFREVQLQQSGPELVKPGASVKMSCK

ASGYTFTSYVIQWVKQKPGQGLEWIGSINPYNDYTKYNEKFKGKATLTSD

KSSITAYMEFSSLTSEDSALYYCARWGDGNYWGRGTTLTVSSGGGGSGGG

GSGGGGSDIEMTQSPAIMSASLGERVTMTCTASSSVSSSYFHWYQQKPGS

SPKLCIYSTSNLASGVPPRFSGSGSGTSYSLTISSMEAEDAATYFCHQYHR

SPTFGGGTKLETKR.

In some embodiments, a single-chain chimeric polypeptide that is a CD3/CD28-binding agent is encoded by a nucleic acid that includes a sequence that is at least 70% identical (e.g., at least 75% identical, at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 99% identical, or 100% identical) to:

(SEQ ID NO: 10)

ATGAAGTGGGTGACCTTCATCAGCTTATTATTTTTATTCAGCTCCGCCTA

TTCCCAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCG

GTGAGAAGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATG

AACTGGTATCAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGA

CACCAGCAAGCTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGAT

CCGGCACCAGCTACTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCT

GCCACCTACTATTGCCAGCAATGGAGCAGCAACCCCTTCACATTCGGATC

TGGCACCAAGCTCGAAATCAATCGTGGAGGAGGTGGCAGCGGCGGCGGTG

GATCCGGCGGAGGAGGAAGCCAAGTTCAACTCCAGCAGAGCGGCGCTGAA

CTGGCCCGGCCCGGCGCCTCCGTCAAGATGAGCTGCAAGGCTTCCGGCTA

TACATTTACTCGTTACACAATGCATTGGGTCAAGCAGAGGCCCGGTCAAG

GTTTAGAGTGGATCGGATATATCAACCCTTCCCGGGGCTACACCAACTAT

AACCAAAAGTTCAAGGATAAAGCCACTTTAACCACTGACAAGAGCTCCTC

CACCGCCTACATGCAGCTGTCCTCTTTAACCAGCGAGGACTCCGCTGTTT

ACTACTGCGCTAGGTATTACGACGACCACTACTGTTTAGACTATTGGGGA

CAAGGTACCACTTTAACCGTCAGCAGCTCCGGCACCACCAATACCGTGGC

CGCTTATAACCTCACATGGAAGAGCACCAACTTCAAGACAATTCTGGAAT

GGGAACCCAAGCCCGTCAATCAAGTTTACACCGTGCAGATCTCCACCAAA

TCCGGAGACTGGAAGAGCAAGTGCTTCTACACAACAGACACCGAGTGTGA

TTTAACCGACGAAATCGTCAAGGACGTCAAGCAAACCTATCTGGCTCGGG

TCTTTTCCTACCCCGCTGGCAATGTCGAGTCCACCGGCTCCGCTGGCGAG

CCTCTCTACGAGAATTCCCCCGAATTCACCCCTTATTTAGAGACCAATTT

AGGCCAGCCTACCATCCAGAGCTTCGAGCAAGTTGGCACCAAGGTGAACG

TCACCGTCGAGGATGAAAGGACTTTAGTGCGGCGGAATAACACATTTTTA

TCCCTCCGGGATGTGTTCGGCAAAGACCTCATCTACACACTGTACTATTG

GAAGTCCAGCTCCTCCGGCAAAAAGACCGCTAAGACCAACACCAACGAGT

TTTTAATTGACGTGGACAAAGGCGAGAACTACTGCTTCAGCGTGCAAGCC

GTGATCCCTTCTCGTACCGTCAACCGGAAGAGCACAGATTCCCCCGTTGA

GTGCATGGGCCAAGAAAAGGGCGAGTTCCGGGAGGTCCAGCTGCAGCAGA

GCGGACCCGAACTCGTGAAACCCGGTGCTTCCGTGAAAATGTCTTGTAAG

GCCAGCGGATACACCTTCACCTCCTATGTGATCCAGTGGGTCAAACAGAA

GCCCGGACAAGGTCTCGAGTGGATCGGCAGCATCAACCCTTACAACGACT

ATACCAAATACAACGAGAAGTTTAAGGGAAAGGCTACTTTAACCTCCGAC

AAAAGCTCCATCACAGCCTACATGGAGTTCAGCTCTTTAACATCCGAGGA

CAGCGCTCTGTACTATTGCGCCCGGTGGGGCGACGGCAATTACTGGGGAC

GGGGCACAACACTGACCGTGAGCAGCGGAGGCGGAGGCTCCGGCGGAGGC

GGATCTGGCGGTGGCGGCTCCGACATCGAGATGACCCAGTCCCCCGCTAT

CATGTCCGCCTCTTTAGGCGAGCGGGTCACAATGACTTGTACAGCCTCCT

CCAGCGTCTCCTCCTCCTACTTCCATTGGTACCAACAGAAACCCGGAAGC

TCCCCTAAACTGTGCATCTACAGCACCAGCAATCTCGCCAGCGGCGTGCC

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CCCTAGGTTTTCCGGAAGCGGAAGCACCAGCTACTCTTTAACCATCTCCT

CCATGGAGGCTGAGGATGCCGCCACCTACTTTTGTCACCAGTACCACCGG

TCCCCCACCTTCGGAGGCGGCACCAAACTGGAGACAAAGAGG.
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Additional Antigen-Binding Domains

Some embodiments of any of the single-chain chimeric polypeptides described herein can further include one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at its N- and/or C-terminus.

In some embodiments, the single-chain chimeric polypeptides can include one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at its N-terminus. In some embodiments, one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the N-terminus of the single-chain chimeric polypeptide can directly abut the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein). In some embodiments, the single-chain chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between one of the at least one additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the N-terminus of the single-chain chimeric polypeptide and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein).

In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide includes one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at its C-terminus. In some embodiments, one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the C-terminus of the single-chain chimeric polypeptide directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein or known in the art). In some embodiments, the single-chain chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between one of the at least one additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the C-terminus of the single-chain chimeric polypeptide and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein).

In some embodiments of any of the single-chain chimeric polypeptides described herein, the single-chain chimeric polypeptide comprises one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at its N-terminus and its C-terminus. In some embodiments, one of the one or more additional antigen binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the N-terminus of the single-chain chimeric polypeptide directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein). In some embodiments, the single-chain chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between one of the one or more additional antigen-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the N-terminus and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains). In some embodiments, one of the one or more additional antigen binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the C-terminus directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains). In some embodiments, the single-chain chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between one of the one or more additional antigen-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the C-terminus and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), or the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein).

In some embodiments of any of the single-chain chimeric polypeptides described herein, two or more (e.g., three, four, five, six, seven, eight, nine, or ten) of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) bind specifically to the same antigen. In some embodiments, two or more (e.g., three, four, five, six, seven, eight, nine, or ten) of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) bind specifically to the same epitope. In some embodiments, two or more (e.g., three, four, five, six, seven, eight, nine, or ten) of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) include the same amino acid sequence.

In some embodiments of any of the single-chain chimeric polypeptides described herein, the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) each bind specifically to the same antigen. In some embodiments, the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) each bind specifically to the same epitope. In some embodiments, the first target-binding domain, the second target-binding domain, and the one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains each comprise the same amino acid sequence.

In some embodiments of any of the single-chain chimeric polypeptides described herein, the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) bind specifically to different antigens.

In some embodiments of any of the single-chain chimeric polypeptides, one or more of the first target-binding domain, the second target-binding domain, and the one or more target-binding domains is an antigen-binding domain (e.g., any of the exemplary antigen-binding domains described herein or known in the art). In some embodiments of any of the single-chain chimeric polypeptides described herein, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains are each an antigen-binding domain (e.g., any of the exemplary antigen-binding domains described herein or known in the art). In some embodiments, the antigen-binding domain can include a scFv or a single domain antibody.

In some embodiments of any of the single-chain chimeric polypeptides described herein, one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) bind specifically to CD28, CD3, or a receptor for IL-2.

In some embodiments of any of the single-chain chimeric polypeptides described herein, one or more of the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), the second target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art), and the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) is a soluble IL-2.

In some embodiments of any of the multi-chain chimeric polypeptides, the first chimeric polypeptide further includes one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domain(s) (e.g., any of the exemplary target-binding domains described herein or known in the art), where at least one of the one or more additional antigen-binding domain(s) is positioned between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein or known in the art) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein). In some embodiments, the first chimeric polypeptide can further include a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the at least one of the one or more additional target-binding domain(s) (e.g., any of the exemplary target-binding domains described herein or known in the art), and/or a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the at least one of the one or more additional target-binding domain(s) (e.g., any of the exemplary target-binding domains described herein or known in the art) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein).

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the first chimeric polypeptide further includes one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains at the N-terminal and/or C-terminal end of the first chimeric polypeptide. In some embodiments, at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abuts the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the first chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein). In some embodiments, the at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) in the first chimeric polypeptide. In some embodiments, the first chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between the at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art).

In some embodiments of any of the multi-chain chimeric polypeptides described herein, at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) is disposed at the N- and/or C-terminus of the first chimeric polypeptide, and at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) is positioned between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein or known in the art) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the at least one additional target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) of the one or more additional target-binding domains disposed at the N-terminus directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) or the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the first chimeric polypeptide further comprises a linker sequence (e.g., any of the linker sequences described herein or known in the art) disposed between the at least one additional target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) or the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the at least one additional target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) of the one or more additional target-binding domains disposed at the C-terminus directly abuts the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) or the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the first chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) disposed between the at least one additional target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) and the first target-binding domain (e.g., any of the exemplary target-binding domains described herein or known in the art) or the first domain of the pair of affinity domains (e.g., any of the exemplary first domains described herein of any of the exemplary pairs of affinity domains described herein) in the first chimeric polypeptide. In some embodiments, the at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) positioned between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the first domain of the pair of affinity domains (e.g., any of the first domains described herein or any of the exemplary pairs of affinity domains described herein), directly abuts the soluble tissue factor domain and/or the first domain of the pair of affinity domains. In some embodiments, the first chimeric polypeptide further comprises a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) disposed (i) between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) positioned between the soluble tissue factor domain (e.g., any of the exemplary soluble tissue factor domains described herein) and the first domain of the pair of affinity domains (e.g., any of the exemplary first domains of any of the exemplary pairs of affinity domains described herein), and/or (ii) between the first domain of the pair of affinity domains and the at least one of the one or more additional target-binding domains positioned between the soluble tissue factor domain and the first domain of the pair of affinity domains.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the second chimeric polypeptide further includes one or more (e.g., two, three, four, five, six, seven, eight, nine, or ten) additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) at the N-terminal end and/or the C-terminal end of the second chimeric polypeptide. In some embodiments, at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abuts the second domain of the pair of affinity domains (e.g., any of the exemplary second domains of any of the exemplary pairs of affinity domains described herein) in the second chimeric polypeptide. In some embodiments, the second chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) and the second domain of the pair of affinity domains (e.g., any of the second domains described herein of any of the exemplary pairs of affinity domains described herein) in the second chimeric polypeptide. In some embodiments, at least one of the one or more additional target-binding domains (e.g., any of the exemplary target-binding domains described herein or known in the art) directly abuts the second target-binding domain (e.g., any of the target-binding domains described herein or known in the art) in the second chimeric polypeptide. In some embodiments, the second chimeric polypeptide further includes a linker sequence (e.g., any of the exemplary linker sequences described herein or known in the art) between at least one of the one or more additional target-binding domains (e.g., any of the exemplary target binding domains described herein or known in the art) and the second target-binding domain (e.g., any of the exemplary target binding domains described herein or known in the art) in the second chimeric polypeptide.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, two or more (e.g., three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or ten or more) of the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains bind specifically to the same antigen. In some embodiments, two or more (e.g., three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or ten or more) of the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains bind specifically to the same epitope. In some embodiments, two or more (e.g., three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or ten or more) of the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains include the same amino acid sequence. In some embodiments, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains each bind specifically to the same antigen. In some embodiments, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains each bind specifically to the same epitope. In some embodiments, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains each include the same amino acid sequence.

In some embodiments of any of the multi-chain chimeric polypeptides described herein, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains bind specifically to different antigens. In some embodiments of any of the multi-chain chimeric polypeptides described herein, one or more (e.g., two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or ten or more) of the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains is an antigen-binding domain. In some embodiments, the first target-binding domain, the second target-binding domain, and the one or more additional target-binding domains are each an antigen-binding domain (e.g., a scFv or a single-domain antibody).

Signal Sequence

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide described herein can further include a signal sequence at its N-terminal end. As will be understood by those of ordinary skill in the art, a signal sequence is an amino acid sequence that is present at the N-terminus of a number of endogenously produced proteins that directs the protein to the secretory pathway (e.g., the protein is directed to reside in certain intracellular organelles, to reside in the cell membrane, or to be secreted from the cell). Signal sequences are heterogeneous and differ greatly in their primary amino acid sequences. However, signal sequences are typically 16 to 30 amino acids in length and include a hydrophilic, usually positively charged N-terminal region, a central hydrophobic domain, and a C-terminal region that contains the cleavage site for signal peptidase.

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence having an amino acid sequence MKWVTFISLLFLFSSAYS (SEQ ID NO: 33). In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence encoded by the nucleic acid sequence:

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                                            (SEQ ID NO: 34)
ATGAAATGGGTGACCTTTATTTCTTTACTGTTCCTCTTTAGCAGCGCCTA

CTCC,

                                            (SEQ ID NO: 35)
ATGAAGTGGGTCACATTTATCTCTTTACTGTTCCTCTTCTCCAGCGCCTA

CAGC,

                                            (SEQ ID NO: 36)
ATGAAATGGGTGACCTTTATTTCTTTACTGTTCCTCTTTAGCAGCGCCTA

CTCC,
or

                                           (SEQ ID NO: 101)
ATGAAATGGGTCACCTTCATCTCTTTACTGTTTTTATTTAGCAGCGCCTA

CAGC.
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In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence having an amino acid sequence MKCLLYLAFLFLGVNC (SEQ ID NO: 37). In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence having an amino acid sequence: MGQIVTMFEALPHIIDEVINIVIIV-LIIITSIKAVYNFATCGILALVSFLFLAGRSCG (SEQ ID NO: 38). In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence having an amino acid sequence:

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                                            (SEQ ID NO: 39)
MPNHQSGSPTGSSDLLLSGKKQRPHLALRRKRRREMRKINRKVRRMNLAP

IKEKTAWQHLQALISEAEEVLKTSQTPQNSLTLFLALLSVLGPPVTG.
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In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence having an amino acid sequence MDSKGSSQKGSRLLLLL-VVSNLLLCQGVVS (SEQ ID NO: 40). Those of ordinary skill in the art will be aware of other appropriate signal sequences for use in a single-chain chimeric polypeptide.

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence that is about 10 to 100 amino acids in length. For example, a signal sequence can be about 10 to 100 amino acids in length, about 15 to 100 amino acids in length, about 20 to 100 amino acids in length, about 25 to 100 amino acids in length, about 30 to 100 amino acids in length, about 35 to 100 amino acids in length, about 40 to 100 amino acids in length, about 45 to 100 amino acids in length, about 50 to 100 amino acids in length, about 55 to 100 amino acids in length, about 60 to 100 amino acids in length, about 65 to 100 amino acids in length, about 70 to 100 amino acids in length, about 75 to 100 amino acids in length, about 80 to 100 amino acids in length, about 85 to 100 amino acids in length, about 90 to 100 amino acids in length, about 95 to 100 amino acids in length, about 10 to 95 amino acids in length, about 10 to 90 amino acids in length, about 10 to 85 amino acids in length, about 10 to 80 amino acids in length, about 10 to 75 amino acids in length, about 10 to 70 amino acids in length, about 10 to 65 amino acids in length, about 10 to 60 amino acids in length, about 10 to 55 amino acids in length, about 10 to 50 amino acids in length, about 10 to 45 amino acids in length, about 10 to 40 amino acids in length, about 10 to 35 amino acids in length, about 10 to 30 amino acids in length, about 10 to 25 amino acids in length, about 10 to 20 amino acids in length, about 10 to 15 amino acids in length, about 20 to 30 amino acids in length, about 30 to 40 amino acids in length, about 40 to 50 amino acids in length, about 50 to 60 amino acids in length, about 60 to 70 amino acids in length, about 70 to 80 amino acids in length, about 80 to 90 amino acids in length, about 90 to 100 amino acids in length, about 20 to 90 amino acids in length, about 30 to 80 amino acids in length, about 40 to 70 amino acids in length, about 50 to 60 amino acids in length, or any range in between. In some embodiments, a signal sequence is about 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 amino acids in length.

In some embodiments, any of the signal sequences disclosed herein can include one or more additional amino acids (e.g., 1, 2, 3, 5, 6, 7, 8, 9, 10, or more amino acids) at its N-terminus and/or C-terminus, so long as the function of the signal sequence remains intact. For example, a signal sequence having the amino acid sequence MKWVTFISLLFLFSSAYS (SEQ ID NO: 33) can include one or more additional amino acids at the N-terminus or C-terminus, while still retaining the ability to direct the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide to the secretory pathway.

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a signal sequence that directs the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide into the extracellular space. Such embodiments are useful in producing single-chain chimeric polypeptides, first chimeric polypeptides, and second chimeric polypeptides that are relatively easy to be isolated and/or purified.

Peptide Tags

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a peptide tag (e.g., at the N-terminal end or the C-terminal end of the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide). In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide includes two or more peptide tags.

Exemplary peptide tags that can be included in a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide include, without limitation: AviTag (GLNDIFEAQKIEWHE; SEQ ID NO: 41), a calmodulin-tag (KRRWKKNFIAVSAANRFKKISSSGAL; SEQ ID NO: 42), a polyglutamate tag (EEEEEE; SEQ ID NO: 43), an E-tag (GAPVPYPDPLEPR; SEQ ID NO: 44), a FLAG-tag (DYKDDDDK; SEQ ID NO: 45), an HA-tag, a peptide from hemagglutinin (YPYDVPDYA; SEQ ID NO: 46), a his-tag (HHHHH (SEQ ID NO: 47); HHHHHH (SEQ ID NO: 48); HHHHHHH (SEQ ID NO: 49); HHHHHHHH (SEQ ID NO: 50); HHHHHHHHH (SEQ ID NO: 51); or HHHHHHHHHH (SEQ ID NO: 52)), a myc-tag (EQKLISEEDL; SEQ ID NO: 53), NE-tag (TKENPRSNQEESYDDNES; SEQ ID NO: 54), S-tag, (KETAAAKFERQHMDS; SEQ ID NO: 55), SBP-tag (MDEKTTGWRGGHVVEGLAGELEQLRARLEHHPQGQREP; SEQ ID NO: 56), Softag 1 (SLAELLNAGLGGS; SEQ ID NO: 57), Softag 3 (TQDPSRVG; SEQ ID NO: 58), Spot-tag (PDRVRAVSHWSS; SEQ ID NO: 59), Strep-tag (WSHPQFEK; SEQ ID NO: 60), TC tag (CCPGCC; SEQ ID NO: 61), Ty tag (EVHTNQDPLD; SEQ ID NO: 62), V5 tag (GKPIPNPLLGLDST; SEQ ID NO: 63), VSV-tag (YTDIEMNRLGK; SEQ ID NO: 64), and Xpress tag (DLYDDDDK; SEQ ID NO: 65). In some embodiments, tissue factor protein is a peptide tag.

Peptide tags that can be included in a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can be used in any of a variety of applications related to the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide. For example, a peptide tag can be used in the purification of a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide. As one non-limiting example, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a myc tag; and can be purified using an antibody that recognizes the myc tag(s). One non-limiting example of an antibody that recognizes a myc tag is 9E10, available from the non-commercial Developmental Studies Hybridoma Bank. As another non-limiting example, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a histidine tag, and can be purified using a nickel or cobalt chelate. Those of ordinary skill in the art will be aware of other suitable tags and agent that bind those tags for use in purifying a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide. In some embodiments, a peptide tag is removed from the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide after purification. In some embodiments, a peptide tag is not removed from the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide after purification.

Peptide tags that can be included in a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can be used, for example, in immunoprecipitation of the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide, imaging of the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide (e.g., via Western blotting, ELISA, flow cytometry, and/or immunocytochemistry), and/or solubilization of the single-chain chimeric polypeptide, the first chimeric polypeptide, or the second chimeric polypeptide.

In some embodiments, a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can include a peptide tag that is about 10 to 100 amino acids in length. For example, a peptide tag can be about 10 to 100 amino acids in length, about 15 to 100 amino acids in length, about 20 to 100 amino acids in length, about 25 to 100 amino acids in length, about 30 to 100 amino acids in length, about 35 to 100 amino acids in length, about 40 to 100 amino acids in length, about 45 to 100 amino acids in length, about 50 to 100 amino acids in length, about 55 to 100 amino acids in length, about 60 to 100 amino acids in length, about 65 to 100 amino acids in length, about 70 to 100 amino acids in length, about 75 to 100 amino acids in length, about 80 to 100 amino acids in length, about 85 to 100 amino acids in length, about 90 to 100 amino acids in length, about 95 to 100 amino acids in length, about 10 to 95 amino acids in length, about 10 to 90 amino acids in length, about 10 to 85 amino acids in length, about 10 to 80 amino acids in length, about 10 to 75 amino acids in length, about 10 to 70 amino acids in length, about 10 to 65 amino acids in length, about 10 to 60 amino acids in length, about 10 to 55 amino acids in length, about 10 to 50 amino acids in length, about 10 to 45 amino acids in length, about 10 to 40 amino acids in length, about 10 to 35 amino acids in length, about 10 to 30 amino acids in length, about 10 to 25 amino acids in length, about 10 to 20 amino acids in length, about 10 to 15 amino acids in length, about 20 to 30 amino acids in length, about 30 to 40 amino acids in length, about 40 to 50 amino acids in length, about 50 to 60 amino acids in length, about 60 to 70 amino acids in length, about 70 to 80 amino acids in length, about 80 to 90 amino acids in length, about 90 to 100 amino acids in length, about 20 to 90 amino acids in length, about 30 to 80 amino acids in length, about 40 to 70 amino acids in length, about 50 to 60 amino acids in length, or any range in between. In some embodiments, a peptide tag is about 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 amino acids in length.

Peptide tags included in a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide can be of any suitable length. For example, peptide tags can be 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, or more amino acids in length. In embodiments in which a single-chain chimeric polypeptide, a first chimeric polypeptide, or a second chimeric polypeptide includes two or more peptide tags, the two or more peptide tags can be of the same or different lengths. In some embodiments, any of the peptide tags disclosed herein may include one or more additional amino acids (e.g., 1, 2, 3, 5, 6, 7, 8, 9, 10, or more amino acids) at the N-terminus and/or C-terminus, so long as the function of the peptide tag remains intact. For example, a myc tag having the amino acid sequence EQKLISEEDL (SEQ ID NO: 53) can include one or more additional amino acids (e.g., at the N-terminus and/or the C-terminus of the peptide tag), while still retaining the ability to be bound by an antibody (e.g., 9E10).

CD3/CD28-Binding Agents

Provided herein are methods that include the use of one or more CD3/CD28-binding agent(s). A CD3/CD28-binding agent can be, for example, a single-chain chimeric polypeptide, a multi-chain chimeric polypeptide, a bi-specific antibody, or an anti-CD3/anti-CD28 bead.

Single-Chain Chimeric Polypeptides

Non-limiting examples of CD3/CD28-binding agents are single-chain chimeric polypeptides that include a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where: the first target-binding domain of the single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the single-chain chimeric polypeptide binds to CD28. Non-limiting examples of single-chain chimeric polypeptides that are CD3/CD28-binding agents are described herein.

Multi-Chain Chimeric Polypeptides

Non-limiting examples of CD3/CD28-binding agents are multi-chain chimeric polypeptides that include a first chimeric polypeptide including (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein: the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3.

In some embodiments, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide. In some embodiments, the IgG1 antibody construct comprises a heavy chain variable domain comprising CDRs of SEQ ID NOs: 66, 67 or 74, and 68, and a light chain variable domain comprising CDRs of SEQ ID NOs: 69, 70, and 71. In some embodiments, the heavy chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 72, and the light chain variable domain comprises a sequence that is at least 90% identical to SEQ ID NO: 73. In some embodiments, the heavy chain variable domain comprises SEQ ID NO: 72, and the light chain variable domain comprises SEQ ID NO: 73.

In some embodiments, the first target-binding domain of the first chimeric polypeptide and the soluble tissue factor domain directly abut each other. In some embodiments, the first chimeric polypeptide further comprises a linker sequence between the first target-binding domain and the soluble tissue factor domain. In some embodiments, the soluble tissue factor domain and the second target-binding domain of the second chimeric polypeptide directly abut each other. In some embodiments, the second chimeric polypeptide further comprises a linker sequence between the soluble tissue factor domain and the second target-binding domain.

In some embodiments, the CD3 is human CD3. In some embodiments, the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 5.

In some embodiments, the soluble tissue factor domain of the first chimeric polypeptide is a soluble human tissue factor domain. In some embodiments, the soluble human tissue factor domain of the first chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 2. In some embodiments, the soluble tissue factor domain of the first chimeric polypeptide comprises or consists of a sequence from a wildtype soluble human tissue factor.

In some embodiments, the soluble tissue factor domain of the first chimeric polypeptide does not include any of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments, the soluble tissue factor domain of the first chimeric polypeptide includes one or more of: a lysine at an amino acid position that corresponds to amino acid position 20 of mature wildtype human tissue factor protein; an isoleucine at an amino acid position that corresponds to amino acid position 22 of mature wildtype human tissue factor protein; a tryptophan at an amino acid position that corresponds to amino acid position 45 of mature wildtype human tissue factor protein; an aspartic acid at an amino acid position that corresponds to amino acid position 58 of mature wildtype human tissue factor protein; a tyrosine at an amino acid position that corresponds to amino acid position 94 of mature wildtype human tissue factor protein; an arginine at an amino acid position that corresponds to amino acid position 135 of mature wildtype human tissue factor protein; and a phenylalanine at an amino acid position that corresponds to amino acid position 140 of mature wildtype human tissue factor protein.

In some embodiments, the soluble tissue factor domain of the first chimeric polypeptide does not initiate blood coagulation. In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 102. In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80%, least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 104.

In some embodiments, the CD28 is human CD28. In some embodiments, the second target-binding domain of the second chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 6. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 106. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80%, at least 85%, at least 90%, at least 95%, at least 99%, or 100% identical to SEQ ID NO: 108. In some embodiments, the multi-chain chimeric polypeptide does not initiate blood coagulation.

In some embodiments, the liquid culture medium comprises about 10 nM to about 1000 nM (or any of the subranges of this range described herein) of the multi-chain chimeric polypeptide. In some embodiments, the liquid culture medium comprises about 50 nM to about 300 nM (or any of the subranges of this range described herein) of the multi-chain chimeric polypeptide.

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Exemplary Mature First Chimeric Polypeptide
                                                (SEQ ID NO: 102)
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDTSKLA

SGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSGTKLEINRGG

GGSGGGGSGGGGSQVQLQQSGAELARPGASVKMSCKASGYTFTRYTMHWVKQ

RPGQGLEWIGYINPSRGYTNYNQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAV

YYCARYYDDHYCLDYWGQGTTLTVSSSGTTNTVAAYNLTWKSTNFKTILEWEP

KPVNQVYTVQISTKSGDWKSKCFYTTDTECDLTDEIVKDVKQTYLARVFSYPAG

NVESTGSAGEPLYENSPEFTPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRR

NNTFLSLRDVFGKDLIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQ

AVIPSRTVNRKSTDSPVECMGQEKGEFRENWVNVISNLKKIEDLIQSMHIDATLY

TESDVHPSCKVTAMKCFLLELQVISLESGDASIHDTVENLIILANNSLSSNGNVTE

SGCKECEELEEKNIKEFLQSFVHIVQMFINTS

Exemplary DNA Encoding Mature First Chimeric Polypeptide
                                                (SEQ ID NO: 103)
CAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCGGTGAGA

AGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATGAACTGGTAT

CAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGACACCAGCAAG

CTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGATCCGGCACCAGCTA

CTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCTGCCACCTACTATTGCC

AGCAATGGAGCAGCAACCCCTTCACATTCGGATCTGGCACCAAGCTCGAAAT

CAATCGTGGAGGAGGTGGCAGCGGCGGCGGTGGATCCGGCGGAGGAGGAAG

CCAAGTTCAACTCCAGCAGAGCGGCGCTGAACTGGCCCGGCCCGGCGCCTCC

GTCAAGATGAGCTGCAAGGCTTCCGGCTATACATTTACTCGTTACACAATGCA

TTGGGTCAAGCAGAGGCCCGGTCAAGGTTTAGAGTGGATCGGATATATCAAC

CCTTCCCGGGGCTACACCAACTATAACCAAAAGTTCAAGGATAAAGCCACTT

TAACCACTGACAAGAGCTCCTCCACCGCCTACATGCAGCTGTCCTCTTTAACC

AGCGAGGACTCCGCTGTTTACTACTGCGCTAGGTATTACGACGACCACTACTG

TTTAGACTATTGGGGACAAGGTACCACTTTAACCGTCAGCAGCTCCGGCACC
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-continued

ACCAATACCGTGGCCGCTTATAACCTCACATGGAAGAGCACCAACTTCAAGA

CAATTCTGGAATGGGAACCCAAGCCCGTCAATCAAGTTTACACCGTGCAGAT

CTCCACCAAATCCGGAGACTGGAAGAGCAAGTGCTTCTACACAACAGACACC

GAGTGTGATTTAACCGACGAAATCGTCAAGGACGTCAAGCAAACCTATCTGG

CTCGGGTCTTTTCCTACCCCGCTGGCAATGTCGAGTCCACCGGCTCCGCTGGC

GAGCCTCTCTACGAGAATTCCCCCGAATTCACCCCTTATTTAGAGACCAATTT

AGGCCAGCCTACCATCCAGAGCTTCGAGCAAGTTGGCACCAAGGTGAACGTC

ACCGTCGAGGATGAAAGGACTTTAGTGCGGCGGAATAACACATTTTTATCCC

TCCGGGATGTGTTCGGCAAAGACCTCATCTACACACTGTACTATTGGAAGTCC

AGCTCCTCCGGCAAAAAGACCGCTAAGACCAACACCAACGAGTTTTTAATTG

ACGTGGACAAAGGCGAGAACTACTGCTTCAGCGTGCAAGCCGTGATCCCTTC

TCGTACCGTCAACCGGAAGAGCACAGATTCCCCCGTTGAGTGCATGGGCCAA

GAAAAGGGCGAGTTCCGGGAGAACTGGGTGAACGTCATCAGCAATTTAAAG

AAGATCGAAGATTTAATTCAGTCCATGCATATCGACGCCACTTTATACACAG

AATCCGACGTGCACCCCTCTTGTAAGGTGACCGCCATGAAATGTTTTTTACTG

GAGCTGCAAGTTATCTCTTTAGAGAGCGGAGACGCTAGCATCCACGACACCG

TGGAGAATTTAATCATTTTAGCCAATAACTCTTTATCCAGCAACGGCAACGTG

ACAGAGTCCGGCTGCAAGGAGTGCGAAGAGCTGGAGGAGAAGAACATCAAG

GAGTTTCTGCAATCCTTTGTGCACATTGTCCAGATGTTCATCAATACCTCC

Exemplary Precursor First Chimeric Polypeptide

(SEQ ID NO: 104)

MKWVTFISLLFLFSSAYSQIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQ

KSGTSPKRWIYDTSKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWS

SNPFTFGSGTKLEINRGGGGSGGGGSGGGGSQVQLQQSGAELARPGASVKMSCK

ASGYTFTRYTMHWVKQRPGQGLEWIGYINPSRGYTNYNQKFKDKATLTTDKSS

STAYMQLSSLTSEDSAVYYCARYYDDHYCLDYWGQGTTLTVSSSGTTNTVAAY

NLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCFYTTDTECDLTDEIVK

DVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEFTPYLETNLGQPTIQSFEQVG

TKVNVTVEDERTLVRRNNTFLSLRDVFGKDLIYTLYYWKSSSSGKKTAKTNTNE

FLIDVDKGENYCFSVQAVIPSRTVNRKSTDSPVECMGQEKGEFRENWVNVISNL

KKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLELQVISLESGDASIHDTVEN

LIILANNSLSSNGNVTESGCKECEELEEKNIKEFLQSFVHIVQMFINTS

Exemplary DNA Encoding Precursor First Chimeric Polypeptide

(SEQ ID NO: 105)

ATGAAGTGGGTGACCTTCATCAGCTTATTATTTTTATTCAGCTCCGCCTATTCC

CAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCGGTGAGA

AGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATGAACTGGTAT

CAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGACACCAGCAAG

CTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGATCCGGCACCAGCTA

CTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCTGCCACCTACTATTGCC

AGCAATGGAGCAGCAACCCCTTCACATTCGGATCTGGCACCAAGCTCGAAAT

CAATCGTGGAGGAGGTGGCAGCGGCGGCGGTGGATCCGGCGGAGGAGGAAG

CCAAGTTCAACTCCAGCAGAGCGGCGCTGAACTGGCCCGGCCCGGCGCCTCC

-continued

GTCAAGATGAGCTGCAAGGCTTCCGGCTATACATTTACTCGTTACACAATGCA

TTGGGTCAAGCAGAGGCCCGGTCAAGGTTTAGAGTGGATCGGATATATCAAC

CCTTCCCGGGGCTACACCAACTATAACCAAAAGTTCAAGGATAAAGCCACTT

TAACCACTGACAAGAGCTCCTCCACCGCCTACATGCAGCTGTCCTCTTTAACC

AGCGAGGACTCCGCTGTTTACTACTGCGCTAGGTATTACGACGACCACTACTG

TTTAGACTATTGGGGACAAGGTACCACTTTAACCGTCAGCAGCTCCGGCACC

ACCAATACCGTGGCCGCTTATAACCTCACATGGAAGAGCACCAACTTCAAGA

CAATTCTGGAATGGGAACCCAAGCCCGTCAATCAAGTTTACACCGTGCAGAT

CTCCACCAAATCCGGAGACTGGAAGAGCAAGTGCTTCTACACAACAGACACC

GAGTGTGATTTAACCGACGAAATCGTCAAGGACGTCAAGCAAACCTATCTGG

CTCGGGTCTTTTCCTACCCCGCTGGCAATGTCGAGTCCACCGGCTCCGCTGGC

GAGCCTCTCTACGAGAATTCCCCCGAATTCACCCCTTATTTAGAGACCAATTT

AGGCCAGCCTACCATCCAGAGCTTCGAGCAAGTTGGCACCAAGGTGAACGTC

ACCGTCGAGGATGAAAGGACTTTAGTGCGGCGGAATAACACATTTTTATCCC

TCCGGGATGTGTTCGGCAAAGACCTCATCTACACACTGTACTATTGGAAGTCC

AGCTCCTCCGGCAAAAAGACCGCTAAGACCAACACCAACGAGTTTTTAATTG

ACGTGGACAAAGGCGAGAACTACTGCTTCAGCGTGCAAGCCGTGATCCCTTC

TCGTACCGTCAACCGGAAGAGCACAGATTCCCCCGTTGAGTGCATGGGCCAA

GAAAAGGGCGAGTTCCGGGAGAACTGGGTGAACGTCATCAGCAATTTAAAG

AAGATCGAAGATTTAATTCAGTCCATGCATATCGACGCCACTTTATACACAG

AATCCGACGTGCACCCCTCTTGTAAGGTGACCGCCATGAAATGTTTTTTACTG

GAGCTGCAAGTTATCTCTTTAGAGAGCGGAGACGCTAGCATCCACGACACCG

TGGAGAATTTAATCATTTTAGCCAATAACTCTTTATCCAGCAACGGCAACGTG

ACAGAGTCCGGCTGCAAGGAGTGCGAAGAGCTGGAGGAGAAGAACATCAAG

GAGTTTCTGCAATCCTTTGTGCACATTGTCCAGATGTTCATCAATACCTCC

Exemplary Mature Second Chimeric Polypeptide
(SEQ ID NO: 106)
VQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKPGQGLEWIGSINPYN

DYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSALYYCARWGDGNYWGRG

TTLTVSSGGGGSGGGGSGGGGSDIEMTQSPAIMSASLGERVTMTCTASSSVSSSY

FHWYQQKPGSSPKLCIYSTSNLASGVPPRFSGSGSTSYSLTISSMEAEDAATYFCH

QYHRSPTFGGGTKLETKRITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKA

GTSSLTECVLNKATNVAHWTTPSLKCIR

Exemplary DNA Encoding Mature Second Chimeric Polypeptide
(SEQ ID NO: 107)
GTGCAGCTGCAGCAGTCCGGACCCGAACTGGTCAAGCCCGGTGCCTCCGTGA

AAATGTCTTGTAAGGCTTCTGGCTACACCTTTACCTCCTACGTCATCCAATGG

GTGAAGCAGAAGCCCGGTCAAGGTCTCGAGTGGATCGGCAGCATCAATCCCT

ACAACGATTACACCAAGTATAACGAAAAGTTTAAGGGCAAGGCCACTCTGAC

AAGCGACAAGAGCTCCATTACCGCCTACATGGAGTTTTCCTCTTTAACTTCTG

AGGACTCCGCTTTATACTATTGCGCTCGTTGGGGCGATGGCAATTATTGGGGC

CGGGGAACTACTTTAACAGTGAGCTCCGGCGGCGGCGGAAGCGGAGGTGGA

-continued

GGATCTGGCGGTGGAGGCAGCGACATCGAGATGACACAGTCCCCCGCTATCA

TGAGCGCCTCTTTAGGAGAACGTGTGACCATGACTTGTACAGCTTCCTCCAGC

GTGAGCAGCTCCTATTTCCACTGGTACCAGCAGAAACCCGGCTCCTCCCCTAA

ACTGTGTATCTACTCCACAAGCAATTTAGCTAGCGGCGTGCCTCCTCGTTTTA

GCGGCTCCGGCAGCACCTCTTACTCTTTAACCATTAGCTCTATGGAGGCCGAA

GATGCCGCCACATACTTTTGCCATCAGTACCACCGGTCCCCTACCTTTGGCGG

AGGCACAAAGCTGGAGACCAAGCGGATTACATGCCCCCCTCCCATGAGCGTG

GAGCACGCCGACATCTGGGTGAAGAGCTATAGCCTCTACAGCCGGGAGAGGT

ATATCTGTAACAGCGGCTTCAAGAGGAAGGCCGGCACCAGCAGCCTCACCGA

GTGCGTGCTGAATAAGGCTACCAACGTGGCTCACTGGACAACACCCTCTTTA

AAGTGCATCCGG

Exemplary Precursor Second Chimeric Polypeptide

(SEQ ID NO: 108)

MKWVTFISLLFLFSSAYSVQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWV

KQKPGQGLEWIGSINPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSA

LYYCARWGDGNYWGRGTTLTVSSGGGGSGGGGSGGGGSDIEMTQSPAIMSASL

GERVTMTCTASSSVSSSYFHWYQQKPGSSPKLCIYSTSNLASGVPPRFSGSGSTSY

SLTISSMEAEDAATYFCHQYHRSPTFGGGTKLETKRITCPPPMSVEHADIWVKSY

SLYSRERYICNSGFKRKAGTSSLTECVLNKATNVAHWTTPSLKCIR

Exemplary DNA Encoding Precursor Second Chimeric Polypeptide

(SEQ ID NO: 109)

ATGAAATGGGTCACCTTCATCTCTTTACTGTTTTTATTTAGCAGCGCCTACAG

CGTGCAGCTGCAGCAGTCCGGACCCGAACTGGTCAAGCCCGGTGCCTCCGTG

AAAATGTCTTGTAAGGCTTCTGGCTACACCTTTACCTCCTACGTCATCCAATG

GGTGAAGCAGAAGCCCGGTCAAGGTCTCGAGTGGATCGGCAGCATCAATCCC

TACAACGATTACACCAAGTATAACGAAAAGTTTAAGGGCAAGGCCACTCTGA

CAAGCGACAAGAGCTCCATTACCGCCTACATGGAGTTTTCCTCTTTAACTTCT

GAGGACTCCGCTTTATACTATTGCGCTCGTTGGGGCGATGGCAATTATTGGGG

CCGGGGAACTACTTTAACAGTGAGCTCCGGCGGCGGCGGAAGCGGAGGTGG

AGGATCTGGCGGTGGAGGCAGCGACATCGAGATGACACAGTCCCCCGCTATC

ATGAGCGCCTCTTTAGGAGAACGTGTGACCATGACTTGTACAGCTTCCTCCAG

CGTGAGCAGCTCCTATTTCCACTGGTACCAGCAGAAACCCGGCTCCTCCCCTA

AACTGTGTATCTACTCCACAAGCAATTTAGCTAGCGGCGTGCCTCCTCGTTTT

AGCGGCTCCGGCAGCACCTCTTACTCTTTAACCATTAGCTCTATGGAGGCCGA

AGATGCCGCCACATACTTTTGCCATCAGTACCACCGGTCCCCTACCTTTGGCG

GAGGCACAAAGCTGGAGACCAAGCGGATTACATGCCCCCCTCCCATGAGCGT

GGAGCACGCCGACATCTGGGTGAAGAGCTATAGCCTCTACAGCCGGGAGAG

GTATATCTGTAACAGCGGCTTCAAGAGGAAGGCCGGCACCAGCAGCCTCACC

GAGTGCGTGCTGAATAAGGCTACCAACGTGGCTCACTGGACAACACCCTCTT

TAAAGTGCATCCGG

Anti-CD3/anti-CD28 Beads

In some embodiments, a CD3/CD28-binding agent can be an anti-CD3/anti-CD28 bead. An anti-CD3/anti-CD28 bead can include an anti-CD3 antibody or an antigen-binding fragment thereof and an anti-CD28 antibody or an antigen-binding fragment thereof, both of which are covalently or non-covalently linked to the beads. In some embodiments, the anti-CD3 antibody or antigen-binding fragment thereof is an anti-CD3 scFv (e.g., an of the exemplary anti-CD3 scFvs described herein). In some embodiments, the anti-CD28 antibody or antigen-binding fragment thereof is an anti-CD28 scFv (e.g., any anti-CD28 scFvs described herein). The beads can be any suitable spherical polymer particles known in the art. An average diameter of the beads can be about 0.5 to about 20 micrometers (e.g., about 0.5 to about 10, about 0.8 to about 8, or about 1 to about 5 micrometers). In some embodiments, the beads can include one or more paramagnetic material, such as but not limited to dynabeads. Non-limiting examples of anti-CD3/anti-CD28 beads include anti-CD3/anti-CD28 dynabeads, such as those developed by ThermoFisher Scientific, Gibco, and Invitrogen. Additional commercial sources of anti-CD3/anti-CD28 beads are known in the art.

IL-2 Receptor-Activating Agents

Provided herein are methods that include the use of one or more IL-2 receptor-activating agent(s). Non-limiting examples of the IL-2 receptor-activating agents include single-chain chimeric polypeptides, multi-chain chimeric polypeptides, soluble IL-2 (e.g., recombinant human IL-2), and an agonistic antibody that binds specifically to an IL-2 receptor (e.g., a human IL-2 receptor).

Single-Chain Chimeric Polypeptides

In some embodiments, the IL-2 receptor-activating agents are single-chain chimeric polypeptides that include a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein the first target-binding domain and the second target-binding domain bind to a receptor for IL-2. Non-limiting examples of single-chain chimeric polypeptides that are IL-2 receptor activating agents are described herein.

Soluble IL-2

In some examples, an IL-2 receptor-activating agent can be a soluble IL-2 (e.g., any of the soluble IL-2 proteins described herein). In some embodiments, a soluble IL-2 protein can include a sequence that is at least 80% identical, at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical to SEQ ID NO: 1.

Agonistic Antibodies

In some examples, an IL-2 receptor activating agent can be an agonistic antibody that binds specifically to an IL-2 receptor (e.g., a human IL-2 receptor) (see, e.g., those described in Gaulton et al., *Clinical Immunology and Immunopathology* 36(1): 18-29, 1985), or an antigen-binding fragment thereof.

IgG1 Antibody Constructs

An IgG1 antibody construct can be an IgG1 antibody (e.g., a monoclonal or a polyclonal IgG1 antibody that binds specifically to a soluble tissue factor domain, e.g., any of the soluble tissue factor domains described herein). In some embodiments, an IgG1 antibody construct can be an antibody or an antibody fragment that includes an IgG1 Fc region (e.g., a human IgG1 Fc region). As is known in the art, the IgG1 Fc region binds to CD16a (FcRgammaIII) (e.g., human CD16a) and induces its intracellular signaling. In some embodiments, an IgG1 antibody construct can be a single-chain or a multi-chain polypeptide that includes an Fc region that is capable of binding specifically to CD16a (FcRgammaIII) (e.g., human CD16a) and is capable of inducing its intracellular signaling in a natural killer cell (e.g., a human natural killer cell), and specifically binds to the soluble tissue factor domain. In some embodiments, an IgG1 antibody construct can be a single chain or a multi-chain polypeptide that includes an Fc region that includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, or at least 99% identical) to a wildtype IgG1 Fc domain (e.g., a wildtype human IgG1 Fc domain, e.g., SEQ ID NO: 98) and is capable of binding specifically to CD16a (FcRgammaIII) (e.g., human CD16a), and is capable of inducing its intracellular signaling in a natural killer cell (e.g., a human natural killer cell), and specifically binds to the soluble tissue factor domain. In some embodiments, the IgG1 antibody construct binds specifically to a soluble tissue factor domain (e.g., any of the soluble tissue factor domains described herein) and includes a non-human Fc region (e.g., a Fc region from a non-human antibody) that has been altered (e.g., by substituting 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 amino acids in the wildtype non-human Fc region) such that the non-human Fc region is capable of binding to human CD16a (human FcRgammaIII) and inducing its intracellular signaling in a natural killer cell (e.g., a human natural killer cell).

```
Wildtype Human IgG1 Fc Region
                                              (SEQ ID NO: 98)
PCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNW

YVDGVEVHNAKTKPREEQYNSTYRVVSVLIVLHQDWINGKEYKCKVSNKA

LPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSITCLVKGFYPSDI

AVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSV

MHEALHNHYTQKSLSLSPGK
```

In some examples, the IgG1 antibody construct is a humanized or fully human anti-tissue factor antibody (see, e.g., the anti-tissue factor antibodies described in U.S. Pat. Nos. 7,968,094 and 8,007,795).

In some embodiments, the IgG1 antibody construct can include a heavy chain variable domain that includes the following set of CDR sequences: a CDR1 including DYNVY (SEQ ID NO: 66); a CDR2 including YIDPYNGITIYDQNFKG (SEQ ID NO: 67); and a CDR3 including DVTTALDF (SEQ ID NO: 68).

In some embodiments, the IgG1 antibody construct can include a heavy chain variable domain that includes the following set of CDR sequences: a CDR1 including DYNVY (SEQ ID NO: 66); a CDR2 including YIDPYNGITIYDQNLKG (SEQ ID NO: 74); and a CDR3 including DVTTALDF (SEQ ID NO: 68).

```
Exemplary heavy chain variable domain
                                              (SEQ ID NO: 72)
QIQLVQSGGEVKKPGASVRVSCKASGYSFTDYNVYWVRQSPGKGLEWIGY

IDPYNGITIYDQNFKGKATLTVDKSTSTAYMELSSLRSEDTAVYFCARDV

TTALDFWGQGTTVTVSS
```

In some embodiments, the IgG1 antibody construct can include a heavy chain variable domain that includes a sequence that is at least 70% identical (e.g., at least 72%, at least 74%, at least 75%, at least 76%, at least 78%, at least 80%, at least 82%, at least 84%, at least 85%, at least 86%, at least 88%, at least 90%, at least 92%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to SEQ ID NO: 72.

In some embodiments, the IgG1 antibody construct can include a light chain variable domain that includes the following set of CDR sequences: a CDR1 including LASQTIDTWLA (SEQ ID NO: 69); a CDR2 including AATNLAD (SEQ ID NO: 70); and a CDR3 including QQVYSSPFT (SEQ ID NO: 71).

```
Exemplary light chain variable domain
                                             (SEQ ID NO: 73)
DIQMTQSPASLSASVGDRVTITCLASQTIDTWLAWYLQKPGKSPQLLIYA

ATNLADGVPSRFSGSGSGTDFSFTISSLQPEDFATYYCQQVYSSPFTFGQ

GTKLEIK
```

In some embodiments, the IgG1 antibody construct can include a light chain variable domain that includes a sequence that is at least 70% identical (e.g., at least 72%, at least 74%, at least 75%, at least 76%, at least 78%, at least 80%, at least 82%, at least 84%, at least 85%, at least 86%, at least 88%, at least 90%, at least 92%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to SEQ ID NO: 73.

Non-limiting examples of the IgG1 antibody construct can include: a heavy chain variable domain including a sequence that is at least 70% identical (e.g., at least 72%, at least 74%, at least 75%, at least 76%, at least 78%, at least 80%, at least 82%, at least 84%, at least 85%, at least 86%, at least 88%, at least 90%, at least 92%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to SEQ ID NO: 72, and a light chain variable domain including a sequence that is at least 70% identical (e.g., at least 72%, at least 74%, at least 75%, at least 76%, at least 78%, at least 80%, at least 82%, at least 84%, at least 85%, at least 86%, at least 88%, at least 90%, at least 92%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical) to SEQ ID NO: 73.

mTOR Inhibitors

Non-limiting examples of mTOR inhibitors include rapamycin (also called sirolimus), Zortress (also called everolimus and RAD001), Torisel (also called temsirolimus and CCI-779), Afinitor (everolimus), dactolisib (also called NVP-BEZ235), GSK2126458, XL765, AZD8055, INK128 (also called MLN0128), OSI027, and RapaLinks.

Exemplary Methods of Stimulating or Increasing Proliferation of a T_{reg} Cell

Provided herein are methods of stimulating or increasing the proliferation of a T_{reg} cell that include culturing a T_{reg} cell in a liquid culture medium over a period of time, where at the beginning of the period of time, the liquid culture medium includes: a CD3/CD28-binding agent (e.g., any of the exemplary CD3/CD28-binding agents described herein); and a single-chain chimeric polypeptide including a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where the first target-binding domain and the second target-binding domain bind to a receptor for IL-2. In some examples, the liquid culture medium further includes an IgG1 antibody construct that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein). In some examples, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 3. In some examples, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 4.

In some embodiments, the liquid culture medium includes about 10 nM to about 500 nM (e.g., about 10 nM to about 450 nM, about 10 nM to about 400 nM, about 10 nM to about 350 nM, about 10 nM to about 300 nM, about 10 nM to about 250 nM, about 10 nM to about 200 nM, about 10 nM to about 150 nM, about 10 nM to about 100 nM, about 10 nM to about 50 nM, about 50 nM to about 500 nM, about 50 nM to about 450 nM, about 50 nM to about 400 nM, about 50 nM to about 350 nM, about 50 nM to about 300 nM, about 50 nM to about 250 nM, about 50 nM to about 200 nM, about 50 nM to about 150 nM, about 50 nM to about 100 nM, about 100 nM to about 500 nM, about 100 nM to about 450 nM, about 100 nM to about 400 nM, about 100 nM to about 350 nM, about 100 nM to about 300 nM, about 100 nM to about 250 nM, about 100 nM to about 200 nM, about 100 nM to about 150 nM, about 150 nM to about 500 nM, about 150 nM to about 450 nM, about 150 nM to about 400 nM, about 150 nM to about 350 nM, about 150 nM to about 300 nM, about 150 nM to about 250 nM, about 150 nM to about 200 nM, about 200 nM to about 500 nM, about 200 nM to about 450 nM, about 200 nM to about 400 nM, about 200 nM to about 350 nM, about 200 nM to about 300 nM, about 200 nM to about 250 nM, about 250 nM to about 500 nM, about 250 nM to about 450 nM, about 250 nM to about 400 nM, about 250 nM to about 350 nM, about 250 nM to about 300 nM, about 300 nM to about 500 nM, about 300 nM to about 450 nM, about 300 nM to about 400 nM, about 300 nM to about 350 nM, about 350 nM to about 500 nM, about 350 nM to about 450 nM, about 350 nM to about 400 nM, about 400 nM to about 500 nM, about 400 nM to about 450 nM, or about 450 nM to about 500 nM), of the single-chain chimeric polypeptide.

In some embodiments, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead (e.g., any of the exemplary anti-CD3/anti-CD28 beads described herein or known in the art). In some embodiments the liquid culture medium includes the anti-CD3/anti-CD28 bead at a ratio of about 1:1 to about 6:1 beads/cell (e.g., about 1:1 to about 5:1, about 1:1 to about 4:1, about 1:1 to about 3:1, about 1:1 to about 2:1, about 2:1 to about 6:1, about 2:1 to about 5:1, about 2:1 to about 4:1, about 2:1 to about 3:1, about 3:1 to about 6:1, about 3:1 to about 5:1, about 3:1 to about 4:1, about 4:1 to about 6:1, about 4:1 to about 5:1, about 5:1 to about 6:1 beads/cell).

In some embodiments, the CD3/CD28-binding agent is an additional single-chain chimeric polypeptide including a first antigen-binding domain, a soluble tissue factor domain, and a second target-binding domain, where the first target-binding domain of the additional single-chain chimeric polypeptide specifically binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide specifically binds to CD28. In some embodiments, the liquid culture medium further includes an IgG1 antibody construct (e.g., any of the exemplary IgG1 antibody constructs described herein) that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional single-chain chimeric polypeptide.

In some embodiments, the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 7. In some embodiments, the additional single-chain chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 8. In some embodiments, the liquid culture medium includes about 10 nM to about 1000 nM (e.g., about 10 nM to about 950 nM, about 10 nM to about 900 nM, about 10 nM to about 850 nM, about 10 nM to about 800 nM, about 10 nM to about 750 nM, about 10 nM to about 700 nM, about 10 nM to about 650 nM, about 10 nM to about 600 nM, about 10 nM to about 550 nM, about 10 nM to about 500 nM, about 10 nM to about 450 nM, about 10 nM to about 400 nM, about 10 nM to about 350 nM, about 10 nM to about 300 nM, about 10 nM to about 250 nM, about 10 nM to about 200 nM, about 10 nM to about 150 nM, about 10 nM to about 100 nM, about 10 nM to about 50 nM, about 25 nM to about 1000 nM, about 25 nM to about 950 nM, about 25 nM to about 900 nM, about 25 nM to about 850 nM, about 25 nM to about 800 nM, about 25 nM to about 750 nM, about 25 nM to about 700 nM, about 25 nM to about 650 nM, about 25 nM to about 600 nM, about 25 nM to about 550 nM, about 25 nM to about 500 nM, about 25 nM to about 450 nM, about 25 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In some embodiments, the CD3/CD28-binding agent is multi-chain chimeric polypeptide that includes a first chimeric polypeptide including (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein: the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3. In some embodiments, the liquid culture medium further includes an IgG1 antibody construct (e.g., any of the exemplary IgG1 antibody constructs described herein) that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide.

In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 102. In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 104. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 106. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 108. In some embodiments, the liquid culture medium includes about 10 nM to about 1000 nM (e.g., or any of the subranges of this range described herein) of the multi-chain chimeric polypeptide. In some embodiments, the liquid culture medium further comprises an mTOR inhibitor (e.g., any of the mTOR inhibitors described herein, e.g., rapamycin). In some embodiments, the liquid culture medium includes about 10 nM to about 500 nM (e.g., or any of the subranges of this range described herein) of the mTOR inhibitor. In some embodiments, the liquid culture medium does not include an mTOR inhibitor.

Some embodiments further include periodically adding to the liquid culture medium, after each 36 hours to about 60 hours (e.g., about 36 to about 56 hours, about 36 to about 52 hours, about 36 to about 48 hours, about 36 to about 44 hours, about 36 to about 40 hours, about 40 to about 60 hours, about 40 to about 56 hours, about 40 to about 52 hours, about 40 to about 48 hours, about 40 to about 44 hours, about 44 to about 60 hours, about 44 to about 56 hours, about 44 to about 52 hours, about 44 to about 48 hours, about 48 to about 60 hours, about 48 to about 56 hours, about 48 to about 52 hours, about 52 to about 60 hours, about 52 to about 56 hours, or about 56 to about 60 hours) after the start of the period of time, the single-chain chimeric polypeptide (e.g., any of the single-chain chimeric polypeptides described herein). In some embodiments, the periodic addition of the single-chain chimeric polypeptide is performed to result in a concentration of about 10 nM to about 500 nM (e.g., or any of the subranges of this range described herein) of the single-chain chimeric polypeptide in the liquid culture medium.

Some embodiments further include periodically adding to the liquid culture medium, after each about 4 days to about 10 days (e.g., after each about 4 days to about 9 days, after each about 4 days to about 8 days, after each about 4 days to about 7 days, after each about 4 days to about 6 days, after each about 4 days to about 5 days, after each about 5 days to about 10 days, after each about 5 days to about 9 days, after each about 5 days to about 8 days, after each about 5 days to about 7 days, after each about 5 days to about 6 days, after each about 6 days to about 10 days, after each about 6 days to about 9 days, after each about 6 days to about 8 days, after each about 6 days to about 7 days, after each about 7 days to about 10 days, after each about 7 days to about 9 days, after each about 7 days to about 8 days, after each about 8 days to about 10 days, after each about 8 days to about 9 days, or after each about 9 days to about 10 days) after the start of the period of time, the CD3/CD28-binding agent.

In some embodiments of any of the methods of stimulating or increasing the proliferation of a T_{reg} cell described herein, the liquid culture medium can further include an mTOR inhibitor (e.g., any of the mTOR inhibitors described herein). In some embodiments, the liquid culture medium includes about 10 nM to about 500 nM (e.g., or any of the subranges of this range described herein) of the mTOR inhibitor (e.g., any of the mTOR inhibitors described herein).

In some examples of these methods, the liquid culture medium is a serum-free liquid culture medium. In some embodiments of any of the methods described herein, the liquid culture medium is a chemically-defined liquid culture medium.

In some embodiments, the period of time is about 7 days to about 56 days (e.g., about 7 days to about 56 days, about 7 days to about 54 days, about 7 days to about 46 days, about 7 days to about 38 days, about 7 days to about 32 days, about 7 days to about 28 days, about 7 days to about 25 days, about 7 days to about 23 days, about 7 days to about 21 days, about 7 days to about 19 days, about 7 days to about 17 days, about 7 days to about 15 days, about 7 days to about 13 days, about 7 days to about 11 days, about 7 days to about 9 days, about 9 days to about 56 days, about 9 days to about 54 days, about 9 days to about 46 days, about 9 days to about 38 days, about 9 days to about 32 days, about 9 days to about 28 days, about 9 days to about 25 days, about 9 days to about 23 days, about 9 days to about 21 days, about 9 days to about 19 days, about 9 days to about 17 days, about 9 days to about 15 days, about 9 days to about 13 days, about 9 days to about 11 days, about 11 days to about 56 days, about 11 days to about 54 days, about 11 days to about 46 days, about 11 days to about 38 days, about 11 days to about 32 days, about 11 days to about 28 days, about 11 days to about 25 days, about 11 days to about 23 days, about 11 days to about 21 days, about 11 days to about 19 days, about 11 days to about 17 days, about 11 days to about 15 days, about 11 days to about 13 days, about 13 days to about 56 days, about 13 days to about 54 days, about 13 days to about 46 days, about 13 days to about 38 days, about 13 days to about 32 days, about 13 days to about 28 days, about 13 days to about 25 days, about 13 days to about 23 days, about 13 days to about 21 days, about 13 days to about 19 days, about 13 days to about 17 days, about 13 days to about 15 days, about 15 days to about 56 days, about 15 days to about 54 days, about 15 days to about 46 days, about 15 days to about 38 days, about 15 days to about 32 days, about 15 days to about 28 days, about 15 days to about 25 days, about 15 days to about 23 days, about 15 days to about 21 days, about 15 days to about 19 days, about 15 days to about 17 days, about 17 days to about 56 days, about 17 days to about 54 days, about 17 days to about 46 days, about 17 days to about 38 days, about 17 days to about 32 days, about 17 days to about 28 days, about 17 days to about 25 days, about 17 days to about 23 days, about 17 days to about 21 days, about 17 days to about 19 days, about 19 days to about 56 days, about 19 days to about 54 days, about 19 days to about 46 days, about 19 days to about 38 days, about 19 days to about 32 days, about 19 days to about 28 days, about 19 days to about 25 days, about 19 days to about 23 days, about 19 days to about 21 days, about 21 days to about 56 days, about 21 days to about 54 days, about 21 days to about 46 days, about 21 days to about 38 days, about 21 days to about 32 days, about 21 days to about 28 days, about 21 days to about 25 days, about 21 days to about 23 days, about 23 days to about 56 days, about 23 days to about 54 days, about 23 days to about 46 days, about 23 days to about 38 days, about 23 days to about 32 days, about 23 days to about 28 days, about 23 days to about 25 days, about 25 days to about 56 days, about 25 days to about 54 days, about 25 days to about 46 days, about 25 days to about 38 days, about 25 days to about 32 days, about 25 days to about 28 days, about 28 days to about 56 days, about 28 days to about 54 days, about 28 days to about 50 days, about 28 days to about 46 days, about 28 days to about 38 days, about 28 days to about 32 days, about 32 days to about 56 days, about 32 days to about 54 days, about 32 days to about 46 days, about 32 days to about 38 days, about 38 days to about 56 days, about 38 days to about 54 days, about 38 days to about 46 days, about 46 days to about 56 days, about 46 days to about 54 days, about 54 days to about 56 days).

Some embodiments further include, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject (e.g., using any of the methods of isolating described herein).

Exemplary Methods of Stimulating or Increasing Proliferation of a T_{reg} Cell

Also provided are methods of stimulating or increasing the proliferation of a T_{reg} cell that include: culturing a T_{reg} cell in a liquid culture medium over a period of time, where at the beginning of the period of time, the liquid culture medium includes: an IL-2 receptor-activating agent (e.g., any of the IL-2 receptor activating agents described herein); and a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where: the first target-binding domain of the single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the single-chain chimeric polypeptide binds to CD28. In some examples, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein).

In some embodiments, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 7. In some embodiments, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 8.

In some embodiments, the liquid culture medium includes about 10 nM to about 1000 nM (e.g., or any of the subranges of this range described herein) of the single-chain chimeric polypeptide (e.g., any of the single-chain chimeric polypeptides described herein)

In some embodiments, the IL-2 receptor-activating agent is a soluble IL-2. In some embodiments, the human soluble IL-2 includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 1. In some embodiments, the liquid culture medium includes about 100 IU/mL to about 800 IU/mL (e.g., about 100 IU/mL to about 800 IU/mL, about 100 IU/mL to about 700 IU/mL, about 100 IU/mL to about 600 IU/mL, about 100 IU/mL to about 500 IU/mL, about 100 IU/mL to about 400 IU/mL, about 100 IU/mL to about 300 IU/mL, about 100 IU/mL to about 200 IU/mL, about 200 IU/mL to about 800 IU/mL, about 200 IU/mL to about 700 IU/mL, about 200 IU/mL to about 600 IU/mL, about 200 IU/mL to about 500 IU/mL, about 200 IU/mL to about 400 IU/mL, about 200 IU/mL to about 300 IU/mL, about 300 IU/mL to about 800 IU/mL, about 300 IU/mL to about 700 IU/mL, about 300 IU/mL to about 600 IU/mL, about 300 IU/mL to about 500 IU/mL, about 300 IU/mL to about 400 IU/mL, about 400 IU/mL to about 800 IU/mL, about 400 IU/mL to about 700 IU/mL, about 400 IU/mL to about 600 IU/mL, about 400 IU/mL to about 500 IU/mL, about 500 IU/mL to about 800 IU/mL, about 500 IU/mL to about 700 IU/mL, about 500 IU/mL to about 600 IU/mL, about 600 IU/mL to about 800 IU/mL, about 600 IU/mL to about 700 IU/mL, or about 700 IU/mL to about 800 IU/mL) of the human soluble IL-2.

In some embodiments, the liquid culture medium further includes an mTOR inhibitor (e.g., any of the exemplary mTOR inhibitors described herein, e.g., rapamycin). In some embodiments, the liquid culture medium includes about 10 nM to about 500 nM (e.g., or any of the subranges of this range described herein) of the mTOR inhibitor.

Some embodiments of these methods further include periodically adding to the liquid culture medium, after each 36 hours to about 60 hours (e.g., about 36 to about 56 hours, about 36 to about 52 hours, about 36 to about 48 hours, about 36 to about 44 hours, about 36 to about 40 hours, about 40 to about 60 hours, about 40 to about 56 hours, about 40 to about 52 hours, about 40 to about 48 hours, about 40 to about 44 hours, about 44 to about 60 hours, about 44 to about 56 hours, about 44 to about 52 hours, about 44 to about 48 hours, about 48 to about 60 hours, about 48 to about 56 hours, about 48 to about 52 hours, about 52 to about 60 hours, about 52 to about 56 hours, or about 56 to about 60 hours) after the start of the period of time, the IL-2 receptor-activating agent (e.g., any of the IL-2 receptor-activating agents described herein). In some embodiments, the periodic addition of the IL-2 receptor activating agent is performed to result in a concentration of about 10 nM to about 1000 nM (e.g., or any of the subranges of this range described herein) of the IL-2 receptor-activating agent in the liquid culture medium.

Some embodiments of these methods further include periodically adding to the liquid culture medium, after each about 4 days to about 10 days (e.g., after each about 4 days to about 9 days, after each about 4 days to about 8 days, after each about 4 days to about 7 days, after each about 4 days to about 6 days, after each about 4 days to about 5 days, after each about 5 days to about 10 days, after each about 5 days to about 9 days, after each about 5 days to about 8 days, after each about 5 days to about 7 days, after each about 5 days to about 6 days, after each about 6 days to about 10 days, after each about 6 days to about 9 days, after each about 6 days to about 8 days, after each about 6 days to about 7 days, after each about 7 days to about 10 days, after each about 7 days to about 9 days, after each about 7 days to about 8 days, after each about 8 days to about 10 days, after each about 8 days to about 9 days, or after each about 9 days to about 10 days) after the start of the period of time, the single-chain chimeric polypeptide (e.g., any of the exemplary single-chain chimeric polypeptides described herein).

In some embodiments of any of these methods, the period of time is about 7 to about 56 days (e.g., or any of the subranges of this range described herein).

Some embodiments further include, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject (e.g., using any of the methods of isolating described herein).

Exemplary Methods of Stimulating or Increasing Proliferation of a T_{reg} Cell

Also provided are methods of stimulating or increasing the proliferation of a T_{reg} cell that include: culturing a T_{reg} cell in a liquid culture medium over a period of time, where at the beginning of the period of time, the liquid culture medium includes: a CD3/CD28-binding agent (e.g., any of the CD3/CD28-binding agents described herein); and a multi-chain chimeric polypeptide that includes: (a) a first chimeric polypeptide including: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide including: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, where the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains (e.g., any of the multi-chain chimeric polypeptides described herein). In some embodiments, the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide (e.g., any of the IgG1 antibody constructs described herein).

In some embodiments, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead (e.g., any of the anti-CD3/anti-CD28 beads described herein). In some embodiments of any of the methods described herein, the liquid culture medium includes the anti-CD3/anti-CD28 bead at a ratio of about 1:1 to about 6:1 beads/cell (e.g., about 1:1 to about 5:1, about 1:1 to about 4:1, about 1:1 to about 3:1, about 1:1 to about 2:1, about 2:1 to about 6:1, about 2:1 to about 5:1, about 2:1 to about 4:1, about 2:1 to about 3:1, about 3:1 to about 6:1, about 3:1 to about 5:1, about 3:1 to about 4:1, about 4:1 to about 6:1, about 4:1 to about 5:1, about 5:1 to about 6:1 beads/cell).

In some embodiments, the CD3/CD28-binding agent is an single-chain chimeric polypeptide including a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where: the first target-binding domain of the single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the single-chain chimeric polypeptide binds to CD28 (e.g., any of the exemplary single-chain chimeric polypeptides described herein). In some embodiments, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 7. In some embodiments, the single-chain chimeric polypeptide includes a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 8. In some embodiments, the liquid culture medium includes about 10 nM to about 1000 nM (e.g., or any of the subranges of this range described herein) of the single-chain chimeric polypeptide.

In some embodiments, the CD3/CD28-binding agent is multi-chain chimeric polypeptide that includes a first chimeric polypeptide including (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, wherein: the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains; and wherein the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28, or the first target-binding domain binds to CD28 and the second target-binding domain to CD3. In some embodiments, the liquid culture medium further includes an IgG1 antibody construct (e.g., any of the exemplary IgG1 antibody constructs described herein) that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide.

In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 102. In some embodiments, the first chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 104. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 106. In some embodiments, the second chimeric polypeptide comprises a sequence that is at least 80% identical (e.g., at least 82% identical, at least 84% identical, at least 86% identical, at least 88% identical, at least 90% identical, at least 92% identical, at least 94% identical, at least 96% identical, at least 98% identical, at least 99% identical, or 100% identical) to SEQ ID NO: 108. In some embodiments, the liquid culture medium includes about 10 nM to about 1000 nM (e.g., or any of the subranges of this range described herein) of the multi-chain chimeric polypeptide.

In some embodiments, the liquid culture medium further includes an mTOR inhibitor (e.g., any of the mTOR inhibitors described herein, e.g., rapamycin). In some embodiments, the liquid culture medium includes about 10 nM to about 500 nM of the mTOR inhibitor (e.g., or any of the subranges of this range described herein). In some embodiments, the liquid culture medium does not include an mTOR inhibitor.

Some embodiments of these methods further include periodically adding to the liquid culture medium, after each 36 hours to about 60 hours (e.g., about 36 to about 56 hours, about 36 to about 52 hours, about 36 to about 48 hours, about 36 to about 44 hours, about 36 to about 40 hours, about 40 to about 60 hours, about 40 to about 56 hours, about 40 to about 52 hours, about 40 to about 48 hours, about 40 to about 44 hours, about 44 to about 60 hours, about 44 to about 56 hours, about 44 to about 52 hours, about 44 to about 48 hours, about 48 to about 60 hours, about 48 to about 56 hours, about 48 to about 52 hours, about 52 to about 60 hours, about 52 to about 56 hours, or about 56 to about 60 hours) after the start of the period of time, the multi-chain chimeric polypeptide (e.g., any of the multi-chain chimeric polypeptides described herein). In some embodiments, the periodic addition of the multi-chain chimeric polypeptide is performed to result in a concentration of about 10 nM to about 500 nM (e.g., or any of the subranges of this range described herein) of the multi-chain chimeric polypeptide in the liquid culture medium.

Some embodiments of these methods further include periodically adding to the liquid culture medium, after each about 4 days to about 10 days (e.g., after each about 4 days to about 9 days, after each about 4 days to about 8 days, after each about 4 days to about 7 days, after each about 4 days to about 6 days, after each about 4 days to about 5 days, after each about 5 days to about 10 days, after each about 5 days to about 9 days, after each about 5 days to about 8 days, after each about 5 days to about 7 days, after each about 5 days to about 6 days, after each about 6 days to about 10 days, after each about 6 days to about 9 days, after each about 6 days to about 8 days, after each about 6 days to about 7 days, after each about 7 days to about 10 days, after each about 7 days to about 9 days, after each about 7 days to about 8 days, after each about 8 days to about 10 days, after each about 8 days to about 9 days, or after each about 9 days to about 10 days) after the start of the period of time, the CD3/CD28-binding agent (e.g., any of the exemplary CD3/CD28-binding agents described herein).

In some embodiments of any of these methods, the period of time is about 7 to about 56 days (e.g., or any of the subranges of this range described herein).

Some embodiments further include, before the culturing step, a step of isolating the Treg cell from a sample obtained from a subject (e.g., using any of the methods of isolating described herein).

Methods of Isolating T_{reg} Cells from a Sample

Any of the methods of stimulating or increasing the proliferation of a T_{reg} cell as described herein can further include, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject.

The sample can be a blood sample (e.g., a peripheral blood sample or a cord blood sample), lymphoid tissue, or a thymus sample (e.g., pediatric thymus sample). Some embodiments of any of the methods described herein further include obtaining a sample from the subject.

Methods of isolating T_{reg} cells from samples obtained from a subject are well known in the art. For example, magnetic-activated cell sorting (MACS) and/or fluorescence-activated cell sorting (FACS) can be used to isolate T_{reg} cells based on marker expression. Non-limiting examples of FACS instruments include FACSAria (Becton Dickinson), FX500 fluidics cell sorter (Sony), and MACSQuant Tyto cell sorter (Miltenyi Biotec). Exemplary cell surface proteins for isolating T_{reg} cells from samples include CD4, CD25, and CD127. An intracellular marker for T_{reg} cells is Foxp3. For example, expression levels of CD4 and CD25 can be used to isolate T_{reg} cells from a sample (e.g., a cord blood sample). Expression levels of CD127 can be used as an additional marker for isolating T_{reg} cells from a sample (e.g., a peripheral blood sample). In some instances, isolating T_{reg} cells from samples include sorting for T_{reg} cells that are $CD4^+CD25^+Foxp3^+$ and/or are $CD4^+CD25^+CD127^{dim}$. Kits for isolating T_{reg} cells include $CD4^+CD25^+CD127^{dim}$ regulatory T cell isolation kit II reagent (Miltenyi Biotec). Additional kits for isolating a T_{reg} cell are commercially available from StemCell Technologies (EasySep™ Human $CD4^+CD127^{low}$, $CD25^+$ Regulatory T cell Isolation Kit), Miltenyi Biotec (Human $CD4^+CD25^+$ Regulatory T Cell Isolation Kit), and R&D Systems (MagCellect Human $CD4^+CD25^+$ Regulatory T Cell Isolation Kit).

Also provided herein are methods of isolating a Treg cell from a sample obtained from a subject that include separating the Treg cell from the sample based on its expression of CD39, thereby isolating the Treg cell. Some embodiments of these methods include mixing the sample with an antibody or ligand capable of binding CD39 under conditions that allow binding of the antibody or ligand to Treg cells expressing CD39; and separating the Treg cell bound to the antibody or ligand from other components in the sample, thereby isolating the Treg cell. In some embodiments of these methods, the antibody is a mouse, a humanized, or a human antibody or antigen-binding fragment thereof; and/or the antibody or the ligand is labeled with at least one of biotin, avidin, streptavidin, or a fluorochrome, or is bound to a particle, bead, resin, or solid support. In some embodiments of these methods, the separating comprises the use of flow cytometry, fluorescence-activated cell sorting (FACS), centrifugation, or column, plate, particle, or bead-based methods. Some embodiments of these methods further include: mixing the sample with a biotinylated antibody or ligand capable of binding CD39 under conditions that allow binding of the antibody or the ligand to the Treg cell; capturing the Treg cell bound to the biotinylated antibody or the ligand using streptavidin-coated magnetic particles; separating the magnetic particle-bound Treg cell using a magnet; washing the magnet particle-bound Treg cell to remove other components in the sample; and releasing the magnetic particle-bound Treg cell from the magnet into a solution, thereby isolating the Treg cell.

In some examples, the Treg cell is an autologous Treg cell, a haploidentical Treg cell, or an allogeneic Treg cell isolated from a sample comprising fresh or frozen peripheral blood, umbilical cord blood, peripheral blood mononuclear cells, lymphocytes, $CD4^+$ T cells or Treg cells. In some examples, the Treg cell is a $CD4^+CD25^+Foxp3^+$ cell. In some examples, the Treg cell is a $CD4^+CD25^+CD127^{dim}$ cell. In some examples, the Treg cells comprises a chimeric antigen receptor. In some examples, the Treg cells are immunosuppressive in vitro and in vivo.

Also provided herein are populations of isolated Treg cells generated and/or isolated using any of the methods described here. In some examples, the isolated population of Treg cells are greater than 70% (e.g., greater than 75%, greater than 80%, greater than 80%, greater than 85%, greater than 90%, greater than 95%, or greater than 99% $CD39^+$ cells). In some examples, the isolated Treg cell or populations of isolated Treg cells provided herein are further expanded in vitro. Also provided are methods of expanding any of the populations of isolated Treg cells described herein that include culturing the population of isolated Treg cells under conditions that allow for proliferation of the isolated Treg cells.

Also provided herein are compositions comprising any of the populations of isolated Treg cells described herein and a pharmaceutically acceptable carrier. Also provided herein are methods of treating a subject in need thereof that include administering to the subject a therapeutically effective amount of any of the compositions described herein. In some examples, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some examples, the aging-related disease is selected from the group of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction. In some examples, the inflammatory disease is selected from the group of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, nonalcoholic steatohepatitis, and mood disorders.

T_{reg} Cells

In some embodiments of any of the methods, compositions, and kits described herein, the T_{reg} cell can be an autologous T_{reg} cell, a haploidentical T_{reg} cell, or an allogeneic T_{reg} cell isolated from peripheral blood or umbilical cord blood. In some embodiments of any of the methods, compositions, and kits described herein, the T_{reg} cell can be a $CD4^+CD25^+Foxp3^+$ cell. In some embodiments of any of the methods, compositions, and kits described herein, the T_{reg} cell can be a $CD4^+CD25^+CD127^{dim}$ cell.

In some embodiments of any of the methods, compositions, and kits described herein, the T_{reg} cell can include a chimeric antigen receptor or a recombinant T-cell receptor. In some examples, the T_{reg} cell has previously been genetically-modified to express a chimeric antigen receptor or a recombinant T-cell receptor. In some examples, the T_{reg} cell has previously been genetically-modified to express a recombinant T-cell receptor recognizing a peptide of interest in the target tissue or a co-stimulatory molecule (e.g., ICOS).

Some embodiments of these methods can further include, after the culturing step, introducing into the T_{reg} cell a nucleic acid encoding a chimeric antigen-receptor or a recombinant T-cell receptor. Some embodiments of these methods can further include, after the culturing step, introducing into the T_{reg} cell a nucleic acid encoding a co-stimulatory molecule (e.g., ICOS).

Some embodiments of these methods can further include, before the culturing step, introducing into the T_{reg} cell a nucleic acid encoding a chimeric antigen-receptor or a recombinant T-cell receptor. Some embodiments of these methods can further include, before the culturing step, introducing into the T_{reg} cell a nucleic acid encoding a co-stimulatory molecule (e.g., ICOS).

Detection of the proliferation of a T_{reg} cell can be performed using methods known in the art, e.g., cytometry (e.g., fluorescence-assisted flow cytometry), microscopy, and immunofluorescence microscopy. For example, proliferation of T_{reg} cells can be determined by staining with dyes (e.g., trypan blue) and counting the number of viable cells. In some examples, proliferation and/or activation of T_{reg} cells can be determined by detecting the levels of surface markers on the T_{reg} cells such as but not limited to: Helios, CTLA-4, CD39, CD62L, CD25, CD103, CD69, PD1 and CD49b. The detection of these surface markers can be performed using FACS. Activation of T_{reg} cells can also be determined by detecting the levels of glucose metabolism in the T_{reg} cells. In some examples, glucose metabolism can be determined by measuring the extracellular acidification rate (ECAR), which involves the stages of non-glycolytic acidification (without drugs), glycolysis (addition of glucose), glycolytic capacity (addition of oligomycin), and glycolytic reserve (addition of 2-deoxyglucose). An extracellular flux analyzer (Agilent) can be used for measuring ECAR.

In some examples, the proliferation and/or activation of T_{reg} cells can be determined by detecting an increase in the level of IL-10 secretion. For example, the methods provided herein can result in an increase of about 1% to about 800% (e.g., about 1% to about 750%, about 1% to about 700%, about 1% to about 650%, about 1% to about 600%, about 1% to about 550%, about 1% to about 500%, about 1% to about 450%, about 1% to about 400%, about 1% to about 350%, about 1% to about 300%, about 1% to about 280%, about 1% to about 260%, about 1% to about 240%, about 1% to about 220%, about 1% to about 200%, about 1% to about 180%, about 1% to about 160%, about 1% to about 140%, about 1% to about 120%, about 1% to about 100%, about 1% to about 90%, about 1% to about 80%, about 1% to about 70%, about 1% to about 60%, about 1% to about 50%, about 1% to about 45%, about 1% to about 40%, about 1% to about 35%, about 1% to about 30%, about 1% to about 25%, about 1% to about 20%, about 1% to about 15%, about 1% to about 10%, about 1% to about 5%, about 5% to about 800%, about 5% to about 750%, about 5% to about 700%, about 5% to about 650%, 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800%, about 650% to about 750%, about 650% to about 700%, about 700% to about 800%, about 700% to about 750%, or about 750% to about 800%) in the level of secreted IL-10 by the T_{reg} cells (e.g., at the end of the period of time as compared to the start of the period of time).

The methods provided herein can result in an increase (e.g., about 1% to about 800% increase, or any of the subranges of this range described herein) in the concentration or number of the T_{reg} cells (e.g., at the end of the period of time as compared to the start of the period of time).

In some embodiments, the methods provided herein can result in an increase in the concentration or number of T_{reg} cells (e.g., about a 0.1-fold increase to about a 100-fold increase, about a 0.1-fold increase to about a 90-fold increase, about a 0.1-fold increase to about a 80-fold increase, about a 0.1-fold increase to about a 70-fold increase, about a 0.1-fold increase to about a 60-fold increase, about a 0.1-fold increase to about a 50-fold increase, about a 0.1-fold increase to about a 45-fold increase, about a 0.1-fold increase to about a 40-fold increase, about a 0.1-fold increase to about a 35-fold increase, about a 0.1-fold increase to about a 30-fold increase, about a 0.1-fold increase to about a 25-fold increase, about a 0.1-fold increase to about a 20-fold increase, about a 0.1-fold increase to about a 15-fold increase, about a 0.1-fold increase to about a 10-fold increase, about a 0.1-fold increase to about a 8-fold increase, about a 0.1-fold increase to about a 6-fold 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Compositions and Kits

Also provided herein populations of T_{reg} cells produced using any of the methods described herein. Also provided herein are compositions (e.g., pharmaceutical compositions) that include a T_{reg} cell produced using any of the methods described herein or a population of T_{reg} cells produced using any of the methods described herein. In some embodiments, the pharmaceutical compositions can include a pharmaceutically acceptable carrier (e.g., phosphate buffered saline). The compositions can be used to treat a subject in need thereof (e.g., any of the exemplary subjects described herein).

Single or multiple administrations of pharmaceutical compositions can be given to a subject in need thereof depending on for example: the dosage and frequency as required and tolerated by the subject. The formulation should provide a sufficient quantity of T_{reg} cells to effectively treat, prevent, or ameliorate conditions, diseases, or symptoms in the subject.

Also provided herein are kits that include (i) a CD3/CD28-binding agent (e.g., any of the CD3/CD28-binding agents described herein); (ii) a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein the first target-binding domain and the second target-binding domain bind to a receptor for IL-2 (e.g., any of such single-chain chimeric polypeptides described herein); and (iii) an mTOR inhibitor (e.g., any mTOR inhibitors described herein). Some embodiments of these kits can further include an IgG1 antibody construct that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein). In some embodiments of these kits, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead (e.g., any of the exemplary anti-CD3/anti-CD28 beads described herein). In some embodiments, the CD3/CD28-binding agent is an additional single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where: the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28 (e.g., any of the examples of such single-chain chimeric polypeptides described herein). In some embodiments, the kit can further include an IgG1 antibody construct that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the additional single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein).

Also provided herein are kits that include (i) an interleukin-2 receptor-activating agent (e.g., any of the interleukin-2 receptor-activating agents described herein); (ii) a single-chain chimeric polypeptide including a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where: the first target-binding domain binds to CD3 and the second target-binding domain binds to CD28; or the first target-binding domain binds to CD28 and the second target-binding domain binds to CD3 (e.g., any of such single-chain chimeric polypeptides described herein); and (iii) an mTOR inhibitor (e.g., any mTOR inhibitors described herein). In some embodiments, the kits can further include an IgG1 antibody construct that includes at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein).

Also provided herein are kits that include: a CD3/CD28-binding agent (e.g., any of the CD3/CD28-binding agents described herein); a multi-chain chimeric polypeptide comprising: (a) a first chimeric polypeptide comprising: (i) a first target-binding domain; (ii) a soluble tissue factor domain; and (iii) a first domain of a pair of affinity domains; and (b) a second chimeric polypeptide comprising: (i) a second domain of a pair of affinity domains; and (ii) a second target-binding domain, where the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains (e.g., any of the exemplary multi-chain chimeric polypeptides described herein); and an mTOR inhibitor (e.g., any of the exemplary mTOR inhibitors described herein). Some embodiments of these kits further include an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the multi-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein). In some embodiments, the CD3/CD28-binding agent is an anti-CD3/anti-CD28 bead (e.g., any of the anti-CD3/anti-CD28 beads described herein). In some embodiments, the CD3/CD28-binding agent is an single-chain chimeric polypeptide including a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, where the first target-binding domain of the additional single-chain chimeric polypeptide binds to CD3 and the second target-binding domain of the additional single-chain chimeric polypeptide binds to CD28 (e.g., any of the examples of such single-chain chimeric polypeptides described herein). In some embodiments, the kit further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide (e.g., any of the exemplary IgG1 antibody constructs described herein).

Nucleic Acids/Vectors

Also provided herein are nucleic acids that encode any of the single-chain chimeric polypeptides or multi-chain chimeric polypeptides described herein. Also provided herein are vectors that include any of the nucleic acids encoding any of the single-chain chimeric polypeptides or multi-chain chimeric polypeptides described herein.

Any of the vectors described herein can be an expression vector. For example, an expression vector can include a promoter sequence operably linked to the sequence encoding the single-chain chimeric polypeptide.

Non-limiting examples of vectors include plasmids, transposons, cosmids, and viral vectors (e.g., any adenoviral vectors (e.g., pSV or pCMV vectors), adeno-associated virus (AAV) vectors, lentivirus vectors, and retroviral vectors), and any Gateway® vectors. A vector can, e.g., include sufficient cis-acting elements for expression; other elements for expression can be supplied by the host mammalian cell or in an in vitro expression system. Skilled practitioners will be capable of selecting suitable vectors and mammalian cells for making any of the single-chain chimeric polypeptides or multi-chain chimeric polypeptides described herein.

Cells

Also provided herein are cells (e.g., any of the exemplary cells described herein or known in the art) comprising any of the nucleic acids described herein that encode any of the single-chain chimeric polypeptides or any of the multi-chain chimeric polypeptides described herein.

Also provided herein are cells (e.g., any of the exemplary cells described herein or known in the art) that include any of the vectors described herein that encode any of the single-chain chimeric polypeptides or any of the multi-chain chimeric polypeptides described herein.

In some embodiments of any of the methods described herein, the cell can be a eukaryotic cell. As used herein, the term "eukaryotic cell" refers to a cell having a distinct, membrane-bound nucleus. Such cells may include, for example, mammalian (e.g., rodent, non-human primate, or human), insect, fungal, or plant cells. In some embodiments, the eukaryotic cell is a yeast cell, such as *Saccharomyces cerevisiae*. In some embodiments, the eukaryotic cell is a higher eukaryote, such as mammalian, avian, plant, or insect cells. Non-limiting examples of mammalian cells include Chinese hamster ovary cells and human embryonic kidney cells (e.g., HEK293 cells).

Methods of introducing nucleic acids and expression vectors into a cell (e.g., an eukaryotic cell) are known in the art. Non-limiting examples of methods that can be used to introduce a nucleic acid into a cell include lipofection, transfection, electroporation, microinjection, calcium phosphate transfection, dendrimer-based transfection, cationic polymer transfection, cell squeezing, sonoporation, optical transfection, impalefection, hydrodynamic delivery, magnetofection, viral transduction (e.g., adenoviral and lentiviral transduction), and nanoparticle transfection.

Methods of Producing Single-Chain or Multi-Chain Chimeric Polypeptides

Also provided herein are methods of producing any of the single-chain chimeric polypeptides or multi-chain chimeric polypeptides described herein that include culturing any of the cells described herein in a culture medium under conditions sufficient to result in the production of the single-chain chimeric polypeptide or a multi-chain chimeric polypeptide; and recovering the single-chain chimeric polypeptide or the multi-chain chimeric polypeptide from the cell and/or the culture medium.

The recovery of the single-chain chimeric polypeptide or the multi-chain chimeric polypeptide from a culture medium or a cell (e.g., a eukaryotic cell) can be performed using techniques well-known in the art (e.g., ammonium sulfate precipitation, polyethylene glycol precipitation, ion-exchange chromatography (anion or cation), chromatography based on hydrophobic interaction, metal-affinity chromatography, ligand-affinity chromatography, and size exclusion chromatography).

Methods of culturing cells are well known in the art. Cells can be maintained in vitro under conditions that favor proliferation, differentiation and growth. Briefly, cells can be cultured by contacting a cell (e.g., any cell) with a cell culture medium that includes the necessary growth factors and supplements to support cell viability and growth.

Also provided herein are single-chain chimeric polypeptides (e.g., any of the single-chain chimeric polypeptides described herein) or multi-chain chimeric polypeptides (e.g., any of the multi-chain chimeric polypeptides) produced by any of the methods described herein.

Methods of Treatment

Also provided herein are methods of treating a subject in need thereof (e.g., any of the exemplary subjects described herein) that include administering to the subject a therapeutically effective amount of a population of T_{reg} cells generated using any of the methods described herein or any of the compositions (e.g., pharmaceutical compositions) comprising a T_{reg} cell or a population of T_{reg} cells generated using any of the methods described herein.

In some embodiments of these methods, the subject has been identified or diagnosed as having an aging-related disease or an inflammatory disease. In some embodiments, the aging-related disease is selected from the group of: Alzheimer's disease, aneurysm, cystic fibrosis, fibrosis in pancreatitis, glaucoma, hypertension, idiopathic pulmonary fibrosis, inflammatory bowel disease, intervertebral disc degeneration, macular degeneration, osteoarthritis, type 2 diabetes mellitus, adipose atrophy, lipodystrophy, atherosclerosis, cataracts, COPD, idiopathic pulmonary fibrosis, kidney transplant failure, liver fibrosis, loss of bone mass, myocardial infarction, sarcopenia, wound healing, alopecia, cardiomyocyte hypertrophy, osteoarthritis, Parkinson's disease, age-associated loss of lung tissue elasticity, macular degeneration, cachexia, glomerulosclerosis, liver cirrhosis, NAFLD, osteoporosis, amyotrophic lateral sclerosis, Huntington's disease, spinocerebellar ataxia, multiple sclerosis, neurodegeneration, stroke, dementia, vascular disease, infection susceptibility, chronic inflammation, and renal dysfunction.

In some embodiments, the inflammatory disease is selected from the group of: rheumatoid arthritis, inflammatory bowel disease, lupus erythematosus, lupus nephritis, diabetic nephropathy, CNS injury, Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Crohn's disease, multiple sclerosis, Guillain-Barre syndrome, psoriasis, Grave's disease, ulcerative colitis, nonalcoholic steatohepatitis, and mood disorders.

In some embodiments, these methods can result in a reduction in the number, severity, or frequency of one or more symptoms of the cancer in the subject (e.g., as compared to the number, severity, or frequency of the one or more symptoms of the aging-related disease or the inflammatory disease in the subject prior to treatment). In some examples, the methods can result in a decrease (e.g., about 1% decrease to about 99% decrease, an about 1% decrease to about 95% decrease, about 1% decrease to about 90% decrease, about 1% decrease to about 85% decrease, about 1% decrease to about 80% decrease, about 1% decrease to about 75% decrease, about 1% to about 70% decrease, about 1% decrease to about 65% decrease, about 1% decrease to about 60% decrease, about 1% decrease to about 55% decrease, about 1% decrease to about 50% decrease, about 1% decrease to about 45% decrease, about 1% decrease to about 40% decrease, about 1% decrease to about 35% decrease, about 1% decrease to about 30% decrease, about 1% decrease to about 25% decrease, about 1% decrease to about 20% decrease, about 1% decrease to about 15% decrease, about 1% decrease to about 10% decrease, about 1% decrease to about 5% decrease, about 5% decrease to about 99% decrease, an about 5% decrease to about 95% decrease, about 5% decrease to about 90% decrease, about 5% decrease to about 85% decrease, about 5% decrease to about 80% decrease, about 5% decrease to about 75% decrease, about 5% to about 70% decrease, about 5% decrease to about 65% decrease, about 5% decrease to about 60% decrease, about 5% decrease to about 55% decrease, about 5% decrease to about 50% decrease, about 5% decrease to about 45% decrease, about 5% decrease to about 40% decrease, about 5% decrease to about 35% decrease, about 5% decrease to about 30% decrease, about 5% decrease to about 25% decrease, about 5% decrease to about 20% decrease, about 5% decrease to about 15% decrease, about 5% decrease to about 10% decrease, about 10% decrease to about 99% decrease, an about 10% decrease to about 95% decrease, about 10% decrease to about 90% decrease, about 10% decrease to about 85% decrease, about 10% decrease to about 80% decrease, about 10% decrease to about 75% decrease, about 10% to about 70% decrease, about 10% decrease to about 65% decrease, about 10% decrease to about 60% decrease, about 10% decrease to about 55% decrease, about 10% decrease to about 50% decrease, about 10% decrease to about 45% decrease, about 10% decrease to about 40% decrease, about 10% decrease to about 35% decrease, about 10% decrease to about 30% decrease, about 10% decrease to about 25% decrease, about 10% decrease to about 20% decrease, about 10% decrease to about 15% decrease, about 15% decrease to about 99% decrease, an about 15% decrease to about 95% decrease, about 15% decrease to about 90% decrease, about 15% decrease to about 85% decrease, about 15% decrease to about 80% decrease, about 15% decrease to about 75% decrease, about 15% to about 70% decrease, about 15% decrease to about 65% decrease, about 15% decrease to about 60% decrease, about 15% decrease to about 55% decrease, about 15% decrease to about 50% decrease, about 15% decrease to about 45% decrease, about 15% decrease to about 40% decrease, about 15% decrease to about 35% decrease, about 15% decrease to about 30% decrease, about 15% decrease to about 25% decrease, about 15% decrease to about 20% decrease, about 20% decrease to about 99% decrease, an about 20% decrease to about 95% decrease, about 20% decrease to about 90% decrease, about 20% decrease to about 85% decrease, about 20% decrease to about 80% decrease, about 20% decrease to about 75% decrease, about 20% to about 70% decrease, about 20% decrease to about 65% decrease, about 20% decrease to about 60% decrease, about 20% decrease to about 55% decrease, about 20% decrease to about 50% decrease, about 20% decrease to about 45% decrease, about 20% decrease to about 40% decrease, about 20% decrease to about 35% decrease, about 20% decrease to about 30% decrease, about 20% decrease to about 25% decrease, about 25% decrease to about 99% decrease, an about 25% decrease to about 95% decrease, about 25% decrease to about 90% decrease, about 25% decrease to about 85% decrease, about 25% decrease to about 80% decrease, about 25% decrease to about 75% decrease, about 25% to about 70% decrease, about 25% decrease to about 65% decrease, about 25% decrease to about 60% decrease, about 25% decrease to about 55% decrease, about 25% decrease to about 50% decrease, about

25% decrease to about 45% decrease, about 25% decrease to about 40% decrease, about 25% decrease to about 35% decrease, about 25% decrease to about 30% decrease, about 30% decrease to about 99% decrease, an about 30% decrease to about 95% decrease, about 30% decrease to about 90% decrease, about 30% decrease to about 85% decrease, about 30% decrease to about 80% decrease, about 30% decrease to about 75% decrease, about 30% to about 70% decrease, about 30% decrease to about 65% decrease, about 30% decrease to about 60% decrease, about 30% decrease to about 55% decrease, about 30% decrease to about 50% decrease, about 30% decrease to about 45% decrease, about 30% decrease to about 40% decrease, about 30% decrease to about 35% decrease, about 35% decrease to about 99% decrease, an about 35% decrease to about 95% decrease, about 35% decrease to about 90% decrease, about 35% decrease to about 85% decrease, about 35% decrease to about 80% decrease, about 35% decrease to about 75% decrease, about 35% to about 70% decrease, about 35% decrease to about 65% decrease, about 35% decrease to about 60% decrease, about 35% decrease to about 55% decrease, about 35% decrease to about 50% decrease, about 35% decrease to about 45% decrease, about 35% decrease to about 40% decrease, about 40% decrease to about 99% decrease, an about 40% decrease to about 95% decrease, about 40% decrease to about 90% decrease, about 40% decrease to about 85% decrease, about 40% decrease to about 80% decrease, about 40% decrease to about 75% decrease, about 40% to about 70% decrease, about 40% decrease to about 65% decrease, about 40% decrease to about 60% decrease, about 40% decrease to about 55% decrease, about 40% decrease to about 50% decrease, about 40% decrease to about 45% decrease, about 45% decrease to about 99% decrease, an about 45% decrease to about 95% decrease, about 45% decrease to about 90% decrease, about 45% decrease to about 85% decrease, about 45% decrease to about 80% decrease, about 45% decrease to about 75% decrease, about 45% to about 70% decrease, about 45% decrease to about 65% decrease, about 45% decrease to about 60% decrease, about 45% decrease to about 55% decrease, about 45% decrease to about 50% decrease, about 50% decrease to about 99% decrease, an about 50% decrease to about 95% decrease, about 50% decrease to about 90% decrease, about 50% decrease to about 85% decrease, about 50% decrease to about 80% decrease, about 50% decrease to about 75% decrease, about 50% to about 70% decrease, about 50% decrease to about 65% decrease, about 50% decrease to about 60% decrease, about 50% decrease to about 55% decrease, about 55% decrease to about 99% decrease, an about 55% decrease to about 95% decrease, about 55% decrease to about 90% decrease, about 55% decrease to about 85% decrease, about 55% decrease to about 80% decrease, about 55% decrease to about 75% decrease, about 55% to about 70% decrease, about 55% decrease to about 65% decrease, about 55% decrease to about 60% decrease, about 60% decrease to about 99% decrease, an about 60% decrease to about 95% decrease, about 60% decrease to about 90% decrease, about 60% decrease to about 85% decrease, about 60% decrease to about 80% decrease, about 60% decrease to about 75% decrease, about 60% to about 70% decrease, about 60% decrease to about 65% decrease, about 65% decrease to about 99% decrease, an about 65% decrease to about 95% decrease, about 65% decrease to about 90% decrease, about 65% decrease to about 85% decrease, about 65% decrease to about 80% decrease, about 65% decrease to about 75% decrease, about 65% to about 70% decrease, about 70% decrease to about 99% decrease, an about 70% decrease to about 95% decrease, about 70% decrease to about 90% decrease, about 70% decrease to about 85% decrease, about 70% decrease to about 80% decrease, about 70% decrease to about 75% decrease, about 75% decrease to about 99% decrease, an about 75% decrease to about 95% decrease, about 75% decrease to about 90% decrease, about 75% decrease to about 85% decrease, about 75% decrease to about 80% decrease, about 80% decrease to about 99% decrease, an about 80% decrease to about 95% decrease, about 80% decrease to about 90% decrease, about 80% decrease to about 85% decrease, about 85% decrease to about 99% decrease, an about 85% decrease to about 95% decrease, about 85% decrease to about 90% decrease, about 90% decrease to about 99% decrease, an about 90% decrease to about 95% decrease, or about 95% decrease to about 99% decrease) in the number of senescent cells in the subject (e.g., a decrease in the number of senescent cells in one or more specific tissues involved and/or implicated in the aging-related disease or the inflammatory disease in the subject), e.g., as compared to the number of senescent cells in the subject prior to treatment.

The term "subject" refers to any mammal. In some embodiments, the subject or "subject in need of treatment" may be a canine (e.g., a dog), feline (e.g., a cat), equine (e.g., a horse), ovine, bovine, porcine, caprine, primate, e.g., a simian (e.g., a monkey (e.g., marmoset, baboon), or an ape (e.g., a gorilla, chimpanzee, orangutan, or gibbon) or a human; or rodent (e.g., a mouse, a guinea pig, a hamster, or a rat). In some embodiments, the subject or "subject in need of treatment" may be a non-human mammal, especially mammals that are conventionally used as models for demonstrating therapeutic efficacy in humans (e.g., murine, lapine, porcine, canine or primate animals) may be employed.

Senescent Cells

Senescence is a form of irreversible growth arrest accompanied by phenotypic changes, resistance to apoptosis and activation of damage-sensing signaling pathways. Cellular senescence was first described in cultured human fibroblast cells that lost their ability to proliferate, reaching permanent arrest after about 50 population doublings (referred to as the Hayflick limit). Senescence is considered a stress response that can be induced by a wide range of intrinsic and extrinsic insults, including oxidative and genotoxic stress, DNA damage, telomere attrition, oncogenic activation, mitochondrial dysfunction, or chemotherapeutic agents.

Senescent cells remain metabolically active and can influence the tissue hemostasis, disease and aging through their secretory phenotype. Senescence is considered as a physiologic process and is important in promoting wound healing, tissue homeostasis, regeneration, and fibrosis regulation. For instance, transient induction of senescent cells is observed during would healing and contributes to wound resolution. Perhaps one of the most important roles of senescence is its role in tumor suppression. However, the accumulation of senescent cells also drives aging- and aging-related diseases and conditions. The senescent phenotype also can trigger chronic inflammatory responses and consequently augment chronic inflammatory conditions to promote tumor growth. The connection between senescence and aging was initially based on observations that senescent cells accumulate in aged tissue. The use of transgenic models has enabled the detection of senescent cells systematically in many age-related pathologies. Strategies to selectively eliminate senescent cells has demonstrated that senescent cells can indeed play a causal role in aging and related pathologies.

Senescent cells display important and unique properties which include changes in morphology, chromatin organization, gene expression, and metabolism. There are several biochemical and functional properties associated with cellular senescence, such as (i) increased expression of p16 and p21, inhibitors of cyclin-dependent kinases, (ii) presence of senescence-associated β-galactosidase, a marker of lysosomal activity, (iii) appearance of senescence-associated heterochromatin foci and downregulation of lamin B1 levels, (iv) resistance to apoptosis caused by an increased expression of anti-apoptotic BCL-family protein, and (v) upregulation of CD26 (DPP4), CD36 (Scavenger receptor), forkhead box 4 (FOXO4), and secretory carrier membrane protein 4 (SCAMP4). Senescent cells also express an inflammatory signature, the so-called senescence-associated secretory phenotype (SASP). Through SASP, the senescent cells produce a wide range of inflammatory cytokines (IL-6, IL-8), growth factors (TGF-β), chemokines (CCL-2), and matrix metalloproteinases (MMP-3, MMP-9) that operate in a cell-autonomous manner to reinforce senescence (autocrine effects) and communicate with and modify the microenvironment (paracrine effects). SASP factors can contribute to tumor suppression by triggering senescence surveillance, an immune-mediated clearance of senescent cells. However, chronic inflammation is also a known driver of tumorigenesis, and accumulating evidence indicates that chronic SASP can also boost cancer and aging-related diseases.

The secretion profile of senescent cells is context dependent. For instance, the mitochondrial dysfunction-associated senescence (MiDAS), induced by different mitochondrial dysfunction in human fibroblasts, led to the appearance of a SASP that was deficient in IL-1-dependent inflammatory factors. A decrease in the NAD+/NADH ratio activated AMPK signaling which induced MiDAS through the activation of p53. As a result, p53 inhibited NF-κB signaling which is a crucial inducer of pro-inflammatory SASP. In contrast, the cellular senescence caused by persistent DNA damage in human cells induced an inflammatory SASP, which was dependent on the activation of ataxia-telangiectasia mutated (ATM) kinase but not on that of p53. In particular, the expression and secretion levels of IL-6 and IL-8 were increased. It was also demonstrated that cellular senescence caused by the ectopic expression p16INK4a and p21CIP1 induced the senescent phenotype in human fibroblasts without an inflammatory SASP indicating that the growth arrest itself did not stimulate SASP.

One of the most defining characteristics of senescence is stable growth arrest. This is achieved by two important pathways, the p16/Rb and the p53/p21, both of which are central in tumor suppression. DNA damage results in: (1) high deposition of γH2Ax (histone coding gene) and 53BP1 (involved in DNA damage response) in chromatin: this leads to activation of a kinase cascade eventually resulting in p53 activation, and (2) activation of p16INK4a and ARF (both encoded by CDKN2A) and P15INK4b (encoded by CDKN2B): p53 induces transcription of cyclin-dependent kinase inhibitor (p21) and along with both p16INK4a and p15INK4b block genes for cell cycle progression (CDK4 and CDK6). This eventually leads to hypophosphorylation of Retinoblastoma protein (Rb) and cell cycle arrest at the G1 phase.

Selectively killing senescent cells has been shown to significantly improve the health span of mice in the context of normal aging and ameliorates the consequences of age-related disease or cancer therapy (Ovadya, *J Clin Invest.* 128(4): 1247-1254, 2018). In nature, the senescent cells are normally removed by the innate immune cells. Induction of senescence not only prevents the potential proliferation and transformation of damaged/altered cells, but also favors tissue repair through the production of SASP factors that function as chemoattractants mainly for Natural Killer (NK) cells (such as IL-15 and CCL2) and macrophages (such as CFS-1 and CCL2). These innate immune cells mediate the immunosurveillance mechanism for eliminating stressed cells. Senescent cells usually up-regulate the NK-cell activating receptor NKG2D and DNAM1 ligands, which belong to a family of stress-inducible ligands: an important component of the frontline immune defense against infectious diseases and malignancies. Upon receptor activation, NK cells can then specifically induce the death of senescent cells through their cytolytic machinery. A role for NK cells in the immune surveillance of senescent cells has been pointed out in liver fibrosis (Sagiv, Oncogene 32(15): 1971-1977, 2013), hepatocellular carcinoma (Iannello, *J Exp Med* 210(10): 2057-2069, 2013), multiple myeloma (Soriani, *Blood* 113 (15): 3503-3511, 2009), and glioma cells stressed by dysfunction of the mevalonate pathway (Ciaglia, *Int J Cancer* 142(1): 176-190, 2018). Endometrial cells undergo acute cellular senescence and do not differentiate into decidual cells. The differentiated decidual cells secrete IL-15 and thereby recruit uterine NK cells to target and eliminate the undifferentiated senescent cells thus helping to re-model and rejuvenate the endometrium (Brighton, *Elife* 6: e31274, 2017). With a similar mechanism, during liver fibrosis, p53-expressing senescent liver satellite cells skewed the polarization of resident Kupfer macrophages and freshly infiltrated macrophages toward the pro-inflammatory M1 phenotype, which display senolytic activity. F4/80+ macrophages have been shown to play a key role in the clearance of mouse uterine senescent cells to maintain postpartum uterine function.

Senescent cells recruit NK cells by mainly upregulating ligands to NKG2D (expressed on NK cells), chemokines, and other SASP factors. In vivo models of liver fibrosis have shown effective clearance of senescent cells by activated NK cells (Krizhanovsky, *Cell* 134(4): 657-667, 2008). Studies have described various models to study senescence including liver fibrosis (Krizhanovsky, *Cell* 134(4): 657-667, 2008), osteoarthritis (Xu, *J Gerontol A Biol Sci Med Sci* 72(6): 780-785, 2017), and Parkinson's disease (Chinta, *Cell Rep* 22(4): 930-940, 2018). Animal models for studying senescent cells are described in: Krizhanovsky, *Cell* 134(4): 657-667, 2008; Baker, *Nature* 479(7372): 232-236, 2011; Farr, *Nat Med* 23(9): 1072-1079, 2017; Bourgeois, *FEBS Lett* 592(12): 2083-2097, 2018; Xu, *Nat Med* 24(8): 1246-1256, 2018).

Additional Therapeutic Agents

Some embodiments of any of the methods described herein can further include administering to a subject (e.g., any of the subjects described herein) a therapeutically effective amount of one or more additional therapeutic agents. The one or more additional therapeutic agents can be administered to the subject at substantially the same time as a T_{reg} cell generated by any of the methods described herein or any of the compositions (e.g., pharmaceutical compositions) comprising a population of T_{reg} cells produced using any of the methods described herein. In some embodiments, one or more additional therapeutic agents can be administered to the subject prior to administration of a T_{reg} cell or a composition (e.g., a pharmaceutical composition) comprising a population of T_{reg} cells produced using any of the methods described herein. In some embodiments, one or more additional therapeutic agents can be administered to the subject after administration of a T_{reg} cell or a composition (e.g., a pharmaceutical composition) comprising a population of T_{reg} cells produced using any of the methods described herein to the subject.

Non-limiting examples of additional therapeutic agents include: anti-cancer drugs, activating receptor agonists, immune checkpoint inhibitors, agents for blocking HLA-specific inhibitory receptors, Glucogen Synthase Kinase (GSK) 3 inhibitors, and antibodies.

Non-limiting examples of anticancer drugs include anti-metabolic drugs (e.g., 5-fluorouracil (5-FU), 6-mercaptopurine (6-MP), capecitabine, cytarabine, floxuridine, fludarabine, gemcitabine, hydroxycarbamide, methotrexate, 6-thioguanine, cladribine, nelarabine, pentostatin, or pemetrexed), plant alkaloids (e.g., vinblastine, vincristine, vindesine, camptothecin, 9-methoxycamptothecin, coronaridine, taxol, naucleaorals, diprenylated indole alkaloid, montamine, schischkiniin, protoberberine, berberine, sanguinarine, chelerythrine, chelidonine, liriodenine, clivorine, β-carboline, antofine, tylophorine, cryptolepine, neocryptolepine, corynoline, sampangine, carbazole, crinamine, montanine, ellipticine, paclitaxel, docetaxel, etoposide, teniposide, irinotecan, topotecan, or acridone alkaloids), proteasome inhibitors (e.g., lactacystin, disulfiram, epigallocatechin-3-gallate, marizomib (salinosporamide A), oprozomib (ONX-0912), delanzomib (CEP-18770), epoxomicin, MG132, beta-hydroxy beta-methylbutyrate, bortezomib, carfilzomib, or ixazomib), antitumor antibiotics (e.g., doxorubicin, daunorubicin, epirubicin, mitoxantrone, idarubicin, actinomycin, plicamycin, mitomycin, or bleomycin), histone deacetylase inhibitors (e.g., vorinostat, panobinostat, belinostat, givinostat, abexinostat, depsipeptide, entinostat, phenyl butyrate, valproic acid, trichostatin A, dacinostat, mocetinostat, pracinostat, nicotinamide, cambinol, tenovin 1, tenovin 6, sirtinol, ricolinostat, tefinostat, kevetrin, quisinostat, resminostat, tacedinaline, chidamide, or selisistat), tyrosine kinase inhibitors (e.g., axitinib, dasatinib, encorafinib, erlotinib, imatinib, nilotinib, pazopanib, and sunitinib), and chemotherapeutic agents (e.g., all-trans retinoic acid, azacitidine, azathioprine, doxifluridine, epothilone, hydroxyurea, imatinib, teniposide, tioguanine, valrubicin, vemurafenib, and lenalidomide). Additional examples of chemotherapeutic agents include alkylating agents, e.g., mechlorethamine, cyclophosphamide, chlorambucil, melphalan, ifosfamide, thiotepa, hexamethylmelamine, busulfan, altretamine, procarbazine, dacarbazine, temozolomide, carmustine, lumustine, streptozocin, carboplatin, cisplatin, and oxaliplatin.

EXAMPLES

The invention is further described in the following examples, which do not limit the scope of the invention described in the claims.

Example 1. Production of αCD3scFv/TF/αCD28scFv

The nucleic acid and amino acid sequences of αCD3scFv/TF/αCD28scFv are shown below.

```
Nucleic Acid Encoding αCD3scFv/TF/αCD28scFv
                                          (SEQ ID NO: 10)
(Signal peptide)
ATGAAGTGGGTGACCTTCATCAGCTTATTATTTTTATTCAGCTCCGCCT

ATTCC
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-continued
(αCD3 light chain variable region)
CAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCGGTGA

GAAGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATGAACT

GGTATCAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGACACC

AGCAAGCTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGATCCGG

CACCAGCTACTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCTGCCA

CCTACTATTGCCAGCAATGGAGCAGCAACCCCTTCACATTCGGATCTGGC

ACCAAGCTCGAAATCAATCGT

(Linker)
GGAGGAGGTGGCAGCGGCGGCGGTGGATCCGGCGGAGGAGGAAGC

(αCD3 heavy chain variable region)
CAAGTTCAACTCCAGCAGAGCGGCGCTGAACTGGCCCGGCCCGGCGCCTC

CGTCAAGATGAGCTGCAAGGCTTCCGGCTATACATTTACTCGTTACACAA

TGCATTGGGTCAAGCAGAGGCCCGGTCAAGGTTTAGAGTGGATCGGATAT

ATCAACCCTTCCCGGGGCTACACCAACTATAACCAAAAGTTCAAGGATAA

AGCCACTTTAACCACTGACAAGAGCTCCTCCACCGCCTACATGCAGCTGT

CCTCTTTAACCAGCGAGGACTCCGCTGTTTACTACTGCGCTAGGTATTAC

GACGACCACTACTGTTTAGACTATTGGGGACAAGGTACCACTTTAACCGT

CAGCAGC

(Human tissue factor 219 form)
TCCGGCACCACCAATACCGTGGCCGCTTATAACCTCACATGGAAGAGCAC

CAACTTCAAGACAATTCTGGAATGGGAACCCAAGCCCGTCAATCAAGTTT

ACACCGTGCAGATCTCCACCAAATCCGGAGACTGGAAGAGCAAGTGCTTC

TACACAACAGACACCGAGTGTGATTTAACCGACGAAATCGTCAAGGACGT

CAAGCAAACCTATCTGGCTCGGGTCTTTTCCTACCCCGCTGGCAATGTCG

AGTCCACCGGCTCCGCTGGCGAGCCTCTCTACGAGAATTCCCCCGAATTC

ACCCCTTATTTAGAGACCAATTTAGGCCAGCCTACCATCCAGAGCTTCGA

GCAAGTTGGCACCAAGGTGAACGTCACCGTCGAGGATGAAAGGACTTTAG

TGCGGCGGAATAACACATTTTTATCCCTCCGGGATGTGTTCGGCAAAGAC

CTCATCTACACACTGTACTATTGGAAGTCCAGCTCCTCCGGCAAAAAGAC

CGCTAAGACCAACACCAACGAGTTTTTAATTGACGTGGACAAAGGCGAGA

ACTACTGCTTCAGCGTGCAAGCCGTGATCCCTTCTCGTACCGTCAACCGG

AAGAGCACAGATTCCCCCGTTGAGTGCATGGGCCAAGAAAAGGGCGAGTT

CCGGGAG

(αCD28 light chain variable region)
GTCCAGCTGCAGCAGAGCGGACCCGAACTCGTGAAACCCGGTGCTTCCGT

GAAAATGTCTTGTAAGGCCAGCGGATACACCTTCACCTCCTATGTGATCC

AGTGGGTCAAACAGAAGCCCGGACAAGGTCTCGAGTGGATCGGCAGCATC

AACCCTTACAACGACTATACCAAATACAACGAGAAGTTTAAGGGAAAGGC

TACTTTAACCTCCGACAAAAGCTCCATCACAGCCTACATGGAGTTCAGCT

CTTTAACATCCGAGGACAGCGCTCTGTACTATTGCGCCCGGTGGGGCGAC

GGCAATTACTGGGGACGGGGCACAACACTGACCGTGAGCAGC
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-continued

(Linker)
GGAGGCGGAGGCTCCGGCGGAGGCGGATCTGGCGGTGGCGGCTCC

(αCD28 light chain variable region)
GACATCGAGATGACCCAGTCCCCCGCTATCATGTCCGCCTCTTTAGGCGA

GCGGGTCACAATGACTTGTACAGCCTCCTCCAGCGTCTCCTCCTCCTACT

TCCATTGGTACCAACAGAAACCCGGAAGCTCCCCTAAACTGTGCATCTAC

AGCACCAGCAATCTCGCCAGCGGCGTGCCCCCTAGGTTTTCCGGAAGCGG

AAGCACCAGCTACTCTTTAACCATCTCCTCCATGGAGGCTGAGGATGCCG

CCACCTACTTTTGTCACCAGTACCACCGGTCCCCCACCTTCGGAGGCGGC

ACCAAACTGGAGACAAAGAGG

αCD3scFv/TF/αCD28scFv
(SEQ ID NO: 8)

(Signal peptide)
MKWVTFISLLFLFSSAYS

(αCD3 light chain variable region)
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDT

SKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSG

TKLEINR

(Linker)
GGGGSGGGGSGGGGS

(αCD3 heavy chain variable region)
QVQLQQSGAELARPGASVKMSCKASGYTFTRYTMHWVKQRPGQGLEWIGY

INPSRGYTNYNQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAVYYCARYY

DDHYCLDYWGQGTTLTVSS

(Human tissue factor 219)
SGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCF

YTTDTECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEF

TPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKD

LIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNR

KSTDSPVECMGQEKGEFRE

(αCD28 light chain variable region)
VQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKPGQGLEWIGSI

NPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSALYYCARWGD

GNYWGRGTTLTVSS

(Linker)
GGGGSGGGGSGGGGS

(αCD28 heavy chain variable region)
DIEMTQSPAIMSASLGERVTMTCTASSSVSSSYFHWYQQKPGSSPKLCIY

STSNLASGVPPRFSGSGSTSYSLTISSMEAEDAATYFCHQYHRSPTFGGG

TKLETKR

The nucleic acid encoding αCD3scFv/TF/αCD28scFv was cloned into a modified retrovirus expression vectors as described previously (Hughes et al., *Hum Gene Ther* 16:457-72, 2005). The expression vector encoding αCD3scFv/TF/αCD28scFv was transfected into CHO-K1 cells. Expression of the expression vector in CHO-K1 cells allowed for secretion of the soluble αCD3scFv/TF/αCD28scFv single-chain chimeric polypeptide (referred to as 3t28), which can be purified by anti-TF antibody affinity and other chromatography methods.

An anti-tissue factor antibody affinity column was used to purify the αCD3scFv/TF/αCD28scFv single-chain chimeric polypeptide. The anti-tissue factor antibody affinity column was connected to a GE Healthcare AKTA Avant system. A flow rate of 4 mL/min was used for all steps except the elution step, which was 2 mL/min.

Cell culture harvest including αCD3scFv/TF/αCD28scFv single-chain chimeric polypeptide was adjusted to pH 7.4 with 1M Tris base and loaded onto the anti-TF antibody affinity column (described above) which was equilibrated with 5 column volumes of PBS. After sample loading, the column was washed with 5 column volumes PBS, followed by elution with 6 column volumes 0.1 M acetic acid, pH 2.9. An A280 elution peak was collected and then neutralized to pH 7.5-8.0 by adding 1 M Tris base. The neutralized sample was then buffer exchanged into PBS using Amicon centrifugal filters with a 30 kDa molecular weight cutoff.

After each elution, the anti-tissue factor antibody affinity column was stripped using 6 column volumes of 0.1 M glycine, pH 2.5. The column was then neutralized using 10 column volumes of PBS, 0.05% NaN_3, and stored at 2-8° C.

Example 2. Production of IL-2/TF/IL-2

The nucleic acid and amino acid sequences of IL-2/TF/IL-2 are shown below.

Nucleic Acid Encoding IL-2/TF/IL-2
(SEQ ID NO: 12)

(Signal peptide)
ATGAAGTGGGTGACCTTCATCAGCCTGCTGTTCCTGTTCTCCAGCGCCT

ACTCC

(First IL-2 fragment)
GCCCCCACCTCCTCCTCCACCAAGAAGACCCAGCTGCAGCTGGAGCATTT

ACTGCTGGATTTACAGATGATTTTAAACGGCATCAACAACTACAAGAACC

CCAAGCTGACTCGTATGCTGACCTTCAAGTTCTACATGCCCAAGAAGGCC

ACCGAGCTGAAGCATTTACAGTGTTTAGAGGAGGAGCTGAAGCCCCTCGA

GGAGGTGCTGAATTTAGCCCAGTCCAAGAATTTCCATTTAAGGCCCCGGG

ATTTAATCAGCAACATCAACGTGATCGTTTTAGAGCTGAAGGGCTCCGAG

ACCACCTTCATGTGCGAGTACGCCGACGAGACCGCCACCATCGTGGAGTT

TTTAAATCGTTGGATCACCTTCTGCCAGTCCATCATCTCCACTTTAACC

(Human tissue factor 219 form)
AGCGGCACAACCAACACAGTCGCTGCCTATAACCTCACTTGGAAGAGCAC

CAACTTCAAAACCATCCTCGAATGGGAACCCAAACCCGTTAACCAAGTTT

ACACCGTGCAGATCAGCACCAAGTCCGGCGACTGGAAGTCCAAATGTTTC

TATACCACCGACACCGAGTGCGATCTCACCGATGAGATCGTGAAAGATGT

GAAACAGACCTACCTCGCCCGGGTGTTTAGCTACCCCGCCGGCAATGTGG

AGAGCACTGGTTCCGCTGGCGAGCCTTTATACGAGAACAGCCCCGAATTT

ACCCCTTACCTCGAGACCAATTTAGGACAGCCCACCATCCAAAGCTTTGA

GCAAGTTGGCACAAAGGTGAATGTGACAGTGGAGGACGAGCGGACTTTAG

TGCGGCGGAACAACACCTTTCTCAGCCTCCGGGATGTGTTCGGCAAAGAT

TTAATCTACACACTGTATTACTGGAAGTCCTCTTCCTCCGGCAAGAAGAC

AGCTAAAACCAACACAAACGAGTTTTTAATCGACGTGGATAAAGGCGAAA

-continued

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ACTACTGTTTCAGCGTGCAAGCTGTGATCCCCTCCCGGACCGTGAATAGG

AAAAGCACCGATAGCCCCGTTGAGTGCATGGGCCAAGAAAAGGGCGAGTT

CCGGGAG

(Second IL-2 fragment)
GCACCTACTTCAAGTTCTACAAAGAAAACACAGCTACAACTGGAGCATTT

ACTGCTGGATTTACAGATGATTTTGAATGGAATTAATAATTACAAGAATC

CCAAACTCACCAGGATGCTCACATTTAAGTTTTACATGCCCAAGAAGGCC

ACAGAACTGAAACATCTTCAGTGTCTAGAAGAAGAACTCAAACCTCTGGA

GGAAGTGCTAAATTTAGCTCAAAGCAAAAACTTTCACTTAAGACCCAGGG

ACTTAATCAGCAATATCAACGTAATAGTTCTGGAACTAAAGGGATCTGAA

ACAACATTCATGTGTGAATATGCTGATGAGACAGCAACCATTGTAGAATT

TCTGAACAGATGGATTACCTTTTGTCAAAGCATCATCTCAACACTAACT

IL-2/TF/IL-2
                                        (SEQ ID NO: 4)
(Signal peptide)
MKWVTFISLLFLFSSAYS

(Human IL-2)
APTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRMLTFKFYMPKKA

TELKHLQCLEEELKPLEEVLNLAQSKNFHLRPRDLISNINVIVLELKGSE

TTFMCEYADETATIVEFLNRWITFCQSIISTLT

(Human Tissue Factor 219)
SGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCF

YTTDTECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEF

TPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKD

LIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNR

KSTDSPVECMGQEKGEFRE

(Human IL-2)
APTSSSTKKTQLQLEHLLLDLQMILNGINNYKNPKLTRMLTFKFYMPKKA

TELKHLQCLEEELKPLEEVLNLAQSKNFHLRPRDLISNINVIVLELKGSE

TTFMCEYADETATIVEFLNRWITFCQSIISTLT
```

The nucleic acid encoding IL-2/TF/IL-2 was cloned into a modified retrovirus expression vector as described previously (Hughes et al., *Hum Gene Ther* 16:457-72, 2005). The expression vector encoding IL-2/TF/IL-2 was transfected into CHO-K1 cells. Expression of the expression vector in CHO-K1 cells allowed for secretion of the soluble IL-2/TF/IL-2 single-chain chimeric polypeptide (referred to as 2t2), which can be purified by anti-TF antibody affinity and other chromatography methods.

Example 3: Stimulation of Foxp3$^+$ T$_{reg}$ Cells by 2t2 with Anti-CD3/Anti-CD28 Beads and Rapamycin Compared with Commercial Recombinant Human IL-2

Fresh human leukocytes (3 donors) were obtained from the blood bank and CD4$^+$CD25$^+$CD127^{dim} T$_{reg}$ cells were isolated with the CD4$^+$CD25$^+$CD127^{dim} regulatory T cell isolation kit II reagent (Miltenyi Biotec). The purity of Foxp3$^+$ T$_{reg}$ cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-APC, CD3 APC-Cy7, Foxp3-PE (Intracellular staining, Invitrogen), and CD127-Alex Fluro 700 (BioLegend). The cells were counted and resuspended in 1×10^6/mL in a 48-well flat-bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone)). The cells were stimulated with either (1) hIL-2 cytokine (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1,Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) and incubated for 21 days at 37° C., 5% CO_2. During the expansion period, the cells were re-stimulated with either (1) hIL-2 cytokine (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma) every alternate day. Every 7 days, the cells were re-stimulated with anti-CD3/anti-CD28 beads (1:1 Beads: Cells) (Dynabeads, Life Technologies). The cells were maintained for 21 days at 1×10^6/mL concentration with fresh complete media containing either (1) hIL-2 cytokine (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma) every alternate day. At Days 7, 14, and 21, the cells were stained with CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7, Foxp3-PE (intracellular staining using Foxp3 staining kit, Invitrogen), and CD127-Alex Fluro 700 (BioLegend) to assess the expansion of Foxp3$^+$ T$_{reg}$ cells by flow cytometry. Fold-expansion of Foxp3$^+$ T$_{reg}$ cells was measured by counting in a trypan blue (0.4%, Invitrogen) viability assay. FIG. **1** shows a 145-fold increase in Foxp3$^+$ T$_{reg}$ cells which were stimulated with 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) compared to a 100-fold increase with hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma).

Example 4: Detection of Helios, CTLA-4, CD39, CD62L, CD25, CD103, CD69, PD1 and CD49b Markers

Fresh human leukocytes (3 donors) were obtained from the blood bank and CD4$^+$CD25$^+$CD127^{dim} T$_{reg}$ cells were isolated with the CD4$^+$CD25$^+$CD127^{dim} regulatory T cell isolation kit II reagent (Miltenyi Biotec). The purity of Foxp3$^+$ T$_{reg}$ cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-APC, CD3 APC-Cy7, Foxp3-PE (Intracellular staining, Invitrogen), and CD127-Alex Fluro 700 (BioLegend). The cells were counted and resuspended in 1×10^6/mL in a 48-well flat-bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone)). The cells were stimulated with either (1) hIL-2 cytokine (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma), then incubated for 21 days at 37° C., 5% CO_2. During the expansion period, the cells were re-stimulated with either (1) hIL-2 (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma) every alternate day. Every 7 days, the cells were re-stimulated with anti-CD3/anti-CD28 beads (1:1,Beads: Cells) (Dynabeads, Life Technologies). The cells were maintained for 21 days at 1×10^6/mL concentration with fresh complete media containing either (1) hIL-2 (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma), every alternate day. The cells were stained with CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7, Foxp3-PE (Intracellular staining using Foxp3 staining kit, Invitrogen), and CD127-Alex Fluro 700 (BioLegend) to assess the expansion of Foxp3$^+$ T_{reg} cells by flow cytometry. Fold-expansion of Foxp3$^+$ T_{reg} cells was measured by counting in a trypan blue (0.4%, Invitrogen) viability assay. On day 21, the cells were harvested, and surface stained for CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7, Foxp3-PE (Intracellular staining using Foxp3 staining kit, Invitrogen), CD127-Alex Fluro 700, Helios PE-Cy7, CTLA-4 Brilliant Violet 605, CD39 APC, CD62L PECy7, CD25 PE, CD103 PerCP.Cy5.5, CD69 APC Fire 750, PD1 Alexa Fluro 750, and CD49b APC, and intracellularly stained using Foxp3-Pacific Blue (BioLegend) (Intracellular staining kit, Invitrogen). For surface staining, the cells were washed (1500 RPM for 5 minutes at room temperature) in FACS buffer (1×PBS (Hyclone) with 0.5% BSA (EMD Millipore) and 0.001% sodium azide (Sigma). After two washes, the cells were analyzed by Flow Cytometry (Celesta-BD Bioscience). For intracellularly cell staining, the cells were fixed in fixative buffer (Invitrogen) for 20 minutes at room temperature. After fixation steps, the cells were washed (1500 RPM for 5 minutes in room temperature) in 1× Permeabilized Buffer (eBioscience) and stained using Foxp3-Pacific Blue (Biolegend) for 30 minutes at room temperature. The cells were washed once again with 1× Permeabilized Buffer and then washed with FACS buffer. The cell pellets were resuspended in 300 μL of FACS Buffer for analysis by Flow Cytometry (Celesta-BD Bioscience). (The cells were gated on CD4$^+$CD25$^+$Foxp3$^+$ cells and plotted Data-%$^+$ of surface markers, 3 healthy donors). FIGS. **2**A-**2**I show comparable surface markers on Foxp3$^+$ T_{reg} cells which were stimulated with either (1) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies) and rapamycin (100 nM) (Sigma) or (2) hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) for 21 days.

Example 5: Activation of Human Regulatory T Cells Displays a High Level of Glucose Metabolism

Fresh human leukocytes (3 donors) were obtained from the blood bank and CD4$^+$CD25$^+$CD127^{dim} T_{reg} cells were isolated using CD4$^+$CD25$^+$CD127^{dim} T_{reg} cell isolation kit II (Miltenyi Biotec). The purity of Foxp3$^+$ T_{reg} cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-APC, CD3 APC-Cy7, Foxp3-PE (Intracellular staining, Invitrogen), and CD127-Alex Fluro 700 (BioLegend). The cells were counted and resuspended in 1×10^6/mL in a 24-well flat-bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone)). The cells were stimulated with either (1) hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma), then incubated overnight at 37° C. at 5% CO_2. The next day, the cells were harvested, counted, and resuspended in 4×10^6/mL in a 96-well plate and 50 μL/well were seeded in Cell-Tak-coated Seahorse Bioanalyzer XFe96 culture plates in Seahorse XF RPMI medium, pH 7.4 supplemented with 2 mM L-glutamine. The cells were allowed to attach to the plate for 30 mins at 37° C. Additional 130 μL of assay medium was added to each well of the plate (also the background wells). The plate was incubated in 37° C., non-CO_2 incubator for 1 hr. For the calibration plate: the 10× solution of glucose/oligomycin/2-deoxyglucose were prepared in Seahorse assay media. 20 μL volume of glucose/oligomycin/2-deoxyglucose were added to ports A, B, and C of the extracellular flux plate that was calibrated overnight. Extracellular acidification rate (ECAR) was measured using an XFe96 Extracellular Flux Analyzer. Complete ECAR analysis consisted of four stages: non-glycolytic acidifications (without drugs), glycolysis (10 mM glucose), glycolytic capacity (2 μM oligomycin), and glycolytic reserve (100 mM 2-deoxyglucose) which was calculated by using Seahorse Xfe96 software, wave. The results (FIGS. **3**A-**3**D) showed that the activated human T_{reg} cells displayed a high level of glucose metabolism.

Example 6: Measurement of Suppression of Autologous T Responder Cells by Expanded Regulatory T Cell

Fresh human leukocytes (3 donors) were obtained from the blood bank and CD4$^+$CD25$^+$CD127^{dim-} T_{reg} cells were isolated with the CD4$^+$CD25$^+$CD127^{dim-} regulatory T cell isolation kit II reagent (Miltenyi Biotec). The purity of Foxp3$^+$ T_{reg} cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-APC, CD3 APC-Cy7, Foxp3-PE (Intracellular staining, Invitrogen), and CD127-Alex Fluro 700 (BioLegend). The cells were counted and resuspended in 1×10^6/mL in a 24-well flat-bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone)). The cells were stimulated with either (1) hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma), and then incubated for 21 days at 37° C., 5% CO_2. During the expansion period, the cells were re-stimulated with either (1) hIL-2 cytokine (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma), every alternate day. Every 7 days, the cells were re-stimulated with anti-CD3/anti-CD28 beads (1:1 Beads: Cells) (Dynabeads, Life Technologies). The cells were maintained for 21 days at 1×10^6/mL concentration with fresh complete media containing either (1) hIL-2 (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma), every alternate day. Fold-expansion of Foxp3$^+$ T_{reg} cells was measured by counting in trypan blue (0.4%, Invitrogen) viability assay. On day 21, frozen autologous T responder (Tresp) cells were thawed and labelled with Cell Trace Violet (Invitrogen). Expanded T_{reg} cells were harvested and counted. T_{reg} cells were cultured with T responder cells at various ratios (5:1, 2:1, 1:1, 0.5:1, T_{reg} cells: T responder cells). The cells were re-stimulated with either (1) anti-CD3/anti-CD28 beads (Beads:Cell 1:70) and 2t2 (25 nM) or (2) anti-CD3/anti-CD28 beads (Beads:Cell 1:70) and IL2 (50 IU/mL), and cultured for 5 days. The cells were harvested and washed (1500 RPM for 5 minutes at room temperature) in FACS buffer (1×PBS (Hyclone) with 0.5% BSA (EMD Millipore) and 0.001% Sodium Azide (Sigma)). After two washes, the cells were analyzed by flow cytometry (Celesta-BD Bioscience). (Plotted %+ of CFSE+ proliferating cells). FIG. **4** shows suppression capacity of Foxp3+ T_{reg} cells which were stimulated with 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) compared to cells stimulated with hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma).

Example 7: Measurement of IL-10 from Suppression Assay

Fresh human leukocytes (3 donors) were obtained from the blood bank and CD4+CD25+CD127^{dim} T_{reg} cells were isolated with the CD4+CD25+CD127^{dim} regulatory T cell isolation kit II reagent (Miltenyi Biotec). The purity of Foxp3+ T_{reg} cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-APC, CD3 APC-Cy7, Foxp3-PE (Intracellular staining, Invitrogen), and CD127-Alex Fluro 700 (BioLegend). The cells were counted and resuspended in 1×10^6/mL in a 24-well flat-bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone)). The cells were stimulated with either (1) hIL-2 (500 IU/mL) (Proleukin), anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2, anti-CD3/anti-CD28 beads (4:1 Beads: Cells) (Dynabeads, Life Technologies), and rapamycin (100 nM) (Sigma), and then incubated for 21 days at 37° C. and 5% CO_2. During the expansion period, the cells were re-stimulated with either (1) hIL-2 cytokine (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma), every alternate day. Every 7 days, the cells were re-stimulated with anti-CD3/anti-CD28 beads (1:1 Beads:Cells) (Dynabeads, Life Technologies). The cells were maintained for 21 days at 1×10^6/mL concentration with fresh complete media containing either (1) hIL-2 (500 IU/mL) (Proleukin) and rapamycin (100 nM) (Sigma) or (2) 100 nM of 2t2 and rapamycin (100 nM) (Sigma), every alternate day. Fold-expansion of Foxp3+ T_{reg} cells was measured by counting in a trypan blue (0.4%, Invitrogen) viability assay. On day 21, frozen autologous T responder (Tresp) cells were thawed and labelled with Cell Trace Violet (Invitrogen). The expanded T_{reg} cells were harvested and counted. The T_{reg} cells were cultured with T responder cells at various ratios (5:1, 1:1, T_{reg} cells: T responder cells). The cells were re-stimulated with (1) anti-CD3/anti-CD28 beads (Beads:Cell 1:70) and 2t2 (25 nM) or (2) anti-CD3/anti-CD28 beads (Beads:Cell 1:70) and IL2 (50 IU/mL), and cultured for 5 days. The cell-free liquid culture medium was collected and analyzed by LEGEND-plex multi-analyte flow assay kit (Biolegend) according to the manufacturer's instructions. FIG. **5** shows IL-10 in the cell-free liquid culture medium from a suppression assay where IL-10 secretion was significantly higher in 2t2-expanded T_{reg} cells compared to IL-2 expanded T_{reg} cells, 10 both in the presences or absence of rapamycin.

Example 8. Generation and Characterization of 3t15*-28s Fusion Protein Complex

Construction and production of 3115*-28s

A fusion protein was generated comprising anti-CD3scFv/TF/IL15D8N mutant and anti-CD28scFv/IL15RαSu fusion proteins. The anti-CD3 antibody (αCD3) variable regions, human tissue factor, IL15D8N mutant, anti-CD28 antibody (αCD28) variable regions, and IL15RαSu sequences were obtained from the UniProt website and DNA for these sequences was synthesized by Genewiz. Specifically, the construct was made linking αCD3 light chain variable region sequence by a (G4S) 3 linker to αCD3 heavy chain variable region sequence to form an αCD3scFv. The αCD3scFv was linked to the N-terminus coding region of tissue factor 219 followed by a linkage to the N-terminus coding region of IL15D8N mutant (FIGS. **6**A-**6**B). The nucleic acid and protein sequences of a construct are shown below.

The nucleic acid sequence of the αCD3scFv/TF/IL15D8N construct (including signal peptide sequence) is as follows (SEQ ID NO: 105):

```
(Signal peptide)
ATGAAGTGGGTGACCTTCATCAGCTTATTATTTTTATTCAGCTCCGCCT

ATTCC

(αCD3 light chain variable region)
CAGATCGTGCTGACCCAAAGCCCCGCCATCATGAGCGCTAGCCCCGGTGA

GAAGGTGACCATGACATGCTCCGCTTCCAGCTCCGTGTCCTACATGAACT

GGTATCAGCAGAAAAGCGGAACCAGCCCCAAAAGGTGGATCTACGACACC

AGCAAGCTGGCCTCCGGAGTGCCCGCTCATTTCCGGGGCTCTGGATCCGG

CACCAGCTACTCTTTAACCATTTCCGGCATGGAAGCTGAAGACGCTGCCA

CCTACTATTGCCAGCAATGGAGCAGCAACCCCTTCACATTCGGATCTGGC

ACCAAGCTCGAAATCAATCGT

(Linker)
GGAGGAGGTGGCAGCGGCGGCGGTGGATCCGGCGGAGGAGGAAGC

(αCD3 heavy chain variable region)
CAAGTTCAACTCCAGCAGAGCGGCGCTGAACTGGCCCGGCCCGGCGCCTC

CGTCAAGATGAGCTGCAAGGCTTCCGGCTATACATTTACTCGTTACACAA

TGCATTGGGTCAAGCAGAGGCCCGGTCAAGGTTTAGAGTGGATCGGATAT

ATCAACCCTTCCCGGGGCTACACCAACTATAACCAAAAGTTCAAGGATAA

AGCCACTTTAACCACTGACAAGAGCTCCTCCACCGCCTACATGCAGCTGT

CCTCTTTAACCAGCGAGGACTCCGCTGTTTACTACTGCGCTAGGTATTAC

GACGACCACTACTGTTTAGACTATTGGGGACAAGGTACCACTTTAACCGT

CAGCAGC

(Human tissue factor 219 form)
TCCGGCACCACCAATACCGTGGCCGCTTATAACCTCACATGGAAGAGCAC

CAACTTCAAGACAATTCTGGAATGGGAACCCAAGCCCGTCAATCAAGTTT

ACACCGTGCAGATCTCCACCAAATCCGGAGACTGGAAGAGCAAGTGCTTC

TACACAACAGACACCGAGTGTGATTTAACCGACGAAATCGTCAAGGACGT

CAAGCAAACCTATCTGGCTCGGGTCTTTTCCTACCCCGCTGGCAATGTCG

AGTCCACCGGCTCCGCTGGCGAGCCTCTCTACGAGAATTCCCCCGAATTC
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ACCCCTTATTTAGAGACCAATTTAGGCCAGCCTACCATCCAGAGCTTCGA

GCAAGTTGGCACCAAGGTGAACGTCACCGTCGAGGATGAAAGGACTTTAG

TGCGGCGGAATAACACATTTTTATCCCTCCGGGATGTGTTCGGCAAAGAC

CTCATCTACACACTGTACTATTGGAAGTCCAGCTCCTCCGGCAAAAAGAC

CGCTAAGACCAACACCAACGAGTTTTTAATTGACGTGGACAAAGGCGAGA

ACTACTGCTTCAGCGTGCAAGCCGTGATCCCTTCTCGTACCGTCAACCGG

AAGAGCACAGATTCCCCCGTTGAGTGCATGGGCCAAGAAAAGGGCGAGTT

CCGGGAG

(Human IL15D8N mutant)
AACTGGGTGAACGTCATCAGCAATTTAAAGAAGATCGAAGATTTAATTCA

GTCCATGCATATCGACGCCACTTTATACACAGAATCCGACGTGCACCCCT

CTTGTAAGGTGACCGCCATGAAATGTTTTTTACTGGAGCTGCAAGTTATC

TCTTTAGAGAGCGGAGACGCTAGCATCCACGACACCGTGGAGAATTTAAT

CATTTTAGCCAATAACTCTTTATCCAGCAACGGCAACGTGACAGAGTCCG

GCTGCAAGGAGTGCGAAGAGCTGGAGGAGAAGAACATCAAGGAGTTTCTG

CAATCCTTTGTGCACATTGTCCAGATGTTCATCAATACCTCC

The amino acid sequence of the αCD3scFv/TF/IL15D8N construct (including signal peptide sequence) is as follows (SEQ ID NO: 104):

(Signal peptide)
MKWVTFISLLFLFSSAYS

(αCD3 light chain variable region)
QIVLTQSPAIMSASPGEKVTMTCSASSSVSYMNWYQQKSGTSPKRWIYDT

SKLASGVPAHFRGSGSGTSYSLTISGMEAEDAATYYCQQWSSNPFTFGSG

TKLEINR

(Linker)
GGGGSGGGGSGGGGS

(αCD3 heavy chain variable region)
QVQLQQSGAELARPGASVKMSCKASGYTFTRYTMHWVKQRPGQGLEWIGY

INPSRGYTNYNQKFKDKATLTTDKSSSTAYMQLSSLTSEDSAVYYCARYY

DDHYCLDYWGQGTTLTVSS

(Human tissue factor 219)
SGTTNTVAAYNLTWKSTNFKTILEWEPKPVNQVYTVQISTKSGDWKSKCF

YTTDTECDLTDEIVKDVKQTYLARVFSYPAGNVESTGSAGEPLYENSPEF

TPYLETNLGQPTIQSFEQVGTKVNVTVEDERTLVRRNNTFLSLRDVFGKD

LIYTLYYWKSSSSGKKTAKTNTNEFLIDVDKGENYCFSVQAVIPSRTVNR

KSTDSPVECMGQEKGEFRE

(Human IL15D8N mutant)
NWVNVISNLKKIEDLIQSMHIDATLYTESDVHPSCKVTAMKCFLLELQVI

SLESGDASIHDTVENLIILANNSLSSNGNVTESGCKECEELEEKNIKEFL

QSFVHIVQMFINTS

Another construct was made by linking αCD28scFv which was constructed by linking αCD28 light chain variable region sequence by a (G4S) 3 linker to αCD28 heavy chain variable region sequence, to the N-terminus coding region of IL15RαSu. The final construct αCD28scFv/IL15RαSu fusion protein sequence was synthesized by Genewiz. The nucleic acid and protein sequences of a construct are shown below.

The nucleic acid sequence of the αCD28scFv/IL15RαSu construct (including signal peptide sequence) is as follows (SEQ ID NO: 109):

(Signal peptide)
ATGAAATGGGTCACCTTCATCTCTTTACTGTTTTTATTTAGCAGCGCCT

ACAGC

(αCD28 light chain variable region)
GTGCAGCTGCAGCAGTCCGGACCCGAACTGGTCAAGCCCGGTGCCTCCGT

GAAAATGTCTTGTAAGGCTTCTGGCTACACCTTTACCTCCTACGTCATCC

AATGGGTGAAGCAGAAGCCCGGTCAAGGTCTCGAGTGGATCGGCAGCATC

AATCCCTACAACGATTACACCAAGTATAACGAAAAGTTTAAGGGCAAGGC

CACTCTGACAAGCGACAAGAGCTCCATTACCGCCTACATGGAGTTTTCCT

CTTTAACTTCTGAGGACTCCGCTTTATACTATTGCGCTCGTTGGGGCGAT

GGCAATTATTGGGGCCGGGGAACTACTTTAACAGTGAGCTCC

(Linker)
GGCGGCGGCGGAAGCGGAGGTGGAGGATCTGGCGGTGGAGGCAGC

(αCD28 heavy chain variable region)
GACATCGAGATGACACAGTCCCCCGCTATCATGAGCGCCTCTTTAGGAGA

ACGTGTGACCATGACTTGTACAGCTTCCTCCAGCGTGAGCAGCTCCTATT

TCCACTGGTACCAGCAGAAACCCGGCTCCTCCCCTAAACTGTGTATCTAC

TCCACAAGCAATTTAGCTAGCGGCGTGCCTCCTCGTTTTAGCGGCTCCGG

CAGCACCTCTTACTCTTTAACCATTAGCTCTATGGAGGCCGAAGATGCCG

CCACATACTTTTGCCATCAGTACCACCGGTCCCCTACCTTTGGCGGAGGC

ACAAAGCTGGAGACCAAGCGG

(Human IL15RαSu)
ATTACATGCCCCCCTCCCATGAGCGTGGAGCACGCCGACATCTGGGTGAA

GAGCTATAGCCTCTACAGCCGGGAGAGGTATATCTGTAACAGCGGCTTCA

AGAGGAAGGCCGGCACCAGCAGCCTCACCGAGTGCGTGCTGAATAAGGCT

ACCAACGTGGCTCACTGGACAACACCCTCTTTAAAGTGCATCCGG

The amino acid sequence of the αCD28scFv/IL15RαSu construct (including signal peptide sequence) is as follows (SEQ ID NO: 108):

(Signal peptide)
MKWVTFISLLFLFSSAYS

(αCD28 light chain variable region)
VQLQQSGPELVKPGASVKMSCKASGYTFTSYVIQWVKQKPGQGLEWIGS

INPYNDYTKYNEKFKGKATLTSDKSSITAYMEFSSLTSEDSALYYCARWG

DGNYWGRGTTLTVSS

(Linker)
GGGGSGGGGSGGGGS

(αCD28 heavy chain variable region)
DIEMTQSPAIMSASLGERVTMTCTASSSVSSSYFHWYQQKPGSSPKLCIY

STSNLASGVPPRFSGSGSGTSYSLTISSMEAEDAATYFCHQYHRSPTFGGG

TKLETKR

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(Human IL-15R α sushi domain)
ITCPPPMSVEHADIWVKSYSLYSRERYICNSGFKRKAGTSSLTECVLNKA

TNVAHWTTPSLKCIR
```

In some cases, the leader peptide is cleaved from the intact polypeptide to generate the mature form that may be soluble or secreted.

The αCD3scFv/TF/IL15D8N and αCD28scFv/IL15RαSu constructs were cloned into a modified retrovirus expression vectors with different antibiotics selection markers as described previously (Hughes et al., *Hum Gene Ther* 16:457-72, 2005), and the expression vectors were co-transfected into CHO-K1 cells. Expression of the construct in CHO-K1 cells allowed for secretion of the soluble αCD3scFv/TF/IL15D8N:αCD28scFv/IL15RαSu fusion protein (referred to as 3t15*-28s), which can be purified by anti-TF antibody affinity chromatography and other chromatographic methods.

Purification of 3115*-28s by immunoaffinity chromatography

The anti-TF antibody affinity column was connected to a GE Healthcare AKTA Avant system. The flow rate was 4 mL/min for all steps except the elution step, which was 2 mL/min.

Cell culture harvest of 3t15*-28s was adjusted to pH 7.4 with 1M Tris base and loaded onto the anti-TF antibody affinity column (described above) which was equilibrated with 5 column volumes of PBS. After sample loading, the column was washed with washed with 5 column volumes PBS, followed by elution with 6 column volumes 0.1M Acetic Acid, pH 2.9. A280 elution peak was collected and then neutralized to pH 7.5-8.0 by adding 1M Tris base. The neutralized sample was then buffer exchanged into PBS using Amicon centrifugal filters with a 30 KDa molecular weight cutoff. As shown in FIG. **7**, it is clear that the anti-TF antibody affinity column can bind 3t15*-28s which contains TF as a fusion partner of 3t15*-28s. The buffer-exchanged protein sample is stored at 2-8° C. for further biochemical analysis and biological activity test.

After each elution, the anti-TF antibody affinity column was then stripped using 6 column volumes 0.1M Glycine, pH 2.5. The column was then neutralized using 10 column volumes PBS, 0.05% NaN_3 and stored at 2-8° C.

Analytical Size Exclusion Chromatography (SEC) Analysis of 3t15*-28s

A Superdex 200 Increase 10/300 GL gel filtration column (from GE Healthcare) was connected to an AKTA Avant system (from GE Healthcare). The column was equilibrated with 2 column volumes of PBS. The flow rate was 0.8 mL/min. A 200 μL sample of 3t15*-28s (1 mg/mL) was injected onto the column using a capillary loop. The injection was then chased with 1.25 column volumes of PBS. The SEC chromatograph was shown in FIG. **8**. The result indicated that there are 4 protein peaks, likely representing a monomer and dimer or other different forms of 3t15*-28s.

SDS PAGE Analysis of 3t15*-28s

To determine the purity and protein molecular weight, the purified 3t15*-28s protein sample from anti-TF antibody affinity chromatography was analyzed by standard sodium dodecyl sulfate polyacrylamide gel (4-12% NuPage Bis-Tris gel) electrophoresis (SDS-PAGE) method under reduced conditions. The gel was stained with InstantBlue for about 30 min, followed by destaining overnight with purified water. FIG. **9** shows the SDS gel of 3t15*-28s purified from anti-TF antibody affinity chromatography. The results indicated that purified 3t15*-28s contains protein bands with expected molecular weight (72 kDa) in reduced SDS PAGE.

In Vitro Characterization of the Activities of 3115*-28s Fusion Protein Complex

ELISA-based methods confirmed the formation of the 3t15*-28s fusion protein complex. To assess whether 3t15*-28s interacts with CD28 and CD3 (FIGS. **10A-10**B), 10 μg/mL of human CD28-Fc (Cat #CD8-H525a, Acro Biosystems) or 10 μg/mL of human CD3E/CD3D heterodimer-Llama Fc/Llama Fc (Cat #CDD-H5258, Acro Biosystems) in 50 mM carbonate buffer pH 9.4 (100 μl/well) was applied to an ELISA plate (Cat #80040LE 0910, ThermoFisher) and incubated overnight at 4° C. Next day, the plate was washed 3 times with ELISA washing buffer (phosphate-buffered saline with 0.05% Tween 20) and blocked with the blocking buffer (1% BSA-PBS) for 1 hour. Descending concentrations (3 to 0.00243 μg/mL in blocking buffer) of 3t15*-28s or a negative control fusion protein (HCW9218 which contains the same TF domain) were added to the plate and the plate was incubated for 1 hour at 25° C. The plate was washed for 3 times with ELISA washing buffer. A detection antibody, biotinylated anti-TF antibody (Cat #BAF2339, R&D Systems) at 0.1 μg/mL was added to the plate and incubated at 25° C. for 1 hour. The plate was washed and Horseradish peroxidase-streptavidin (code #016-030-084, Jackson ImmunoResearch) at 0.25 μg/mL was added to the plate and incubated at 25° C. for 30 minutes. The plate was washed and a substrate of HRP, ABTS (Cat #ABTS-1000-01, Surmodics) was added to the plate and incubated for 20 minutes at 25° C. The plate was read with a microplate reader (Multiscan Sky, Thermo Scientific) at OD405 nm. As shown in FIGS. **10A-10**B, 3t15*-28s interacted with CD28 and CD3 while the control fusion protein HCW9218 did not. The αCD3scFv and αCD28scFv domains of 3t15*-28s fusion protein complex were capable of binding CD3 and CD28. Binding of the αCD28scFv and TF domains to CD28 and anti-TF antibody, respectively (FIG. **10**A), confirmed the purified protein was comprised of the 3t15*-28s heterodimeric complex.

Example 9. Expansion of $Foxp3^+$ Regulatory T Cells Using 2t2 and 3t15*-28s

Fresh human leukocytes (3 donors) were obtained from the blood bank and $CD4^+CD25^+CD127^{low}$ T regulatory cells were isolated with the EasySep Human $CD4^+CD25^+CD127^{low}$ regulatory T cells isolation kit (StemCell Technology). The purity of Foxp3+ regulatory T cells was >35% and confirmed by staining with CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7 and Foxp3-PacBlue antibodies (Abs) (BioLegend). Cells were counted and resuspended in 1×10^6/mL in 24 well flat bottom plates in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone). Cells were stimulated with 2t2 (100 nM) and 3t15*-28s (200 nM) and incubated for 11 days at 37° C., 5% CO_2. During the expansion period, cells were maintained for 11 days at 1×10^6/mL concentration with fresh complete media containing 2t2 (100 nM) and 3t28 (200 nM) every alternate day. Cells were then stained with CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7 and Foxp3-PacBlue Abs (BioLegend) to assess the expansion of $Foxp3^+$ regulatory T cells by flow cytometry. Fold expansion of $Foxp3^+$ regulatory T cells was measured by counting using trypan blue (0.4%, Invitrogen) exclusion. FIG. **11** shows 2.5-fold increase in Foxp3$^+$ regulatory T cells at 11 days.

Example 10. Expansion of Human Immune Cells Using 2t2 and 3t15*-28s Plus Anti-Tissue Factor Antibody

Fresh human leukocytes (2 donors) were obtained from the blood bank and PBMC (peripheral blood mononuclear cells) were isolated with Ficoll separation. Cells were counted and resuspended at a density of 1×10^7 cells/mL in PBS. 2 μM of final concentration of Cell-Trace Violet Dye (Life Technologies) was added in the cells and incubated for 20 minutes in 37° C. under an atmosphere of 5% CO_2. Cells were resuspended at 10 times volume in fresh pre-warmed RPMI 1640 medium (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), antibiotics (penicillin, 10,000 units/mL; streptomycin, 10,000 μg/mL; Thermo Life Technologies), and 10% FBS (Hyclone) for 5 minutes at room temperature. Cells were washed 2 times at 1200 rpm for 10 minutes at counted and resuspended at a density of 2×10^6 cells/mL in RPMI 1640 medium (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), antibiotics (penicillin, 10,000 units/mL; streptomycin, 10,000 μg/mL; Thermo Life Technologies), and 10% FBS (Hyclone). Cells (0.1 mL) were transferred in a 96 well flat bottom plate and incubated with 2t2 (100 nM) and 3t15*-28s (0.01-1000 nM) with or without the anti-tissue factor antibody (αTF Ab, 0.01-100 nM). Complex of 3t15*-28s and anti-tissue factor antibody was made in 2:1 ratio, 15 minutes prior to stimulation. Half of the media was removed and replaced with fresh 2t2 (100 nM) and 3t15*-28s with or without anti-tissue factor antibody every 48 to 72 hrs. Cells were stained with CD4-Alexa Fluro 488, CD3-PE, CD8-PerCP Cy5.5 Abs (BioLegend) to assess dilution of Cell Trace Violet by flow cytometry.

FIG. **12** shows the percent of CD4$^+$ T cells undergoing six or more cell division under different culture conditions. Incubation with 3t15*-28s:αTF Ab complex plus 2t2 was capable of supporting cell proliferation at a lower 3t15*-28s concentration than was observed with 3t15*-28s plus 2t2 (without αTF Ab). This result suggests that immune cells derived from PBMC could be optimally stimulated to proliferate using a complex of anti-tissue factor antibody and 3t28.

Example 11. Purification of CD39$^+$Foxp3$^+$ Regulatory T Cells Using an Anti-CD39 IgG1$_k$ Reagent

Fresh human leukocytes were obtained from the blood bank. Half of the leukocytes were used to purify CD4$^+$CD25$^+$CD127^{low} T regulatory cells with the EasySep Human CD4$^+$CD25$^+$CD127^{low} regulatory T cells isolation kit (StemCell Technology). The purity of Foxp3$^+$ regulatory T cells was >60% and confirmed by staining with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue (BioLegend). The other half of the leukocytes were used to purify CD39$^+$ Foxp3$^+$ regulatory T cells by two step process. First CD4$^+$ T cells were isolated with the RosetteSep Human CD4$^+$ cells isolation kit (StemCell Technology). After enriching CD4$^+$ T cells, CD39$^+$CD4$^+$ cells were isolated using a novel biotinylated anti CD39 antibody reagent (HCW Biologic) and magnetic particles per manufacturer's instructions (EasySep Human Biotin Positive Selection Kit, StemCell Technology). Cells were stained with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue Abs (BioLegend) to assess the purity of CD39$^+$ Foxp3$^+$ regulatory T cells by flow cytometry (Cells were gated on CD3$^+$CD4$^+$Foxp3$^+$). Post-purification purity analysis by flow cytometer showed that cell purified using biotinylated anti CD39 Ab were 84.8% CD39$^+$ compared to 62% for cells isolated with the EasySep Human CD4$^+$CD25$^+$CD127^{low} regulatory T cells isolation kit (StemCell Technology). This data demonstrates a novel strategy to isolated highly pure and phenotypical active regulatory T cells (FIG. **13**A). These cells can be expanded in media containing hIL-2, anti-CD3/anti-CD28 beads, rapamycin, 2t2, 3t15*, and or 3t15*-28s: αTF Ab complex by the methods described above.

In a second study, Foxp3$^+$ regulatory T cells were first expanded with 2t2 from purified CD4$^+$CD25$^+$CD127^{low} T regulatory cells, and then frozen. The cells were thawed and maintained at 2×10^6/mL in T25 flask with 2t2 (100 nM) in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone). CD39$^+$CD4$^+$ cells were isolated using a novel biotinylated anti CD39 Ab reagent (HCW Biologic) and magnetic particles per manufacturer's instructions (EasySep Human Biotin Positive Selection Kit, StemCell Technology). Cells were stained with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue Abs (BioLegend) to assess the purity of CD39$^+$ Foxp3$^+$ regulatory T cells by flow cytometry (Cells were gated on CD3$^+$CD4$^+$Foxp3$^+$). Post-purification purity analysis by flow cytometer shows that cell purified using biotinylated anti CD39 antibody yielded a Foxp3$^+$ regulatory T cells that are 99.9% enriched for the highly suppressive CD39$^+$ subset (FIG. **13**B). These cells can be expanded in media containing hIL-2, anti-CD3/anti-CD28 beads, rapamycin, 2t2, 3t15*, and or 3t15*-28s:αTF Ab complex by the methods described above.

Example 12. Suppression of Autologous T Responder Cells by Purified CD39$^+$ Regulatory T Cells

Fresh human leukocytes were obtained from the blood bank. Half of the leukocytes were used to purify CD4$^+$CD25$^+$CD127^{low} T regulatory cells with the Easy Sep Human CD4$^+$CD25$^+$CD127^{low} regulatory T cells isolation kit (StemCell Technology). The purity of Foxp3$^+$ regulatory T cells was >60% and confirmed by staining with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue (BioLegend). The other half of the leukocytes were used to purify CD39$^+$ Foxp3$^+$ regulatory T cells by two step process. First CD4$^+$ T cells were isolated with the RosetteSep Human CD4$^+$ cells isolation kit (StemCell Technology). After enriching CD4$^+$ T cells, CD39$^+$ CD4$^+$ cells were isolated using a novel biotinylated anti CD39 antibody reagent (HCW Biologics) and magnetic particles per manufacturer's instructions (EasySep Human Biotin Positive Selection Kit, StemCell Technology). Cells were stained with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue Abs (BioLegend) to assess the purity of CD39$^+$ Foxp3$^+$ regulatory T cells by flow cytometry (Cells gated on CD3$^+$CD4$^+$Foxp3$^+$). Autologous T responder (Tresp) cells were labelled with Cell Trace Violet (Invitrogen). The purified Foxp3$^+$ T cells were cultured with T responder cells (1:1 regulatory T cells: T responder cells). Cells were re-stimulated with 2t2 (10 nM) and cultured for 5 days. Cells were harvested, washed (1500 RPM for 5 minutes at room temperature) in FACS buffer (1×PBS (Hyclone) with 0.5% BSA (EMD

Millipore) and 0.001% Sodium Azide (Sigma)). After two washes, cells were analyzed by flow cytometry (Celesta-BD Bioscience) and % of dividing Tresp cells were determined based on Cell Trace Violet dilution. FIG. **14** shows that purified $CD39^+Foxp3^+$ regulatory T cells were more effective at suppressing T responder cell proliferation when compared to cells isolated based on $CD4^+CD25^+CD127^{low}$ T regulatory cells.

Example 13. Suppression of THP-1 Monocyte Cells by Expanded and $CD39^+$-Purified T Regulatory Cells

Fresh human leukocytes were obtained from the blood bank. $CD4^+CD25^+CD127^{low}$ T regulatory cells with the EasySep Human CD4+CD25+CD127low regulatory T cells isolation kit (StemCell Technology). The purity of $Foxp3^+$ regulatory T cells was >60% and confirmed by staining with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue (BioLegend). Cells were counted and resuspended in 1×10^6/mL in 24 wells flat bottom plate in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone). Cells were stimulated with 100 nM of 2t2 with CD3/CD28 beads (4:1,Beads: Cells) (Dynabeads, Life Technologies) and incubated for 21 days at 37°, 5% CO_2. During the expansion period, cells were re-stimulated with 100 nM of 2t2 with every alternate day and every 7 days cells were re-stimulated with CD3/CD28 beads (1:1, Beads: Cells) (Dynabeads, Life Technologies). Cells were maintained up to 21 days at 1×10^6/mL concentration with fresh complete media containing 100 nM of 2t2 every alternate day.

On day 21 cells were stained with CD4-Alexa Fluro 488, CD25-PE, CD3 APC-Cy7, Foxp3-PE (Intracellular staining using Foxp3 staining kit, Invitrogen), CD127-Alex Fluro 700 (BioLegend) to assess the expansion of $Foxp3^+$ regulatory T cells by flow cytometry and frozen at 20×10^6/mL. Previously expanded cells were thawed and maintained at 2×10^6/mL in T25 flask in complete media (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone) for an hour and washed before isolating $CD39^+$ $CD4^+$T reg cells. CD39 CD4+ cells were isolated using a novel biotinylated anti-CD39 Ab reagent (HCW Biologics; EasySep Human Biotin Positive Selection Kit, StemCell Technology). Cells were stained with CD4-Alexa Fluro 488, CD39-APC, CD3 PE, Foxp3-PacBlue Abs (BioLegend) to assess the purity of $CD39^+$ $Foxp3^+$ regulatory T cells by flow cytometry (cells were gated on $CD3^+CD4^+Foxp3^+$). Post-purification purity analysis by flow cytometer shows that cell purified using biotinylated anti CD39 antibody yielded a $Foxp3^+$ regulatory T cells that are 99.9% enriched for the highly suppressive $CD39^+$ subset. Suppression assay was setup with THP-1 (Human Monocyte Cells) and purified $CD39^+$ $Foxp3^+$ T Reg cells or CD39− $Foxp3^+$ T Reg cells or Total $CD4^+$ T Reg cells (1:1 ratio). Cells were cultured for 3 days in (RPMI 1640 (Gibco) supplemented with 2 mM L-glutamine (Thermo Life Technologies), penicillin (Thermo Life Technologies), streptomycin (Thermo Life Technologies), and 10% FBS (Hyclone) supplemented with 50 ng/mL LPS. Cells supernatant was collected on day 3 to analyze IL-10 by ELISA (R&D System).

FIG. **15** shows that purified $CD39^+Foxp3^+$ regulatory T cells secret significantly more IL-10 compare to CD39− $Foxp3^+$ T Reg cells or Total CD4+ T Reg cells. This data shows that $CD39^+$ $Foxp3^+$ T Reg has more suppressive capacity compare to CD39− $Foxp3^+$ T Reg cells or Total $CD4^+$ T Reg cells.

OTHER EMBODIMENTS

It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

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Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys
        35                  40                  45

Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
    50                  55                  60

Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
65                  70                  75                  80

Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
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Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
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            20                  25                  30

Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp Lys Ser Lys
        35                  40                  45

Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp Glu Ile Val
    50                  55                  60

Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser Tyr Pro Ala
65                  70                  75                  80

Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu Tyr Glu Asn
                85                  90                  95

Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly Gln Pro Thr
            100                 105                 110

Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val Thr Val Glu
        115                 120                 125

Asp Glu Arg Thr Leu Val Arg Arg Asn Asn Thr Phe Leu Ser Leu Arg
    130                 135                 140

Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr Trp Lys Ser
145                 150                 155                 160

Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn Glu Phe Leu
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Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val Gln Ala Val
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            20                  25                  30

Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys
        35                  40                  45
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Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
    50                  55                  60

Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
65                  70                  75                  80

Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
                85                  90                  95

Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
            100                 105                 110

Thr Ile Val Glu Phe Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile
        115                 120                 125

Ile Ser Thr Leu Thr Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn
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Leu Thr Trp Lys Ser Thr Asn Phe Lys Thr Ile Leu Glu Trp Glu Pro
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Asp Trp Lys Ser Lys Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu
            180                 185                 190

Thr Asp Glu Ile Val Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val
        195                 200                 205

Phe Ser Tyr Pro Ala Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu
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Pro Leu Tyr Glu Asn Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn
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                245                 250                 255

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            260                 265                 270

Phe Leu Ser Leu Arg Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu
        275                 280                 285

Tyr Tyr Trp Lys Ser Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn
    290                 295                 300

Thr Asn Glu Phe Leu Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe
305                 310                 315                 320

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            340                 345                 350

Ala Pro Thr Ser Ser Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His
        355                 360                 365

Leu Leu Leu Asp Leu Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys
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Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
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50 55 60

Pro Lys Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu
65 70 75 80

Leu Lys Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe
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His Leu Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu
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Tyr Lys Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met
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Pro Lys Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu
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Leu Lys Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe
        435                 440                 445

His Leu Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu
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        35                  40                  45

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Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu Ala Glu
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Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe Thr
                85                  90                  95

Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg Gly Gly Gly Gly Ser
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Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser
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Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe Lys Asp Lys Ala Thr Leu
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Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr Met Gln Leu Ser Ser Leu
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Ser

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Ile Gln Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile Gly
        35   40   45

Ser Ile Asn Pro Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys Phe Lys
    50   55   60

Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr Met
65   70   75   80

Glu Phe Ser Ser Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr Cys Ala
                85   90   95

Arg Trp Gly Asp Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu Thr Val
            100   105   110

Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
        115   120   125

Ser Asp Ile Glu Met Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu
    130   135   140

Gly Glu Arg Val Thr Met Thr Cys Thr Ala Ser Ser Ser Val Ser Ser
145   150   155   160

Ser Tyr Phe His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu
                165   170   175

Cys Ile Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Pro Arg Phe
            180   185   190

Ser Gly Ser Gly Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu
        195   200   205

Ala Glu Asp Ala Ala Thr Tyr Phe Cys His Gln Tyr His Arg Ser Pro
    210   215   220

Thr Phe Gly Gly Gly Thr Lys Leu Glu Thr Lys Arg
225   230   235

<210> SEQ ID NO 7
<211> LENGTH: 696
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 7

-continued

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Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1               5                   10                  15

Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
            20                  25                  30

Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Tyr
        35                  40                  45

Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala His Phe Arg Gly Ser
    50                  55                  60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu Ala Glu
65                  70                  75                  80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe Thr
                85                  90                  95

Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg Gly Gly Gly Gly Ser
            100                 105                 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Gln
        115                 120                 125

Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys
    130                 135                 140

Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr Thr Met His Trp Val Lys
145                 150                 155                 160

Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser
                165                 170                 175

Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe Lys Asp Lys Ala Thr Leu
            180                 185                 190

Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr Met Gln Leu Ser Ser Leu
        195                 200                 205

Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys Ala Arg Tyr Tyr Asp Asp
    210                 215                 220

His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr Val Ser
225                 230                 235                 240

Ser Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr Trp Lys
                245                 250                 255

Ser Thr Asn Phe Lys Thr Ile Leu Glu Trp Glu Pro Lys Pro Val Asn
            260                 265                 270

Gln Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp Lys Ser
        275                 280                 285

Lys Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp Glu Ile
    290                 295                 300

Val Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser Tyr Pro
305                 310                 315                 320

Ala Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu Tyr Glu
                325                 330                 335

Asn Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly Gln Pro
            340                 345                 350

Thr Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val Thr Val
        355                 360                 365

Glu Asp Glu Arg Thr Leu Val Arg Arg Asn Asn Thr Phe Leu Ser Leu
    370                 375                 380

Arg Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr Trp Lys
385                 390                 395                 400

Ser Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn Glu Phe
                405                 410                 415

Leu Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val Gln Ala
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            420                 425                 430

Val Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser Pro Val
        435                 440                 445

Glu Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu Val Gln Leu Gln
    450                 455                 460

Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala Ser Val Lys Met Ser
465                 470                 475                 480

Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr Val Ile Gln Trp Val
                485                 490                 495

Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile Gly Ser Ile Asn Pro
            500                 505                 510

Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys Phe Lys Gly Lys Ala Thr
        515                 520                 525

Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr Met Glu Phe Ser Ser
    530                 535                 540

Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr Cys Ala Arg Trp Gly Asp
545                 550                 555                 560

Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu Thr Val Ser Ser Gly Gly
                565                 570                 575

Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp Ile Glu
            580                 585                 590

Met Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu Gly Glu Arg Val
        595                 600                 605

Thr Met Thr Cys Thr Ala Ser Ser Ser Val Ser Ser Ser Tyr Phe His
    610                 615                 620

Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu Cys Ile Tyr Ser
625                 630                 635                 640

Thr Ser Asn Leu Ala Ser Gly Val Pro Pro Arg Phe Ser Gly Ser Gly
                645                 650                 655

Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu Asp Ala
            660                 665                 670

Ala Thr Tyr Phe Cys His Gln Tyr His Arg Ser Pro Thr Phe Gly Gly
        675                 680                 685

Gly Thr Lys Leu Glu Thr Lys Arg
    690                 695


<210> SEQ ID NO 8
<211> LENGTH: 714
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 8

Met Lys Trp Val Thr Phe Ile Ser Leu Leu Phe Leu Phe Ser Ser Ala
1               5                   10                  15

Tyr Ser Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser
            20                  25                  30

Pro Gly Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser
        35                  40                  45

Tyr Met Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp
    50                  55                  60

Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala His Phe Arg
65                  70                  75                  80
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Gly Ser Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu
                85                  90                  95

Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro
            100                 105                 110

Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg Gly Gly Gly
        115                 120                 125

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu
    130                 135                 140

Gln Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met
145                 150                 155                 160

Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr Thr Met His Trp
                165                 170                 175

Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Ile Asn
            180                 185                 190

Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe Lys Asp Lys Ala
        195                 200                 205

Thr Leu Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr Met Gln Leu Ser
    210                 215                 220

Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys Ala Arg Tyr Tyr
225                 230                 235                 240

Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr
                245                 250                 255

Val Ser Ser Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr
            260                 265                 270

Trp Lys Ser Thr Asn Phe Lys Thr Ile Leu Glu Trp Glu Pro Lys Pro
        275                 280                 285

Val Asn Gln Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp
    290                 295                 300

Lys Ser Lys Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp
305                 310                 315                 320

Glu Ile Val Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser
                325                 330                 335

Tyr Pro Ala Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu
            340                 345                 350

Tyr Glu Asn Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly
        355                 360                 365

Gln Pro Thr Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val
    370                 375                 380

Thr Val Glu Asp Glu Arg Thr Leu Val Arg Arg Asn Asn Thr Phe Leu
385                 390                 395                 400

Ser Leu Arg Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr
                405                 410                 415

Trp Lys Ser Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn
            420                 425                 430

Glu Phe Leu Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val
        435                 440                 445

Gln Ala Val Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser
    450                 455                 460

Pro Val Glu Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu Val Gln
465                 470                 475                 480

Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala Ser Val Lys
                485                 490                 495

Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr Val Ile Gln
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            500                 505                 510

Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile Gly Ser Ile
        515                 520                 525

Asn Pro Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys Phe Lys Gly Lys
    530                 535                 540

Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr Met Glu Phe
545                 550                 555                 560

Ser Ser Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr Cys Ala Arg Trp
                565                 570                 575

Gly Asp Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu Thr Val Ser Ser
            580                 585                 590

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Asp
        595                 600                 605

Ile Glu Met Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu Gly Glu
    610                 615                 620

Arg Val Thr Met Thr Cys Thr Ala Ser Ser Ser Val Ser Ser Ser Tyr
625                 630                 635                 640

Phe His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu Cys Ile
                645                 650                 655

Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Pro Arg Phe Ser Gly
            660                 665                 670

Ser Gly Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala Glu
        675                 680                 685

Asp Ala Ala Thr Tyr Phe Cys His Gln Tyr His Arg Ser Pro Thr Phe
    690                 695                 700

Gly Gly Gly Thr Lys Leu Glu Thr Lys Arg
705                 710


<210> SEQ ID NO 9
<211> LENGTH: 2088
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 9

cagatcgtgc tgacccaaag ccccgccatc atgagcgcta gccccggtga gaaggtgacc     60

atgacatgct ccgcttccag ctccgtgtcc tacatgaact ggtatcagca gaaaagcgga    120

accagcccca aaaggtggat ctacgacacc agcaagctgg cctccggagt gcccgctcat    180

ttccggggct ctggatccgg caccagctac tctttaacca tttccggcat ggaagctgaa    240

gacgctgcca cctactattg ccagcaatgg agcagcaacc ccttcacatt cggatctggc    300

accaagctcg aaatcaatcg tggaggaggt ggcagcggcg gcggtggatc cggcggagga    360

ggaagccaag ttcaactcca gcagagcggc gctgaactgg cccggcccgg cgcctccgtc    420

aagatgagct gcaaggcttc cggctataca tttactcgtt acacaatgca ttgggtcaag    480

cagaggcccg gtcaaggttt agagtggatc ggatatatca acccttcccg gggctacacc    540

aactataacc aaaagttcaa ggataaagcc actttaacca ctgacaagag ctcctccacc    600

gcctacatgc agctgtcctc tttaaccagc gaggactccg ctgtttacta ctgcgctagg    660

tattacgacg accactactg tttagactat tggggacaag gtaccacttt aaccgtcagc    720

agctccggca ccaccaatac cgtggccgct tataacctca catggaagag caccaacttc    780

aagacaattc tggaatggga acccaagccc gtcaatcaag tttacaccgt gcagatctcc    840
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accaaatccg gagactggaa gagcaagtgc ttctacacaa cagacaccga gtgtgattta 900

accgacgaaa tcgtcaagga cgtcaagcaa acctatctgg ctcgggtctt ttcctacccc 960

gctggcaatg tcgagtccac cggctccgct ggcgagcctc tctacgagaa ttcccccgaa 1020

ttcacccctt atttagagac caatttaggc cagcctacca tccagagctt cgagcaagtt 1080

ggcaccaagg tgaacgtcac cgtcgaggat gaaaggactt tagtgcggcg gaataacaca 1140

tttttatccc tccgggatgt gttcggcaaa gacctcatct acacactgta ctattggaag 1200

tccagctcct ccggcaaaaa gaccgctaag accaacacca acgagttttt aattgacgtg 1260

gacaaaggcg agaactactg cttcagcgtg caagccgtga tcccttctcg taccgtcaac 1320

cggaagagca cagattcccc cgttgagtgc atgggccaag aaaagggcga gttccgggag 1380

gtccagctgc agcagagcgg acccgaactc gtgaaacccg gtgcttccgt gaaaatgtct 1440

tgtaaggcca gcggatacac cttcacctcc tatgtgatcc agtgggtcaa acagaagccc 1500

ggacaaggtc tcgagtggat cggcagcatc aacccttaca acgactatac caaatacaac 1560

gagaagttta agggaaaggc tactttaacc tccgacaaaa gctccatcac agcctacatg 1620

gagttcagct ctttaacatc cgaggacagc gctctgtact attgcgcccg gtggggcgac 1680

ggcaattact ggggacgggg cacaacactg accgtgagca gcggaggcgg aggctccggc 1740

ggaggcggat ctggcggtgg cggctccgac atcgagatga cccagtcccc cgctatcatg 1800

tccgcctctt taggcgagcg ggtcacaatg acttgtacag cctcctccag cgtctcctcc 1860

tcctacttcc attggtacca acagaaaccc ggaagctccc ctaaactgtg catctacagc 1920

accagcaatc tcgccagcgg cgtgccccct aggttttccg gaagcggaag caccagctac 1980

tctttaacca tctcctccat ggaggctgag gatgccgcca cctacttttg tcaccagtac 2040

caccggtccc ccaccttcgg aggcggcacc aaactggaga caaagagg 2088

<210> SEQ ID NO 10
<211> LENGTH: 2142
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 10

atgaagtggg tgaccttcat cagcttatta tttttattca gctccgccta ttcccagatc 60

gtgctgaccc aaagccccgc catcatgagc gctagccccg gtgagaaggt gaccatgaca 120

tgctccgctt ccagctccgt gtcctacatg aactggtatc agcagaaaag cggaaccagc 180

cccaaaaggt ggatctacga caccagcaag ctggcctccg gagtgcccgc tcatttccgg 240

ggctctggat ccggcaccag ctactcttta accatttccg gcatggaagc tgaagacgct 300

gccacctact attgccagca atggagcagc aacccccttca cattcggatc tggcaccaag 360

ctcgaaatca atcgtggagg aggtggcagc ggcggcggtg gatccggcgg aggaggaagc 420

caagttcaac tccagcagag cggcgctgaa ctggcccggc ccggcgcctc cgtcaagatg 480

agctgcaagg cttccggcta tacatttact cgttacacaa tgcattgggt caagcagagg 540

cccggtcaag gtttagagtg gatcggatat atcaaccctt cccggggcta caccaactat 600

aaccaaaagt tcaaggataa agccacttta accactgaca agagctcctc caccgcctac 660

atgcagctgt cctctttaac cagcgaggac tccgctgttt actactgcgc taggtattac 720

gacgaccact actgtttaga ctattgggga caaggtacca ctttaaccgt cagcagctcc 780

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ggcaccacca ataccgtggc cgcttataac ctcacatgga agagcaccaa cttcaagaca    840
attctggaat gggaacccaa gcccgtcaat caagtttaca ccgtgcagat ctccaccaaa    900
tccggagact ggaagagcaa gtgcttctac acaacagaca ccgagtgtga tttaaccgac    960
gaaatcgtca aggacgtcaa gcaaacctat ctggctcggg tcttttccta ccccgctggc   1020
aatgtcgagt ccaccggctc cgctggcgag cctctctacg agaattcccc cgaattcacc   1080
ccttatttag agaccaattt aggccagcct accatccaga gcttcgagca agttggcacc   1140
aaggtgaacg tcaccgtcga ggatgaaagg actttagtgc ggcggaataa cacattttta   1200
tccctccggg atgtgttcgg caaagacctc atctacacac tgtactattg gaagtccagc   1260
tcctccggca aaaagaccgc taagaccaac accaacgagt ttttaattga cgtggacaaa   1320
ggcgagaact actgcttcag cgtgcaagcc gtgatccctt ctcgtaccgt caaccggaag   1380
agcacagatt cccccgttga gtgcatgggc caagaaaagg gcgagttccg ggaggtccag   1440
ctgcagcaga gcggacccga actcgtgaaa cccggtgctt ccgtgaaaat gtcttgtaag   1500
gccagcggat acaccttcac ctcctatgtg atccagtggg tcaaacagaa gcccggacaa   1560
ggtctcgagt ggatcggcag catcaaccct tacaacgact ataccaaata caacgagaag   1620
tttaagggaa aggctacttt aacctccgac aaaagctcca tcacagccta catggagttc   1680
agctctttaa catccgagga cagcgctctg tactattgcg cccggtgggg cgacggcaat   1740
tactggggac ggggcacaac actgaccgtg agcagcggag gcggaggctc cggcggaggc   1800
ggatctggcg gtggcggctc cgacatcgag atgacccagt cccccgctat catgtccgcc   1860
tctttaggcg agcgggtcac aatgacttgt acagcctcct ccagcgtctc ctcctcctac   1920
ttccattggt accaacagaa acccggaagc tcccctaaac tgtgcatcta cagcaccagc   1980
aatctcgcca gcggcgtgcc ccctaggttt tccggaagcg gaagcaccag ctactcttta   2040
accatctcct ccatggaggc tgaggatgcc gccacctact tttgtcacca gtaccaccgg   2100
tcccccacct tcggaggcgg caccaaactg gagacaaaga gg                      2142
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<210> SEQ ID NO 11
<211> LENGTH: 1455
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 11

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gcccccacct cctcctccac caagaagacc cagctgcagc tggagcattt actgctggat     60
ttacagatga ttttaaacgg catcaacaac tacaagaacc ccaagctgac tcgtatgctg    120
accttcaagt tctacatgcc caagaaggcc accgagctga agcatttaca gtgtttagag    180
gaggagctga agcccctcga ggaggtgctg aatttagccc agtccaagaa tttccattta    240
aggccccggg atttaatcag caacatcaac gtgatcgttt tagagctgaa gggctccgag    300
accaccttca tgtgcgagta cgccgacgag accgccacca tcgtggagtt tttaaatcgt    360
tggatcacct tctgccagtc catcatctcc actttaacca gcggcacaac caacacagtc    420
gctgcctata acctcacttg gaagagcacc aacttcaaaa ccatcctcga atgggaaccc    480
aaacccgtta accaagttta caccgtgcag atcagcacca agtccggcga ctggaagtcc    540
aaatgtttct ataccaccga caccgagtgc gatctcaccg atgagatcgt gaaagatgtg    600
aaacagacct acctcgcccg ggtgtttagc taccccgccg gcaatgtgga gagcactggt    660
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tccgctggcg agcctttata cgagaacagc cccgaattta ccccttacct cgagaccaat    720

ttaggacagc ccaccatcca aagctttgag caagttggca caaaggtgaa tgtgacagtg    780

gaggacgagc ggactttagt gcggcggaac aacacctttc tcagcctccg ggatgtgttc    840

ggcaaagatt taatctacac actgtattac tggaagtcct cttcctccgg caagaagaca    900

gctaaaacca acacaaacga gtttttaatc gacgtggata aaggcgaaaa ctactgtttc    960

agcgtgcaag ctgtgatccc ctcccggacc gtgaatagga aaagcaccga tagccccgtt   1020

gagtgcatgg gccaagaaaa gggcgagttc cgggaggcac ctacttcaag ttctacaaag   1080

aaaacacagc tacaactgga gcatttactg ctggatttac agatgatttt gaatggaatt   1140

aataattaca agaatcccaa actcaccagg atgctcacat ttaagtttta catgcccaag   1200

aaggccacag aactgaaaca tcttcagtgt ctagaagaag aactcaaacc tctggaggaa   1260

gtgctaaatt tagctcaaag caaaaacttt cacttaagac ccagggactt aatcagcaat   1320

atcaacgtaa tagttctgga actaaaggga tctgaaacaa cattcatgtg tgaatatgct   1380

gatgagacag caaccattgt agaatttctg aacagatgga ttaccttttg tcaaagcatc   1440

atctcaacac taact                                                    1455
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<210> SEQ ID NO 12
<211> LENGTH: 1509
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 12

atgaagtggg tgaccttcat cagcctgctg ttcctgttct ccagcgccta ctccgccccc     60

acctcctcct ccaccaagaa gacccagctg cagctggagc atttactgct ggatttacag    120

atgattttaa acggcatcaa caactacaag aaccccaagc tgactcgtat gctgaccttc    180

aagttctaca tgcccaagaa ggccaccgag ctgaagcatt tacagtgttt agaggaggag    240

ctgaagcccc tcgaggaggt gctgaattta gcccagtcca agaatttcca tttaaggccc    300

cgggatttaa tcagcaacat caacgtgatc gttttagagc tgaagggctc cgagaccacc    360

ttcatgtgcg agtacgccga cgagaccgcc accatcgtgg agtttttaaa tcgttggatc    420

accttctgcc agtccatcat ctccacttta accagcggca caaccaacac agtcgctgcc    480

tataacctca cttggaagag caccaacttc aaaaccatcc tcgaatggga acccaaaccc    540

gttaaccaag tttacaccgt gcagatcagc accaagtccg gcgactggaa gtccaaatgt    600

ttctatacca ccgacaccga gtgcgatctc accgatgaga tcgtgaaaga tgtgaaacag    660

acctacctcg cccgggtgtt tagctacccc gccggcaatg tggagagcac tggttccgct    720

ggcgagcctt tatacgagaa cagccccgaa tttacccctt acctcgagac caatttagga    780

cagcccacca tccaaagctt tgagcaagtt ggcacaaagg tgaatgtgac agtggaggac    840

gagcggactt tagtgcggcg gaacaacacc tttctcagcc tccgggatgt gttcggcaaa    900

gatttaatct acacactgta ttactggaag tcctcttcct ccggcaagaa gacagctaaa    960

accaacacaa acgagttttt aatcgacgtg gataaaggcg aaaactactg tttcagcgtg   1020

caagctgtga tcccctcccg gaccgtgaat aggaaaagca ccgatagccc cgttgagtgc   1080

atgggccaag aaaagggcga gttccgggag gcacctactt caagttctac aaagaaaaca   1140

cagctacaac tggagcattt actgctggat ttacagatga ttttgaatgg aattaataat   1200
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tacaagaatc ccaaactcac caggatgctc acatttaagt tttacatgcc caagaaggcc   1260

acagaactga aacatcttca gtgtctagaa gaagaactca aacctctgga ggaagtgcta   1320

aatttagctc aaagcaaaaa ctttcactta agacccaggg acttaatcag caatatcaac   1380

gtaatagttc tggaactaaa gggatctgaa acaacattca tgtgtgaata tgctgatgag   1440

acagcaacca ttgtagaatt tctgaacaga tggattacct tttgtcaaag catcatctca   1500

acactaact                                                           1509
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<210> SEQ ID NO 13
<211> LENGTH: 657
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 13

agcggcacaa ccaacacagt cgctgcctat aacctcactt ggaagagcac caacttcaaa     60

accatcctcg aatgggaacc caaacccgtt aaccaagttt acaccgtgca gatcagcacc    120

aagtccggcg actggaagtc caaatgtttc tataccaccg acaccgagtg cgatctcacc    180

gatgagatcg tgaaagatgt gaaacagacc tacctcgccc gggtgtttag ctaccccgcc    240

ggcaatgtgg agagcactgg ttccgctggc gagcctttat acgagaacag ccccgaattt    300

acccccttacc tcgagaccaa tttaggacag cccaccatcc aaagctttga gcaagttggc    360

acaaaggtga atgtgacagt ggaggacgag cggactttag tgcggcggaa caacaccttt    420

ctcagcctcc gggatgtgtt cggcaaagat ttaatctaca cactgtatta ctggaagtcc    480

tcttcctccg gcaagaagac agctaaaacc aacacaaacg agtttttaat cgacgtggat    540

aaaggcgaaa actactgttt cagcgtgcaa gctgtgatcc cctcccggac cgtgaatagg    600

aaaagcaccg atagccccgt tgagtgcatg ggccaagaaa agggcgagtt ccgggag       657
```

```
<210> SEQ ID NO 14
<211> LENGTH: 223
<212> TYPE: PRT
<213> ORGANISM: Mus musculus

<400> SEQUENCE: 14

Ala Gly Ile Pro Glu Lys Ala Phe Asn Leu Thr Trp Ile Ser Thr Asp
1               5                   10                  15

Phe Lys Thr Ile Leu Glu Trp Gln Pro Lys Pro Thr Asn Tyr Thr Tyr
            20                  25                  30

Thr Val Gln Ile Ser Asp Arg Ser Arg Asn Trp Lys Asn Lys Cys Phe
        35                  40                  45

Ser Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp Glu Ile Val Lys Asp
    50                  55                  60

Val Thr Trp Ala Tyr Glu Ala Lys Val Leu Ser Val Pro Arg Arg Asn
65                  70                  75                  80

Ser Val His Gly Asp Gly Asp Gln Leu Val Ile His Gly Glu Glu Pro
                85                  90                  95

Pro Phe Thr Asn Ala Pro Lys Phe Leu Pro Tyr Arg Asp Thr Asn Leu
            100                 105                 110

Gly Gln Pro Val Ile Gln Gln Phe Glu Gln Asp Gly Arg Lys Leu Asn
        115                 120                 125

Val Val Val Lys Asp Ser Leu Thr Leu Val Arg Lys Asn Gly Thr Phe
    130                 135                 140

Leu Thr Leu Arg Gln Val Phe Gly Lys Asp Leu Gly Tyr Ile Ile Thr
```

145 150 155 160

Tyr Arg Lys Gly Ser Ser Thr Gly Lys Lys Thr Asn Ile Thr Asn Thr
165 170 175

Asn Glu Phe Ser Ile Asp Val Glu Glu Gly Val Ser Tyr Cys Phe Phe
180 185 190

Val Gln Ala Met Ile Phe Ser Arg Lys Thr Asn Gln Asn Ser Pro Gly
195 200 205

Ser Ser Thr Val Cys Thr Glu Gln Trp Lys Ser Phe Leu Gly Glu
210 215 220

<210> SEQ ID NO 15
<211> LENGTH: 224
<212> TYPE: PRT
<213> ORGANISM: Rattus rattus

<400> SEQUENCE: 15

Ala Gly Thr Pro Pro Gly Lys Ala Phe Asn Leu Thr Trp Ile Ser Thr
1 5 10 15

Asp Phe Lys Thr Ile Leu Glu Trp Gln Pro Lys Pro Thr Asn Tyr Thr
20 25 30

Tyr Thr Val Gln Ile Ser Asp Arg Ser Arg Asn Trp Lys Tyr Lys Cys
35 40 45

Thr Gly Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp Glu Ile Val Lys
50 55 60

Asp Val Asn Trp Thr Tyr Glu Ala Arg Val Leu Ser Val Pro Trp Arg
65 70 75 80

Asn Ser Thr His Gly Lys Glu Thr Leu Phe Gly Thr His Gly Glu Glu
85 90 95

Pro Pro Phe Thr Asn Ala Arg Lys Phe Leu Pro Tyr Arg Asp Thr Lys
100 105 110

Ile Gly Gln Pro Val Ile Gln Lys Tyr Glu Gln Gly Gly Thr Lys Leu
115 120 125

Lys Val Thr Val Lys Asp Ser Phe Thr Leu Val Arg Lys Asn Gly Thr
130 135 140

Phe Leu Thr Leu Arg Gln Val Phe Gly Asn Asp Leu Gly Tyr Ile Leu
145 150 155 160

Thr Tyr Arg Lys Asp Ser Ser Thr Gly Arg Lys Thr Asn Thr Thr His
165 170 175

Thr Asn Glu Phe Leu Ile Asp Val Glu Lys Gly Val Ser Tyr Cys Phe
180 185 190

Phe Ala Gln Ala Val Ile Phe Ser Arg Lys Thr Asn His Lys Ser Pro
195 200 205

Glu Ser Ile Thr Lys Cys Thr Glu Gln Trp Lys Ser Val Leu Gly Glu
210 215 220

<210> SEQ ID NO 16
<211> LENGTH: 219
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 16

Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr Trp Lys Ser
1 5 10 15

Thr Asn Phe Ala Thr Ala Leu Glu Trp Glu Pro Lys Pro Val Asn Gln

```
            20                  25                  30

Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp Lys Ser Lys
        35                  40                  45

Cys Phe Tyr Thr Thr Asp Thr Glu Cys Ala Leu Thr Asp Glu Ile Val
    50                  55                  60

Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser Tyr Pro Ala
65                  70                  75                  80

Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu Tyr Glu Asn
                85                  90                  95

Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly Gln Pro Thr
            100                 105                 110

Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val Thr Val Glu
        115                 120                 125

Asp Glu Arg Thr Leu Val Ala Arg Asn Asn Thr Ala Leu Ser Leu Arg
    130                 135                 140

Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr Trp Lys Ser
145                 150                 155                 160

Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn Glu Phe Leu
                165                 170                 175

Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val Gln Ala Val
            180                 185                 190

Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser Pro Val Glu
        195                 200                 205

Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu
    210                 215
```

<210> SEQ ID NO 17
<211> LENGTH: 219
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 17

```
Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr Trp Lys Ser
1               5                   10                  15

Thr Asn Phe Ala Thr Ala Leu Glu Trp Glu Pro Lys Pro Val Asn Gln
            20                  25                  30

Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Ala Lys Ser Lys
        35                  40                  45

Cys Phe Tyr Thr Thr Asp Thr Glu Cys Ala Leu Thr Asp Glu Ile Val
    50                  55                  60

Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser Tyr Pro Ala
65                  70                  75                  80

Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu Ala Glu Asn
                85                  90                  95

Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly Gln Pro Thr
            100                 105                 110

Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val Thr Val Glu
        115                 120                 125

Asp Glu Arg Thr Leu Val Ala Arg Asn Asn Thr Ala Leu Ser Leu Arg
    130                 135                 140

Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr Trp Lys Ser
145                 150                 155                 160
```

```
Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn Glu Phe Leu
                165                 170                 175

Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val Gln Ala Val
            180                 185                 190

Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser Pro Val Glu
        195                 200                 205

Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu
    210                 215


<210> SEQ ID NO 18
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 18

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1               5                   10                  15


<210> SEQ ID NO 19
<211> LENGTH: 45
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      oligonucleotide

<400> SEQUENCE: 19

ggcggtggag gatccggagg aggtggctcc ggcggcggag gatct                      45


<210> SEQ ID NO 20
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 20

Gly Gly Gly Ser Gly Gly Gly Ser
1               5


<210> SEQ ID NO 21
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 21

Gln Val Gln Leu Gln Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala
1               5                   10                  15

Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr
            20                  25                  30

Thr Met His Trp Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile
        35                  40                  45

Gly Tyr Ile Asn Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe
    50                  55                  60

Lys Asp Lys Ala Thr Leu Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr
65                  70                  75                  80
```

```
Met Gln Leu Ser Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys
                85                  90                  95

Ala Arg Tyr Tyr Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly
            100                 105                 110

Thr Thr Leu Thr Val Ser Ser
        115


<210> SEQ ID NO 22
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 22

Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1               5                   10                  15

Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
            20                  25                  30

Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Tyr
        35                  40                  45

Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala His Phe Arg Gly Ser
    50                  55                  60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu Ala Glu
65                  70                  75                  80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe Thr
                85                  90                  95

Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg
            100                 105


<210> SEQ ID NO 23
<211> LENGTH: 357
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 23

caagttcagc tccagcaaag cggcgccgaa ctcgctcggc ccggcgcttc cgtgaagatg      60

tcttgtaagg cctccggcta taccttcacc cggtacacaa tgcactgggt caagcaacgg     120

cccggtcaag gtttagagtg gattggctat atcaacccct cccggggcta taccaactac     180

aaccagaagt tcaaggacaa agccaccctc accaccgaca agtccagcag caccgcttac     240

atgcagctga gctctttaac atccgaggat tccgccgtgt actactgcgc tcggtactac     300

gacgatcatt actgcctcga ttactggggc caaggtacca ccttaacagt ctcctcc        357


<210> SEQ ID NO 24
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 24

cagatcgtgc tgacccagtc ccccgctatt atgagcgcta gccccggtga aaaggtgact      60

atgacatgca gcgccagctc ttccgtgagc tacatgaact ggtatcagca gaagtccggc     120
```

```
accagcccta aaaggtggat ctacgacacc agcaagctgg ccagcggcgt ccccgctcac    180

tttcggggct ccggctccgg aacaagctac tctctgacca tcagcggcat ggaagccgag    240

gatgccgcta cctattactg tcagcagtgg agctccaacc ccttcacctt tggatccggc    300

accaagctcg agattaatcg t                                              321


<210> SEQ ID NO 25
<211> LENGTH: 723
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 25

cagatcgtgc tgacccagtc ccccgctatt atgagcgcta gccccggtga aaaggtgact     60

atgacatgca gcgccagctc ttccgtgagc tacatgaact ggtatcagca gaagtccggc    120

accagcccta aaaggtggat ctacgacacc agcaagctgg ccagcggcgt ccccgctcac    180

tttcggggct ccggctccgg aacaagctac tctctgacca tcagcggcat ggaagccgag    240

gatgccgcta cctattactg tcagcagtgg agctccaacc ccttcacctt tggatccggc    300

accaagctcg agattaatcg tggaggcgga ggtagcggag gaggcggatc cggcggtgga    360

ggtagccaag ttcagctcca gcaaagcggc gccgaactcg ctcggcccgg cgcttccgtg    420

aagatgtctt gtaaggcctc cggctatacc ttcacccggt acacaatgca ctgggtcaag    480

caacggcccg gtcaaggttt agagtggatt ggctatatca acccctcccg gggctatacc    540

aactacaacc agaagttcaa ggacaaagcc accctcacca ccgacaagtc cagcagcacc    600

gcttacatgc agctgagctc tttaacatcc gaggattccg ccgtgtacta ctgcgctcgg    660

tactacgacg atcattactg cctcgattac tggggccaag gtaccacctt aacagtctcc    720

tcc                                                                  723


<210> SEQ ID NO 26
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 26

Asp Ile Glu Met Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu Gly
1               5                   10                  15

Glu Arg Val Thr Met Thr Cys Thr Ala Ser Ser Ser Val Ser Ser Ser
            20                  25                  30

Tyr Phe His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu Cys
        35                  40                  45

Ile Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Pro Arg Phe Ser
    50                  55                  60

Gly Ser Gly Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu Ala
65                  70                  75                  80

Glu Asp Ala Ala Thr Tyr Phe Cys His Gln Tyr His Arg Ser Pro Thr
                85                  90                  95

Phe Gly Gly Gly Thr Lys Leu Glu Thr Lys Arg
            100                 105


<210> SEQ ID NO 27
```

<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 27

Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala Ser
1 5 10 15

Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr Val
20 25 30

Ile Gln Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile Gly
35 40 45

Ser Ile Asn Pro Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys Phe Lys
50 55 60

Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr Met
65 70 75 80

Glu Phe Ser Ser Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr Cys Ala
85 90 95

Arg Trp Gly Asp Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu Thr Val
100 105 110

Ser Ser

<210> SEQ ID NO 28
<211> LENGTH: 321
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 28

gacatcgaga tgacacagtc ccccgctatc atgagcgcct ctttaggaga acgtgtgacc 60

atgacttgta cagcttcctc cagcgtgagc agctcctatt tccactggta ccagcagaaa 120

cccggctcct cccctaaact gtgtatctac tccacaagca atttagctag cggcgtgcct 180

cctcgtttta gcggctccgg cagcacctct tactctttaa ccattagctc tatggaggcc 240

gaagatgccg ccacatactt ttgccatcag taccaccggt cccctacctt tggcggaggc 300

acaaagctgg agaccaagcg g 321

<210> SEQ ID NO 29
<211> LENGTH: 342
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 29

gtgcagctgc agcagtccgg acccgaactg gtcaagcccg gtgcctccgt gaaaatgtct 60

tgtaaggctt ctggctacac ctttacctcc tacgtcatcc aatgggtgaa gcagaagccc 120

ggtcaaggtc tcgagtggat cggcagcatc aatccctaca acgattacac caagtataac 180

gaaaagttta agggcaaggc cactctgaca agcgacaaga gctccattac cgcctacatg 240

gagttttcct ctttaacttc tgaggactcc gctttatact attgcgctcg ttggggcgat 300

ggcaattatt ggggccgggg aactacttta acagtgagct cc 342

```
<210> SEQ ID NO 30
<211> LENGTH: 708
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 30

gtgcagctgc agcagtccgg acccgaactg gtcaagcccg gtgcctccgt gaaaatgtct     60

tgtaaggctt ctggctacac ctttacctcc tacgtcatcc aatgggtgaa gcagaagccc    120

ggtcaaggtc tcgagtggat cggcagcatc aatccctaca acgattacac caagtataac    180

gaaaagttta agggcaaggc cactctgaca agcgacaaga gctccattac cgcctacatg    240

gagttttcct ctttaacttc tgaggactcc gctttatact attgcgctcg ttggggcgat    300

ggcaattatt ggggccgggg aactacttta acagtgagct ccggcggcgg cggaagcgga    360

ggtggaggat ctggcggtgg aggcagcgac atcgagatga cacagtcccc cgctatcatg    420

agcgcctctt taggagaacg tgtgaccatg acttgtacag cttcctccag cgtgagcagc    480

tcctatttcc actggtacca gcagaaaccc ggctcctccc ctaaactgtg tatctactcc    540

acaagcaatt tagctagcgg cgtgcctcct cgttttagcg gctccggcag cacctcttac    600

tctttaacca ttagctctat ggaggccgaa gatgccgcca catacttttg ccatcagtac    660

caccggtccc ctacctttgg cggaggcaca aagctggaga ccaagcgg                 708


<210> SEQ ID NO 31
<211> LENGTH: 399
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 31

gcacctactt caagttctac aaagaaaaca cagctacaac tggagcattt actgctggat     60

ttacagatga ttttgaatgg aattaataat tacaagaatc ccaaactcac caggatgctc    120

acatttaagt tttacatgcc caagaaggcc acagaactga aacatcttca gtgtctagaa    180

gaagaactca aacctctgga ggaagtgcta aatttagctc aaagcaaaaa ctttcactta    240

agacccaggg acttaatcag caatatcaac gtaatagttc tggaactaaa gggatctgaa    300

acaacattca tgtgtgaata tgctgatgag acagcaacca ttgtagaatt tctgaacaga    360

tggattacct tttgtcaaag catcatctca acactaact                           399


<210> SEQ ID NO 32
<211> LENGTH: 399
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 32

gcccccacct cctcctccac caagaagacc cagctgcagc tggagcattt actgctggat     60

ttacagatga ttttaaacgg catcaacaac tacaagaacc ccaagctgac tcgtatgctg    120

accttcaagt tctacatgcc caagaaggcc accgagctga agcatttaca gtgtttagag    180

gaggagctga agcccctcga ggaggtgctg aatttagccc agtccaagaa tttccattta    240

aggccccggg atttaatcag caacatcaac gtgatcgttt tagagctgaa gggctccgag    300

accaccttca tgtgcgagta cgccgacgag accgccacca tcgtggagtt tttaaatcgt    360

tggatcacct tctgccagtc catcatctcc actttaacc                           399
```

<210> SEQ ID NO 33
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 33

Met Lys Trp Val Thr Phe Ile Ser Leu Leu Phe Leu Phe Ser Ser Ala
1 5 10 15

Tyr Ser

<210> SEQ ID NO 34
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 34

atgaaatggg tgacctttat ttctttactg ttcctcttta gcagcgccta ctcc 54

<210> SEQ ID NO 35
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 35

atgaagtggg tcacatttat ctctttactg ttcctcttct ccagcgccta cagc 54

<210> SEQ ID NO 36
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 36

atgaaatggg tgacctttat ttctttactg ttcctcttta gcagcgccta ctcc 54

<210> SEQ ID NO 37
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 37

Met Lys Cys Leu Leu Tyr Leu Ala Phe Leu Phe Leu Gly Val Asn Cys
1 5 10 15

<210> SEQ ID NO 38
<211> LENGTH: 58
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 38

-continued

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Met Gly Gln Ile Val Thr Met Phe Glu Ala Leu Pro His Ile Ile Asp
1               5                   10                  15

Glu Val Ile Asn Ile Val Ile Ile Val Leu Ile Ile Ile Thr Ser Ile
            20                  25                  30

Lys Ala Val Tyr Asn Phe Ala Thr Cys Gly Ile Leu Ala Leu Val Ser
        35                  40                  45

Phe Leu Phe Leu Ala Gly Arg Ser Cys Gly
    50                  55


<210> SEQ ID NO 39
<211> LENGTH: 97
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 39

Met Pro Asn His Gln Ser Gly Ser Pro Thr Gly Ser Ser Asp Leu Leu
1               5                   10                  15

Leu Ser Gly Lys Lys Gln Arg Pro His Leu Ala Leu Arg Arg Lys Arg
            20                  25                  30

Arg Arg Glu Met Arg Lys Ile Asn Arg Lys Val Arg Arg Met Asn Leu
        35                  40                  45

Ala Pro Ile Lys Glu Lys Thr Ala Trp Gln His Leu Gln Ala Leu Ile
    50                  55                  60

Ser Glu Ala Glu Glu Val Leu Lys Thr Ser Gln Thr Pro Gln Asn Ser
65                  70                  75                  80

Leu Thr Leu Phe Leu Ala Leu Leu Ser Val Leu Gly Pro Pro Val Thr
                85                  90                  95

Gly


<210> SEQ ID NO 40
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 40

Met Asp Ser Lys Gly Ser Ser Gln Lys Gly Ser Arg Leu Leu Leu Leu
1               5                   10                  15

Leu Val Val Ser Asn Leu Leu Leu Cys Gln Gly Val Val Ser
            20                  25                  30


<210> SEQ ID NO 41
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 41

Gly Leu Asn Asp Ile Phe Glu Ala Gln Lys Ile Glu Trp His Glu
1               5                   10                  15


<210> SEQ ID NO 42
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
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<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 42

Lys Arg Arg Trp Lys Lys Asn Phe Ile Ala Val Ser Ala Ala Asn Arg
1               5                   10                  15

Phe Lys Lys Ile Ser Ser Ser Gly Ala Leu
            20                  25


<210> SEQ ID NO 43
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 43

Glu Glu Glu Glu Glu Glu
1               5


<210> SEQ ID NO 44
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 44

Gly Ala Pro Val Pro Tyr Pro Asp Pro Leu Glu Pro Arg
1               5                   10


<210> SEQ ID NO 45
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 45

Asp Tyr Lys Asp Asp Asp Asp Lys
1               5


<210> SEQ ID NO 46
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 46

Tyr Pro Tyr Asp Val Pro Asp Tyr Ala
1               5


<210> SEQ ID NO 47
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 47
```

```
His His His His His
1               5


<210> SEQ ID NO 48
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 48

His His His His His His
1               5


<210> SEQ ID NO 49
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 49

His His His His His His His
1               5


<210> SEQ ID NO 50
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 50

His His His His His His His His
1               5


<210> SEQ ID NO 51
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 51

His His His His His His His His His
1               5


<210> SEQ ID NO 52
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 52

His His His His His His His His His His
1               5                   10


<210> SEQ ID NO 53
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
```

<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 53

```
Glu Gln Lys Leu Ile Ser Glu Glu Asp Leu
1               5                   10
```

<210> SEQ ID NO 54
<211> LENGTH: 18
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 54

```
Thr Lys Glu Asn Pro Arg Ser Asn Gln Glu Glu Ser Tyr Asp Asp Asn
1               5                   10                  15

Glu Ser
```

<210> SEQ ID NO 55
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 55

```
Lys Glu Thr Ala Ala Ala Lys Phe Glu Arg Gln His Met Asp Ser
1               5                   10                  15
```

<210> SEQ ID NO 56
<211> LENGTH: 38
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 56

```
Met Asp Glu Lys Thr Thr Gly Trp Arg Gly Gly His Val Val Glu Gly
1               5                   10                  15

Leu Ala Gly Glu Leu Glu Gln Leu Arg Ala Arg Leu Glu His His Pro
            20                  25                  30

Gln Gly Gln Arg Glu Pro
        35
```

<210> SEQ ID NO 57
<211> LENGTH: 13
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 57

```
Ser Leu Ala Glu Leu Leu Asn Ala Gly Leu Gly Gly Ser
1               5                   10
```

<210> SEQ ID NO 58
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

peptide

<400> SEQUENCE: 58

Thr Gln Asp Pro Ser Arg Val Gly
1 5

<210> SEQ ID NO 59
<211> LENGTH: 12
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 59

Pro Asp Arg Val Arg Ala Val Ser His Trp Ser Ser
1 5 10

<210> SEQ ID NO 60
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 60

Trp Ser His Pro Gln Phe Glu Lys
1 5

<210> SEQ ID NO 61
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 61

Cys Cys Pro Gly Cys Cys
1 5

<210> SEQ ID NO 62
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 62

Glu Val His Thr Asn Gln Asp Pro Leu Asp
1 5 10

<210> SEQ ID NO 63
<211> LENGTH: 14
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 63

Gly Lys Pro Ile Pro Asn Pro Leu Leu Gly Leu Asp Ser Thr
1 5 10

<210> SEQ ID NO 64

<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 64

Tyr Thr Asp Ile Glu Met Asn Arg Leu Gly Lys
1 5 10

<210> SEQ ID NO 65
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 65

Asp Leu Tyr Asp Asp Asp Asp Lys
1 5

<210> SEQ ID NO 66
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 66

Asp Tyr Asn Val Tyr
1 5

<210> SEQ ID NO 67
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 67

Tyr Ile Asp Pro Tyr Asn Gly Ile Thr Ile Tyr Asp Gln Asn Phe Lys
1 5 10 15

Gly

<210> SEQ ID NO 68
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 68

Asp Val Thr Thr Ala Leu Asp Phe
1 5

<210> SEQ ID NO 69
<211> LENGTH: 11
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
peptide

<400> SEQUENCE: 69

Leu Ala Ser Gln Thr Ile Asp Thr Trp Leu Ala
1 5 10

<210> SEQ ID NO 70
<211> LENGTH: 7
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 70

Ala Ala Thr Asn Leu Ala Asp
1 5

<210> SEQ ID NO 71
<211> LENGTH: 9
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic peptide

<400> SEQUENCE: 71

Gln Gln Val Tyr Ser Ser Pro Phe Thr
1 5

<210> SEQ ID NO 72
<211> LENGTH: 117
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 72

Gln Ile Gln Leu Val Gln Ser Gly Gly Glu Val Lys Lys Pro Gly Ala
1 5 10 15

Ser Val Arg Val Ser Cys Lys Ala Ser Gly Tyr Ser Phe Thr Asp Tyr
20 25 30

Asn Val Tyr Trp Val Arg Gln Ser Pro Gly Lys Gly Leu Glu Trp Ile
35 40 45

Gly Tyr Ile Asp Pro Tyr Asn Gly Ile Thr Ile Tyr Asp Gln Asn Phe
50 55 60

Lys Gly Lys Ala Thr Leu Thr Val Asp Lys Ser Thr Ser Thr Ala Tyr
65 70 75 80

Met Glu Leu Ser Ser Leu Arg Ser Glu Asp Thr Ala Val Tyr Phe Cys
85 90 95

Ala Arg Asp Val Thr Thr Ala Leu Asp Phe Trp Gly Gln Gly Thr Thr
100 105 110

Val Thr Val Ser Ser
115

<210> SEQ ID NO 73
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 73

```
Asp Ile Gln Met Thr Gln Ser Pro Ala Ser Leu Ser Ala Ser Val Gly
1               5                   10                  15

Asp Arg Val Thr Ile Thr Cys Leu Ala Ser Gln Thr Ile Asp Thr Trp
            20                  25                  30

Leu Ala Trp Tyr Leu Gln Lys Pro Gly Lys Ser Pro Gln Leu Leu Ile
        35                  40                  45

Tyr Ala Ala Thr Asn Leu Ala Asp Gly Val Pro Ser Arg Phe Ser Gly
    50                  55                  60

Ser Gly Ser Gly Thr Asp Phe Ser Phe Thr Ile Ser Ser Leu Gln Pro
65                  70                  75                  80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Val Tyr Ser Ser Pro Phe
                85                  90                  95

Thr Phe Gly Gln Gly Thr Lys Leu Glu Ile Lys
            100                 105


<210> SEQ ID NO 74
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      peptide

<400> SEQUENCE: 74

Tyr Ile Asp Pro Tyr Asn Gly Ile Thr Ile Tyr Asp Gln Asn Leu Lys
1               5                   10                  15

Gly


<210> SEQ ID NO 75
<211> LENGTH: 133
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 75

Ala Pro Thr Ser Ser Ser Thr Lys Lys Thr Gln Leu Gln Leu Glu His
1               5                   10                  15

Leu Leu Leu Asp Leu Gln Met Ile Leu Asn Gly Ile Asn Asn Tyr Lys
            20                  25                  30

Asn Pro Lys Leu Thr Arg Met Leu Thr Phe Lys Phe Tyr Met Pro Lys
        35                  40                  45

Lys Ala Thr Glu Leu Lys His Leu Gln Cys Leu Glu Glu Glu Leu Lys
    50                  55                  60

Pro Leu Glu Glu Val Leu Asn Leu Ala Gln Ser Lys Asn Phe His Leu
65                  70                  75                  80

Arg Pro Arg Asp Leu Ile Ser Asn Ile Asn Val Ile Val Leu Glu Leu
                85                  90                  95

Lys Gly Ser Glu Thr Thr Phe Met Cys Glu Tyr Ala Asp Glu Thr Ala
            100                 105                 110

Thr Ile Val Glu Phe Leu Asn Arg Trp Ile Thr Phe Cys Gln Ser Ile
        115                 120                 125

Ile Ser Thr Leu Thr
    130


<210> SEQ ID NO 76
<211> LENGTH: 133
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 76
```

Ala Pro Met Thr Gln Thr Thr Pro Leu Lys Thr Ser Trp Val Asn Cys
1   5   10   15

Ser Asn Met Ile Asp Glu Ile Ile Thr His Leu Lys Gln Pro Pro Leu
            20   25   30

Pro Leu Leu Asp Phe Asn Asn Leu Asn Gly Glu Asp Gln Asp Ile Leu
        35   40   45

Met Glu Asn Asn Leu Arg Arg Pro Asn Leu Glu Ala Phe Asn Arg Ala
    50   55   60

Val Lys Ser Leu Gln Asn Ala Ser Ala Ile Glu Ser Ile Leu Lys Asn
65   70   75   80

Leu Leu Pro Cys Leu Pro Leu Ala Thr Ala Ala Pro Thr Arg His Pro
                85   90   95

Ile His Ile Lys Asp Gly Asp Trp Asn Glu Phe Arg Arg Lys Leu Thr
            100   105   110

Phe Tyr Leu Lys Thr Leu Glu Asn Ala Gln Ala Gln Gln Thr Thr Leu
        115   120   125

Ser Leu Ala Ile Phe
    130

<210> SEQ ID NO 77
<211> LENGTH: 152
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 77

Asp Cys Asp Ile Glu Gly Lys Asp Gly Lys Gln Tyr Glu Ser Val Leu
1   5   10   15

Met Val Ser Ile Asp Gln Leu Leu Asp Ser Met Lys Glu Ile Gly Ser
            20   25   30

Asn Cys Leu Asn Asn Glu Phe Asn Phe Phe Lys Arg His Ile Cys Asp
        35   40   45

Ala Asn Lys Glu Gly Met Phe Leu Phe Arg Ala Ala Arg Lys Leu Arg
    50   55   60

Gln Phe Leu Lys Met Asn Ser Thr Gly Asp Phe Asp Leu His Leu Leu
65   70   75   80

Lys Val Ser Glu Gly Thr Thr Ile Leu Leu Asn Cys Thr Gly Gln Val
                85   90   95

Lys Gly Arg Lys Pro Ala Ala Leu Gly Glu Ala Gln Pro Thr Lys Ser
            100   105   110

Leu Glu Glu Asn Lys Ser Leu Lys Glu Gln Lys Lys Leu Asn Asp Leu
        115   120   125

Cys Phe Leu Lys Arg Leu Leu Gln Glu Ile Lys Thr Cys Trp Asn Lys
    130   135   140

Ile Leu Met Gly Thr Lys Glu His
145   150

<210> SEQ ID NO 78
<211> LENGTH: 79
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 78

Glu Gly Ala Val Leu Pro Arg Ser Ala Lys Glu Leu Arg Cys Gln Cys
1   5   10   15

Ile Lys Thr Tyr Ser Lys Pro Phe His Pro Lys Phe Ile Lys Glu Leu
            20   25   30

```
Arg Val Ile Glu Ser Gly Pro His Cys Ala Asn Thr Glu Ile Ile Val
        35                  40                  45

Lys Leu Ser Asp Gly Arg Glu Leu Cys Leu Asp Pro Lys Glu Asn Trp
    50                  55                  60

Val Gln Arg Val Val Glu Lys Phe Leu Lys Arg Ala Glu Asn Ser
65                  70                  75


<210> SEQ ID NO 79
<211> LENGTH: 160
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 79

Ser Pro Gly Gln Gly Thr Gln Ser Glu Asn Ser Cys Thr His Phe Pro
1               5                   10                  15

Gly Asn Leu Pro Asn Met Leu Arg Asp Leu Arg Asp Ala Phe Ser Arg
            20                  25                  30

Val Lys Thr Phe Phe Gln Met Lys Asp Gln Leu Asp Asn Leu Leu Leu
        35                  40                  45

Lys Glu Ser Leu Leu Glu Asp Phe Lys Gly Tyr Leu Gly Cys Gln Ala
    50                  55                  60

Leu Ser Glu Met Ile Gln Phe Tyr Leu Glu Glu Val Met Pro Gln Ala
65                  70                  75                  80

Glu Asn Gln Asp Pro Asp Ile Lys Ala His Val Asn Ser Leu Gly Glu
                85                  90                  95

Asn Leu Lys Thr Leu Arg Leu Arg Leu Arg Arg Cys His Arg Phe Leu
            100                 105                 110

Pro Cys Glu Asn Lys Ser Lys Ala Val Glu Gln Val Lys Asn Ala Phe
        115                 120                 125

Asn Lys Leu Gln Glu Lys Gly Ile Tyr Lys Ala Met Ser Glu Phe Asp
    130                 135                 140

Ile Phe Ile Asn Tyr Ile Glu Ala Tyr Met Thr Met Lys Ile Arg Asn
145                 150                 155                 160


<210> SEQ ID NO 80
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 80

Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
1               5                   10                  15

Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
            20                  25                  30

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
        35                  40                  45

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu
    50                  55                  60

Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
65                  70                  75                  80

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
                85                  90                  95

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
            100                 105                 110

Thr Ser
```

<210> SEQ ID NO 81
<211> LENGTH: 132
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 81

Gly Ile Thr Ile Pro Arg Asn Pro Gly Cys Pro Asn Ser Glu Asp Lys
1   5   10   15

Asn Phe Pro Arg Thr Val Met Val Asn Leu Asn Ile His Asn Arg Asn
            20   25   30

Thr Asn Thr Asn Pro Lys Arg Ser Ser Asp Tyr Tyr Asn Arg Ser Thr
        35   40   45

Ser Pro Trp Asn Leu His Arg Asn Glu Asp Pro Glu Arg Tyr Pro Ser
    50   55   60

Val Ile Trp Glu Ala Lys Cys Arg His Leu Gly Cys Ile Asn Ala Asp
65   70   75   80

Gly Asn Val Asp Tyr His Met Asn Ser Val Pro Ile Gln Gln Glu Ile
                85   90   95

Leu Val Leu Arg Arg Glu Pro Pro His Cys Pro Asn Ser Phe Arg Leu
            100   105   110

Glu Lys Ile Leu Val Ser Val Gly Cys Thr Cys Val Thr Pro Ile Val
        115   120   125

His His Val Ala
    130

<210> SEQ ID NO 82
<211> LENGTH: 157
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 82

Tyr Phe Gly Lys Leu Glu Ser Lys Leu Ser Val Ile Arg Asn Leu Asn
1   5   10   15

Asp Gln Val Leu Phe Ile Asp Gln Gly Asn Arg Pro Leu Phe Glu Asp
            20   25   30

Met Thr Asp Ser Asp Cys Arg Asp Asn Ala Pro Arg Thr Ile Phe Ile
        35   40   45

Ile Ser Met Tyr Lys Asp Ser Gln Pro Arg Gly Met Ala Val Thr Ile
    50   55   60

Ser Val Lys Cys Glu Lys Ile Ser Thr Leu Ser Cys Glu Asn Lys Ile
65   70   75   80

Ile Ser Phe Lys Glu Met Asn Pro Pro Asp Asn Ile Lys Asp Thr Lys
                85   90   95

Ser Asp Ile Ile Phe Phe Gln Arg Ser Val Pro Gly His Asp Asn Lys
            100   105   110

Met Gln Phe Glu Ser Ser Ser Tyr Glu Gly Tyr Phe Leu Ala Cys Glu
        115   120   125

Lys Glu Arg Asp Leu Phe Lys Leu Ile Leu Lys Lys Glu Asp Glu Leu
    130   135   140

Gly Asp Arg Ser Ile Met Phe Thr Val Gln Asn Glu Asp
145   150   155

<210> SEQ ID NO 83
<211> LENGTH: 352
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 83

```
Arg Asp Thr Ser Ala Thr Pro Gln Ser Ala Ser Ile Lys Ala Leu Arg
1               5                   10                  15

Asn Ala Asn Leu Arg Arg Asp Glu Ser Asn His Leu Thr Asp Leu Tyr
            20                  25                  30

Arg Arg Asp Glu Thr Ile Gln Val Lys Gly Asn Gly Tyr Val Gln Ser
        35                  40                  45

Pro Arg Phe Pro Asn Ser Tyr Pro Arg Asn Leu Leu Leu Thr Trp Arg
    50                  55                  60

Leu His Ser Gln Glu Asn Thr Arg Ile Gln Leu Val Phe Asp Asn Gln
65                  70                  75                  80

Phe Gly Leu Glu Glu Ala Glu Asn Asp Ile Cys Arg Tyr Asp Phe Val
                85                  90                  95

Glu Val Glu Asp Ile Ser Glu Thr Ser Thr Ile Ile Arg Gly Arg Trp
            100                 105                 110

Cys Gly His Lys Glu Val Pro Pro Arg Ile Lys Ser Arg Thr Asn Gln
        115                 120                 125

Ile Lys Ile Thr Phe Lys Ser Asp Asp Tyr Phe Val Ala Lys Pro Gly
    130                 135                 140

Phe Lys Ile Tyr Tyr Ser Leu Leu Glu Asp Phe Gln Pro Ala Ala Ala
145                 150                 155                 160

Ser Glu Thr Asn Trp Glu Ser Val Thr Ser Ser Ile Ser Gly Val Ser
                165                 170                 175

Tyr Asn Ser Pro Ser Val Thr Asp Pro Thr Leu Ile Ala Asp Ala Leu
            180                 185                 190

Asp Lys Lys Ile Ala Glu Phe Asp Thr Val Glu Asp Leu Leu Lys Tyr
        195                 200                 205

Phe Asn Pro Glu Ser Trp Gln Glu Asp Leu Glu Asn Met Tyr Leu Asp
    210                 215                 220

Thr Pro Arg Tyr Arg Gly Arg Ser Tyr His Asp Arg Lys Ser Lys Val
225                 230                 235                 240

Asp Leu Asp Arg Leu Asn Asp Asp Ala Lys Arg Tyr Ser Cys Thr Pro
                245                 250                 255

Arg Asn Tyr Ser Val Asn Ile Arg Glu Glu Leu Lys Leu Ala Asn Val
            260                 265                 270

Val Phe Phe Pro Arg Cys Leu Leu Val Gln Arg Cys Gly Gly Asn Cys
        275                 280                 285

Gly Cys Gly Thr Val Asn Trp Arg Ser Cys Thr Cys Asn Ser Gly Lys
    290                 295                 300

Thr Val Lys Lys Tyr His Glu Val Leu Gln Phe Glu Pro Gly His Ile
305                 310                 315                 320

Lys Arg Arg Gly Arg Ala Lys Thr Met Ala Leu Val Asp Ile Gln Leu
                325                 330                 335

Asp His His Glu Arg Cys Asp Cys Ile Cys Ser Ser Arg Pro Pro Arg
            340                 345                 350
```

<210> SEQ ID NO 84
<211> LENGTH: 248
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 84

```
Glu Gly Ile Cys Arg Asn Arg Val Thr Asn Asn Val Lys Asp Val Thr
1               5                   10                  15
```

```
Lys Leu Val Ala Asn Leu Pro Lys Asp Tyr Met Ile Thr Leu Lys Tyr
            20                  25                  30

Val Pro Gly Met Asp Val Leu Pro Ser His Cys Trp Ile Ser Glu Met
        35                  40                  45

Val Val Gln Leu Ser Asp Ser Leu Thr Asp Leu Leu Asp Lys Phe Ser
    50                  55                  60

Asn Ile Ser Glu Gly Leu Ser Asn Tyr Ser Ile Ile Asp Lys Leu Val
65                  70                  75                  80

Asn Ile Val Asp Asp Leu Val Glu Cys Val Lys Glu Asn Ser Ser Lys
                85                  90                  95

Asp Leu Lys Lys Ser Phe Lys Ser Pro Glu Pro Arg Leu Phe Thr Pro
            100                 105                 110

Glu Glu Phe Phe Arg Ile Phe Asn Arg Ser Ile Asp Ala Phe Lys Asp
        115                 120                 125

Phe Val Val Ala Ser Glu Thr Ser Asp Cys Val Val Ser Ser Thr Leu
    130                 135                 140

Ser Pro Glu Lys Asp Ser Arg Val Ser Val Thr Lys Pro Phe Met Leu
145                 150                 155                 160

Pro Pro Val Ala Ala Ser Ser Leu Arg Asn Asp Ser Ser Ser Ser Asn
                165                 170                 175

Arg Lys Ala Lys Asn Pro Pro Gly Asp Ser Ser Leu His Trp Ala Ala
            180                 185                 190

Met Ala Leu Pro Ala Leu Phe Ser Leu Ile Ile Gly Phe Ala Phe Gly
        195                 200                 205

Ala Leu Tyr Trp Lys Lys Arg Gln Pro Ser Leu Thr Arg Ala Val Glu
    210                 215                 220

Asn Ile Gln Ile Asn Glu Glu Asp Asn Glu Ile Ser Met Leu Gln Glu
225                 230                 235                 240

Lys Glu Arg Glu Phe Gln Glu Val
                245


<210> SEQ ID NO 85
<211> LENGTH: 209
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 85

Thr Gln Asp Cys Ser Phe Gln His Ser Pro Ile Ser Ser Asp Phe Ala
1               5                   10                  15

Val Lys Ile Arg Glu Leu Ser Asp Tyr Leu Leu Gln Asp Tyr Pro Val
            20                  25                  30

Thr Val Ala Ser Asn Leu Gln Asp Glu Glu Leu Cys Gly Gly Leu Trp
        35                  40                  45

Arg Leu Val Leu Ala Gln Arg Trp Met Glu Arg Leu Lys Thr Val Ala
    50                  55                  60

Gly Ser Lys Met Gln Gly Leu Leu Glu Arg Val Asn Thr Glu Ile His
65                  70                  75                  80

Phe Val Thr Lys Cys Ala Phe Gln Pro Pro Pro Ser Cys Leu Arg Phe
                85                  90                  95

Val Gln Thr Asn Ile Ser Arg Leu Leu Gln Glu Thr Ser Glu Gln Leu
            100                 105                 110

Val Ala Leu Lys Pro Trp Ile Thr Arg Gln Asn Phe Ser Arg Cys Leu
        115                 120                 125

Glu Leu Gln Cys Gln Pro Asp Ser Ser Thr Leu Pro Pro Pro Trp Ser
    130                 135                 140
```

Pro Arg Pro Leu Glu Ala Thr Ala Pro Thr Ala Pro Gln Pro Pro Leu
145 150 155 160

Leu Leu Leu Leu Leu Leu Pro Val Gly Leu Leu Leu Leu Ala Ala Ala
165 170 175

Trp Cys Leu His Trp Gln Arg Thr Arg Arg Arg Thr Pro Arg Pro Gly
180 185 190

Glu Gln Val Pro Pro Val Pro Ser Pro Gln Asp Leu Leu Leu Val Glu
195 200 205

His

<210> SEQ ID NO 86
<211> LENGTH: 360
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 86

Glu Pro His Ser Leu Arg Tyr Asn Leu Thr Val Leu Ser Trp Asp Gly
1 5 10 15

Ser Val Gln Ser Gly Phe Leu Thr Glu Val His Leu Asp Gly Gln Pro
20 25 30

Phe Leu Arg Cys Asp Arg Gln Lys Cys Arg Ala Lys Pro Gln Gly Gln
35 40 45

Trp Ala Glu Asp Val Leu Gly Asn Lys Thr Trp Asp Arg Glu Thr Arg
50 55 60

Asp Leu Thr Gly Asn Gly Lys Asp Leu Arg Met Thr Leu Ala His Ile
65 70 75 80

Lys Asp Gln Lys Glu Gly Leu His Ser Leu Gln Glu Ile Arg Val Cys
85 90 95

Glu Ile His Glu Asp Asn Ser Thr Arg Ser Ser Gln His Phe Tyr Tyr
100 105 110

Asp Gly Glu Leu Phe Leu Ser Gln Asn Leu Glu Thr Lys Glu Trp Thr
115 120 125

Met Pro Gln Ser Ser Arg Ala Gln Thr Leu Ala Met Asn Val Arg Asn
130 135 140

Phe Leu Lys Glu Asp Ala Met Lys Thr Lys Thr His Tyr His Ala Met
145 150 155 160

His Ala Asp Cys Leu Gln Glu Leu Arg Arg Tyr Leu Lys Ser Gly Val
165 170 175

Val Leu Arg Arg Thr Val Pro Pro Met Val Asn Val Thr Arg Ser Glu
180 185 190

Ala Ser Glu Gly Asn Ile Thr Val Thr Cys Arg Ala Ser Gly Phe Tyr
195 200 205

Pro Trp Asn Ile Thr Leu Ser Trp Arg Gln Asp Gly Val Ser Leu Ser
210 215 220

His Asp Thr Gln Gln Trp Gly Asp Val Leu Pro Asp Gly Asn Gly Thr
225 230 235 240

Tyr Gln Thr Trp Val Ala Thr Arg Ile Cys Gln Gly Glu Glu Gln Arg
245 250 255

Phe Thr Cys Tyr Met Glu His Ser Gly Asn His Ser Thr His Pro Val
260 265 270

Pro Ser Gly Lys Val Leu Val Leu Gln Ser His Trp Gln Thr Phe His
275 280 285

Val Ser Ala Val Ala Ala Ala Ala Ile Phe Val Ile Ile Ile Phe Tyr
290 295 300

Val Arg Cys Cys Lys Lys Lys Thr Ser Ala Ala Glu Gly Pro Glu Leu
305 310 315 320

Val Ser Leu Gln Val Leu Asp Gln His Pro Val Gly Thr Ser Asp His
325 330 335

Arg Asp Ala Thr Gln Leu Gly Phe Gln Pro Leu Met Ser Asp Leu Gly
340 345 350

Ser Thr Gly Ser Thr Glu Gly Ala
355 360

<210> SEQ ID NO 87
<211> LENGTH: 361
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 87

Ala Glu Pro His Ser Leu Arg Tyr Asn Leu Met Val Leu Ser Gln Asp
1 5 10 15

Glu Ser Val Gln Ser Gly Phe Leu Ala Glu Gly His Leu Asp Gly Gln
20 25 30

Pro Phe Leu Arg Tyr Asp Arg Gln Lys Arg Arg Ala Lys Pro Gln Gly
35 40 45

Gln Trp Ala Glu Asp Val Leu Gly Ala Lys Thr Trp Asp Thr Glu Thr
50 55 60

Glu Asp Leu Thr Glu Asn Gly Gln Asp Leu Arg Arg Thr Leu Thr His
65 70 75 80

Ile Lys Asp Gln Lys Gly Gly Leu His Ser Leu Gln Glu Ile Arg Val
85 90 95

Cys Glu Ile His Glu Asp Ser Ser Thr Arg Gly Ser Arg His Phe Tyr
100 105 110

Tyr Asp Gly Glu Leu Phe Leu Ser Gln Asn Leu Glu Thr Gln Glu Ser
115 120 125

Thr Val Pro Gln Ser Ser Arg Ala Gln Thr Leu Ala Met Asn Val Thr
130 135 140

Asn Phe Trp Lys Glu Asp Ala Met Lys Thr Lys Thr His Tyr Arg Ala
145 150 155 160

Met Gln Ala Asp Cys Leu Gln Lys Leu Gln Arg Tyr Leu Lys Ser Gly
165 170 175

Val Ala Ile Arg Arg Thr Val Pro Pro Met Val Asn Val Thr Cys Ser
180 185 190

Glu Val Ser Glu Gly Asn Ile Thr Val Thr Cys Arg Ala Ser Ser Phe
195 200 205

Tyr Pro Arg Asn Ile Thr Leu Thr Trp Arg Gln Asp Gly Val Ser Leu
210 215 220

Ser His Asn Thr Gln Gln Trp Gly Asp Val Leu Pro Asp Gly Asn Gly
225 230 235 240

Thr Tyr Gln Thr Trp Val Ala Thr Arg Ile Arg Gln Gly Glu Glu Gln
245 250 255

Arg Phe Thr Cys Tyr Met Glu His Ser Gly Asn His Gly Thr His Pro
260 265 270

Val Pro Ser Gly Lys Val Leu Val Leu Gln Ser Gln Arg Thr Asp Phe
275 280 285

Pro Tyr Val Ser Ala Ala Met Pro Cys Phe Val Ile Ile Ile Ile Leu
290 295 300

Cys Val Pro Cys Cys Lys Lys Lys Thr Ser Ala Ala Glu Gly Pro Glu

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305                 310                 315                 320

Leu Val Ser Leu Gln Val Leu Asp Gln His Pro Val Gly Thr Gly Asp
                325                 330                 335

His Arg Asp Ala Ala Gln Leu Gly Phe Gln Pro Leu Met Ser Ala Thr
            340                 345                 350

Gly Ser Thr Gly Ser Thr Glu Gly Ala
        355                 360


<210> SEQ ID NO 88
<211> LENGTH: 190
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 88

Trp Val Asp Thr His Cys Leu Cys Tyr Asp Phe Ile Ile Thr Pro Lys
1               5                   10                  15

Ser Arg Pro Glu Pro Gln Trp Cys Glu Val Gln Gly Leu Val Asp Glu
            20                  25                  30

Arg Pro Phe Leu His Tyr Asp Cys Val Asn His Lys Ala Lys Ala Phe
        35                  40                  45

Ala Ser Leu Gly Lys Lys Val Asn Val Thr Lys Thr Trp Glu Glu Gln
    50                  55                  60

Thr Glu Thr Leu Arg Asp Val Val Asp Phe Leu Lys Gly Gln Leu Leu
65                  70                  75                  80

Asp Ile Gln Val Glu Asn Leu Ile Pro Ile Glu Pro Leu Thr Leu Gln
                85                  90                  95

Ala Arg Met Ser Cys Glu His Glu Ala His Gly His Gly Arg Gly Ser
            100                 105                 110

Trp Gln Phe Leu Phe Asn Gly Gln Lys Phe Leu Leu Phe Asp Ser Asn
        115                 120                 125

Asn Arg Lys Trp Thr Ala Leu His Pro Gly Ala Lys Lys Met Thr Glu
    130                 135                 140

Lys Trp Glu Lys Asn Arg Asp Val Thr Met Phe Phe Gln Lys Ile Ser
145                 150                 155                 160

Leu Gly Asp Cys Lys Met Trp Leu Glu Glu Phe Leu Met Tyr Trp Glu
                165                 170                 175

Gln Met Leu Asp Pro Thr Lys Pro Pro Ser Leu Ala Pro Gly
            180                 185                 190


<210> SEQ ID NO 89
<211> LENGTH: 191
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 89

Gly Arg Ala Asp Pro His Ser Leu Cys Tyr Asp Ile Thr Val Ile Pro
1               5                   10                  15

Lys Phe Arg Pro Gly Pro Arg Trp Cys Ala Val Gln Gly Gln Val Asp
            20                  25                  30

Glu Lys Thr Phe Leu His Tyr Asp Cys Gly Asn Lys Thr Val Thr Pro
        35                  40                  45

Val Ser Pro Leu Gly Lys Lys Leu Asn Val Thr Thr Ala Trp Lys Ala
    50                  55                  60

Gln Asn Pro Val Leu Arg Glu Val Val Asp Ile Leu Thr Glu Gln Leu
65                  70                  75                  80

Arg Asp Ile Gln Leu Glu Asn Tyr Thr Pro Lys Glu Pro Leu Thr Leu
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```
                85                  90                  95

Gln Ala Arg Met Ser Cys Glu Gln Lys Ala Glu Gly His Ser Ser Gly
            100                 105                 110

Ser Trp Gln Phe Ser Phe Asp Gly Gln Ile Phe Leu Leu Phe Asp Ser
        115                 120                 125

Glu Lys Arg Met Trp Thr Thr Val His Pro Gly Ala Arg Lys Met Lys
    130                 135                 140

Glu Lys Trp Glu Asn Asp Lys Val Val Ala Met Ser Phe His Tyr Phe
145                 150                 155                 160

Ser Met Gly Asp Cys Ile Gly Trp Leu Glu Asp Phe Leu Met Gly Met
                165                 170                 175

Asp Ser Thr Leu Glu Pro Ser Ala Gly Ala Pro Leu Ala Met Ser
            180                 185                 190


<210> SEQ ID NO 90
<211> LENGTH: 188
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 90

Asp Ala His Ser Leu Trp Tyr Asn Phe Thr Ile Ile His Leu Pro Arg
1               5                   10                  15

His Gly Gln Gln Trp Cys Glu Val Gln Ser Gln Val Asp Gln Lys Asn
            20                  25                  30

Phe Leu Ser Tyr Asp Cys Gly Ser Asp Lys Val Leu Ser Met Gly His
        35                  40                  45

Leu Glu Glu Gln Leu Tyr Ala Thr Asp Ala Trp Gly Lys Gln Leu Glu
    50                  55                  60

Met Leu Arg Glu Val Gly Gln Arg Leu Arg Leu Glu Leu Ala Asp Thr
65                  70                  75                  80

Glu Leu Glu Asp Phe Thr Pro Ser Gly Pro Leu Thr Leu Gln Val Arg
                85                  90                  95

Met Ser Cys Glu Cys Glu Ala Asp Gly Tyr Ile Arg Gly Ser Trp Gln
            100                 105                 110

Phe Ser Phe Asp Gly Arg Lys Phe Leu Leu Phe Asp Ser Asn Asn Arg
        115                 120                 125

Lys Trp Thr Val Val His Ala Gly Ala Arg Arg Met Lys Glu Lys Trp
    130                 135                 140

Glu Lys Asp Ser Gly Leu Thr Thr Phe Phe Lys Met Val Ser Met Arg
145                 150                 155                 160

Asp Cys Lys Ser Trp Leu Arg Asp Phe Leu Met His Arg Lys Lys Arg
                165                 170                 175

Leu Glu Pro Thr Ala Pro Pro Thr Met Ala Pro Gly
            180                 185


<210> SEQ ID NO 91
<211> LENGTH: 233
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 91

His Ser Leu Cys Phe Asn Phe Thr Ile Lys Ser Leu Ser Arg Pro Gly
1               5                   10                  15

Gln Pro Trp Cys Glu Ala Gln Val Phe Leu Asn Lys Asn Leu Phe Leu
            20                  25                  30

Gln Tyr Asn Ser Asp Asn Asn Met Val Lys Pro Leu Gly Leu Leu Gly
```

```
        35                  40                  45

Lys Lys Val Tyr Ala Thr Ser Thr Trp Gly Glu Leu Thr Gln Thr Leu
    50                  55                  60

Gly Glu Val Gly Arg Asp Leu Arg Met Leu Leu Cys Asp Ile Lys Pro
65                  70                  75                  80

Gln Ile Lys Thr Ser Asp Pro Ser Thr Leu Gln Val Glu Met Phe Cys
                85                  90                  95

Gln Arg Glu Ala Glu Arg Cys Thr Gly Ala Ser Trp Gln Phe Ala Thr
            100                 105                 110

Asn Gly Glu Lys Ser Leu Leu Phe Asp Ala Met Asn Met Thr Trp Thr
        115                 120                 125

Val Ile Asn His Glu Ala Ser Lys Ile Lys Glu Thr Trp Lys Lys Asp
    130                 135                 140

Arg Gly Leu Glu Lys Tyr Phe Arg Lys Leu Ser Lys Gly Asp Cys Asp
145                 150                 155                 160

His Trp Leu Arg Glu Phe Leu Gly His Trp Glu Ala Met Pro Glu Pro
                165                 170                 175

Thr Val Ser Pro Val Asn Ala Ser Asp Ile His Trp Ser Ser Ser Ser
            180                 185                 190

Leu Pro Asp Arg Trp Ile Ile Leu Gly Ala Phe Ile Leu Leu Val Leu
        195                 200                 205

Met Gly Ile Val Leu Ile Cys Val Trp Trp Gln Asn Gly Glu Trp Gln
    210                 215                 220

Ala Gly Leu Trp Pro Leu Arg Thr Ser
225                 230


<210> SEQ ID NO 92
<211> LENGTH: 193
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 92

Gly Leu Ala Asp Pro His Ser Leu Cys Tyr Asp Ile Thr Val Ile Pro
1               5                   10                  15

Lys Phe Arg Pro Gly Pro Arg Trp Cys Ala Val Gln Gly Gln Val Asp
            20                  25                  30

Glu Lys Thr Phe Leu His Tyr Asp Cys Gly Ser Lys Thr Val Thr Pro
        35                  40                  45

Val Ser Pro Leu Gly Lys Lys Leu Asn Val Thr Thr Ala Trp Lys Ala
    50                  55                  60

Gln Asn Pro Val Leu Arg Glu Val Val Asp Ile Leu Thr Glu Gln Leu
65                  70                  75                  80

Leu Asp Ile Gln Leu Glu Asn Tyr Ile Pro Lys Glu Pro Leu Thr Leu
                85                  90                  95

Gln Ala Arg Met Ser Cys Glu Gln Lys Ala Glu Gly His Gly Ser Gly
            100                 105                 110

Ser Trp Gln Leu Ser Phe Asp Gly Gln Ile Phe Leu Leu Phe Asp Ser
        115                 120                 125

Glu Asn Arg Met Trp Thr Thr Val His Pro Gly Ala Arg Lys Met Lys
    130                 135                 140

Glu Lys Trp Glu Asn Asp Lys Asp Met Thr Met Ser Phe His Tyr Ile
145                 150                 155                 160

Ser Met Gly Asp Cys Thr Gly Trp Leu Glu Asp Phe Leu Met Gly Met
                165                 170                 175
```

```
Asp Ser Thr Leu Glu Pro Ser Ala Gly Ala Pro Pro Thr Met Ser Ser
            180                 185                 190

Gly


<210> SEQ ID NO 93
<211> LENGTH: 193
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 93

Arg Arg Asp Asp Pro His Ser Leu Cys Tyr Asp Ile Thr Val Ile Pro
1               5                   10                  15

Lys Phe Arg Pro Gly Pro Arg Trp Cys Ala Val Gln Gly Gln Val Asp
            20                  25                  30

Glu Lys Thr Phe Leu His Tyr Asp Cys Gly Asn Lys Thr Val Thr Pro
        35                  40                  45

Val Ser Pro Leu Gly Lys Lys Leu Asn Val Thr Met Ala Trp Lys Ala
    50                  55                  60

Gln Asn Pro Val Leu Arg Glu Val Val Asp Ile Leu Thr Glu Gln Leu
65                  70                  75                  80

Leu Asp Ile Gln Leu Glu Asn Tyr Thr Pro Lys Glu Pro Leu Thr Leu
                85                  90                  95

Gln Ala Arg Met Ser Cys Glu Gln Lys Ala Glu Gly His Ser Ser Gly
            100                 105                 110

Ser Trp Gln Phe Ser Ile Asp Gly Gln Thr Phe Leu Leu Phe Asp Ser
        115                 120                 125

Glu Lys Arg Met Trp Thr Thr Val His Pro Gly Ala Arg Lys Met Lys
    130                 135                 140

Glu Lys Trp Glu Asn Asp Lys Asp Val Ala Met Ser Phe His Tyr Ile
145                 150                 155                 160

Ser Met Gly Asp Cys Ile Gly Trp Leu Glu Asp Phe Leu Met Gly Met
                165                 170                 175

Asp Ser Thr Leu Glu Pro Ser Ala Gly Ala Pro Leu Ala Met Ser Ser
            180                 185                 190

Gly


<210> SEQ ID NO 94
<211> LENGTH: 65
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 94

Ile Thr Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val
1               5                   10                  15

Lys Ser Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly
            20                  25                  30

Phe Lys Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn
        35                  40                  45

Lys Ala Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile
    50                  55                  60

Arg
65


<210> SEQ ID NO 95
```

<211> LENGTH: 195
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 95

attacatgcc cccctcccat gagcgtggag cacgccgaca tctgggtgaa gagctatagc   60

ctctacagcc gggagaggta tatctgtaac agcggcttca agaggaaggc cggcaccagc   120

agcctcaccg agtgcgtgct gaataaggct accaacgtgg ctcactggac aacaccctct   180

ttaaagtgca tccgg   195

<210> SEQ ID NO 96
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 96

Asn Trp Val Asn Val Ile Ser Asp Leu Lys Lys Ile Glu Asp Leu Ile
1   5   10   15

Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
            20   25   30

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
        35   40   45

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu
    50   55   60

Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
65   70   75   80

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
                85   90   95

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
            100   105   110

Thr Ser

<210> SEQ ID NO 97
<211> LENGTH: 342
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 97

aactgggtga acgtcatcag cgatttaaag aagatcgaag atttaattca gtccatgcat   60

atcgacgcca ctttatacac agaatccgac gtgcacccct cttgtaaggt gaccgccatg   120

aaatgttttt tactggagct gcaagttatc tctttagaga gcggagacgc tagcatccac   180

gacaccgtgg agaatttaat cattttagcc aataactctt tatccagcaa cggcaacgtg   240

acagagtccg gctgcaagga gtgcgaagag ctggaggaga agaacatcaa ggagtttctg   300

caatcctttg tgcacattgt ccagatgttc atcaatacct cc   342

<210> SEQ ID NO 98
<211> LENGTH: 220
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 98

```
Pro Cys Pro Ala Pro Glu Leu Leu Gly Gly Pro Ser Val Phe Leu Phe
1               5                   10                  15

Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
            20                  25                  30

Thr Cys Val Val Val Asp Val Ser His Glu Asp Pro Glu Val Lys Phe
        35                  40                  45

Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
    50                  55                  60

Arg Glu Glu Gln Tyr Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr
65                  70                  75                  80

Val Leu His Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val
                85                  90                  95

Ser Asn Lys Ala Leu Pro Ala Pro Ile Glu Lys Thr Ile Ser Lys Ala
            100                 105                 110

Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Arg
        115                 120                 125

Asp Glu Leu Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly
    130                 135                 140

Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro
145                 150                 155                 160

Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser
                165                 170                 175

Phe Phe Leu Tyr Ser Lys Leu Thr Val Asp Lys Ser Arg Trp Gln Gln
            180                 185                 190

Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His
        195                 200                 205

Tyr Thr Gln Lys Ser Leu Ser Leu Ser Pro Gly Lys
    210                 215                 220
```

<210> SEQ ID NO 99
<211> LENGTH: 114
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 99

```
Asn Trp Val Asn Val Ile Ser Asn Leu Lys Lys Ile Glu Asp Leu Ile
1               5                   10                  15

Gln Ser Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His
            20                  25                  30

Pro Ser Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln
        35                  40                  45

Val Ile Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu
    50                  55                  60

Asn Leu Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val
65                  70                  75                  80

Thr Glu Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile
                85                  90                  95

Lys Glu Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn
            100                 105                 110

Thr Ser
```

<210> SEQ ID NO 100
<211> LENGTH: 342
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 100

aactgggtga acgtcatcag caatttaaag aagatcgaag atttaattca gtccatgcat 60

atcgacgcca ctttatacac agaatccgac gtgcacccct cttgtaaggt gaccgccatg 120

aaatgttttt tactggagct gcaagttatc tctttagaga gcggagacgc tagcatccac 180

gacaccgtgg agaatttaat cattttagcc aataactctt tatccagcaa cggcaacgtg 240

acagagtccg gctgcaagga gtgcgaagag ctggaggaga agaacatcaa ggagtttctg 300

caatcctttg tgcacattgt ccagatgttc atcaatacct cc 342

<210> SEQ ID NO 101
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
oligonucleotide

<400> SEQUENCE: 101

atgaaatggg tcaccttcat ctctttactg tttttattta gcagcgccta cagc 54

<210> SEQ ID NO 102
<211> LENGTH: 574
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 102

Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Pro Gly
1 5 10 15

Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser Tyr Met
20 25 30

Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp Ile Tyr
35 40 45

Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala His Phe Arg Gly Ser
50 55 60

Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu Ala Glu
65 70 75 80

Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro Phe Thr
85 90 95

Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg Gly Gly Gly Gly Ser
100 105 110

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu Gln Gln
115 120 125

Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met Ser Cys
130 135 140

Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr Thr Met His Trp Val Lys
145 150 155 160

Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Ile Asn Pro Ser
165 170 175

```
Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe Lys Asp Lys Ala Thr Leu
            180                 185                 190

Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr Met Gln Leu Ser Ser Leu
        195                 200                 205

Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys Ala Arg Tyr Tyr Asp Asp
    210                 215                 220

His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr Val Ser
225                 230                 235                 240

Ser Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr Trp Lys
                245                 250                 255

Ser Thr Asn Phe Lys Thr Ile Leu Glu Trp Glu Pro Lys Pro Val Asn
            260                 265                 270

Gln Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp Lys Ser
        275                 280                 285

Lys Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp Glu Ile
    290                 295                 300

Val Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser Tyr Pro
305                 310                 315                 320

Ala Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu Tyr Glu
                325                 330                 335

Asn Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly Gln Pro
            340                 345                 350

Thr Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val Thr Val
        355                 360                 365

Glu Asp Glu Arg Thr Leu Val Arg Arg Asn Asn Thr Phe Leu Ser Leu
    370                 375                 380

Arg Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr Trp Lys
385                 390                 395                 400

Ser Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn Glu Phe
                405                 410                 415

Leu Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val Gln Ala
            420                 425                 430

Val Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser Pro Val
        435                 440                 445

Glu Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu Asn Trp Val Asn
    450                 455                 460

Val Ile Ser Asn Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser Met His
465                 470                 475                 480

Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His Pro Ser Cys Lys
                485                 490                 495

Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln Val Ile Ser Leu
            500                 505                 510

Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu Ile Ile
        515                 520                 525

Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu Ser Gly
    530                 535                 540

Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu Phe Leu
545                 550                 555                 560

Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
                565                 570
```

<210> SEQ ID NO 103
<211> LENGTH: 1722

<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 103

```
cagatcgtgc tgacccaaag ccccgccatc atgagcgcta gccccggtga gaaggtgacc     60
atgacatgct ccgcttccag ctccgtgtcc tacatgaact ggtatcagca gaaaagcgga    120
accagcccca aaaggtggat ctacgacacc agcaagctgg cctccggagt gcccgctcat    180
ttccggggct ctggatccgg caccagctac tctttaacca tttccggcat ggaagctgaa    240
gacgctgcca cctactattg ccagcaatgg agcagcaacc ccttcacatt cggatctggc    300
accaagctcg aaatcaatcg tggaggaggt ggcagcggcg gcggtggatc cggcggagga    360
ggaagccaag ttcaactcca gcagagcggc gctgaactgg cccggcccgg cgcctccgtc    420
aagatgagct gcaaggcttc cggctataca tttactcgtt acacaatgca ttgggtcaag    480
cagaggcccg gtcaaggttt agagtggatc ggatatatca acccttcccg gggctacacc    540
aactataacc aaaagttcaa ggataaagcc actttaacca ctgacaagag ctcctccacc    600
gcctacatgc agctgtcctc tttaaccagc gaggactccg ctgtttacta ctgcgctagg    660
tattacgacg accactactg tttagactat tggggacaag gtaccacttt aaccgtcagc    720
agctccggca ccaccaatac cgtggccgct tataacctca catggaagag caccaacttc    780
aagacaattc tggaatggga acccaagccc gtcaatcaag tttacaccgt gcagatctcc    840
accaaatccg gagactggaa gagcaagtgc ttctacacaa cagacaccga gtgtgattta    900
accgacgaaa tcgtcaagga cgtcaagcaa acctatctgg ctcgggtctt ttcctacccc    960
gctggcaatg tcgagtccac cggctccgct ggcgagcctc tctacgagaa ttcccccgaa   1020
ttcacccctt atttagagac caatttaggc cagcctacca tccagagctt cgagcaagtt   1080
ggcaccaagg tgaacgtcac cgtcgaggat gaaaggactt tagtgcggcg gaataacaca   1140
tttttatccc tccgggatgt gttcggcaaa gacctcatct acacactgta ctattggaag   1200
tccagctcct ccggcaaaaa gaccgctaag accaacacca acgagttttt aattgacgtg   1260
gacaaaggcg agaactactg cttcagcgtg caagccgtga tcccttctcg taccgtcaac   1320
cggaagagca cagattcccc cgttgagtgc atgggccaag aaaagggcga gttccgggag   1380
aactgggtga acgtcatcag caatttaaag aagatcgaag atttaattca gtccatgcat   1440
atcgacgcca ctttatacac agaatccgac gtgcacccct cttgtaaggt gaccgccatg   1500
aaatgttttt tactggagct gcaagttatc tctttagaga gcggagacgc tagcatccac   1560
gacaccgtgg agaatttaat cattttagcc aataactctt tatccagcaa cggcaacgtg   1620
acagagtccg gctgcaagga gtgcgaagag ctggaggaga agaacatcaa ggagtttctg   1680
caatcctttg tgcacattgt ccagatgttc atcaatacct cc                       1722
```

<210> SEQ ID NO 104
<211> LENGTH: 592
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 104

```
Met Lys Trp Val Thr Phe Ile Ser Leu Leu Phe Leu Phe Ser Ser Ala
1               5                   10                  15
```

```
Tyr Ser Gln Ile Val Leu Thr Gln Ser Pro Ala Ile Met Ser Ala Ser
            20                  25                  30

Pro Gly Glu Lys Val Thr Met Thr Cys Ser Ala Ser Ser Ser Val Ser
        35                  40                  45

Tyr Met Asn Trp Tyr Gln Gln Lys Ser Gly Thr Ser Pro Lys Arg Trp
    50                  55                  60

Ile Tyr Asp Thr Ser Lys Leu Ala Ser Gly Val Pro Ala His Phe Arg
65                  70                  75                  80

Gly Ser Gly Ser Gly Thr Ser Tyr Ser Leu Thr Ile Ser Gly Met Glu
                85                  90                  95

Ala Glu Asp Ala Ala Thr Tyr Tyr Cys Gln Gln Trp Ser Ser Asn Pro
            100                 105                 110

Phe Thr Phe Gly Ser Gly Thr Lys Leu Glu Ile Asn Arg Gly Gly Gly
        115                 120                 125

Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gln Val Gln Leu
    130                 135                 140

Gln Gln Ser Gly Ala Glu Leu Ala Arg Pro Gly Ala Ser Val Lys Met
145                 150                 155                 160

Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Arg Tyr Thr Met His Trp
                165                 170                 175

Val Lys Gln Arg Pro Gly Gln Gly Leu Glu Trp Ile Gly Tyr Ile Asn
            180                 185                 190

Pro Ser Arg Gly Tyr Thr Asn Tyr Asn Gln Lys Phe Lys Asp Lys Ala
        195                 200                 205

Thr Leu Thr Thr Asp Lys Ser Ser Ser Thr Ala Tyr Met Gln Leu Ser
    210                 215                 220

Ser Leu Thr Ser Glu Asp Ser Ala Val Tyr Tyr Cys Ala Arg Tyr Tyr
225                 230                 235                 240

Asp Asp His Tyr Cys Leu Asp Tyr Trp Gly Gln Gly Thr Thr Leu Thr
                245                 250                 255

Val Ser Ser Ser Gly Thr Thr Asn Thr Val Ala Ala Tyr Asn Leu Thr
            260                 265                 270

Trp Lys Ser Thr Asn Phe Lys Thr Ile Leu Glu Trp Glu Pro Lys Pro
        275                 280                 285

Val Asn Gln Val Tyr Thr Val Gln Ile Ser Thr Lys Ser Gly Asp Trp
    290                 295                 300

Lys Ser Lys Cys Phe Tyr Thr Thr Asp Thr Glu Cys Asp Leu Thr Asp
305                 310                 315                 320

Glu Ile Val Lys Asp Val Lys Gln Thr Tyr Leu Ala Arg Val Phe Ser
                325                 330                 335

Tyr Pro Ala Gly Asn Val Glu Ser Thr Gly Ser Ala Gly Glu Pro Leu
            340                 345                 350

Tyr Glu Asn Ser Pro Glu Phe Thr Pro Tyr Leu Glu Thr Asn Leu Gly
        355                 360                 365

Gln Pro Thr Ile Gln Ser Phe Glu Gln Val Gly Thr Lys Val Asn Val
    370                 375                 380

Thr Val Glu Asp Glu Arg Thr Leu Val Arg Arg Asn Asn Thr Phe Leu
385                 390                 395                 400

Ser Leu Arg Asp Val Phe Gly Lys Asp Leu Ile Tyr Thr Leu Tyr Tyr
                405                 410                 415

Trp Lys Ser Ser Ser Ser Gly Lys Lys Thr Ala Lys Thr Asn Thr Asn
            420                 425                 430
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Glu Phe Leu Ile Asp Val Asp Lys Gly Glu Asn Tyr Cys Phe Ser Val
        435                 440                 445

Gln Ala Val Ile Pro Ser Arg Thr Val Asn Arg Lys Ser Thr Asp Ser
    450                 455                 460

Pro Val Glu Cys Met Gly Gln Glu Lys Gly Glu Phe Arg Glu Asn Trp
465                 470                 475                 480

Val Asn Val Ile Ser Asn Leu Lys Lys Ile Glu Asp Leu Ile Gln Ser
                485                 490                 495

Met His Ile Asp Ala Thr Leu Tyr Thr Glu Ser Asp Val His Pro Ser
            500                 505                 510

Cys Lys Val Thr Ala Met Lys Cys Phe Leu Leu Glu Leu Gln Val Ile
        515                 520                 525

Ser Leu Glu Ser Gly Asp Ala Ser Ile His Asp Thr Val Glu Asn Leu
    530                 535                 540

Ile Ile Leu Ala Asn Asn Ser Leu Ser Ser Asn Gly Asn Val Thr Glu
545                 550                 555                 560

Ser Gly Cys Lys Glu Cys Glu Glu Leu Glu Glu Lys Asn Ile Lys Glu
                565                 570                 575

Phe Leu Gln Ser Phe Val His Ile Val Gln Met Phe Ile Asn Thr Ser
            580                 585                 590


<210> SEQ ID NO 105
<211> LENGTH: 1776
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 105

atgaagtggg tgaccttcat cagcttatta tttttattca gctccgccta ttcccagatc     60

gtgctgaccc aaagccccgc catcatgagc gctagccccg gtgagaaggt gaccatgaca    120

tgctccgctt ccagctccgt gtcctacatg aactggtatc agcagaaaag cggaaccagc    180

cccaaaaggt ggatctacga caccagcaag ctggcctccg gagtgcccgc tcatttccgg    240

ggctctggat ccggcaccag ctactcttta accatttccg gcatggaagc tgaagacgct    300

gccacctact attgccagca atggagcagc aaccccttca cattcggatc tggcaccaag    360

ctcgaaatca atcgtggagg aggtggcagc ggcggcggtg gatccggcgg aggaggaagc    420

caagttcaac tccagcagag cggcgctgaa ctggcccggc ccggcgcctc cgtcaagatg    480

agctgcaagg cttccggcta tacatttact cgttacacaa tgcattgggt caagcagagg    540

cccggtcaag gtttagagtg gatcggatat atcaaccctt cccggggcta caccaactat    600

aaccaaaagt tcaaggataa agccacttta accactgaca agagctcctc caccgcctac    660

atgcagctgt cctctttaac cagcgaggac tccgctgttt actactgcgc taggtattac    720

gacgaccact actgtttaga ctattgggga caaggtacca ctttaaccgt cagcagctcc    780

ggcaccacca ataccgtggc cgcttataac ctcacatgga agagcaccaa cttcaagaca    840

attctggaat gggaacccaa gcccgtcaat caagtttaca ccgtgcagat ctccaccaaa    900

tccggagact ggaagagcaa gtgcttctac acaacagaca ccgagtgtga tttaaccgac    960

gaaatcgtca aggacgtcaa gcaaacctat ctggctcggg tcttttccta ccccgctggc   1020

aatgtcgagt ccaccggctc cgctggcgag cctctctacg agaattcccc cgaattcacc   1080

ccttatttag agaccaattt aggccagcct accatccaga gcttcgagca agttggcacc   1140
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aaggtgaacg tcaccgtcga ggatgaaagg actttagtgc ggcggaataa cacattttta   1200

tccctccggg atgtgttcgg caaagacctc atctacacac tgtactattg gaagtccagc   1260

tcctccggca aaaagaccgc taagaccaac accaacgagt ttttaattga cgtggacaaa   1320

ggcgagaact actgcttcag cgtgcaagcc gtgatccctt ctcgtaccgt caaccggaag   1380

agcacagatt cccccgttga gtgcatgggc caagaaaagg gcgagttccg ggagaactgg   1440

gtgaacgtca tcagcaattt aaagaagatc gaagatttaa ttcagtccat gcatatcgac   1500

gccactttat acacagaatc cgacgtgcac ccctcttgta aggtgaccgc catgaaatgt   1560

tttttactgg agctgcaagt tatctcttta gagagcggag acgctagcat ccacgacacc   1620

gtggagaatt taatcatttt agccaataac tctttatcca gcaacggcaa cgtgacagag   1680

tccggctgca aggagtgcga agagctggag gagaagaaca tcaaggagtt tctgcaatcc   1740

tttgtgcaca ttgtccagat gttcatcaat acctcc                               1776


<210> SEQ ID NO 106
<211> LENGTH: 301
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 106

Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly Ala Ser
1               5                   10                  15

Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser Tyr Val
            20                  25                  30

Ile Gln Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp Ile Gly
        35                  40                  45

Ser Ile Asn Pro Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys Phe Lys
    50                  55                  60

Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala Tyr Met
65                  70                  75                  80

Glu Phe Ser Ser Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr Cys Ala
                85                  90                  95

Arg Trp Gly Asp Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu Thr Val
            100                 105                 110

Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly
        115                 120                 125

Ser Asp Ile Glu Met Thr Gln Ser Pro Ala Ile Met Ser Ala Ser Leu
    130                 135                 140

Gly Glu Arg Val Thr Met Thr Cys Thr Ala Ser Ser Ser Val Ser Ser
145                 150                 155                 160

Ser Tyr Phe His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro Lys Leu
                165                 170                 175

Cys Ile Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Pro Arg Phe
            180                 185                 190

Ser Gly Ser Gly Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser Met Glu
        195                 200                 205

Ala Glu Asp Ala Ala Thr Tyr Phe Cys His Gln Tyr His Arg Ser Pro
    210                 215                 220

Thr Phe Gly Gly Gly Thr Lys Leu Glu Thr Lys Arg Ile Thr Cys Pro
225                 230                 235                 240

Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser Tyr Ser
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                245                 250                 255

Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys Arg Lys
            260                 265                 270

Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala Thr Asn
        275                 280                 285

Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg
    290                 295                 300


<210> SEQ ID NO 107
<211> LENGTH: 903
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 107

gtgcagctgc agcagtccgg acccgaactg gtcaagcccg gtgcctccgt gaaaatgtct     60

tgtaaggctt ctggctacac ctttacctcc tacgtcatcc aatgggtgaa gcagaagccc    120

ggtcaaggtc tcgagtggat cggcagcatc aatccctaca acgattacac caagtataac    180

gaaaagttta agggcaaggc cactctgaca agcgacaaga gctccattac cgcctacatg    240

gagttttcct ctttaacttc tgaggactcc gctttatact attgcgctcg ttggggcgat    300

ggcaattatt ggggccgggg aactacttta acagtgagct ccggcggcgg cggaagcgga    360

ggtggaggat ctggcggtgg aggcagcgac atcgagatga cacagtcccc cgctatcatg    420

agcgcctctt taggagaacg tgtgaccatg acttgtacag cttcctccag cgtgagcagc    480

tcctatttcc actggtacca gcagaaaccc ggctcctccc ctaaactgtg tatctactcc    540

acaagcaatt tagctagcgg cgtgcctcct cgttttagcg gctccggcag cacctcttac    600

tctttaacca ttagctctat ggaggccgaa gatgccgcca catacttttg ccatcagtac    660

caccggtccc ctacctttgg cggaggcaca aagctggaga ccaagcggat tacatgcccc    720

cctcccatga gcgtggagca cgccgacatc tgggtgaaga gctatagcct ctacagccgg    780

gagaggtata tctgtaacag cggcttcaag aggaaggccg gcaccagcag cctcaccgag    840

tgcgtgctga ataaggctac caacgtggct cactggacaa caccctcttt aaagtgcatc    900

cgg                                                                  903


<210> SEQ ID NO 108
<211> LENGTH: 319
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 108

Met Lys Trp Val Thr Phe Ile Ser Leu Leu Phe Leu Phe Ser Ser Ala
1               5                   10                  15

Tyr Ser Val Gln Leu Gln Gln Ser Gly Pro Glu Leu Val Lys Pro Gly
            20                  25                  30

Ala Ser Val Lys Met Ser Cys Lys Ala Ser Gly Tyr Thr Phe Thr Ser
        35                  40                  45

Tyr Val Ile Gln Trp Val Lys Gln Lys Pro Gly Gln Gly Leu Glu Trp
    50                  55                  60

Ile Gly Ser Ile Asn Pro Tyr Asn Asp Tyr Thr Lys Tyr Asn Glu Lys
65                  70                  75                  80
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Phe Lys Gly Lys Ala Thr Leu Thr Ser Asp Lys Ser Ser Ile Thr Ala
                85                  90                  95

Tyr Met Glu Phe Ser Ser Leu Thr Ser Glu Asp Ser Ala Leu Tyr Tyr
            100                 105                 110

Cys Ala Arg Trp Gly Asp Gly Asn Tyr Trp Gly Arg Gly Thr Thr Leu
        115                 120                 125

Thr Val Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly
    130                 135                 140

Gly Gly Ser Asp Ile Glu Met Thr Gln Ser Pro Ala Ile Met Ser Ala
145                 150                 155                 160

Ser Leu Gly Glu Arg Val Thr Met Thr Cys Thr Ala Ser Ser Ser Val
                165                 170                 175

Ser Ser Ser Tyr Phe His Trp Tyr Gln Gln Lys Pro Gly Ser Ser Pro
            180                 185                 190

Lys Leu Cys Ile Tyr Ser Thr Ser Asn Leu Ala Ser Gly Val Pro Pro
        195                 200                 205

Arg Phe Ser Gly Ser Gly Ser Thr Ser Tyr Ser Leu Thr Ile Ser Ser
    210                 215                 220

Met Glu Ala Glu Asp Ala Ala Thr Tyr Phe Cys His Gln Tyr His Arg
225                 230                 235                 240

Ser Pro Thr Phe Gly Gly Gly Thr Lys Leu Glu Thr Lys Arg Ile Thr
                245                 250                 255

Cys Pro Pro Pro Met Ser Val Glu His Ala Asp Ile Trp Val Lys Ser
            260                 265                 270

Tyr Ser Leu Tyr Ser Arg Glu Arg Tyr Ile Cys Asn Ser Gly Phe Lys
        275                 280                 285

Arg Lys Ala Gly Thr Ser Ser Leu Thr Glu Cys Val Leu Asn Lys Ala
    290                 295                 300

Thr Asn Val Ala His Trp Thr Thr Pro Ser Leu Lys Cys Ile Arg
305                 310                 315
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<210> SEQ ID NO 109
<211> LENGTH: 957
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 109

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atgaaatggg tcaccttcat ctctttactg tttttattta gcagcgccta cagcgtgcag     60

ctgcagcagt ccggacccga actggtcaag cccggtgcct ccgtgaaaat gtcttgtaag    120

gcttctggct acacctttac ctcctacgtc atccaatggg tgaagcagaa gcccggtcaa    180

ggtctcgagt ggatcggcag catcaatccc tacaacgatt acaccaagta taacgaaaag    240

tttaagggca aggccactct gacaagcgac aagagctcca ttaccgccta catggagttt    300

tcctctttaa cttctgagga ctccgcttta tactattgcg ctcgttgggg cgatggcaat    360

tattggggcc ggggaactac tttaacagtg agctccggcg gcggcggaag cggaggtgga    420

ggatctggcg gtggaggcag cgacatcgag atgacacagt cccccgctat catgagcgcc    480

tctttaggag aacgtgtgac catgacttgt acagcttcct ccagcgtgag cagctcctat    540

ttccactggt accagcagaa acccggctcc tcccctaaac tgtgtatcta ctccacaagc    600

aatttagcta gcggcgtgcc tcctcgtttt agcggctccg gcagcacctc ttactcttta    660
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accattagct ctatggaggc cgaagatgcc gccacatact tttgccatca gtaccaccgg    720

tcccctacct ttggcggagg cacaaagctg gagaccaagc ggattacatg ccccctccc     780

atgagcgtgg agcacgccga catctgggtg aagagctata gcctctacag ccgggagagg    840

tatatctgta acagcggctt caagaggaag gccggcacca gcagcctcac cgagtgcgtg    900

ctgaataagg ctaccaacgt ggctcactgg acaacaccct ctttaaagtg catccgg       957


<210> SEQ ID NO 110
<211> LENGTH: 5
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

<400> SEQUENCE: 110

Gly Gly Gly Gly Ser
1               5


<210> SEQ ID NO 111
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

<400> SEQUENCE: 111

Gly Gly Gly Ser
1


<210> SEQ ID NO 112
<211> LENGTH: 4
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Linker

<400> SEQUENCE: 112

Gly Gly Ser Gly
1
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What is claimed is:

1. A method of stimulating or increasing the proliferation of a T_{reg} cell, wherein the method comprises:

culturing a T_{reg} cell in a liquid culture medium over a period of time, wherein the period of time is 7 days to 56 days and at the beginning of the period of time, the liquid culture medium comprises:

a CD3/CD28-binding agent; and

a single-chain chimeric polypeptide comprising a first target-binding domain, a soluble tissue factor domain, and a second target-binding domain, wherein:

the first target-binding domain comprises a sequence that is at least 90% identical to SEQ ID NO: 1;

the soluble tissue factor domain comprises a sequence that is at least 90% identical to SEQ ID NO: 2;

the second target-binding domain comprises a sequence that is at least 90% identical to SEQ ID NO: 1; and

the single-chain chimeric polypeptide does not stimulate blood coagulation in a mammal; and

wherein the CD3/CD28-binding agent is:

(1) an additional single-chain chimeric polypeptide comprising a sequence that is at least 90% identical to SEQ ID NO: 7; or

(2) a multi-chain chimeric polypeptide comprising:

(a) a first chimeric polypeptide comprising:

(i) a first target-binding domain;

(ii) a soluble tissue factor domain comprising a sequence that is at least 90% identical to SEQ ID NO: 2; and

(iii) a first domain of a pair of affinity domains comprising a sequence that is at least 90% identical to SEQ ID NO: 96; and

(b) a second chimeric polypeptide comprising:

(i) a second domain of a pair of affinity domains comprising a sequence that is at least 90% identical to SEQ ID NO: 94; and

(ii) a second target-binding domain,

wherein:

the first chimeric polypeptide and the second chimeric polypeptide associate through the binding of the first domain and the second domain of the pair of affinity domains;

the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5 and the second target-binding domain of the second chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6, or the first target-binding domain of the first chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 6 and the second target-binding domain of the second chimeric polypeptide comprises a sequence that is at least 90% identical to SEQ ID NO: 5; and

the multi-chain chimeric polypeptide would not stimulate blood coagulation if administered to a mammal.

2. The method of claim **1**, wherein the liquid culture medium further includes an IgG1 antibody construct that comprises at least one antigen-binding domain that binds specifically to the soluble tissue factor domain of the single-chain chimeric polypeptide.

3. The method of claim **1**, wherein the CD3/CD28 binding agent is the multi-chain chimeric polypeptide.

4. The method of claim **1**, wherein the method further comprises, before the culturing step, a step of isolating the T_{reg} cell from a sample obtained from a subject.

5. The method of claim **1**, wherein the method further comprises, after the culturing step, a step of isolating the T_{reg} cell from the liquid culture medium.

6. The method of claim **1**, wherein the T_{reg} cells comprises a chimeric antigen receptor.

7. The method of claim **1**, wherein:

the first target-binding domain of the single-chain chimeric polypeptide comprises SEQ ID NO: 1;

the soluble tissue factor domain of the single-chain chimeric polypeptide comprises SEQ ID NO: 2; and

the second target-binding domain of the single-chain chimeric polypeptide comprises SEQ ID NO: 1.

8. The method of claim **1**, wherein the single-chain chimeric polypeptide comprises a sequence at least 90% identical to SEQ ID NO: 3.

9. The method of claim **1**, wherein the single-chain chimeric polypeptide comprises SEQ ID NO: 3.

10. The method of claim **1**, wherein the CD3/CD28-binding agent is the additional single-chain chimeric polypeptide.

11. The method of claim **3**, wherein:

the first target-binding domain of the first chimeric polypeptide comprises SEQ ID NO: 5;

the soluble tissue factor domain of the first chimeric polypeptide comprises SEQ ID NO: 2;

the first domain of the pair of affinity domains comprises SEQ ID NO: 96;

the second domain of the pair of affinity domains comprises SEQ ID NO: 94; and

the second target-binding domain of the second chimeric polypeptide comprises SEQ ID NO: 6.

12. The method of claim **3**, wherein:

the first chimeric polypeptide comprises a sequence at least 90% identical to SEQ ID NO: 102; and

the second chimeric polypeptide comprises a sequence at least 90% identical to SEQ ID NO: 106.

13. The method of claim **10**, wherein the additional single-chain chimeric polypeptide comprises SEQ ID NO: 7.

14. The method of claim **12**, wherein:

the first chimeric polypeptide comprises SEQ ID NO: 102; and

the second chimeric polypeptide comprises SEQ ID NO: 106.

* * * * *